

# Algebraic Genomics

The Combinatorics of Sequence Reconstruction  
and Representation

Ruqing Chen

Exact Eulerian reconstruction, double-stranded spectral fibres,  
sandpile groups, and the computational complexity of sequence space

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# Contents

|                                                                         |            |
|-------------------------------------------------------------------------|------------|
| <b>Preface</b>                                                          | <b>vii</b> |
| <b>I Classical foundations: the BEST theorem and the sandpile group</b> | <b>1</b>   |
| <b>1 Critical groups of de Bruijn graphs</b>                            | <b>3</b>   |
| 1.1 Introduction . . . . .                                              | 3          |
| 1.2 Setting . . . . .                                                   | 4          |
| 1.3 The local and global factors . . . . .                              | 5          |
| 1.4 Contraction, and a single repeat . . . . .                          | 5          |
| 1.5 Two repeats: the group sees interleaving . . . . .                  | 6          |
| 1.6 Branch statistics are insufficient . . . . .                        | 8          |
| 1.7 Computation . . . . .                                               | 8          |
| 1.8 A real genome . . . . .                                             | 8          |
| 1.9 What is open, and what is false . . . . .                           | 9          |
| 1.10 Verification . . . . .                                             | 9          |
| <b>2 The sandpile group as a coordinate system</b>                      | <b>11</b>  |
| 2.1 Introduction . . . . .                                              | 11         |
| 2.2 Setting . . . . .                                                   | 12         |
| 2.3 The torsor . . . . .                                                | 13         |
| 2.4 A chip is a rearrangement . . . . .                                 | 13         |
| 2.5 Sampling, and what the group adds . . . . .                         | 14         |
| 2.6 $\varphi_{X174}$ : one global choice . . . . .                      | 14         |
| 2.7 Verification . . . . .                                              | 15         |
| 2.8 Scope . . . . .                                                     | 16         |
| <b>3 The sandpile group splits along the repeats</b>                    | <b>17</b>  |
| 3.1 Introduction . . . . .                                              | 17         |
| 3.2 Setting . . . . .                                                   | 18         |
| 3.3 The lemma . . . . .                                                 | 18         |
| 3.4 The splitting theorem . . . . .                                     | 19         |
| 3.5 $\varphi_{X174}$ . . . . .                                          | 22         |
| 3.6 Verification . . . . .                                              | 22         |
| 3.7 Scope . . . . .                                                     | 23         |

|           |                                                                                |           |
|-----------|--------------------------------------------------------------------------------|-----------|
| <b>4</b>  | <b>Multi-copy repeats</b>                                                      | <b>25</b> |
| 4.1       | Introduction . . . . .                                                         | 25        |
| 4.2       | Setting . . . . .                                                              | 26        |
| 4.3       | Two families . . . . .                                                         | 26        |
| 4.4       | Beyond two families nothing pairwise survives . . . . .                        | 27        |
| 4.5       | Census . . . . .                                                               | 28        |
| 4.6       | $\phi$ X174 . . . . .                                                          | 28        |
| 4.7       | Verification . . . . .                                                         | 29        |
| 4.8       | Scope . . . . .                                                                | 29        |
| <br>      |                                                                                |           |
| <b>II</b> | <b>The double-stranded frontier and its complexity</b>                         | <b>31</b> |
| <br>      |                                                                                |           |
| <b>5</b>  | <b>The reverse-complement double cover</b>                                     | <b>33</b> |
| 5.1       | Introduction . . . . .                                                         | 33        |
| 5.2       | The double cover . . . . .                                                     | 34        |
| 5.3       | Multiplicities add across strands . . . . .                                    | 34        |
| 5.4       | The involution dualises, it does not split . . . . .                           | 35        |
| 5.5       | Computations . . . . .                                                         | 36        |
| 5.6       | Verification . . . . .                                                         | 36        |
| 5.7       | Scope . . . . .                                                                | 37        |
| <br>      |                                                                                |           |
| <b>6</b>  | <b>The quotient is a signed graph</b>                                          | <b>39</b> |
| 6.1       | What was asked, and what is true . . . . .                                     | 40        |
| 6.2       | The quotient, and when it exists . . . . .                                     | 40        |
| 6.3       | Balance is an inverted repeat . . . . .                                        | 42        |
| 6.4       | The covering theory that does exist . . . . .                                  | 42        |
| 6.5       | The deck involution is an anti-automorphism . . . . .                          | 43        |
| 6.6       | Signed bundle chains do not contract . . . . .                                 | 44        |
| 6.7       | <i>E. coli</i> K-12: what is left of the seven operons . . . . .               | 46        |
| 6.8       | Scope, provenance, and what is open . . . . .                                  | 47        |
| <br>      |                                                                                |           |
| <b>7</b>  | <b>Double-stranded assembly as a lattice sum</b>                               | <b>49</b> |
| 7.1       | What was asked, and what is true . . . . .                                     | 49        |
| 7.2       | The double cover, its contents, and the spectrum of a reconstruction . . . . . | 50        |
| 7.3       | The lattice sum . . . . .                                                      | 52        |
| 7.4       | The inversion flip . . . . .                                                   | 55        |
| 7.5       | Scope, provenance, and what is open . . . . .                                  | 56        |
| <br>      |                                                                                |           |
| <b>8</b>  | <b>Frontier elimination</b>                                                    | <b>59</b> |
| 8.1       | What is being counted . . . . .                                                | 59        |
| 8.2       | The constraint system and its frontier . . . . .                               | 60        |
| 8.3       | Connectivity is the matrix-tree determinant . . . . .                          | 61        |
| 8.4       | The genome's own coordinate is not the good order . . . . .                    | 62        |
| 8.5       | $\phi$ X174 at $k = 10$ . . . . .                                              | 63        |
| 8.6       | Verification . . . . .                                                         | 64        |
| 8.7       | Scope, provenance, and what is open . . . . .                                  | 64        |

|            |                                                                            |            |
|------------|----------------------------------------------------------------------------|------------|
| <b>9</b>   | <b>A fermionic partition function</b>                                      | <b>67</b>  |
| 9.1        | What is left to count . . . . .                                            | 67         |
| 9.2        | The determinant as a product over arcs . . . . .                           | 68         |
| 9.3        | The root, and why the box is halved . . . . .                              | 69         |
| 9.4        | What the sweep has to carry besides the fermions . . . . .                 | 69         |
| 9.5        | The sweep . . . . .                                                        | 70         |
| 9.6        | Why the frontier stops short of $k = 10$ . . . . .                         | 70         |
| 9.7        | Point by point . . . . .                                                   | 71         |
| 9.8        | $\phi$ X174 at $k = 10$ . . . . .                                          | 72         |
| <b>10</b>  | <b>Double-stranded reconstruction is #P-hard</b>                           | <b>75</b>  |
| 10.1       | The question . . . . .                                                     | 75         |
| 10.2       | Eulerian circuits and their orientations . . . . .                         | 76         |
| 10.3       | Genomes whose double cover is a given graph . . . . .                      | 76         |
| 10.4       | The count . . . . .                                                        | 77         |
| 10.5       | Two by-products . . . . .                                                  | 78         |
| 10.6       | What this does and does not exclude . . . . .                              | 79         |
| <b>III</b> | <b>Spectral fibres and rearrangements</b>                                  | <b>81</b>  |
| <b>11</b>  | <b>Inversions at inverted repeats</b>                                      | <b>83</b>  |
| 11.1       | The question . . . . .                                                     | 83         |
| 11.2       | Setting . . . . .                                                          | 84         |
| 11.3       | Inversions at inverted repeats . . . . .                                   | 84         |
| 11.4       | Which moves generate the fibre . . . . .                                   | 85         |
| 11.5       | The inversion lattice . . . . .                                            | 86         |
| 11.6       | $\phi$ X174 at $k = 10$ . . . . .                                          | 87         |
| 11.7       | Biological reading . . . . .                                               | 87         |
| 11.8       | On the proposed inversion-group programme . . . . .                        | 88         |
| <b>12</b>  | <b>What a local encoder cannot see</b>                                     | <b>89</b>  |
| 12.1       | Introduction . . . . .                                                     | 90         |
| 12.2       | Local encoders and spectral fibres . . . . .                               | 90         |
| 12.3       | The human proteome . . . . .                                               | 92         |
| 12.4       | Where the unresolved order lives . . . . .                                 | 93         |
| 12.5       | Explicit collisions . . . . .                                              | 94         |
| 12.6       | The chemical side: ECFP . . . . .                                          | 95         |
| 12.7       | DNA perturbagens and the double cover . . . . .                            | 96         |
| 12.8       | What this does and does not imply . . . . .                                | 97         |
| <b>13</b>  | <b>Polyplod phasing</b>                                                    | <b>101</b> |
| 13.1       | What was asked, and what is true . . . . .                                 | 101        |
| 13.2       | The pooled multigraph and the phasing lattice . . . . .                    | 102        |
| 13.3       | The homogeneous fibre, and linkage . . . . .                               | 104        |
| 13.4       | The unlabelled multiset formula and the $\mathfrak{S}_p$ -action . . . . . | 105        |
| 13.5       | Copy-number variation across repeats . . . . .                             | 106        |
| 13.6       | Verification and data . . . . .                                            | 107        |

|           |                                                                 |            |
|-----------|-----------------------------------------------------------------|------------|
| 13.7      | Scope, provenance, and what is open . . . . .                   | 108        |
| <b>14</b> | <b>The recombination group of a pan-genome</b>                  | <b>109</b> |
| 14.1      | What was asked, and what is true . . . . .                      | 109        |
| 14.2      | The pan-genome sheaf . . . . .                                  | 110        |
| 14.3      | The recombination group . . . . .                               | 111        |
| 14.4      | Verification and data . . . . .                                 | 112        |
| 14.5      | A real pan-genome . . . . .                                     | 113        |
| 14.6      | Scope, provenance, and what is open . . . . .                   | 115        |
| <b>IV</b> | <b>Algorithmic implementations on genomes</b>                   | <b>117</b> |
| <b>15</b> | <b>The bundle-chain decomposition of <i>E. coli</i></b>         | <b>119</b> |
| 15.1      | Introduction . . . . .                                          | 119        |
| 15.2      | Why a 4.6 Mbp genome is tractable . . . . .                     | 120        |
| 15.3      | Results . . . . .                                               | 120        |
| 15.4      | The ribosomal operons, and a correction . . . . .               | 121        |
| 15.5      | Verification . . . . .                                          | 122        |
| 15.6      | Scope . . . . .                                                 | 122        |
| <b>16</b> | <b>The BEST determinant along the snarl tree</b>                | <b>125</b> |
| 16.1      | The question . . . . .                                          | 125        |
| 16.2      | Schur complements of Laplacians . . . . .                       | 126        |
| 16.3      | Elimination along the snarl tree . . . . .                      | 126        |
| 16.4      | Computations . . . . .                                          | 127        |
| 16.5      | What this does not do . . . . .                                 | 128        |
| <b>17</b> | <b>The isoform decomposition polytope</b>                       | <b>129</b> |
| 17.1      | What was asked, and what is true . . . . .                      | 129        |
| 17.2      | The splicing graph, its flows, and the fibre polytope . . . . . | 130        |
| 17.3      | Vertices, minimum decompositions, and integrality . . . . .     | 131        |
| 17.4      | The BEST identity . . . . .                                     | 132        |
| 17.5      | Verification and data . . . . .                                 | 133        |
| 17.6      | Scope, provenance, and what is open . . . . .                   | 134        |
|           | <b>Epilogue: what is open</b>                                   | <b>135</b> |
|           | <b>Guide to chapters and verification scripts</b>               | <b>137</b> |
|           | <b>Build record</b>                                             | <b>139</b> |

# Preface

**What this book is about.** Sequencing does not read a genome; it reads its words. From short reads one learns, at best, the multiset of  $k$ -mers of the genome and of its reverse complement, and everything an assembler, a variant caller or a sequence encoder does is constrained by what that multiset determines. This book is about the exact size and shape of that constraint. Given the data, the set of genomes compatible with them is a *fibre*. We count fibres exactly, describe the moves that connect their members, prove where exact counting stops being tractable, and show where the same fibres reappear in modern machine-learning representations of biological sequences.

**One strand (Part I).** For a single strand the fibre is the set of Eulerian circuits of a de Bruijn multigraph, and its size is a determinant, by the theorem of van Aardenne-Ehrenfest, de Bruijn, Smith and Tutte. The determinant factors into a local part, fixed by the branch statistics, and a global part: the order of the graph's sandpile group. Chapter 1 shows that the global part is not determined by the branch statistics. Chapter 2 turns the group into a coordinate system on reconstructions, one chip per rearrangement. Chapter 3 splits it along the repeats into a part that longer words remove and a part that no word length removes. Chapter 4 shows where pairwise repeat data stop describing it.

**Two strands (Part II).** Real data do not know which strand a word came from. The reverse-complement double cover keeps every single-strand lemma (Chapter 5). But its involution is an anti-automorphism of the Laplacian (Chapter 6), and the double-stranded count is no longer one determinant but a sum of determinants over the lattice points of a box (Chapter 7). For the phage  $\phi$ X174 at word length 10 the box has 2 481 687 900 points (Chapter 8), and the count is 31 925 753 246 212 (Chapter 9). Chapter 10 proves that this is not a failure of ingenuity: the double-stranded count is  $\#P$ -hard, so no closed formula computes it unless  $FP = \#P$ .

**Fibres and moves (Part III).** The members of a fibre are connected by explicit moves: transpositions between direct repeats and inversions between inverted repeats. Chapter 11 computes the lattice that the inversions span, by a Tate-cohomology dichotomy, and states a conjecture on generation. Chapter 12, written with Zhengyi Chen, shows that the fibres are exactly what local sequence encoders cannot see. It computes them for the whole human proteome and constructs explicit proteins that a widely used composition encoder cannot tell apart. Chapters 13 and 14 treat pooled data: polyploid phasing as a decomposition of one flow, and pan-genomes through their recombination group.

**Algorithms (Part IV).** Hardness is a statement about the worst case. Real genomes are tractable when their tangled parts are small, and the book names the reason each time. Chapter 15 reduces the 4.6-million-vertex graph of *E. coli* to 115 vertices by contracting everything except

repeats. Chapter 16 computes the single-strand determinant in linear time along the snarl tree of a genome graph. Chapter 17 describes the polytope of isoform abundances compatible with splicing data.

**How the results are checked.** Every chapter comes with a verification script, and every number in the text, the tables and the figures is printed by one of them. The scripts check three kinds of statement, and the book keeps them apart:

- *theorems*, which are proved in the text; the scripts test them on exhaustive or random instances, and a failure would indicate an error in the proof or in the code;
- *computations*, such as  $\#DS(\phi X174, 10)$ , which the scripts produce and cross-check by independent methods;
- *conjectures and empirical statements*, which are labelled as such and for which the scripts report the evidence and nothing more.

The repository <https://github.com/Ruqing1963/algebraic-genomics>, archived under DOI 10.5281/zenodo.22943487, contains the papers, the scripts, the input data and the reference transcripts. The script `build_all.py` compiles every chapter and reruns every check; `--quick` takes about half an hour, `--full` an afternoon. The state of that build for this release is recorded at the end of the book.

**What the book corrects.** Several chapters began as specifications that turned out to be wrong in part: a rank that was not the number of inverted repeats, a normalisation that was missing, a tensor network that was a Schur complement, a polynomial formula that cannot exist. Each chapter says which parts of its specification failed and why. We have kept these corrections in the text because they are part of the result.

**How to read it.** Part I is self-contained. Part II assumes Chapters 1 and 3. Part III can be read after Chapter 7. Part IV uses Chapter 3 (Chapter 15) and the BEST theorem (Chapters 16, 17). The chapters were written as papers; we have unified the notation, numbering and bibliography, replaced repeated definitions throughout by references to the chapter where each is first set up, and each chapter still opens with a summary that can be read alone.

**Acknowledgements.** Chapter 12 is joint work with Zhengyi Chen (Guangxi Key Laboratory of Drug Discovery and Optimization, School of Pharmacy, Guilin Medical University), and is included with the co-author's agreement. The phage, bacterial and human data are those of NCBI RefSeq, GenBank and UniProtKB.

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Rivière-Beaudette, Québec, 24 September 2026

## Part I

# Classical foundations: the BEST theorem and the sandpile group



# Chapter 1

## Assembly ambiguity is not determined by branch statistics: critical groups of de Bruijn graphs

**Summary.** The number of arc-rooted circular sequences sharing a given  $k$ -mer spectrum is, by the BEST theorem, the product of a *local* factor  $\prod_v (d^+(v) - 1)!$ , which depends only on the out-degree multiset of the de Bruijn graph, and a *global* factor, which is the order of the graph's critical (sandpile) group. We make four points about the global factor. First, it is not determined by the degree data: we exhibit two sequences of length 60 whose de Bruijn graphs at  $k = 4$  have identical vertex counts, identical out-degree multisets and hence identical local factors, but admit 4 and 256 reconstructions — a factor of 64. This is a strict sharpening of the observation of Kingsford, Schatz and Pop that repeat counting is insufficient. Second, the group structure reads off repeat parameters: a word of length  $L$  occurring  $r$  times in an otherwise  $k$ -unique sequence contributes  $\mathcal{K}(G_k) \cong (\mathbb{Z}/r)^{L-k}$ . Third, the group detects how repeats *interleave*: for two repeats of length  $L$  occurring twice each, the disjoint and nested arrangements  $A A B B$  and  $A B B A$  have isomorphic compacted graphs and  $\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{2s-2}$ , the direct sum of the two single-repeat contributions, while the interleaved arrangement  $A B A B$  has  $\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{2s-1}$ , one extra  $\mathbb{Z}/2$  that survives to  $k = L$  after both individual contributions have died. The degree data are identical in all three cases. Fourth, the invariant is cheap where it matters: contracting unitigs leaves the critical group unchanged, so it is computed on the compacted graph. We demonstrate this on a real genome: for bacteriophage  $\varphi$ X174 (5386 bp) the  $k = 10$  graph has 5339 vertices but 47 branch points, and its critical group, of order  $4.4 \times 10^{12}$ , is computed in ten milliseconds; the spectrum collapses through 132, 2, 1 at  $k = 11, 12, 13$ , one past the longest exact repeat. All claims are verified by machine: 30 checks, exact integer arithmetic except where labelled.

### 1.1 Introduction

Given a collection of  $k$ -mers with multiplicities — an idealised sequencing experiment — how many genomes are consistent with it? The  $k$ -mers and their overlaps form a de Bruijn multigraph  $G_k$  in which the genome is an Eulerian circuit, so the answer is the number of Eulerian circuits, which the BEST theorem [10, 78] evaluates as

$$\text{ec}(G_k) = |\mathcal{K}(G_k)| \cdot \prod_v (d^+(v) - 1)!, \quad (1.1)$$

$\mathcal{K}(G_k)$  being the group of spanning arborescences — the *critical* or *sandpile* group. Kingsford, Schatz and Pop [51] computed this for real prokaryotic genomes as a measure of assembly complexity; Loukides, Pissis and co-authors [56] gave fast algorithms for assessing it.

Equation (1.1) splits the ambiguity into two factors of different character. The product of factorials is a function of the out-degree multiset alone — the *branch statistics*, read off the graph by inspection. The other factor is not. This chapter is about what the other factor knows; Chapters 2–4 are about the group behind it.

**What is already known.** Two literatures meet here, and we claim nothing in either. On the combinatorial side the critical groups of *complete* de Bruijn graphs are completely understood: Knuth [48] gave the line-digraph recursion, Levine [55] the group structure  $\mathcal{K}(\text{DB}(2, n)) \cong \bigoplus_{j=1}^{n-1} (\mathbb{Z}/2^j)^{2^{n-1-j}}$ , and Chan, Hollmann and Pasechnik [24] the generalised de Bruijn and Kautz cases. On the assembly side, counting reconstructions via BEST is established [51, 56], as are uniqueness and safety criteria [65] and feasibility thresholds [8, 63]. What appears not to have happened is that the two ever met: we have found no work applying the critical group of a graph to assembly, and none that looks at the *structure* of  $\mathcal{K}(G_k)$  for a genome de Bruijn graph, which is a subgraph of a complete one and not covered by [55, 24].

**What is new here.**

- (i) **Theorem 1.10:** the global factor is not a function of the degree data. Two explicit sequences have identical branch statistics and reconstruction counts differing by  $64\times$ .
- (ii) **Theorem 1.5:** for a single repeat of length  $L$  and multiplicity  $r$ ,  $\mathcal{K}(G_k) \cong (\mathbb{Z}/r)^{L-k}$ .
- (iii) **Theorem 1.6:** for two repeats the group is additive in the disjoint and nested arrangements and picks up exactly one extra  $\mathbb{Z}/2$  in the interleaved arrangement. Since interleaved repeats are the notoriously hard case for assembly, this is the first place the algebra and the practice visibly agree. Proposition 1.8 then shows that for three repeats the correction is the sandpile group of a small digraph and not a chord-diagram rank — it can be  $\mathbb{Z}/3$ .
- (iv) **Lemma 1.4** makes the invariant computable on the compacted graph, and Section 1.8 computes it on a real genome.

**Relation to [51].** Kingsford, Schatz and Pop already argue that repeat counting is insufficient, with the example that three repeats in two copies each (8 reconstructions) is easier than one repeat in six copies (720). Their two situations have *different* out-degree multisets,  $(2, 2, 2)$  against  $(6)$ , so the local factor already distinguishes them. Theorem 1.10 fixes the out-degree multiset and still finds a 64-fold spread, and Theorem 1.6 says what the residual information is: not how many repeats or how many copies, but how they are wired to one another.

## 1.2 Setting

Let  $S$  be a circular sequence of length  $N$  over an alphabet of size  $\sigma$ . For  $k \geq 1$  the *de Bruijn multigraph*  $G_k = G_k(S)$  has as vertices the distinct  $k$ -mers of  $S$  and one arc for each of the  $N$  occurrences of a  $(k+1)$ -mer, from its length- $k$  prefix to its length- $k$  suffix.  $S$  traverses every arc once, so  $G_k$  is connected and balanced, and  $S$  is an Eulerian circuit; every Eulerian circuit is a circular sequence with the same  $k$ -mer spectrum, counted with multiplicities. We write  $n_k = |V(G_k)|$ ,  $d^+(v)$  for out-degree (the number of occurrences of  $v$ ), and  $\text{Loc}(G) := \prod_v (d^+(v) - 1)!$ .

*Remark 1.1* (Counting convention). Throughout, reconstruction counts are *arc-rooted*: circuits are counted as beginning with a fixed arc, which is the convention under which (1.1) holds [10, 78]. Circular sequences up to cyclic shift are counted by dividing by  $N$  when  $S$  is aperiodic, and by Burnside's lemma otherwise; none of the algebra below depends on which convention is chosen, since the correction is a function of  $N$  alone.

The *critical group* is  $\mathcal{K}(G) := \text{coker}(\Delta^\top)$ ,  $\Delta$  the out-Laplacian  $D^+ - A$  with the row and column of a fixed root deleted; for a connected balanced digraph the isomorphism class does not depend on the root, and  $|\mathcal{K}(G)|$  is the number of spanning arborescences. We compute it as a Smith normal form.

*Remark 1.2* (The tower).  $G_{k+1}(S)$  is a spanning subgraph of the line digraph  $L(G_k(S))$ , with equality for the complete de Bruijn graph; the prefix and suffix maps are the tail and head maps of  $G_k$ . The line-digraph recursion for arborescences [48] gives Knuth's formula and reproduces  $|\mathcal{K}(\text{DB}(\sigma, k))| = \sigma^{\sigma^k - k - 1}$  (check (T)); check (L) reproduces Levine's decomposition. These are consistency checks on our conventions, not results.

### 1.3 The local and global factors

**Proposition 1.3** (Decomposition). *The number of arc-rooted circular sequences with the same  $k$ -mer spectrum as  $S$  equals  $|\mathcal{K}(G_k)| \cdot \text{Loc}(G_k)$ . The factor  $\text{Loc}(G_k)$  depends only on the multiset  $\{d^+(v)\}$ ; the factor  $|\mathcal{K}(G_k)|$  does not.*

The first sentence is (1.1). The second is Theorem 1.10.

### 1.4 Contraction, and a single repeat

**Lemma 1.4** (Unitig contraction). *Let  $v$  be a vertex of a connected balanced multidigraph  $G$  with  $d^-(v) = d^+(v) = 1$ , whose incoming and outgoing arcs are distinct, say  $u \rightarrow v$  and  $v \rightarrow w$ . Let  $G'$  be obtained by deleting  $v$  and replacing those two arcs by a single arc  $u \rightarrow w$ . Then  $\mathcal{K}(G') \cong \mathcal{K}(G)$ . Self-loops may be deleted freely: a loop at  $v$  adds 1 to both  $D_{vv}^+$  and  $A_{vv}$  and so does not appear in  $\Delta$ .*

*Proof.* Root away from  $v$  and  $u$ . The row of  $v$  in  $\Delta$  is  $e_v - e_w$  and the column of  $v$  has a single off-diagonal entry, in the row of  $u$ ; adding the row of  $v$  to the row of  $u$  clears it and turns the row of  $u$  into its row in  $\Delta(G')$ , after which the row and column of  $v$  form a unit block. Both operations are unimodular. The statement about loops is immediate from  $\Delta = D^+ - A$  and is check (P).  $\square$

Iterating the lemma yields the *compacted* (unitig) de Bruijn graph, in which every maximal non-branching path is one arc and only branch vertices survive.

**Theorem 1.5** (A repeat of length  $L$  and multiplicity  $r$ ). *Let  $W$  have length  $L$  and let  $S$  be circular with every  $k$ -mer occurring once except the  $L - k + 1$   $k$ -mers of  $W$ , each occurring exactly  $r \geq 2$  times. Then for  $1 \leq k \leq L$*

$$\mathcal{K}(G_k(S)) \cong (\mathbb{Z}/r)^{L-k}.$$

*Proof.* Put  $s = L - k + 1$  and let  $v_1, \dots, v_s$  be the  $k$ -mers of  $W$  in order. Every other vertex has in- and out-degree 1; contract them. What remains is the spine  $v_i \rightarrow v_{i+1}$  with multiplicity  $r$  and  $r$  arcs  $v_s \rightarrow v_1$ . Rooted at  $v_1$ , on  $v_2, \dots, v_s$  the out-Laplacian is  $r(I - N)$  with  $N$  the nilpotent shift;  $I - N$  is unimodular, so the cokernel is  $(\mathbb{Z}/r)^{s-1}$ .  $\square$

## 1.5 Two repeats: the group sees interleaving

Fix two distinct words  $A, B$  of length  $L$  and a circular sequence in which each occurs exactly twice, non-overlapping, with every other  $k$ -mer unique; put  $s = L - k + 1$ . The four occurrences can be arranged in three ways around the circle:

$$\text{disjoint } A A B B, \quad \text{nested } A B B A, \quad \text{interleaved } A B A B.$$

The  $k$ -mer spectra of the three sequences have identical multiplicity profiles, so  $n_k$ , the out-degree multiset, and hence  $\text{Loc}(G_k)$  agree. This is Theorem 1.10's setting one level up, and the question is whether  $\mathcal{K}$  can tell them apart.

**Theorem 1.6** (Two repeats). *With  $a_1, \dots, a_s$  and  $b_1, \dots, b_s$  the  $k$ -mers of  $A$  and  $B$ :*

(i) *the disjoint and nested arrangements have isomorphic compacted graphs, and*

$$\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{2s-2} = (\mathbb{Z}/2)^{s-1} \oplus (\mathbb{Z}/2)^{s-1},$$

*the direct sum of the two single-repeat groups of Theorem 1.5;*

(ii) *the interleaved arrangement has*

$$\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{2s-1}.$$

*In particular the arrangement is detected by  $\mathcal{K}$  and not by the degree data, and at  $k = L$  ( $s = 1$ ) the interleaved arrangement retains a  $\mathbb{Z}/2$  after both single-repeat contributions have become trivial.*

*Proof.* Contract by Lemma 1.4. The surviving vertices are the  $2s$  shared  $k$ -mers, each of in- and out-degree 2. The spine arcs  $a_i \rightarrow a_{i+1}$  and  $b_i \rightarrow b_{i+1}$  have multiplicity 2. The four spacer paths contract to four connector arcs from  $\{a_s, b_s\}$  to  $\{a_1, b_1\}$ , one per transition between consecutive occurrences around the circle. Reading the cyclic order: disjoint gives  $a_s \rightarrow a_1$ ,  $a_s \rightarrow b_1$ ,  $b_s \rightarrow b_1$ ,  $b_s \rightarrow a_1$ ; nested gives  $a_s \rightarrow b_1$ ,  $b_s \rightarrow b_1$ ,  $b_s \rightarrow a_1$ ,  $a_s \rightarrow a_1$  — the same multiset, so the two compacted graphs coincide; interleaved gives  $a_s \rightarrow b_1$  twice and  $b_s \rightarrow a_1$  twice.

(ii) Root at  $a_1$ . On  $a_2, \dots, a_s, b_1, \dots, b_s$  in this order the reduced out-Laplacian is  $2(I - N)$ ,  $N$  the shift along that path (both arcs of  $b_s$  go to the deleted root).  $I - N$  is unimodular, so  $\mathcal{K} \cong (\mathbb{Z}/2)^{2s-1}$ .

(i) Root at  $a_1$ ; write  $x_v$  for the generator of the cokernel presentation at  $v$ . The rows of  $a_s$  and  $b_s$  read  $2x_{a_s} - x_{b_1}$  and  $2x_{b_s} - x_{b_1}$ , so  $x_{b_1} = 2x_{a_s}$  and  $x_{b_1}$  can be eliminated. On the remaining  $2s - 2$  generators  $a_2, \dots, a_s, b_2, \dots, b_s$  every relation is now even — the spine relations  $2x_{a_i} - 2x_{a_{i+1}}$  and  $2x_{b_i} - 2x_{b_{i+1}}$ , the relation  $2x_{b_s} - 2x_{a_s}$ , and the former row of  $b_1$ , which becomes  $4x_{a_s} - 2x_{b_2}$  — so the relation matrix is  $2M'$  with

$$M' = \begin{pmatrix} \text{rows } e_{a_i} - e_{a_{i+1}}, & 2 \leq i \leq s-1 \\ & e_{b_s} - e_{a_s} \\ & 2e_{a_s} - e_{b_2} \\ \text{rows } e_{b_i} - e_{b_{i+1}}, & 2 \leq i \leq s-1 \end{pmatrix}.$$

Expanding  $\det M'$  along the columns  $a_2, \dots, a_{s-1}$  in turn (each has a single entry,  $+1$ ) leaves the  $s \times s$  block on columns  $(a_s, b_2, \dots, b_s)$  with rows  $2e_{a_s} - e_{b_2}$ ,  $e_{b_i} - e_{b_{i+1}}$  ( $2 \leq i \leq s-1$ ),  $e_{b_s} - e_{a_s}$ . Expanding along the first row with the cofactor signs written out,  $\det M' = 2(-1)^{1+1}M_{11} + (-1)(-1)^{1+2}M_{12} = 2M_{11} + M_{12}$ . The minor  $M_{11}$  (the remaining rows on columns  $b_2, \dots, b_s$ ) is upper bidiagonal with

unit diagonal, so  $M_{11} = 1$ ; the minor  $M_{12}$  (the same rows on columns  $a_s, b_3, \dots, b_s$ ) has a single entry  $-1$  in its first column, in its last row, and expanding along that column leaves a lower bidiagonal block with diagonal  $-1$ , the signs combining to  $M_{12} = -1$ . Hence  $\det M' = 2 - 1 = 1$  in this column order,  $|\det M'| = 1$  in any order,  $M'$  is unimodular and  $\mathcal{K} \cong (\mathbb{Z}/2)^{2s-2}$ .  $\square$

*Remark 1.7* (Reading). For a single repeat the group is trivial at  $k = L$ : the repeated  $L$ -mer is a branch vertex but its two exits lead to different places and there is only one way to thread it. For two interleaved repeats at  $k = L$  there is a genuine choice — leave  $A$  for the first  $B$  or the second — and  $\mathbb{Z}/2$  records it. Nested and disjoint repeats offer no such choice: whichever way one threads  $A$ , the two  $B$ 's are traversed consecutively. This is the algebraic form of the folklore that interleaved repeats are what makes assembly hard. Additivity of  $\mathcal{K}$  over two repeats therefore holds exactly when they do not interleave; the general correction term is indexed by the interleaving pattern, Theorem 1.6 is its two-repeat case, and Chapter 3 identifies it in general.

Check (I) verifies the theorem three ways: on clean random sequence instances (the prediction is exact in every one), on the explicit compacted graphs (six branch vertices; connector multiplicities  $\{1, 1, 1, 1\}$  for disjoint and nested and  $\{2, 2\}$  for interleaved), and by symbolic Smith normal form of the matrices in the proof for  $s \leq 7$ .

### Three repeats, and what the correction is not

With three repeats the natural guess is that the interleaving correction is a function of the chord diagram of the six occurrences through the rank over  $\mathbb{F}_2$  of its intersection matrix. It is not.

**Proposition 1.8** (Three repeats). *Let  $A, B, C$  be distinct words of length  $L$ , each occurring exactly twice, with every other  $k$ -mer unique and  $s = L - k + 1$ . Let  $D_w$  be the digraph on the three symbols with one arc from the symbol of each occurrence to the symbol of the next occurrence around the circle, loops deleted. Then  $\mathcal{K}(G_L) \cong \mathcal{K}(D_w)$ , and according to which pairs of the three chords cross:*

| crossing pairs  | $\mathbb{F}_2$ -rank of intersection matrix | $\mathcal{K}(G_k), s = 1$ | $\mathcal{K}(G_k), s = 2$              |
|-----------------|---------------------------------------------|---------------------------|----------------------------------------|
| none            | 0                                           | 0                         | $(\mathbb{Z}/2)^3$                     |
| one pair        | 2                                           | $\mathbb{Z}/2$            | $(\mathbb{Z}/2)^4$                     |
| a path of two   | 2                                           | $\mathbb{Z}/3$            | $(\mathbb{Z}/2)^3 \oplus \mathbb{Z}/3$ |
| all three pairs | 2                                           | $(\mathbb{Z}/2)^2$        | $(\mathbb{Z}/2)^5$                     |

In every case  $\mathcal{K}(G_k) \cong (\mathbb{Z}/2)^{3(s-1)} \oplus \mathcal{K}(D_w)$  for  $s = 1, 2$ .

*Proof.* At  $s = 1$  the compacted graph is  $D_w$  by construction (Lemma 1.4 deletes the loops), which gives  $\mathcal{K}(G_L) = \mathcal{K}(D_w)$ ; the four values are the sandpile groups of the four 2-regular digraphs on three vertices that occur. For instance the path pattern  $A B A C B C$  gives the complete symmetric digraph on three vertices, whose sandpile group is  $\mathbb{Z}/3$ . The  $s = 2$  column and the splitting are verified exactly on all 90 circular words with two of each letter (check (M)); the splitting for general  $s$ , and for any number of repeats, is Corollary 3.7.  $\square$

*Remark 1.9.* Three of the four patterns have intersection rank 2 and give three different groups, one of them of odd order: multiplicity-two repeats can interleave into 3-torsion. So the correction is not a chord-diagram rank, nor even a 2-group. The right object is the sandpile group of the connector digraph  $D_w$ , which for two repeats is trivial or  $\mathbb{Z}/2$  (Theorem 1.6) and for three is the table above.

## 1.6 Branch statistics are insufficient

**Theorem 1.10** (The global factor is not degree-determined). *There exist circular sequences  $S_{\text{lo}}, S_{\text{hi}}$  of length 60 over  $\{A, C, G, T\}$  such that at  $k = 4$  their de Bruijn graphs have the same number of vertices  $n_4 = 51$  and the same out-degree multiset (seven branch vertices of out-degree 2 and one of out-degree 3), hence the same local factor  $\text{Loc} = 2$ , but*

$$\mathcal{K}(G_4(S_{\text{lo}})) = \mathbb{Z}/2, \quad \mathcal{K}(G_4(S_{\text{hi}})) = (\mathbb{Z}/2)^4 \oplus \mathbb{Z}/8,$$

so that the numbers of reconstructions are 4 and 256. Explicit witnesses:

$S_{\text{lo}}:$     **GAACGAGAGAAAAACACGTACCAGCGCGGGGGCGGGACGGCAGGTGACGCTCTGGGAAG**  
 $S_{\text{hi}}:$     **GCACAGGGTCCAGTCTGCGAACGCGCTACAGTCTGGGTCAAGCGGTCTCTGTGCATCG**

*Proof.* Exact computation; check (N). □

**Corollary 1.11.** *No function of the out-degree multiset of  $G_k$  determines the number of reconstructions.*

In a randomised search over 1500 sequences of length 60 at  $k \in \{4, 5\}$ , 67 of the 98 degree-classes containing more than one sequence contained more than one critical group (check (N), sampled).

## 1.7 Computation

Computing  $\mathcal{K}(G_k)$  directly is one Smith normal form of an  $(n_k - 1) \times (n_k - 1)$  integer matrix,  $O(n_k^3)$  arithmetic operations, which for a genome is impractical. Lemma 1.4 removes the difficulty: the critical group of  $G_k$  equals that of its compacted graph, whose vertices are the branch points. The compaction is linear in  $N$  — one pass to find the branch vertices, one walk per outgoing arc — and assemblers already build this graph. The cost is therefore governed by repeat content, not genome length; the next section shows what this means in practice.

## 1.8 A real genome

Bacteriophage  $\varphi\text{X174}$  (GenBank NC\_001422.1, 5386 bp, circular) was the first genome ever sequenced. Its longest exact circular repeat has length  $R = 12$ . Table 1.1 gives the critical-group spectrum around that scale. The compaction is dramatic: at  $k = 10$  the graph has 5339 vertices and 47 branch points, and the Smith normal form takes ten milliseconds. The group order at  $k = 10$ ,  $4.4 \times 10^{12}$ , is the global part of the number of 10-mer reconstructions of the phage; it falls to 132 at  $k = 11$ , to 2 at  $k = 12$ , and the graph becomes a single cycle at  $k = 13 = R + 1$ .

Two remarks. The levels  $k \leq 9$  have compacted graphs of 146 to 1384 vertices; they are within reach of a sparse or  $p$ -adic Smith normal form but not of the dense routine used here; Chapter 3 reaches  $k = 9$  by a different route (Table 3.1). And  $k^* = R + 1$  for this genome. A single isolated repeat collapses at  $k = R$  (Theorem 1.5), so a surviving  $\mathbb{Z}/2$  at  $k = 12$  is *consistent with* Theorem 1.6 rather than implied by it; Proposition 2.5 settles the question by exhibiting the two reconstructions: the longest 12-mer repeat is interleaved with a second repeated 12-mer, and the residual bit is exactly the one of Theorem 1.6 at  $s = 1$ .

| $k$      | $n_k$       | branch vertices | $ \mathcal{K}(G_k) $ | invariant factors                 |
|----------|-------------|-----------------|----------------------|-----------------------------------|
| $\leq 9$ | $\leq 5233$ | $\geq 146$      | —                    | above the exact-SNF cap used here |
| 10       | 5339        | 47              | 4 408 682 819 584    | $2^8, \dots, 4305354316$          |
| 11       | 5376        | 10              | 132                  | 2, 66                             |
| 12       | 5384        | 2               | 2                    | 2                                 |
| 13       | 5386        | 0               | 1                    | — (single cycle)                  |

Table 1.1:  $\varphi$ X174, NC\_001422.1. Branch vertices are the vertices of the compacted graph. All entries exact (check (F)); the sequence file is distributed with the script and no network access is needed.

## 1.9 What is open, and what is false

**False.** The collapse order is not one more than the longest repeat. With  $R(S)$  the longest repeated substring and  $k^*(S)$  the least  $k$  with  $\mathcal{K}(G_k)$  trivial, over 40 random sequences of length 100 we find  $k^* = R$  in 26 cases and  $k^* = R + 1$  in 13 (check (C)). Exhaustive search over short binary sequences gives minimal witnesses: AACAC has  $R = 3$ ,  $k^* = 3$ ; AACACC has  $R = 2$ ,  $k^* = 3$ ; and homopolymer runs such as AAAA have  $k^* < R$ , because a repeat that always extends the same way branches nowhere. Theorem 1.5 explains the first family and Theorem 1.6 the second, but we have no closed form for  $k^*$  in general.

**Open.** (a) *Additivity beyond two repeats.* Theorem 1.6 and Proposition 1.8 identify the correction at  $s = 1$  with the sandpile group of the connector digraph  $D_w$  of the circular word of occurrences, and show it is not a function of the chord diagram’s intersection rank. The general form,  $\mathcal{K}(G_k) \cong \bigoplus_i (\mathbb{Z}/r_i)^{s_i-1} \oplus \mathcal{K}(D_w)$  for  $m$  repeats of multiplicities  $r_i$  (with  $D_w$  then carrying multiplicities), is proved in Chapter 3 (Corollary 3.7); what stays open is the structure of  $\mathcal{K}(D_w)$  itself. (b) *Higher multiplicity.* For repeats of multiplicity  $r > 2$  the two-copy description of  $\mathcal{K}(D_w)$  does not extend (Chapter 4). (c) *Double-strandedness.* Real assembly graphs are bidirected. This is a different combinatorial object, and we warn against the tempting decomposition of  $\mathcal{K}$  into  $\pm 1$ -eigenspaces of the reverse-complement involution: an involution splits an abelian group only after 2 is inverted, and the 2-torsion — which is where all of our examples live, and where inverted repeats would show up — is exactly what such a splitting discards. The tools that survive are the signed and  $\mathbb{Z}/2$ -voltage Laplacians [88, 75], with one caveat: reverse complementation reverses arc direction, so the passage to the bidirected graph is not a covering of digraphs and the covering theory of [75] does not apply as it stands. Part II starts from this point. (d) *Errors and coverage.* We assume an exact  $k$ -mer spectrum with true multiplicities.

## 1.10 Verification

The script `dbg_sandpile.py` runs 30 checks in about two minutes, all passing; its output is archived as `dbg_sandpile_output.txt`. Arithmetic is exact except in the two rows marked *sampled*.

## Provenance

The classical inputs are the BEST theorem [10, 78], Knuth’s recursion [48] and the matrix–tree theorem for digraphs; the critical groups of complete de Bruijn graphs [55, 11, 24] are used only

| key   | check                                                                                         | #     | mode            |
|-------|-----------------------------------------------------------------------------------------------|-------|-----------------|
| (T)   | $\text{DB}(\sigma, k+1) = L(\text{DB}(\sigma, k));  \mathcal{K}  = \sigma^{\sigma^k - k - 1}$ | 8/8   | exact           |
| (L)   | $\mathcal{K}(\text{DB}(2, n))$ reproduces Levine [55], $n \leq 5$                             | 4/4   | exact           |
| (E)   | BEST (1.1) against brute-force arc-rooted enumeration                                         | 6/6   | exact           |
| (R)   | Theorem 1.5, 54 parameter settings                                                            | 1/1   | exact           |
| (I)   | Theorem 1.6: sequences, compacted graphs, symbolic SNF                                        | 3/3   | exact           |
| (M)   | Proposition 1.8: all 90 circular words of three repeats, $s = 1, 2$                           | 1/1   | exact           |
| (N)   | Theorem 1.10 witnesses; degree-class splitting rate                                           | 2/2   | exact / sampled |
| (C)   | collapse order: statistics; minimal binary witnesses                                          | 2/2   | sampled / exact |
| (P)   | Lemma 1.4: contraction and self-loops                                                         | 2/2   | exact           |
| (F)   | Table 1.1: $\varphi\text{X174}$                                                               | 1/1   | exact           |
| total |                                                                                               | 30/30 |                 |

Table 1.2: The verification battery (`dbg_sandpile.py`).

as checks. Proposition 1.3 is a reading of (1.1); the use of BEST to count genome reconstructions is due to [51]. What we claim as new is Theorems 1.5, 1.6 and 1.10 with Corollary 1.11, and Lemma 1.4 with its use in Sections 1.7–1.8.

A detailed reading of an earlier version of this chapter made four corrections; all were right and all are incorporated, and we are grateful for them.

## Chapter 2

# The sandpile group as a coordinate system on genome reconstructions

**Summary.** The reconstructions of a circular genome from its  $k$ -mer spectrum are the Eulerian circuits of its de Bruijn multigraph  $G$ , and by the BEST theorem they correspond bijectively to pairs (spanning arborescence, choice of exit order at every branch vertex). The sandpile group  $\mathcal{K}(G)$  acts simply transitively on the arborescences by rotor-routing (Holroyd, Levine, Mészáros, Peres, Propp and Wilson), whose proof of BEST is itself a rotor walk. We put the two facts to a use we have not found in the literature:  $\mathcal{K}(G)$  is a *torsor of coordinates on the global part of the reconstruction space*, one chip placed on a  $k$ -mer is an explicit rearrangement carrying one candidate genome to another with the same spectrum, and the difference of two candidates is a group element. We exhibit the moves, verify the torsor structure and the bijection on assembly graphs by exhaustive enumeration, and show that the classical uniform sampler of Eulerian circuits — known since Altschul and Erickson (1985) and implemented in uShuffle — is the torsor in Smith-normal-form coordinates, which is what makes the distance between two samples computable. The centrepiece is a real genome: for bacteriophage  $\varphi$ X174 the 12-mer spectrum leaves exactly one global choice,  $\mathcal{K} = \mathbb{Z}/2$ ; the two reconstructions are exhibited, they are  $AXBYAZBU$  and  $AXBUAZBY$  for two interleaved repeated 12-mers, and a single chip exchanges the 433 and 181 bp segments. This is the  $\mathbb{Z}/2$  of the interleaving theorem of Chapter 1, realised in the first genome ever sequenced. Seven checks, exact except the two sampling rows.

## 2.1 Introduction

Chapter 1 showed that the number of reconstructions of a circular sequence from its  $k$ -mer spectrum factors, by the BEST theorem, into a local part  $\prod_v (d^+(v) - 1)!$  determined by branch statistics and a global part  $|\mathcal{K}(G_k)|$ , the order of the sandpile group, which branch statistics do not determine; and that the group structure of  $\mathcal{K}(G_k)$  records repeat length, multiplicity and interleaving. This chapter is about the group itself rather than its order: what does the group *do* to the set of reconstructions?

**What is known.** Three literatures bear on the question, and we claim nothing in any of them.

- *Counting and sampling.* The BEST theorem [10, 78] counts Eulerian circuits of a directed Eulerian graph as  $(\text{arborescences}) \times \prod (d - 1)!$ , and since arborescences can be sampled uniformly in polynomial time [25, 73, 86], so can Eulerian circuits; Creed and Cryan state

this explicitly [18]. (For *undirected* graphs counting is #P-complete [17]; the directed case, which is ours, is easy.) Bioinformatics has used exactly this since 1985 to shuffle sequences preserving  $k$ -let counts [3, 47, 84], and to count reconstructions from  $k$ -mer spectra [1, 51].

- *Rotor-routing*. Holroyd et al. [40] prove that the sandpile group of a digraph with a sink acts freely and transitively on the spanning arborescences toward the sink (their §3), and in their §4 give a bijective proof of BEST by a rotor walk: an Eulerian tour is exactly one full turn of every rotor (Cor. 4.10). The rotor-router model itself is the “Eulerian walkers” model of Priezzhev et al. [67].
- *Sandpile groups of de Bruijn graphs* are known for the complete graphs [55, 11, 24] (Section 1.1). None concern genome graphs.

**What is here.** We compose these into a statement about assembly (Theorem 2.1): the set of reconstructions, modulo the local exit choices, is a torsor for  $\mathcal{K}(G)$ . Consequences: a chip placed on a  $k$ -mer is an explicit rearrangement between two candidate genomes (Section 2.4); the difference of two candidates is a group element, computable as a chip word; and the classical uniform sampler is the torsor written in Smith-normal-form coordinates (Section 2.5). Section 2.6 applies this to  $\varphi$ X174, where the 12-mer spectrum leaves one binary choice and we show what that choice is.

**What we do not claim.** Not the torsor theorem, which is [40, §3]; not the bijection between tours and (arborescence, rotor order), which is [40, Cor. 4.10] and which our check (B) merely re-implements; not uniform sampling, which is [25, 73, 18] and, in bioinformatics, [3, 47, 84]. The contribution is the reading of these as a coordinate system on assemblies, the identification of chip-firing with rearrangement, and the  $\varphi$ X174 computation.

## 2.2 Setting

$S$  is a circular sequence and  $G = G_k(S)$  its de Bruijn multigraph, as in Section 1.2;  $\text{Rec}(G)$  denotes the set of arc-rooted Eulerian circuits, the reconstructions in the convention of Remark 1.1. Fix a sink  $s \in V$  and let  $\text{Arb}(G)$  be the set of spanning arborescences oriented toward  $s$ ; for balanced  $G$  the count  $|\text{Arb}(G)|$  and the group  $\mathcal{K}(G) = \text{coker } \Delta^\top$  do not depend on  $s$  [40, Lem. 4.12]. Here  $\Delta = D^+ - A$  is the out-Laplacian with the row and column of  $s$  deleted; its determinant counts arborescences *toward*  $s$  because each non-sink vertex chooses one of its out-arcs, and the transpose enters because firing  $v$  sends one chip along each out-arc,  $\sigma \mapsto \sigma - \Delta^\top e_v$ , so the firing lattice is the column span of  $\Delta^\top$ . For balanced  $G$ ,  $\Delta^\top = D^+ - A^\top$  is the in-Laplacian, so the choice is between two presentations of one group. A *rotor configuration* is a choice of one out-arc at every non-sink vertex; the acyclic ones are exactly  $\text{Arb}(G)$ . A chip placed at  $v$  is routed to the sink by the rule “advance the rotor at the current vertex by one, then move along it” [40, §3]; the map  $\text{Arb}(G) \rightarrow \text{Arb}(G)$  this induces is the action of the class of  $e_v$  in  $\mathcal{K}(G)$ . The same action can be realised algebraically, by adding  $e_v$  to a recurrent chip configuration and toppling back to the recurrent representative, since the rotor-router group and the sandpile group coincide [40, Lem. 3.19]; we use the dynamical description because it delivers the new arborescence directly.

By Lemma 1.4, both  $\mathcal{K}(G)$  and, arc for arc,  $\text{Arb}(G)$  are unchanged when every vertex of in- and out-degree 1 is smoothed away, so all of the following may be computed on the compacted (unitig) graph; the contracted arcs carry the concatenated labels and a circuit on the compacted graph is spelled back into a full-length sequence.

## 2.3 The torsor

**Theorem 2.1** (Coordinates on reconstructions). *Let  $G$  be the de Bruijn multigraph of a circular sequence. Then*

- (i)  $\mathcal{K}(G)$  acts freely and transitively on  $\text{Arb}(G)$  by rotor-routing [40, Lem. 3.12, 3.13, 3.17];
- (ii) the map  $(T, \rho) \mapsto$  “walk from the sink taking, at each vertex, the next unused out-arc in the cyclic order  $\rho_v$  with the tree arc of  $T$  last” is a bijection from pairs (arborescence, exit orders) onto  $\text{Rec}(G)$  [40, Cor. 4.10];
- (iii) consequently  $\text{Rec}(G)/\sim$ , the reconstructions modulo the local exit choices, is a  $\mathcal{K}(G)$ -torsor: after fixing one reconstruction, every other one is labelled by a unique group element, and  $|\text{Rec}(G)| = |\mathcal{K}(G)| \cdot \prod_v (d^+(v) - 1)!$ .

The theorem is a composition of two results of [40]; what we add is (iii) as a statement about genomes, and the two verifications of Section 2.7: check (A) enumerates  $\text{Arb}(G)$  for six assembly graphs, builds the permutation representation of the chip generators, and confirms that the generated group has order  $|\mathcal{K}(G)|$ , a single orbit, and no fixed points; check (B) implements the walk of (ii) and confirms it is a bijection onto the brute-force enumeration of Eulerian circuits (288, 576, 144, 8 circuits on four graphs).

*Remark 2.2* (Why the quotient). The local choices at a branch vertex of out-degree  $d$  — which of the  $d - 1$  non-tree exits to take first — are what the local factor  $\prod (d - 1)!$  counts, and they are independent of one another and of the arborescence. They are the ambiguity that branch statistics *do* see. The torsor lives on what remains, which Theorem 1.10 showed is invisible to branch statistics; Theorem 2.1 says it nevertheless has the structure of a finite abelian group, given explicitly by the Smith normal form of the out-Laplacian.

## 2.4 A chip is a rearrangement

Fix exit orders once and for all (say, in arc order). Then each arborescence determines one reconstruction, and adding a chip at a  $k$ -mer  $v$  moves the arborescence and hence the reconstruction. Check (M) does this on  $S = \text{ACACCAAAACCA}$ ,  $k = 2$ , where  $\mathcal{K}(G) = \mathbb{Z}/2 \oplus \mathbb{Z}/6$ :

|                     |              |                                               |
|---------------------|--------------|-----------------------------------------------|
| base reconstruction | AAAACCAACCAC |                                               |
| + one chip at CA    | ACCAAAACCAAC | (6 positions changed)                         |
| + one chip at CC    | ACCAAAACCAAC | (6 positions changed)                         |
| + one chip at AA    | AAAACCAACCAC | (unchanged: $[e_{AA}] = 0$ in $\mathcal{K}$ ) |

Each row has the 2-mer spectrum of  $S$ . The chip at **AA** is trivial in the group and moves nothing, which is the correct behaviour: not every  $k$ -mer carries a nonzero coordinate.

**Proposition 2.3** (Distance). *For two reconstructions with the same exit orders there is a unique  $g \in \mathcal{K}(G)$  carrying one arborescence to the other. Writing  $g$  as a sum of chip classes  $[e_v]$  with non-negative coefficients — always possible in a finite group — gives a sequence of physically realisable moves converting the first genome into the second. The minimal number of chips  $\delta(T, T')$  is a directed distance on  $\text{Rec}(G)/\sim$ : a quasi-metric, since  $\delta(T, T')$  and  $\delta(T', T)$  need not agree. Admitting chips of both signs, i.e. the symmetric generating set  $\{\pm[e_v]\}$ , gives the Cayley word metric of  $\mathcal{K}(G)$ .*

This is the sense in which the group is a coordinate system rather than merely a count. Check (W) computes the word for two independently sampled reconstructions of a 27-element torsor: three chips.

One thing a chip is *not* is a single transposition. The rotor walk re-threads every branch vertex it passes on its way to the sink, so one chip can change the tree arc at several junctions at once. On the compacted  $\varphi$ X174 graph at  $k = 11$ , which has ten branch vertices, the nine generators change 6, 5, 3, 2, 4, 2, 3, 1, 0 tree arcs respectively (check (J)); only one of them is a single re-threading. The directed distance of Proposition 2.3 is therefore not the transposition or reversal distance of comparative genomics, and we see no immediate inequality between them.

## 2.5 Sampling, and what the group adds

Uniform sampling of  $\text{Rec}(G)$  is classical: sample an arborescence uniformly by a loop-erased random walk [86, 73] or by unranking [25], then choose exit orders uniformly [18]; this is what [3, 47] do and what uShuffle implements [84]. Theorem 2.1 gives the same distribution in a different form: with  $\mathcal{K}(G) \cong \bigoplus \mathbb{Z}/d_i$  from the Smith normal form  $U\Delta^\top V = D$ , draw  $r_i$  uniformly mod  $d_i$ , take  $U^{-1}r$  as a chip configuration, and route it from a fixed base arborescence. Check (U) draws 2700 samples this way on a torsor of order 27 and finds all 27 arborescences with  $\chi^2 = 31.2$  on 26 degrees of freedom ( $p = 0.22$ ); Wilson’s sampler on the same graph gives  $\chi^2 = 22.3$  ( $p = 0.67$ , check (W)). The two are the same distribution. What the coordinate form adds is not speed — Wilson’s algorithm is faster — but the ability to say *where* a sample is. A shuffler such as uShuffle returns independent draws and nothing else: two of its outputs stand in no computable relation, and the sample space has no structure beyond its cardinality. In the torsor two samples differ by a computable group element, a sample can be moved by a prescribed chip, the whole space can be enumerated by stepping through Smith coordinates, and a sample can be conditioned on a prescribed value of some coordinates. What it does *not* give is a linear way to impose local constraints, as the next remark records.

*Remark 2.4* (Local constraints are not cosets). A long read that spans a branch vertex  $v$  fixes which out-arc of  $v$  is the tree arc, selecting  $S_e = \{T \in \text{Arb}(G) : T(v) = e\}$ . If several such constraints could be combined by linear algebra in  $\mathcal{K}(G)$ , each  $S_e$ , transported to the group through the torsor, would have to be a coset of a subgroup. It is not: on the graph of ACACCAAAACCA at  $k = 2$  ( $|\mathcal{K}| = 12$ ) two of the seven non-empty constraint sets are not cosets, on CCAACAACCAAC at  $k = 3$  three of nine, on AABABBAABABB two of eight; only the single-repeat graph AABBAABBAABB ( $\mathcal{K} = (\mathbb{Z}/3)^2$ ) has every constraint a coset (check (N)). The group structure is global; the local structure of arborescences is not linear in it, and a “congruence solver” for long-read disambiguation would have to be a search over the torsor, not a system of congruences.

## 2.6 $\varphi$ X174: one global choice

Table 1.1 gives the sandpile spectrum of bacteriophage  $\varphi$ X174 (NC\_001422.1, 5386 bp):  $\mathcal{K}(G_{12}) = \mathbb{Z}/2$  and  $\mathcal{K}(G_{11}) = \mathbb{Z}/2 \oplus \mathbb{Z}/66$ . Theorem 2.1 says that the 12-mer spectrum of the phage leaves exactly *two* global reconstructions. Check (F) exhibits them.

**Proposition 2.5** ( $\varphi$ X174 at  $k = 12$ ). *The compacted 12-mer graph of  $\varphi$ X174 has two branch vertices, the repeated 12-mers*

$$A = \text{CTTCTGCCGTTT}, \quad B = \text{CGTCAAGGACTG},$$

at positions 457 and 3205 (A) and 264 and 2760 (B) of NC\_001422.1. Reading the reference from the second A, the genome is

$$\begin{aligned} & A X B U A Z B Y, \\ & X = 3217..263 \text{ (2433 nt, wrapping),} \quad U = 276..456 \text{ (181 nt),} \\ & Z = 469..2759 \text{ (2291 nt),} \quad Y = 2772..3204 \text{ (433 nt),} \end{aligned}$$

an interleaved pair. Its two reconstructions, both of length 5386 and both with the phage's 12-mer spectrum, are the reference  $A X B U A Z B Y$  and  $A X B Y A Z B U$ , the segments following the two copies of  $B$  exchanged; one chip at a branch vertex carries one to the other.

The coordinates account for the whole genome:  $2433 + 181 + 2291 + 433 + 4 \cdot 12 = 5386$ , the four unique segments and the four occurrences of the two 12-mers; the exchange permutes segments, so both reconstructions have the same length and, segment by segment, the same content.

This is precisely the situation of Theorem 1.6 at  $s = 1$ : a single repeat of multiplicity two leaves no circular ambiguity (its two threadings are rotations of one another), so a surviving  $\mathbb{Z}/2$  at  $k = L$  must come from two interleaved repeats, and the one bit it records is which of the two  $B$ -exits follows which  $A$ . The first genome ever sequenced has exactly one such pair of 12-mers, and the group finds it. At  $k = 11$  the 132 classes of  $\mathbb{Z}/2 \oplus \mathbb{Z}/66$  are the corresponding statement one level down; Chapter 3 splits the group as  $(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/11$  into a part that longer  $k$ -mers remove and a part that they do not (Table 3.1).

The biology of the one bit is worth stating, because it is not what one might hope. By the feature table of NC\_001422.1, the exchanged segment  $U$  (276..456) spans the junction of genes  $C$  (133..393) and  $D$  (390..848), and  $Y$  (2772..3204) spans the junction of the spike genes  $G$  (2395..2922) and  $H$  (2931..3917); the four repeat occurrences lie in  $C$ ,  $D$ ,  $G$  and  $H$  respectively. Exchanging a  $C/D$  junction for a  $G/H$  junction breaks the reading frames on both sides, so the alternative reconstruction is not a viable isomer of the phage. It is the misassembly that a 12-mer assembler is at risk of producing, and the group locates it exactly: one bit, two named 12-mers, two named segments.

## 2.7 Verification

The script `rotor_assembly.py` runs the eight checks of Table 2.1 in about five seconds, all passing; its output is archived as `rotor_assembly_output.txt`, and the  $\varphi$ X174 sequence is distributed with it.

| key | check                                                                                | mode    |
|-----|--------------------------------------------------------------------------------------|---------|
| (A) | Theorem 2.1(i) on six assembly graphs, $ \mathcal{K}  \leq 27$ : free and transitive | exact   |
| (B) | Theorem 2.1(ii): the walk is a bijection onto the enumerated circuits                | exact   |
| (M) | Section 2.4: chips are spectrum-preserving rearrangements                            | exact   |
| (U) | Smith-coordinate sampler uniform, $\chi^2 = 31.2$ on 26 df                           | sampled |
| (W) | Wilson's sampler agrees; the chip word between two samples                           | sampled |
| (F) | Proposition 2.5                                                                      | exact   |
| (J) | junctions re-threaded per chip on $\varphi$ X174 at $k = 11$ : 0 to 6                | exact   |
| (N) | Remark 2.4: constraint sets are not cosets on 3 of 4 graphs                          | exact   |

Table 2.1: The verification battery (`rotor_assembly.py`), 8/8.

## 2.8 Scope

*Quoted.* The torsor and the rotor proof of BEST [40]; uniform sampling [25, 73, 86, 18] and its bioinformatic use [3, 47, 84]; counting [1, 51]; the spectrum of  $\varphi$ X174 and the interleaving theorem (Table 1.1, Theorem 1.6).

*New here, as far as we can determine.* Theorem 2.1(iii) as a statement about assemblies; chip-firing as rearrangement and the word metric of Proposition 2.3; the coordinate form of the sampler; and Proposition 2.5.

*Open.* (a) *Double strands.* Real assembly graphs are bidirected [46, 59]. The reverse-complement map sends an arc  $u \rightarrow v$  to  $\bar{v} \rightarrow \bar{u}$ , reversing direction, so the passage from the  $k$ -mer digraph to the bidirected graph is not a covering of digraphs and the covering theory of critical groups [75] does not apply as it stands; a sandpile theory for bidirected de Bruijn graphs would have to start from the signed or bidirected Laplacian [88]; Part II takes this up. (b) What the invariant factors of a real genome's group mean position by position; Chapter 3 separates the length-driven factors from the arrangement factors, and goes no further. (c) Whether the directed distance of Proposition 2.3 relates to any edit distance between assemblies; since a single chip can re-thread many junctions (Section 2.4), it is not the transposition distance, and any relation would have to pass through the structure of rotor walks on the compacted graph.

## Provenance

Theorem 2.1 is a composition of published results and is verified, not reproved; Proposition 2.3 is immediate from it; Proposition 2.5 is an exact computation reproducible in seconds. We searched for prior use of the sandpile torsor on genome graphs and for any critical-group treatment of bidirected de Bruijn graphs, and found neither; a search cannot establish a negative.

## Chapter 3

# The sandpile group of a de Bruijn graph splits along its repeats

**Summary.** Chapter 1 showed that the sandpile group  $\mathcal{K}(G_k)$  of a genome’s de Bruijn graph records repeat length, multiplicity and interleaving — a single repeat of length  $L$  and multiplicity  $r$  contributes  $(\mathbb{Z}/r)^{L-k}$ , two interleaved repeats add a  $\mathbb{Z}/2$ , three can add a  $\mathbb{Z}/3$  — and showed that the correction is not a rank of the chord diagram. Here we prove the general statement. *Bundle-chain lemma:* in any balanced multidigraph, a chain  $v_1 \rightarrow \cdots \rightarrow v_s$  in which every vertex sends all  $r$  of its arcs to the next can be contracted to a point, and the sandpile group splits off a direct summand  $(\mathbb{Z}/r)^{s-1}$  exactly. The proof is a unimodular change of variables and takes ten lines; the  $r = 1$  case is the familiar unitig contraction. Contracting every bundle chain leaves a *skeleton* with one vertex per repeat family and one arc per adjacency of repeat occurrences around the genome — a line-digraph dual of the repeat graph of Pevzner, Tang and Tesler — and

$$\mathcal{K}(G_k) \cong \bigoplus_{\text{chains } c} (\mathbb{Z}/r_c)^{\ell_c-1} \oplus \mathcal{K}(\text{Sk}).$$

The skeleton is canonical, and the repeat graph itself is the case in which every chain has length two. The first summand is what longer  $k$ -mers remove; the second, the sandpile group of the repeat-adjacency skeleton, is what no  $k$  removes. For repeats of multiplicity two the second summand is the critical group of the one-vertex map whose edges are the repeats: it is  $\text{coker}(A + I)$  for Bouchet’s signed interlace matrix  $A$  (Merino–Moffatt–Noble), its order is the interlace polynomial at 1 (Arratia–Bollobás–Sorkin), and — as their example shows and we recompute — it is not a function of the unsigned chord-crossing graph, although it happens to be for up to five repeats. For  $\varphi\text{X174}$  the decomposition is verified against direct computation at  $k = 10, 11, 12$  and extends the computable spectrum to  $k = 9$ , where the compacted graph (146 branch vertices) exceeded our exact Smith-form range but its skeleton does not:  $\mathcal{K}(G_9) = (\mathbb{Z}/2)^{42} \oplus (\mathbb{Z}/4)^4 \oplus \mathbb{Z}/7 \oplus \cdots \oplus \mathbb{Z}/2707831823$ , of which  $(\mathbb{Z}/2)^{36}$  is length-driven and the rest is the arrangement of repeats. Seven checks, all exact.

### 3.1 Introduction

Chapter 1 computed the sandpile group  $\mathcal{K}(G_k)$  in three model situations: a single repeat of length  $L$  and multiplicity  $r$  gives  $(\mathbb{Z}/r)^{L-k}$  (Theorem 1.5); two repeats of multiplicity two give  $(\mathbb{Z}/2)^{2s-2}$  if they are disjoint or nested and  $(\mathbb{Z}/2)^{2s-1}$  if interleaved,  $s = L - k + 1$  (Theorem 1.6); three repeats give a correction that is  $\mathbb{Z}/2$ ,  $\mathbb{Z}/3$  or  $(\mathbb{Z}/2)^2$  according to their crossing pattern

(Proposition 1.8); since the three non-trivial patterns have intersection matrices of the same rank over  $\mathbb{F}_2$ , the correction is not a function of that rank (Remark 3.10 sharpens this). Chapter 2 used the group as a coordinate system on reconstructions and found that for  $\varphi$ X174 at  $k = 12$  the single surviving  $\mathbb{Z}/2$  is a pair of interleaved 12-mers (Proposition 2.5).

Those three model computations are special cases of one structural fact, which this chapter proves. The fact is local and holds in any balanced multidigraph; its consequence for genomes is that the sandpile group splits as a direct sum of a *length-driven* part, one cyclic factor per shared  $k$ -mer of each repeat, and an *arrangement* part, the sandpile group of a repeat-adjacency skeleton.

**What is known.** Contracting a vertex of in- and out-degree one leaves the sandpile group unchanged (Lemma 1.4); this is the  $r = 1$  case of Lemma 3.2 and is the reason assemblers' compacted (unitig) graphs [9, 46] carry the same group. For *undirected* graphs with multiple edges the sandpile groups of thick trees [21] and thick cycles [5] are known, and the thick-tree answer,  $\bigoplus \mathbb{Z}/m_e$  over the edge multiplicities, looks like our formula; but the undirected analogue of Lemma 3.2 is false (Remark 3.4), so the lemma is not a corollary of those results. The nearest directed relative is Levine's theorem that  $\mathcal{K}(G)$  is a quotient of  $\mathcal{K}(LG)$  for the line digraph [55, 70]. The Eulerian-path formulation of assembly is due to Pevzner, Tang and Waterman [72], and the *repeat graph*, in which repeat families are edges of multiplicity  $> 1$  and vertices are repeat boundaries, to Pevzner, Tang and Tesler [71]; it is the central object of the long-read assembler Flye [31]. Our skeleton has one *vertex* per repeat family and one arc per adjacency of occurrences, so it stands to the repeat graph roughly as a line digraph to its graph; we keep the two apart by name. For two-copy repeats the skeleton is the directed medial graph of a one-vertex map, and its sandpile group is the critical group of that map in the sense of Merino, Moffatt and Noble [61], which is where the question of what determines it has already been answered (Remark 3.10). We have not found the  $r > 1$  splitting of Lemma 3.2 in the sandpile literature, nor the sandpile group of the repeat graph considered as such; a search cannot establish a negative.

## 3.2 Setting

All digraphs are finite multidigraphs, balanced ( $d^-(v) = d^+(v)$  at every vertex) and connected;  $\Delta = D^+ - A$  is the out-Laplacian and  $\mathcal{K}(G)$  the sandpile group of Section 1.2, with the sink conventions of Section 2.2. We compare finite abelian groups by their elementary divisors.

**Definition 3.1** (Bundle chain). A *bundle chain* of multiplicity  $r$  and length  $s \geq 2$  is a sequence of distinct vertices  $v_1, \dots, v_s$  such that for each  $t < s$  the vertex  $v_t$  has out-degree  $r$  and all  $r$  of its out-arcs go to  $v_{t+1}$ , and  $v_s$  has out-degree  $r$ . Since  $G$  is balanced, every in-arc of  $v_{t+1}$  ( $t < s$ ) comes from  $v_t$ . The *contraction*  $G/C$  replaces the chain by a single vertex carrying the in-arcs of  $v_1$  and the out-arcs of  $v_s$ ; it is again balanced, and an arc  $v_s \rightarrow v_1$  becomes a loop, which does not affect  $\Delta$ .

## 3.3 The lemma

**Lemma 3.2** (Bundle-chain contraction). *Let  $C = (v_1, \dots, v_s)$  be a bundle chain of multiplicity  $r$  in a balanced multidigraph  $G$  with at least one vertex outside  $C$ . Then*

$$\mathcal{K}(G) \cong (\mathbb{Z}/r)^{s-1} \oplus \mathcal{K}(G/C).$$

*Proof.* Root at a vertex outside  $C$  and write  $x_v$  for the generator of the cokernel presentation at each non-root vertex, the relations being the rows of  $\Delta$ . Change generators on the chain:

$$z := x_{v_1}, \quad y_t := x_{v_t} - x_{v_{t+1}} \quad (1 \leq t < s),$$

so that  $x_{v_t} = z - \sum_{u < t} y_u$  and in particular  $x_{v_s} = z - \sum_{u < s} y_u$ ; this is a unimodular (triangular) change of basis. The rows of  $\Delta$  fall into three kinds.

- The row of  $v_t$  for  $t < s$  is  $r x_{v_t} - r x_{v_{t+1}} = r y_t$ .
- The row of  $v_s$  is  $r x_{v_s} - \sum_w c_{v_s w} x_w = r z - r \sum_{u < s} y_u - \sum_w c_{v_s w} x_w$ . Subtracting the rows  $r y_u$  — a unimodular row operation — leaves  $r z - \sum_w c_{v_s w} x_w$ , which is the row of the contracted vertex in  $\Delta(G/C)$ .
- The row of a vertex  $u \notin C$  involves  $x_{v_t}$  only through arcs from  $u$  into the chain, and all of those land on  $v_1$ ; so it reads  $d^+(u)x_u - \sum_w c_{uw} x_w$  with  $x_{v_1} = z$ , which is its row in  $\Delta(G/C)$ .

The relation lattice is therefore generated by  $\{r y_t\}_{t < s}$  together with the rows of  $\Delta(G/C)$  in the variables  $z$  and  $x_u$ , and the generators separate accordingly. Hence  $\mathcal{K}(G) \cong \bigoplus_{t < s} \mathbb{Z}/r \oplus \mathcal{K}(G/C)$ .  $\square$

If  $G$  consists of the chain alone — the only vertices are  $v_1, \dots, v_s$  and the  $r$  arcs of  $v_s$  return to  $v_1$  — the same computation rooted at  $v_1$  gives  $\mathcal{K}(G) \cong (\mathbb{Z}/r)^{s-1}$ ; this is Theorem 1.5.

*Remark 3.3.* For  $r = 1$  the summand is trivial and the lemma says that smoothing a vertex of in- and out-degree one preserves the group, which is Lemma 1.4. The content of the lemma for  $r > 1$  is that a vertex which *cannot* be smoothed — it has  $r$  parallel exits — can nevertheless be removed at the exact cost of a  $\mathbb{Z}/r$ .

*Remark 3.4* (The direction is essential). The undirected statement one might guess — a path of doubled edges contributes  $(\mathbb{Z}/2)^{\text{length}-1}$  — is false. The undirected 4-cycle with every edge doubled has reduced Laplacian  $2M$  with  $M$  the reduced Laplacian of the simple 4-cycle, whose Smith form is  $\text{diag}(1, 1, 4)$ ; so its sandpile group is  $(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/8$ , whereas the directed doubled 4-cycle is a single bundle chain and has  $(\mathbb{Z}/2)^3$ . What makes the directed lemma work is that every arc *into* the chain lands on its head  $v_1$  and every arc *out* of it leaves from  $v_s$  with the full multiplicity  $r$ ; an undirected doubled edge carries flow both ways and the change of variables in the proof no longer separates.

### 3.4 The splitting theorem

Contracting bundle chains one after another terminates, since each contraction reduces the number of vertices, and the result contains no bundle chain of length  $\geq 2$ . We call it the *skeleton*  $\text{Sk}(G)$ . Its sandpile group is well defined by the lemma; the skeleton itself is too.

**Lemma 3.5** (The skeleton is canonical). *Call  $v \rightarrow w$  a chain step if  $v \neq w$ , all out-arcs of  $v$  go to  $w$ , and  $d^+(w) = d^+(v)$ . In a balanced multidigraph the chain steps form a partial injection on the vertex set, so the maximal bundle chains are its maximal paths: they are pairwise vertex-disjoint and determined by  $G$ . Contracting them in any order gives the same digraph up to isomorphism, which is chain-free.*

*Proof.* If  $v \rightarrow w$  is a chain step then  $w$  receives  $d^+(v) = d^+(w) = d^-(w)$  arcs from  $v$ , hence all of its in-arcs, so no  $v' \neq v$  steps to  $w$ ; and  $v$  has a single target. Thus the step relation is a partial injection and its maximal paths partition the vertices it touches (a path may close into a cycle only if  $G$  is a single closed chain, when the skeleton is one vertex with  $r$  loops). Contracting a maximal chain  $C$  to a vertex  $c$  changes no degree outside  $C$ , gives  $c$  the in-arcs of  $v_1$  and the out-arcs of  $v_s$ , and so creates no new chain step:  $u \rightarrow c$  would require  $u \rightarrow v_1$  to have been a step and  $c \rightarrow w$  would require  $v_s \rightarrow w$  to have been one, both excluded by maximality. Nor does it destroy a step outside  $C$ . Hence the contractions of distinct maximal chains commute, and the result is  $G$  with every maximal chain contracted.  $\square$

Check (E) verifies this on 100 random balanced digraphs with planted chains: the partition into maximal chains, the multiset of  $(r_c, \ell_c)$  and the skeleton's group are invariant under random relabelling, and the skeleton contains no chain step.

**Theorem 3.6** (Splitting). *For any balanced connected multidigraph  $G$ , with  $\{(r_c, \ell_c)\}$  the multiplicities and lengths of the chains contracted on the way to a skeleton,*

$$\mathcal{K}(G) \cong \bigoplus_c (\mathbb{Z}/r_c)^{\ell_c-1} \oplus \mathcal{K}(\text{Sk}(G)), \quad |\mathcal{K}(G)| = \prod_c r_c^{\ell_c-1} \cdot |\mathcal{K}(\text{Sk}(G))|.$$

*Proof.* Induction on the number of contractions, using Lemma 3.2 at each step.  $\square$

Now let  $S$  be a sequence with repeats  $W_1, \dots, W_m$  of lengths  $L_i$  and multiplicities  $r_i$ , and suppose the *clean* hypothesis of Theorem 1.5: the  $k$ -mers inside  $W_i$  occur exactly  $r_i$  times each and every other  $k$ -mer once. Write  $w$  for the circular word of repeat occurrences and  $D_w$  for the *connector digraph*: one vertex per repeat, one arc  $i \rightarrow j$  for each occurrence of  $W_i$  immediately followed around the circle by an occurrence of  $W_j$  (loops deleted).  $D_w$  is balanced with out-degree  $r_i$  at vertex  $i$ .

**Corollary 3.7** (Clean sequences). *Under the clean hypothesis, with  $s_i = L_i - k + 1$ ,*

$$\mathcal{K}(G_k(S)) \cong \bigoplus_{i=1}^m (\mathbb{Z}/r_i)^{s_i-1} \oplus \mathcal{K}(D_w).$$

*Proof.* After unitig contraction the graph consists of the  $m$  spines — bundle chains of multiplicity  $r_i$  and length  $s_i$  — joined by the connector arcs; contracting the spines gives  $D_w$ . Apply Theorem 3.6.  $\square$

**Corollary 3.8** (The cases of Chapter 1). *For  $m = 1$ ,  $D_w$  is a single vertex and  $\mathcal{K} = (\mathbb{Z}/r)^{s-1}$  (Theorem 1.5). For  $m = 2$ ,  $r_1 = r_2 = 2$ :  $D_w$  is trivial for  $A A B B$  and  $A B B A$  and is the doubled 2-cycle, with  $\mathcal{K}(D_w) = \mathbb{Z}/2$ , for  $A B A B$  (Theorem 1.6). For  $m = 3$  the four crossing patterns give  $\mathcal{K}(D_w) = 0, \mathbb{Z}/2, \mathbb{Z}/3, (\mathbb{Z}/2)^2$  (Proposition 1.8); the  $\mathbb{Z}/3$  is the sandpile group of the complete symmetric digraph on three vertices, which is  $D_w$  for  $A B A C B C$ .*

**Remark 3.9** (The skeleton and the repeat graph). The duality with the repeat graph of [71, 31] can be made exact. Under the clean hypothesis, collapse each spine of the compacted graph to a single arc: the result is the multidigraph  $\text{RG}_w$  with two vertices  $h_i, t_i$  per family,  $r_i$  parallel arcs  $h_i \rightarrow t_i$ , and one arc  $t_i \rightarrow h_j$  per adjacency  $ij$  in  $w$  — repeats as multi-arcs between their boundary vertices, unique segments as single arcs, which is the repeat graph. Each pair  $(h_i, t_i)$  is a bundle chain of length two, and contracting it identifies the multi-arc with a vertex:  $D_w$  is

the transition digraph of the repeat arcs of  $\text{RG}_w$  (the line digraph, restricted to repeat arcs, with unique segments as the adjacencies). By Lemma 3.2,

$$\mathcal{K}(\text{RG}_w) \cong \bigoplus_{i=1}^m \mathbb{Z}/r_i \oplus \mathcal{K}(D_w),$$

so the sandpile group of the repeat graph is the arrangement part plus one  $\mathbb{Z}/r_i$  per family — the length-driven part at  $s_i = 2$  (check (E), 200 random words). The two objects carry the same information; we work with  $D_w$  because it has no length-driven residue.

*Remark 3.10* (Which invariant of the chord diagram). For  $r_i = 2$  the  $2m$  occurrences form a chord diagram with  $m$  chords — a double-occurrence word, or a one-vertex map (bouquet) with the repeats as edges — and  $D_w$  is a 2-in 2-out digraph. On what invariant of the diagram does  $\mathcal{K}(D_w)$  depend? Not on the rank of the unsigned intersection matrix, over  $\mathbb{F}_2$  or over  $\mathbb{Z}$ , and not on its integer Smith form: for  $m = 3$  the three non-trivial patterns all have  $\mathbb{F}_2$ -rank 2, and  $A A B C B C$  (one crossing) and  $A B A C B C$  (path) share the Smith form  $(1, 1, 0)$  while their groups are  $\mathbb{Z}/2$  and  $\mathbb{Z}/3$ ; for  $m = 5$  the rank-2 diagrams carry nine different groups, among them  $\mathbb{Z}/7$  and  $\mathbb{Z}/9$  (check (D)). The *order*  $|\mathcal{K}(D_w)|$  is the number of Eulerian circuits of  $D_w$ , and this is a function of the intersection graph  $H$  alone (the *interlace graph* of [2], whose Definition 1 is exactly chord crossing): it is the interlace polynomial  $q(H; 1)$  [2, Thm. 9, Cor. 13], with roots in Martin’s and Las Vergnas’ circuit-partition polynomials [52]. Exhaustively for  $m \leq 5$  the group is also constant on the 4, 11 and 34 isomorphism classes of  $H$  — but this is an accident of small  $m$ . Merino, Moffatt and Noble [61, Rem. 6.7] give two 6-chord bouquets,  $1\,2\,3\,4\,5\,6\,1\,3\,2\,4\,6\,5$  and  $1\,2\,3\,6\,5\,4\,1\,3\,2\,5\,6\,4$ , with the same interlace graph ( $K_6$  minus two disjoint edges, so  $|\mathcal{K}| = 18$  for both) and critical groups  $\mathbb{Z}/3 \oplus \mathbb{Z}/6$  and  $\mathbb{Z}/18$ ; check (D) recomputes both from  $D_w$  and exhibits two further such pairs. What does determine the group is the *signed* interlace matrix  $A$  of Bouchet ( $A_{ef} = \pm 1$  according as the occurrences run  $e\,f\,e\,f$  or  $f\,e\,f\,e$ , 0 if not interlaced): by [61, Thm. 7.3], whose directed medial graph of a bouquet is our  $D_w$ ,

$$\mathcal{K}(D_w) \cong \text{coker}(A + I),$$

the critical group of the map; its order  $\det(A + I)$  as an Euler-circuit count is older [58, 43]. Check (D) confirms the isomorphism on every double-occurrence word with  $m \leq 5$  and on 1200 random words with  $6 \leq m \leq 9$ . So the arrangement part of an assembly’s ambiguity, for two-copy repeats, is the critical group of the bouquet whose edges are the repeats, computable from a signed  $m \times m$  matrix rather than from the de Bruijn graph; and the reason it is not a function of the chord-crossing graph is the reason [61] record: the signs matter. For  $r_i > 2$  the occurrences form a hypergraph rather than a chord diagram, and Chapter 4 shows that no pairwise analogue exists.

*Remark 3.11* (What  $k$  can and cannot remove). Increasing  $k$  shortens every spine:  $s_i$  decreases by one per unit of  $k$ , and the length-driven summand  $\bigoplus (\mathbb{Z}/r_i)^{s_i-1}$  shrinks accordingly, vanishing at  $k = \max L_i$ . The skeleton summand  $\mathcal{K}(D_w)$  does not depend on  $k$  at all, as long as  $k \leq \min_i L_i$ . So the global ambiguity of an assembly splits into a part that longer  $k$ -mers resolve and a part that only information beyond the repeats — long reads, pairs, a reference — can resolve, and the second part is exactly the sandpile group of the repeat-adjacency skeleton. By BEST applied to  $D_w$ ,  $|\mathcal{K}(D_w)| \cdot \prod_i (r_i - 1)!$  is the number of Eulerian circuits of  $D_w$ , i.e. of circular words with the same adjacency counts as  $w$ : the number of ways to thread the repeats, ignoring their lengths. That this is  $|\mathcal{K}(D_w)|$  up to the local factor is the repeat-level form of Proposition 1.3.

### 3.5 $\varphi$ X174

Table 3.1 applies the theorem to the first sequenced genome (NC\_001422.1, 5386 bp). At each  $k$  the compacted graph is bundle-contracted; the chains, the skeleton, and the two summands are listed, and where the compacted graph is small enough the group is also computed directly for comparison. Two things happen. The direct and decomposed groups agree wherever both exist ( $k = 10, 11, 12$ ). And at  $k = 9$ , where the compacted graph has 146 branch vertices and was beyond the dense exact Smith form used for Table 1.1, the skeleton has 110 and is not: the spectrum extends by one level, with the length-driven part  $(\mathbb{Z}/2)^{36}$  split off exactly and the arrangement part read off the skeleton.

| $k$ | branch | chains | skeleton | $\mathcal{K}(\text{Sk})$ (arrangement part)                                                                                                                                                                        | length-driven part    |
|-----|--------|--------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| 12  | 2      | 1      | 1        | 0                                                                                                                                                                                                                  | $\mathbb{Z}/2$        |
| 11  | 10     | 2      | 8        | $\mathbb{Z}/3 \oplus \mathbb{Z}/11$                                                                                                                                                                                | $(\mathbb{Z}/2)^2$    |
| 10  | 47     | 8      | 37       | $\mathbb{Z}/4 \oplus \mathbb{Z}/1076338579$                                                                                                                                                                        | $(\mathbb{Z}/2)^{10}$ |
| 9   | 146    | 28     | 110      | $(\mathbb{Z}/2)^6 \oplus (\mathbb{Z}/4)^4 \oplus \mathbb{Z}/7 \oplus \mathbb{Z}/523 \oplus \mathbb{Z}/3779$<br>$\oplus \mathbb{Z}/5297 \oplus \mathbb{Z}/6561 \oplus \mathbb{Z}/7669 \oplus \mathbb{Z}/2707831823$ | $(\mathbb{Z}/2)^{36}$ |
| 8   | 401    | 65     | 312      | (skeleton beyond the exact-SNF range used)                                                                                                                                                                         |                       |

Table 3.1:  $\varphi$ X174 under bundle-chain contraction: number of branch vertices of the compacted graph, number of bundle chains contracted, number of skeleton vertices, and the two summands.  $\mathcal{K}(G_k)$  is their direct sum; at  $k = 10, 11, 12$  this agrees with the group computed directly on the compacted graph (check (F)). All chains have multiplicity 2. All entries exact.

The reading at  $k = 11$  is the cleanest:  $\mathcal{K}(G_{11}) = (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/11$ , and the theorem says which is which — the two  $\mathbb{Z}/2$ 's are two repeat families each with one extra shared 11-mer beyond the minimum, and  $\mathbb{Z}/3 \oplus \mathbb{Z}/11$ , of order 33, is the arrangement ambiguity of the eight-vertex skeleton, which no choice of  $k \leq 11$  can reduce.

### 3.6 Verification

The script `repeat_splitting.py` runs the seven checks of Table 3.2 in about thirty seconds, all passing; its output is archived as `repeat_splitting_output.txt`. All arithmetic is exact; groups are compared by elementary divisors.

| key | check                                                                                  | mode  |
|-----|----------------------------------------------------------------------------------------|-------|
| (A) | Corollary 3.7 on 120 clean random sequences, $m \leq 4$ , $r_i \leq 4$                 | exact |
| (B) | Lemma 3.2 on 60 random balanced digraphs with a planted chain, $r \leq 6$ , $s \leq 4$ | exact |
| (C) | Corollary 3.8: the nine values of Chapter 1 are the formula on $D_w$                   | exact |
| (D) | Remark 3.10: all double-occurrence words, $m \leq 5$ ; rank, Smith form, graph         | exact |
| (E) | Lemma 3.5 on 100 relabelled digraphs; Remark 3.9 on 200 words                          | exact |
| (F) | Table 3.1: decomposition vs. direct computation; extension to $k = 9$                  | exact |

Table 3.2: The verification battery, 7/7 (check (E) has two rows).

### 3.7 Scope

*Proved.* Lemma 3.2, Theorem 3.6 and its corollaries, for arbitrary balanced multidigraphs and arbitrary multiplicities; the clean hypothesis enters only in identifying the skeleton with  $D_w$ .

*Quoted.* Sink-independence [40]; BEST [10, 78]; the Eulerian formulation and the repeat graph [72, 71, 31]; the thick-graph and line-digraph results that are the lemma’s nearest relatives [21, 5, 55, 70]; the special cases and the  $\varphi$ X174 spectrum (Chapters 1 and 2).

*Open.* (a) A structure theorem for  $\mathcal{K}(D_w)$  itself — for which circular words is it trivial, and what bounds its exponent? For  $r_i = 2$  this is the critical group of a bouquet (Remark 3.10), and [61] note that for 3-connected regular delta-matroids the group is well defined on the delta-matroid; whether  $\mathcal{K}(D_w)$  is determined by the interlace graph whenever that graph is prime (no split) we have not examined. For  $r_i > 2$  nothing of the kind is available (Chapter 4). (b) Real genomes are not clean: the skeleton of  $\varphi$ X174 at  $k = 9$  has 110 vertices and carries factors as large as 2707831823. All six factors above 100 in Table 3.1 (523, 3779, 5297, 7669, 1076338579, 2707831823) are prime, which is what the sandpile group of a generic digraph looks like (typically cyclic, or nearly so); we see no reason to expect a topological reading of a large prime cyclic factor and claim none. (c) Double strands: reverse complementation reverses arc direction, so a bidirected version of Lemma 3.2 must start from the signed Laplacian of Chapter 6; we have not attempted it.

### Provenance

Lemmas 3.2 and 3.5 are proved in full above and are the only new mathematics; Remark 3.10 is a translation of published results [2, 61] into the assembly setting, verified computationally — an earlier draft of this chapter conjectured that the group was an invariant of the unsigned interlace graph, which [61, Rem. 6.7] had already shown to be false; everything else is either quoted or an exact computation reproducible in seconds. We searched for the  $r > 1$  splitting in the sandpile literature and for any use of the sandpile group of a repeat graph or repeat-adjacency skeleton and found neither; the undirected thick-graph results [21, 5] are the closest, and Remark 3.4 shows they do not imply the lemma.



## Chapter 4

# Multi-copy repeats: where the two-copy theory of assembly ambiguity stops

**Summary.** The sandpile group of a genome’s de Bruijn graph splits into a length-driven part and the group  $\mathcal{K}(D_w)$  of the transition digraph of the circular word of repeat occurrences Chapter 3. For repeats of multiplicity two,  $\mathcal{K}(D_w)$  is the critical group of a one-vertex map:  $\text{coker}(A + I)$  for Bouchet’s signed interlace matrix, by a theorem of Merino, Moffatt and Noble — a presentation whose entries are *pairwise* data about the repeats. This chapter asks what survives for repeats with three or more copies. Two families of any multiplicities  $r, s$  give  $\mathcal{K}(D_w) = \mathbb{Z}/c$ , where  $c$  is the number of transitions from one family to the other,  $1 \leq c \leq \min(r, s)$ ; this contains the interleaving theorem for two-copy repeats. Beyond two families nothing pairwise survives: for three-copy repeats, two words with identical pairwise occurrence patterns already have groups  $\mathbb{Z}/3$  and  $\mathbb{Z}/4$  at  $m = 3$ , so no matrix with pairwise entries can present  $\mathcal{K}(D_w)$ ; detaching a three-copy vertex into two-copy vertices does not control the group; and the literature has no interlacement formula for  $r$ -in  $r$ -out digraphs with  $r \geq 3$ . A census of the groups realised by three- and four-copy arrangements is given (9, 17 and 28 groups for  $(3, 3, 3)$ ,  $(4, 4, 4)$  and  $(3, 3, 3, 3)$ , up to  $(\mathbb{Z}/3)^3$ ). On  $\varphi$ X174 the skeleton is 2-in 2-out at  $k = 11, 10$ , where the signed interlace matrix reproduces the group exactly, and acquires vertices of out-degree 3 to 6 from  $k = 9$  down: the two-copy presentation stops exactly where the arrangement group becomes large. Four checks, all exact.

### 4.1 Introduction

Let  $S$  be a circular sequence with repeat families  $W_1, \dots, W_m$  of multiplicities  $r_i$ , clean in the sense of Corollary 3.7, and let  $w$  be the circular word of their occurrences. The splitting theorem (Corollary 3.7) gives

$$\mathcal{K}(G_k(S)) \cong \bigoplus_i (\mathbb{Z}/r_i)^{s_i-1} \oplus \mathcal{K}(D_w),$$

where  $D_w$  has one vertex per family and one arc per adjacency of occurrences. The first summand is what longer  $k$ -mers remove. The second, the *arrangement group*, is what no  $k$  removes, and for  $r_i = 2$  it has a striking description:  $D_w$  is the directed medial graph of a bouquet whose edges are the repeats, and by [61, Thm. 7.3]  $\mathcal{K}(D_w) \cong \text{coker}(A + I)$  with  $A$  the skew-symmetric  $\{0, \pm 1\}$  matrix recording, for each pair of repeats, whether and in which order their occurrences interlace.

The entries of  $A$  are pairwise data; its determinant is the interlace polynomial at 1 [2], the number of Euler circuits of  $D_w$  [58, 43].

Real genomes have repeats with more than two copies. In  $\varphi$ X174 the skeleton of Chapter 3 is 2-in 2-out at  $k = 11$  and  $k = 10$  but has vertices of out-degree 3 and 4 at  $k = 9$  and up to 6 at  $k = 7$  (Table 4.2). This chapter asks whether the two-copy description extends. The answer is: only in the smallest case.

**What is known.** For 2-in 2-out digraphs the determinant formula  $\# \text{Euler circuits} = \det(I + A)$  is due to Macris and Pulé [58], with elementary proofs by Lauri and Jonsson [43]; the group statement is [61]. Jonsson [43, p. 192] refers to an unpublished report generalising the intersection matrix to arbitrary Eulerian digraphs, where the determinant counts arborescences rather than circuits; we could not locate it. Lauri’s own attempt at extensions [53] is described by its reviewer as inconclusive. For rotor-routing the picture is different: Tóthmérés [79] constructs a canonical simply transitive action of the sandpile group on compatible Euler tours of any Eulerian ribbon digraph, whatever the degrees, so the torsor of Theorem 2.1 generalises even though, as we show, the matrix presentation does not.

## 4.2 Setting

As in Section 3.4,  $D_w$  is the transition digraph of the circular word  $w$ : vertices the letters, one arc  $e \rightarrow f$  per occurrence of  $e$  immediately followed by  $f$  around the circle; loops are deleted (they do not affect the Laplacian). It is balanced with  $d^\pm(e) = r_e$ . Its sandpile group is  $\mathcal{K}(D_w) = \text{coker}(\Delta_s^T)$  with  $\Delta = D^+ - T$ ,  $T_{ef}$  the number of transitions  $e \rightarrow f$ ; as throughout Part I, the sink does not matter and groups are compared by elementary divisors. The *pairwise pattern* of two letters  $e, f$  is the circular binary word obtained by deleting all other letters from  $w$ , taken up to rotation.

## 4.3 Two families

**Theorem 4.1** (Two families). *Let  $w$  be a circular word over  $\{A, B\}$  with  $r$  copies of  $A$  and  $s$  of  $B$ , and let  $c$  be the number of transitions  $A \rightarrow B$  (equivalently the number of maximal blocks of  $A$ ’s). Then  $1 \leq c \leq \min(r, s)$ , every such  $c$  occurs, and*

$$\mathcal{K}(D_w) \cong \mathbb{Z}/c.$$

*Proof.*  $D_w$  has two vertices; the number of arcs  $A \rightarrow B$  equals the number of arcs  $B \rightarrow A$  (each block of  $A$ ’s is entered once and left once), namely  $c$ , and the remaining arcs are loops. Reduced at  $B$ , the Laplacian is the  $1 \times 1$  matrix  $(c)$ . The word  $A^{r-c+1}B^{s-c+1}(AB)^{c-1}$  realises  $c$ , and  $c \leq \min(r, s)$  since each block contains at least one letter.  $\square$

For  $r = s = 2$  this is Theorem 1.6:  $c = 1$  for  $A A B B$  and  $A B B A$  (disjoint or nested, no arrangement ambiguity) and  $c = 2$  for  $A B A B$ . For two families the arrangement ambiguity is thus a single cyclic factor whose order is a crossing number, and no sign is needed. Check (A) verifies the theorem over all 119 circular words with  $r, s \leq 5$ .

## 4.4 Beyond two families nothing pairwise survives

**Proposition 4.2** (Pairwise patterns do not determine the group). *The circular words*

$$w_1 = A A B A C C B B C, \quad w_2 = A A B A B C B C C,$$

*each letter three times, have the same pairwise pattern for every pair of letters, and*

$$\mathcal{K}(D_{w_1}) = \mathbb{Z}/3, \quad \mathcal{K}(D_{w_2}) = \mathbb{Z}/4.$$

*Consequently no square matrix whose  $(e, f)$  entry is a function of the pairwise pattern of  $e$  and  $f$  can present  $\mathcal{K}(D_w)$  for three-copy repeats, even for  $m = 3$ .*

*Proof.* Restricting to  $\{A, B\}$ ,  $\{A, C\}$  and  $\{B, C\}$  gives, for both words, the cyclic patterns 001101, 000111 and 001101 (with the first letter of the pair written as 1). In  $w_1$  all six off-diagonal transition counts are 1, so the Laplacian reduced at  $C$  is  $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ , of determinant 3 and content 1:  $\mathbb{Z}/3$ . In  $w_2$  the transitions are  $A \rightarrow B$  twice,  $A \rightarrow C$  never,  $B \rightarrow A$  once,  $B \rightarrow C$  twice, giving  $\begin{pmatrix} 2 & -2 \\ -1 & 3 \end{pmatrix}$ , of determinant 4 and content 1:  $\mathbb{Z}/4$ .  $\square$

**Remark 4.3** (Why this is sharper than the two-copy failure). For two-copy repeats the pairwise pattern of  $e, f$  is a single bit (interlaced or not), and the group is likewise not a function of these bits alone — but the first failure is at  $m = 6$  [61, Rem. 6.7], and it is repaired by a sign: reading from a base point, record whether the occurrences run  $e f e f$  or  $f e f e$ . The sign depends on the base point, and the cokernel does not, because changing base point acts by pivots [61, Lem. 4.8]. For three or more copies the analogous base-pointed pairwise data — the linear restriction of  $w$  to each pair — determine the whole word for every  $m$  (they fix the relative order of any two occurrences), so “the group is a function of them” is vacuous, and the only genuine pairwise invariants are the rotation classes of Proposition 4.2, which fail already at  $m = 3$ . There is no room between the two for a sign.

**Proposition 4.4** (No reduction to two copies). *Let  $G$  be a balanced digraph with a vertex  $v$  of in- and out-degree 3, and let  $G'$  be obtained by detaching  $v$  into  $v'$  (two of the in-arcs, one of the out-arcs) and  $v''$  (the remaining in-arc, the other two out-arcs) joined by a new arc  $v' \rightarrow v''$ , so that  $G'$  is 2-in 2-out at both. Then  $\mathcal{K}(G)$  is in general neither a quotient nor a subgroup of  $\mathcal{K}(G')$ : there is a  $G$  with  $\mathcal{K}(G) = \mathbb{Z}/4 \oplus \mathbb{Z}/5$  whose nine detachments have groups  $\mathbb{Z}/23, \mathbb{Z}/25, \mathbb{Z}/31, \mathbb{Z}/32, \mathbb{Z}/2 \oplus \mathbb{Z}/13, \mathbb{Z}/3 \oplus \mathbb{Z}/8, \mathbb{Z}/2 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/4$  and  $\mathbb{Z}/2 \oplus \mathbb{Z}/3 \oplus \mathbb{Z}/5$ .*

Over 150 random balanced digraphs and all nine detachments each,  $\mathcal{K}(G)$  is a quotient of  $\mathcal{K}(G')$  in 140 of 1350 cases (check (C)). Euler circuits of  $G'$  are those of  $G$  compatible with the chosen transition at  $v$ , a subset of no fixed proportion, and the group follows the count. So the arrangement group of a multi-copy skeleton cannot be computed by splitting it into a two-copy one and applying [61].

**Remark 4.5** (What does determine it). Trivially, the transition matrix  $T$ : the arrangement group is the sandpile group of the transition digraph, which is Corollary 3.7. The content of the two-copy theory is that  $T$  can be traded for interlacement — a different, topological parametrisation of the same group. Propositions 4.2 and 4.4 say that for three or more copies no such trade is available through pairwise data or through two-copy surrogates. A Gram-matrix route (the form  $x^T(I - S)y$  of the position cycle restricted to consecutive-occurrence differences) was also tried and is not the Merino–Moffatt–Noble matrix even for two copies; it is recorded in the README of the chapter’s folder in the repository so that it is not retried.

## 4.5 Census

Table 4.1 lists, for each multiplicity vector, the number of circular words (one per rotation class), the number of distinct transition digraphs up to relabelling, the number of distinct groups, the largest order, and the non-cyclic groups that occur. The last column counts the classes of pairwise patterns and, in parentheses, how many of them carry more than one group. For two families the groups are exactly  $\mathbb{Z}/c$ ,  $c \leq \min(r, s)$ , as Theorem 4.1 predicts, and every pairwise class is pure. From three families on the arrangement group is far richer than for two-copy repeats — three two-copy repeats give only  $0, \mathbb{Z}/2, \mathbb{Z}/3, (\mathbb{Z}/2)^2$  (Proposition 1.8), three three-copy repeats give nine groups up to order 9 — and most pairwise classes are mixed.

| mult.        | words | $D_w$ | groups | $\max  \mathcal{K} $ | non-cyclic groups                                                                                                              | pairwise (mixed) |
|--------------|-------|-------|--------|----------------------|--------------------------------------------------------------------------------------------------------------------------------|------------------|
| (3, 3)       | 4     | 3     | 3      | 3                    | —                                                                                                                              | 3 (0)            |
| (4, 4)       | 10    | 4     | 4      | 4                    | —                                                                                                                              | 7 (0)            |
| (3, 3, 3)    | 188   | 12    | 9      | 9                    | $(\mathbb{Z}/2)^2, (\mathbb{Z}/3)^2$                                                                                           | 15 (6)           |
| (3, 3, 2)    | 70    | 15    | 7      | 6                    | $(\mathbb{Z}/2)^2$                                                                                                             | 10 (4)           |
| (2, 2, 3)    | 30    | 10    | 5      | 4                    | $(\mathbb{Z}/2)^2$                                                                                                             | 6 (2)            |
| (4, 4, 4)    | 2896  | 26    | 17     | 16                   | $(\mathbb{Z}/2)^2, (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/3, (\mathbb{Z}/3)^2, (\mathbb{Z}/4)^2, \mathbb{Z}/2 \oplus \mathbb{Z}/4$ | 174 (86)         |
| (3, 3, 3, 3) | 30804 | 96    | 28     | 27                   | those, and $(\mathbb{Z}/2)^3, (\mathbb{Z}/3)^3, \mathbb{Z}/2 \oplus (\mathbb{Z}/3)^2$                                          | 180 (132)        |

Table 4.1: Arrangement groups of multi-copy repeats, exhaustive over circular words (check (B)).  $D_w$ : distinct transition digraphs. All entries exact.

## 4.6 $\varphi$ X174

Table 4.2 gives the out-degree profile of the bundle-contracted skeleton of  $\varphi$ X174 (NC\_001422.1) at  $k = 12, \dots, 7$ . At  $k = 11$  and  $k = 10$  every skeleton vertex has in- and out-degree 2, and the signed interlace matrix of an Euler circuit found by Hierholzer’s algorithm gives  $\text{coker}(A + I) = \mathbb{Z}/3 \oplus \mathbb{Z}/11$  and  $\mathbb{Z}/4 \oplus \mathbb{Z}/1076338579$ , the groups of Table 3.1, computed there from the Laplacian (check (D)) — the theorem of [61] on a real genome. From  $k = 9$  down the skeleton has vertices of out-degree 3 (5 of them at  $k = 9$ , 42 at  $k = 8$ , 147 at  $k = 7$ ), 4, 5 and 6: the two-copy presentation stops applying at the same  $k$  at which the skeleton grows from 37 to 110 vertices and the group acquires its large prime factors.

| $k$ | skeleton | arcs | out-degree multiset                 |
|-----|----------|------|-------------------------------------|
| 12  | 1        | 0    | (single vertex)                     |
| 11  | 8        | 16   | $2^8$                               |
| 10  | 37       | 74   | $2^{37}$                            |
| 9   | 110      | 227  | $2^{104} 3^5 4^1$                   |
| 8   | 312      | 679  | $2^{264} 3^{42} 4^5 5^1$            |
| 7   | 826      | 1945 | $2^{619} 3^{147} 4^{41} 5^{12} 6^7$ |

Table 4.2:  $\varphi$ X174: size and out-degree profile of the skeleton ( $d^n = n$  vertices of out-degree  $d$ ). At  $k = 11, 10$  the skeleton is 2-in 2-out and  $\text{coker}(A + I)$  equals  $\mathcal{K}(\text{Sk})$ ; below, multi-copy vertices appear. Exact.

## 4.7 Verification

The script `multicopy.py` runs the four checks of Table 4.3 in about forty seconds, all passing; its output is archived as `multicopy_output.txt`. All arithmetic is exact; groups are compared by elementary divisors.

| key | check                                                                     | mode  |
|-----|---------------------------------------------------------------------------|-------|
| (A) | Theorem 4.1 on all 119 two-letter circular words, $r, s \leq 5$           | exact |
| (B) | Table 4.1; the witness of Proposition 4.2                                 | exact |
| (C) | Proposition 4.4: 150 digraphs, 1350 detachments; the witness              | exact |
| (D) | Table 4.2; $\text{coker}(A + I) = \mathcal{K}(\text{Sk})$ at $k = 11, 10$ | exact |

Table 4.3: The verification battery, 4/4.

## 4.8 Scope

*Proved.* Theorem 4.1; Propositions 4.2 and 4.4 by explicit witness.

*Quoted.* The splitting theorem (Theorem 3.6); the two-copy presentation [61]; the determinant formula [58, 43] and interlace polynomial [2]; sink-independence [40]; the rotor torsor for Eulerian ribbon digraphs [79].

*Open.* (a) Whether any presentation of  $\mathcal{K}(D_w)$  for  $r \geq 3$  exists whose entries are “local” in a weaker sense than pairwise — e.g. functions of the patterns of triples — and whether Jonsson’s unpublished generalised intersection matrix, which counts arborescences, has a cokernel equal to  $\mathcal{K}$ . Theorem 4.1 shows that at least for two families the group is a crossing number, so a multi-chord crossing calculus is not ruled out; Proposition 4.2 shows it cannot be pairwise. (b) The census suggests questions we have not pursued: which groups occur for  $(r, r, r)$  as  $r$  grows, and whether the largest order is always attained by an elementary abelian group. (c) The  $\varphi X_{174}$  skeleton at  $k = 8$  (312 vertices) is beyond the dense exact Smith form used here; a modular Smith form would extend Table 3.1.

## Provenance

Theorem 4.1 is elementary and, as far as we can find, unstated in this form; the propositions are negative results established by explicit computation and are the reason this chapter exists: the natural extension of [61] to multi-copy repeats was the obvious next step after Chapter 3, and it does not exist in any pairwise form. We searched for an interlacement or determinant formula for Euler circuits of  $r$ -in  $r$ -out digraphs with  $r \geq 3$  and found none; a search cannot establish a negative.



## Part II

# The double-stranded frontier and its complexity



## Chapter 5

# Double strands without a new lemma: the reverse-complement double cover of a de Bruijn graph

**Summary.** The single-strand theory of Part I sees a repeat only between co-oriented copies, so the seven ribosomal operons of *E. coli* — five on one strand, two on the other — appear as a five-copy family and a two-copy family and never as a seven-copy one. Repairing this was expected to need a sandpile theory for bidirected graphs, which does not exist. It does not need one. Let  $D_k(S)$  be the multidigraph whose arcs are the  $(k+1)$ -mer occurrences in  $S$  and in its reverse complement  $\bar{S}$ . Then  $\text{mult}_D(v) = \text{mult}_S(v) + \text{mult}_S(\bar{v})$  and  $d^+(v) = d^-(v) = \text{mult}_D(v)$ :  $D_k$  is balanced, so the unitig lemma, the bundle-chain lemma and the splitting theorem apply to it verbatim. Three consequences. *One:* multiplicities add across strands, so a family with  $a$  forward and  $b$  reverse copies has multiplicity  $a + b$  in  $D_k$ ; for *E. coli* the longest multiplicity-seven chain has span 728 bp and its seven occurrences are the seven operons, one each, the two minus-strand ones on the reverse strand of  $D_k$ . *Two:*  $D_k$  is connected exactly when  $S$  contains an inverted repeat of length  $\geq k$ , so the inverted repeats are precisely what glues the strands;  $\phi$ X174 already qualifies at  $k = 12$ , through ATTAAGCTCATT at 441 and its reverse complement at 1448. *Three:* the involution  $\rho(v) = \bar{v}$  reverses arcs, hence carries  $D_k$  to its opposite; since  $\Delta^{\text{op}} = \Delta^T$  for a balanced digraph and the two Bowen–Franks cokernels are dual,  $\rho$  dualises  $\mathcal{K}(D_k)$  instead of splitting it — which is what the warning against  $\pm 1$ -eigenspace decompositions was pointing at. The phage’s celebrated single global bit is a single-strand phenomenon:  $\mathcal{K}(G_{12}) = \mathbb{Z}/2$  but  $\mathcal{K}(D_{12}) = \mathbb{Z}/33$ . Six checks, all exact. What is *not* done:  $D_k$  is the double cover, not the bidirected quotient, and  $\mathcal{K}(D_k)$  does not count double-stranded reconstructions; that problem is left exactly where it was.

### 5.1 Introduction

The theory of Part I is single-strand throughout:  $G_k(S)$  is the de Bruijn multigraph of Section 1.2, and its sandpile group splits along the repeats by Theorem 3.6. Chapter 15 measures what that costs on a real genome: of the seven *rrn* operons of *E. coli* K-12, five lie on the  $+$  strand and two on the  $-$ , and since a repeat of  $G_k(S)$  requires co-oriented copies, the operons split into a multiplicity-five family and a multiplicity-two family, with multiplicity seven occurring nowhere.

The obvious repair is to work with the bidirected graph that assemblers actually build, whose vertices are the classes  $\{v, \bar{v}\}$ ; Chapters 1 and 2 left this open (Sections 1.9 and 2.8) and warned

against the tempting shortcut of decomposing  $\mathcal{K}$  into  $\pm 1$ -eigenspaces of the reverse-complement involution, since an involution splits an abelian group only after 2 is inverted and the 2-torsion is where the interesting phenomena live. That warning stands. What this chapter observes is that most of what one wants from the bidirected picture is already available, without any new lemma, from a balanced digraph.

## 5.2 The double cover

**Definition 5.1** (Reverse-complement double cover). Let  $\bar{S}$  be the reverse complement of  $S$ , again circular of length  $N$ . Let  $D_k = D_k(S)$  be the multidigraph whose vertices are the  $k$ -mers occurring in  $S$  or in  $\bar{S}$  and whose arcs are the occurrences of  $(k+1)$ -mers in  $S$  together with those in  $\bar{S}$ ; it has  $2N$  arcs. Write  $\rho(v) = \bar{v}$ .

**Lemma 5.2** (Balance).  $\text{mult}_D(v) = \text{mult}_S(v) + \text{mult}_S(\bar{v})$  and  $d^+(v) = d^-(v) = \text{mult}_D(v)$  for every vertex. Hence  $D_k$  is a balanced multidigraph.

*Proof.*  $\text{mult}_{\bar{S}}(v) = \text{mult}_S(\bar{v})$ , because a copy of  $v$  in  $\bar{S}$  is a copy of  $\bar{v}$  in  $S$ ; the first formula follows. In each of the two circular sequences separately, every occurrence of  $v$  contributes one arc out of  $v$  and one arc into  $v$ , as in Lemma 15.1; summing the two gives the second.  $\square$

**Corollary 5.3** (Nothing new is needed). *The unitig contraction lemma (Lemma 1.4), the bundle-chain lemma (Lemma 3.2), the canonicity of the skeleton (Lemma 3.5) and the splitting theorem (Theorem 3.6) apply to  $D_k$  verbatim, and a vertex of  $D_k$  survives unitig contraction if and only if  $\text{mult}_S(v) + \text{mult}_S(\bar{v}) \geq 2$  (Lemma 15.1).*

That last clause is the practical point: the branch vertices of  $D_k$  are the  $k$ -mers that are repeated *when both strands are counted*, so the entire fast pipeline of Chapter 15 runs on  $D_k$  unchanged, with the positions of  $\bar{S}$  appended to those of  $S$ .

**Proposition 5.4** (The strands are glued by the inverted repeats).  *$D_k$  is connected if and only if some  $k$ -mer  $w$  occurs in  $S$  together with  $\bar{w}$ ; otherwise  $D_k = G_k(S) \sqcup G_k(\bar{S})$ , two isomorphic components.*

*Proof.* The arcs coming from  $S$  form a closed walk through every  $k$ -mer of  $S$ , so those vertices lie in one component, and likewise for  $\bar{S}$ ; the two vertex sets are the  $k$ -mers of  $S$  and their reverse complements. They meet exactly when some  $w$  and  $\bar{w}$  both occur in  $S$ .  $\square$

So the double cover is interesting precisely to the extent that the genome has inverted repeats of length at least  $k$ . Both of our genomes qualify:  $\phi\text{X174}$  at  $k = 12$  through **ATTAAGCTCATT** at position 441, whose reverse complement sits at 1448; *E. coli* at  $k = 150$  through a 150-mer at 19 796 whose reverse complement is at 3 584 046 (check (C)).

## 5.3 Multiplicities add across strands

**Theorem 5.5** (Strand merging). *Let  $W$  be a sequence occurring  $a$  times on the forward strand of  $S$  and  $b$  times on the reverse, with  $a + b \geq 1$  and  $W$  not equal to  $\bar{W}$ . In  $G_k(S)$  the  $k$ -mers of  $W$  have multiplicity  $a$  and those of  $\bar{W}$  multiplicity  $b$ . In  $D_k$  both have multiplicity  $a + b$ , and the two families are exchanged by  $\rho$ . If, moreover, the  $a + b$  copies agree over a window of  $s + k - 1$  letters, that window contributes a bundle chain of multiplicity  $a + b$  and length  $s$  to  $D_k$ , hence a summand  $(\mathbb{Z}/(a + b))^{s-1}$  to  $\mathcal{K}(D_k)$ , and another from its  $\rho$ -image.*

*Proof.* A reverse-strand copy of  $W$  is a forward occurrence of  $\bar{W}$ , so  $\text{mult}_S(w) = a$  and  $\text{mult}_S(\bar{w}) = b$  for a  $k$ -mer  $w$  of  $W$  in general position; by Lemma 5.2,  $\text{mult}_D(w) = a + b$ , and symmetrically for  $\bar{w}$ . The last sentence is Lemma 5.2 followed by Lemma 3.2 applied to the balanced digraph  $D_k$ .  $\square$

**Corollary 5.6** (The seven operons). *For E. coli K-12 the seven rrn operons carry, in  $D_k$ , bundle chains of multiplicity seven. At  $k = 150, 100, 51$  the longest such chain has span 728 bp and its seven occurrences are the seven operons, one each:*

$$\begin{aligned} &+ 224\,044 \rightarrow 223\,771, & - 2\,730\,735 \rightarrow 2\,726\,069, & - 3\,428\,340 \rightarrow 3\,423\,423, \\ &+ 3\,942\,081 \rightarrow 3\,941\,808, & + 4\,035\,804 \rightarrow 4\,035\,531, & + 4\,166\,932 \rightarrow 4\,166\,659, \\ &+ 4\,208\,420 \rightarrow 4\,208\,147, \end{aligned}$$

*the sign giving the strand of  $D_k$  on which the occurrence lies and the arrow pointing to the operon it meets. The multiplicity-seven chains carry more  $k$ -mers than the chains of any other multiplicity over the operons: 4154 at  $k = 150$ , 5254 at  $k = 100$ , 6660 at  $k = 51$  (check  $(M)$ ).*

*Remark 5.7* (What was wrong with  $5 + 2$ , exactly). Nothing was wrong with it; it was the correct answer to a single-strand question. The content of Corollary 5.6 is that the question has a balanced-digraph answer too, and that the object which answers it is not the bidirected quotient but the double cover, on which the existing theorems already hold. The residual multiplicities 2, 3, 4, 5, 6 that still appear over the operons in  $D_k$  are the sub-families on which only some of the seven copies agree — the operons are not identical to one another, on either strand.

## 5.4 The involution dualises, it does not split

**Proposition 5.8** ( $\rho$  and the opposite digraph).  *$\rho$  is an involution of the vertex set of  $D_k$  carrying the arc for a  $(k+1)$ -mer  $x$  to the arc for  $\bar{x}$  with its direction reversed. Hence  $\rho$  is an isomorphism  $D_k \rightarrow D_k^{\text{op}}$ , and it induces an isomorphism*

$$\mathcal{K}(D_k) \xrightarrow{\sim} \mathcal{K}(D_k)^\vee$$

*onto the Pontryagin dual, i.e. a perfect pairing on  $\mathcal{K}(D_k)$ .*

*Proof.* The prefix of  $\bar{x}$  is the reverse complement of the suffix of  $x$  and vice versa, so the arc of  $\bar{x}$  runs  $\rho(\text{suffix}) \rightarrow \rho(\text{prefix})$ : direction reverses, and  $\rho$  is an isomorphism onto the opposite digraph. For a balanced digraph  $\Delta^{\text{op}} = D^+ - A^\top = \Delta^\top$ , so  $\mathcal{K}(D_k^{\text{op}}) = \text{Tor coker } \Delta_s$  while  $\mathcal{K}(D_k) = \text{Tor coker } \Delta_s^\top$ , and these two are canonically dual. This is a standard fact about any square integer matrix  $M$ . If  $M$  is nonsingular,  $(x, y) \mapsto x^\top M^{-1} y \bmod \mathbb{Z}$  is a well-defined pairing  $\text{coker } M^\top \times \text{coker } M \rightarrow \mathbb{Q}/\mathbb{Z}$ , and it is perfect because it is diagonal in the Smith coordinates of  $M$ , with the invariant factors on the diagonal. For singular  $M$  the same holds for the torsion parts, applied to the nonsingular block of a Smith decomposition. Composing gives the stated isomorphism.  $\square$

*Remark 5.9* (Against the eigenspace decomposition). Section 1.9, item (c), warned that  $\mathcal{K}$  should not be split into  $\pm 1$ -eigenspaces of  $\rho$ , because that discards the 2-torsion. Proposition 5.8 is what happens instead: an involution that reverses arcs cannot act on  $\mathcal{K}(D_k)$  at all — it does not preserve the presentation — but it identifies  $\mathcal{K}(D_k)$  with its dual. No inversion of 2 occurs anywhere, and the 2-torsion is untouched. Remark 6.10 shows that the resulting pairing is symmetric; whether it is independent of the choice of sink is left open.

| $k$ | graph | branch | chains | skeleton | $\mathcal{K}(\text{Sk})$                                        | length-driven         |
|-----|-------|--------|--------|----------|-----------------------------------------------------------------|-----------------------|
| 12  | $G_k$ | 2      | 1      | 1        | 0                                                               | $\mathbb{Z}/2$        |
|     | $D_k$ | 7      | 0      | 7        | $\mathbb{Z}/3 \oplus \mathbb{Z}/11$                             | 0                     |
| 11  | $G_k$ | 10     | 2      | 8        | $\mathbb{Z}/3 \oplus \mathbb{Z}/11$                             | $(\mathbb{Z}/2)^2$    |
|     | $D_k$ | 30     | 7      | 23       | $(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/49451$                      | $(\mathbb{Z}/2)^7$    |
| 10  | $G_k$ | 47     | 8      | 37       | $\mathbb{Z}/4 \oplus \mathbb{Z}/1076338579$                     | $(\mathbb{Z}/2)^{10}$ |
|     | $D_k$ | 125    | 23     | 97       | $(\mathbb{Z}/4)^2 \oplus \mathbb{Z}/79624452132275189516228869$ | $(\mathbb{Z}/2)^{28}$ |

Table 5.1:  $\varphi\text{X174}$ : the single-strand graph and its double cover. All entries exact (check (K)); the single-strand rows reproduce Table 3.1.

## 5.5 Computations

*Remark 5.10* (The phage’s single bit is single-strand).  $\mathcal{K}(G_{12}) = \mathbb{Z}/2$ : the 12-mer spectrum of  $\varphi\text{X174}$  leaves exactly two global reconstructions, and Proposition 2.5 located the bit as a pair of interleaved 12-mers. On the double-stranded spectrum the answer is different:  $\mathcal{K}(D_{12}) = \mathbb{Z}/3 \oplus \mathbb{Z}/11 = \mathbb{Z}/33$ , and it is entirely arrangement —  $D_{12}$  has no bundle chain at all, so the length-driven summand is trivial. The two statements do not conflict; they answer different questions, and the contrast is a fair measure of how much of the single-strand picture is an artefact of ignoring one strand.

| $k$ | branch | chains | skeleton | $\log_{10}$ length-driven | $\log_{10}  \mathcal{K}(\text{Sk}) $ | $\log_{10}  \mathcal{K}(D_k) $ |
|-----|--------|--------|----------|---------------------------|--------------------------------------|--------------------------------|
| 150 | 39 768 | 278    | 312      | 20 987.91                 | 156.51                               | 21 144.42                      |
| 100 | 44 808 | 396    | 434      | 23 394.31                 | 208.92                               | 23 603.24                      |
| 51  | 53 380 | 798    | 886      | 27 405.96                 | 395.80                               | 27 801.75                      |

Table 5.2: *E. coli* K-12 MG1655, the double cover. Structural entries exact;  $\log_{10} |\mathcal{K}(\text{Sk})|$  by float LU. Compare Table 15.1: on one strand the same  $k$  give skeletons of 115, 175 and 366 vertices and arrangement exponents 49.71, 71.11 and 143.17.

Two readings of Table 5.2. The skeleton roughly triples ( $115 \rightarrow 312$  at  $k = 150$ ) and the arrangement exponent does too ( $49.71 \rightarrow 156.51$ ): inverted repeats are a large part of the irreducible ambiguity of this genome and the single-strand theory sees none of them. And  $\log_{10} |\mathcal{K}(D_k)|$  exceeds twice the single-strand value by 166.6, 222.1 and 368.1 at  $k = 150, 100, 51$  — a crude but honest measure of what double-strandedness adds beyond simply having two strands to reconstruct.

## 5.6 Verification

| key | check                                                                               | mode          |
|-----|-------------------------------------------------------------------------------------|---------------|
| (B) | Lemma 5.2: $D_k$ balanced, multiplicities add across strands                        | exact         |
| (C) | Proposition 5.4: connected iff an inverted repeat of length $\geq k$ , with witness | exact         |
| (R) | Proposition 5.8: $\rho$ is an involution with $A(\rho u, \rho v) = A(v, u)$         | exact         |
| (M) | Corollary 5.6: the multiplicity-seven chains and their seven operons                | exact         |
| (K) | Table 5.1: $\mathcal{K}(D_k)$ exactly, beside $\mathcal{K}(G_k)$                    | exact         |
| (E) | Table 5.2                                                                           | exact/numeric |

Table 5.3: The verification battery (`rc_double_cover.py`), 6/6, about 33 seconds.

## 5.7 Scope

*Proved.* Lemma 5.2, Corollary 5.3, Proposition 5.4, Theorem 5.5, Corollary 5.6 and Proposition 5.8. All are short; the work is in noticing that the double cover is balanced and that this is enough.

*What this is not.*  $D_k$  is the *double cover*, not the bidirected quotient. The two are different objects and only the first is a digraph. In particular:

- $\mathcal{K}(D_k)$  does *not* count double-stranded reconstructions. By BEST,  $|\mathcal{K}(D_k)| \prod_v (d^+(v) - 1)!$  is the number of arc-rooted Eulerian circuits of  $D_k$ , and such a circuit traverses all  $2N$  arcs in one closed walk, which is not a genome. A double-stranded circular molecule corresponds instead to a decomposition of  $D_k$  into two  $\rho$ -conjugate circuits. Counting those is the subject of Chapter 7.
- No sandpile theory for the bidirected quotient is offered here; Chapter 6 takes it up. The tools [88] named in Chapters 1 and 2 are the place to start, and the obstruction noted there — that reverse complementation reverses arc direction, so the quotient map is not a covering of digraphs and [75] does not apply — is untouched.
- The pairing of Proposition 5.8 is identified as symmetric only in Chapter 6, and we have not checked that it is independent of the sink.

*Numeric.*  $\log_{10} |\mathcal{K}(\text{Sk})|$  in Table 5.2 is a float LU determinant of a 311- to 885-dimensional matrix; everything else is exact integer work.

*Open.* (a) The two questions just listed. (b) Whether Theorem 5.5 has a converse: is every multiplicity- $r$  family of  $D_k$  a strand-split family of  $S$ , and can  $a$  and  $b$  be recovered from  $D_k$  alone? (c) The 26-digit prime in Table 5.1 we do not interpret, as in Section 3.7.

## Provenance

Every statement above is proved in full and every number is produced by `rc_double_cover.py` from the two public genomes. The specification that prompted this chapter asked for a bidirected bundle-chain lemma yielding summands  $(\mathbb{Z}/2r)^{s-1}$ ; no such lemma is needed for the double cover, and Theorem 5.5 shows the correct multiplicity is  $a + b$ , the number of copies on both strands together, not  $2r$ . The same task asked for “the algebraic proof that the seven operons participate in the skeleton as a single seven-copy family”; that is Corollary 5.6, with the caveat of Remark 5.7 — they do so in the double cover, and there are two such families exchanged by  $\rho$ , not one.



## Chapter 6

# The reverse-complement quotient is a signed graph: what Zaslavsky's theory gives, and what it cannot

**Summary.** Chapter 5 built the reverse-complement double cover  $D_k(S)$  of a de Bruijn multidigraph and found that the involution  $\rho(v) = \bar{v}$  carries  $D_k$  to its *opposite* digraph, so that  $\rho$  induces no endomorphism of  $\mathcal{K}(D_k)$  and the cover does not descend. The obvious repair is to pass to the quotient  $\Sigma_k = D_k/\rho$  and to use Zaslavsky's signed Laplacian there. Half of that works, and the half that fails is the interesting half. *Works:*  $\rho$  is free exactly when no repeated  $k$ -mer is its own reverse complement, which is automatic for odd  $k$  and genuinely fails for even  $k$  ( $\phi$ X174 has three palindromic repeated 10-mers and one palindromic repeated 12-mer); every bundle of  $\Sigma_k$  is monochromatic, so  $\Sigma_k$  carries honest  $r$ -bundles;  $\Sigma_k$  is *unbalanced* precisely when  $S$  carries an inverted repeat of length  $\geq k$ , which is Zaslavsky balance read through the connectivity criterion of Chapter 5; and the Reiner–Tseng sequence  $0 \rightarrow \mathcal{K}(\Sigma_k) \rightarrow \mathcal{K}(|D_k|) \rightarrow \mathcal{K}(|\Sigma_k|) \rightarrow 0$  holds on the nose, with a non-split 2-primary part. *Fails, twice.* First, if  $P$  is the permutation matrix of  $\rho$  then  $P\Delta(D_k)P = \Delta(D_k)^\top$  and never  $\Delta(D_k)$ : the deck involution is an *anti*-automorphism of the Laplacian, the  $\pm 1$  eigenlattices of  $P$  are not  $\Delta$ -stable, and the *directed* sandpile group — the one the BEST theorem counts reconstructions with — lies in no covering sequence. What it gives instead is a perfect *symmetric* linking form  $\mathcal{K}(D_k) \times \mathcal{K}(D_k) \rightarrow \mathbb{Q}/\mathbb{Z}$ , so the relevant notion is a metabolizer rather than a Lagrangian; and since  $|\mathcal{K}(D_{11})| = 25\,318\,912$  is not a square,  $\phi$ X174 has none. Second, a signed  $r$ -bundle chain does not contract: the presentation column at the chain's endpoint differs between  $\Sigma$  and  $\Sigma/C$  by exactly  $sr d_0$ , the series law, so the splitting theorem that makes the genome-scale computation feasible on one strand has no bidirected analogue. What does survive is the  $r$ -rank, and exactly so: modulo a prime  $p$  dividing  $r$  an interior vertex of the chain has degree and adjacencies all  $\equiv 0$ , so its row and column of the signed Laplacian vanish and  $\Sigma$  and  $\Sigma/C$  reduce to the same matrix. That one observation — a *p-divisible subgraph is invisible mod p* — proves the  $p$ -rank gain to be exactly  $s$  for every prime  $p \mid r$ , composite  $r$  included, and it is also what the  $\mathbb{F}_7$ -nullity below measures. The seven ribosomal operons of *E. coli* K-12 merge into genuine 7-bundles of  $\Sigma_k$  whose longest block spans 728 bp at  $k = 31, 51$  and  $75$  alike — the same block Chapter 5 matched one-to-one onto the seven operons — and the only trace the quotient keeps of them is the  $\mathbb{F}_7$ -nullity of the signed Laplacian.

## 6.1 What was asked, and what is true

Part I reads the reconstruction count of a sequence off the critical group  $\mathcal{K}(G_k)$  through the BEST theorem. Chapter 5 handled double-strandedness by passing to the *double cover*  $D_k(S)$  of Definition 5.1, to which the unitig lemma and the splitting theorem apply verbatim (Corollary 5.3), and on which the seven ribosomal operons of *E. coli* — five on one strand, two on the other — appear as one seven-copy family (Corollary 5.6).

What Chapter 5 could not do was quotient. The map  $\rho(v) = \bar{v}$  satisfies

$$\rho: D_k \longrightarrow D_k^{\text{op}}, \quad \rho(u \rightarrow v) = \bar{v} \rightarrow \bar{u},$$

so it reverses arcs; the quotient of a digraph by a direction-reversing involution is not a digraph, and  $\rho$  induces a map  $\mathcal{K}(D_k) \rightarrow \mathcal{K}(D_k^{\text{op}}) \cong \mathcal{K}(D_k)^\vee$  rather than an endomorphism (Proposition 5.8). The standard repair in combinatorics is to forget the directions, to record instead a *sign* on each edge of the quotient, and to work with Zaslavsky’s signed Laplacian  $\Delta_\Sigma = D - A_\sigma$  [88]. This chapter carries that out, and reports both what it yields and the two places where it provably cannot yield what was wanted.

The one-paragraph summary is this. The quotient is a bona fide signed graph, its critical group is a switching invariant of the genome, its balance has an exact biological meaning, and the integral covering theory of Reiner and Tseng [75] applies to it word for word. But that theory computes an invariant of the *undirected shadow*  $|D_k|$ , which by the matrix-tree theorem counts spanning trees; the quantity of interest counts Eulerian circuits and lives on the digraph, and Theorem 6.9 says the digraph admits no such theory. Separately, Theorem 6.11 says the chain contraction that reduces a bacterial genome from tens of thousands of vertices to a few hundred has no signed analogue. Together these explain why the “unified bidirected sandpile group” that one would like to write down does not exist, and what the correct residue of the seven operons is instead.

Throughout,  $S$  is a circular sequence of length  $N$  over  $\{A, C, G, T\}$ ,  $\bar{S}$  its reverse complement, and  $\bar{w}$  the reverse complement of a word  $w$ . All computations are exact integer arithmetic; the script `bidirected_sandpile.py` reproduces every number below and prints a seven-line pass/fail battery.

## 6.2 The quotient, and when it exists

**Definition 6.1.**  $D_k = D_k(S)$  is the double cover of Definition 5.1, unitig-contracted as in Corollary 5.3, so that its vertices are the  $k$ -mers occurring at least twice in  $S$  and  $\bar{S}$  together. The involution  $\rho$  sends the vertex  $w$  to  $\bar{w}$  and the arc  $u \rightarrow v$  to  $\bar{v} \rightarrow \bar{u}$ .

The first question is whether  $\rho$  acts freely on vertices, because the whole signed-graph dictionary presupposes a free  $\mathbb{Z}/2$  action.

**Proposition 6.2.**  $\rho$  fixes a vertex of  $D_k$  if and only if some repeated  $k$ -mer  $w$  satisfies  $\bar{w} = w$ . For odd  $k$  no such word exists over any alphabet with a fixed-point-free complement, so  $\rho$  is free. For even  $k$  such words exist and do occur.

*Proof.* A word  $w$  of odd length  $k$  with  $\bar{w} = w$  would have its middle letter equal to its own complement, and complementation on  $\{A, C, G, T\}$  has no fixed point. For even  $k$  the condition  $\bar{w} = w$  is the condition that  $w$  be a palindromic restriction-site-like word, and these are common.  $\square$

| $k$ | parity | $ V(D_k) $ | $\rho$ -fixed | $A_\sigma$ symmetric |
|-----|--------|------------|---------------|----------------------|
| 9   | odd    | 396        | 0             | yes                  |
| 10  | even   | 125        | 3             | <b>no</b>            |
| 11  | odd    | 30         | 0             | yes                  |
| 12  | even   | 7          | 1             | <b>no</b>            |

Table 6.1:  $\phi X174$  (NC\_001422.1). Palindromic repeated  $k$ -mers exist at  $k = 10$  and  $k = 12$ , and at exactly those  $k$  the quotient adjacency matrix fails to be symmetric, so no signed graph is defined. Check (P).

Table 6.1 makes the point concretely and shows that the obstruction is not hypothetical: it is precisely at the even  $k$  with palindromic repeats that the construction below breaks, and it breaks visibly, in the symmetry of the adjacency matrix. From here on  $k$  is odd, or even with no palindromic repeated  $k$ -mer.

*Remark 6.3* (What the fixed points are, topologically). The failure at even  $k$  is not a technicality but a change of category. A vertex  $w = \bar{w}$  is a branch point of the projection  $D_k \rightarrow D_k/\rho$ , so the quotient is not a covering space and the fibre over  $w$  is a single point rather than two. In Zaslavsky’s language the image acquires a *half-edge* — an edge with one endpoint — at  $w$ : an arc  $u \rightarrow w$  and its partner  $w \rightarrow \bar{u}$  descend to the same edge of the quotient but attach to  $[w]$  with opposite end-signs, since  $\varepsilon(w) = +1$  forces the two signs  $\varepsilon(u)\varepsilon(w)$  and  $\varepsilon(w)\varepsilon(\bar{u})$  to differ. That is exactly why the quotient adjacency matrix in Table 6.1 fails to be symmetric:  $A_\sigma$  records one sign per edge, and a half-edge has two. Reiner and Tseng do treat signed graphs with half-edges [75, §9], so the object is not outside the theory, but its critical group is no longer the one computed below, and the double cover it belongs to is branched. Biologically the fixed points are the palindromic restriction-site-like words, so the branch locus of the quotient is precisely the repeated restriction sites of the genome.

**Definition 6.4.** Choose a representative of each  $\rho$ -orbit  $\{v, \bar{v}\}$  and set  $\varepsilon(v) = +1$  if  $v$  is the chosen one and  $-1$  otherwise. The signed graph  $\Sigma_k = D_k/\rho$  has one vertex per  $\rho$ -orbit of vertices, one edge per  $\rho$ -orbit of arcs, and the edge coming from  $u \rightarrow v$  carries the sign  $\varepsilon(u)\varepsilon(v)$ .

The arcs do fall into orbits of size two, and here no parity hypothesis is needed. An arc of  $D_k$  is an *occurrence*: a position in  $S$  or a position in  $\bar{S}$ . The involution  $\rho$  carries occurrences in  $S$  to occurrences in  $\bar{S}$  and back, and no occurrence lies in both, so  $\rho$  is free on arcs outright. The script nevertheless constructs the pairing explicitly and fails loudly if any arc lacks a partner.

Two facts make Definition 6.4 usable.

**Lemma 6.5.** *Every bundle of parallel edges of  $\Sigma_k$  is monochromatic.*

*Proof.* The sign  $\varepsilon(u)\varepsilon(v)$  depends only on the endpoints of the arc, not on the arc. All  $r$  arcs  $u \rightarrow v$  therefore descend to  $r$  edges of the same sign.  $\square$

This is what licenses the phrase “ $r$ -bundle” below, and it is not automatic for a general signed multigraph.

**Lemma 6.6.** *Changing the choice of representatives is a Zaslavsky switching, and  $\Delta_{\Sigma_k}$  changes by conjugation by a diagonal  $\pm 1$  matrix. Hence coker  $\Delta_{\Sigma_k}$ , and with it the critical group of Section 6.4, depends only on  $(S, k)$ .*

Replacing the representative at one orbit multiplies  $\varepsilon$  by  $-1$  there, which multiplies the corresponding row and column of  $A_\sigma$  by  $-1$  and leaves  $D$  alone. The script verifies invariance directly on eight random representative sets at  $k = 9$  and  $k = 11$ , obtaining for instance

$$\mathcal{K}(\Sigma_{11}) = (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4204481$$

every time. Check (W).

### 6.3 Balance is an inverted repeat

**Theorem 6.7.**  $\Sigma_k$  is unbalanced if and only if  $S$  carries an inverted repeat of length at least  $k$ .

*Proof.* A signed graph is balanced if and only if its signed double cover is disconnected; for the  $\mathbb{Z}/2$ -voltage double cover this is Reiner and Tseng, Propositions 11.4 and 11.5 [75]. The underlying multigraph of  $D_k$  is that double cover, by Definition 6.4. Proposition 5.4 says that  $D_k$  is connected exactly when  $S$  carries an inverted repeat of length at least  $k$ , the two components otherwise being the plus-strand and minus-strand halves.  $\square$

For  $\phi X174$  the longest inverted repeat has length 12, realised by ATTAAGCTCATT at position 441 against its reverse complement at 1448. The script confirms the trichotomy at  $k = 9, 11, 13, 15$ : unbalanced and connected at 9 and 11, balanced and disconnected at 13 and 15. Check (B).

Theorem 6.7 is worth stating because it converts a purely combinatorial dichotomy into a statement about the sequence. A genome with no inverted repeat as long as  $k$  gives a balanced  $\Sigma_k$ ; balanced means switching-equivalent to all-positive, which means the signs carry no information, which means the whole construction has collapsed back to the single-strand one. The sign structure is *exactly* the record of inverted repeats, nothing more and nothing less.

### 6.4 The covering theory that does exist

Reiner and Tseng define the critical group of a signed graph not as  $\text{coker } \Delta_\Sigma$  but as  $\mathcal{K}(\Sigma) = \text{im } \partial / \text{im } \Delta_\Sigma$ , where  $\partial$  is Zaslavsky's signed incidence matrix [75, eq. (1.3), §9]. For a connected unbalanced  $\Sigma$  one has  $\text{im } \partial = \{x \in \mathbb{Z}^V : \sum_v x_v \text{ even}\}$ , an index-two sublattice of  $\mathbb{Z}^V$ , so  $\mathcal{K}(\Sigma)$  is an index-two subgroup of  $\text{coker } \Delta_\Sigma$ ; ignoring the distinction doubles every order and corrupts the 2-primary part. (The competing convention of Cho, Dochtermann, Inagaki, Oh, Snustad and Zacovic [19] deletes a sink row and column instead, which is the chip-firing-natural choice but a different group.) We use the Reiner–Tseng convention, computed from the explicit basis  $2e_0, e_0 + e_1, \dots, e_0 + e_{n-1}$  of  $\text{im } \partial$ .

**Theorem 6.8** (Reiner–Tseng [75], Thm. 1.2, verified here). *For every odd  $k$  with  $\Sigma_k$  unbalanced there is a short exact sequence*

$$0 \longrightarrow \mathcal{K}(\Sigma_k) \longrightarrow \mathcal{K}(|D_k|) \longrightarrow \mathcal{K}(|\Sigma_k|) \longrightarrow 0,$$

*split on  $p$ -primary components for odd  $p$  and in general not split at  $p = 2$ .*

The  $\phi X174$  instance at  $k = 11$  is a textbook illustration:

$$\begin{aligned} \mathcal{K}(\Sigma_{11}) &= (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4204481, & \text{order } 16\,817\,924, \\ \mathcal{K}(|D_{11}|) &= (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/23 \oplus \mathbb{Z}/12757 \oplus \mathbb{Z}/4204481, & \text{order } 19\,738\,255\,595\,056, \\ \mathcal{K}(|\Sigma_{11}|) &= (\mathbb{Z}/2)^2 \oplus \mathbb{Z}/23 \oplus \mathbb{Z}/12757, & \text{order } 1\,173\,644. \end{aligned}$$

The orders multiply exactly. The odd parts split visibly: 4204481 comes from the signed graph,  $23 \cdot 12757$  from the base. The 2-primary parts do *not*: an extension of  $(\mathbb{Z}/2)^2$  by  $(\mathbb{Z}/2)^2$  has produced  $(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/4$ , of the right order 16 and the wrong isomorphism type. This is precisely the failure Reiner and Tseng exhibit with  $0 \rightarrow \mathbb{Z}_2 \rightarrow \mathbb{Z}_4 \rightarrow \mathbb{Z}_2 \rightarrow 0$ , arising here from a real bacteriophage. Check (R).

## 6.5 The deck involution is an anti-automorphism

Theorem 6.8 is a statement about  $|D_k|$ , the undirected shadow. The quantity the single-strand theory actually computes is  $\mathcal{K}(D_k)$ , the sandpile group of the *digraph*, because that is what the BEST theorem multiplies by  $\prod_v (d^+(v) - 1)!$  to count reconstructions. The following says the two are unrelated, and says why.

**Theorem 6.9.** *Let  $\Delta = \Delta(D_k)$  be the out-degree Laplacian of  $D_k$  and  $P$  the permutation matrix of  $\rho$ . Then*

$$P \Delta P = \Delta^\top,$$

*and  $P \Delta P \neq \Delta$  whenever  $D_k$  has an arc that is not part of a  $\rho$ -symmetric digon. Consequently  $\Delta$  does not commute with  $P$ , the  $\pm 1$  eigenlattices of  $P$  are not  $\Delta$ -stable, and there is no decomposition of  $\Delta$  over  $\mathbb{Z}$ , or over  $\mathbb{Z}[1/2]$ , or over any ring, into a part on the invariants and a part on the anti-invariants.*

*Proof.*  $\rho$  is a bijection from the arc set of  $D_k$  to itself carrying  $u \rightarrow v$  to  $\bar{v} \rightarrow \bar{u}$ . Hence the adjacency matrix satisfies  $A_{\bar{v}\bar{u}} = A_{uv}$ , that is  $PAP = A^\top$ , and  $d^+(\bar{v}) = d^-(v) = d^+(v)$  since  $D_k$  is balanced, so  $PDP = D$ . Subtracting gives  $P\Delta P = \Delta^\top$ . If  $P\Delta P$  were also  $\Delta$  then  $\Delta = \Delta^\top$ , forcing  $A = A^\top$ , i.e. every arc paired with a reverse arc of the same multiplicity.

For the consequence: a decomposition along the eigenlattices of  $P$  requires  $\Delta P = P\Delta$ . Here  $\Delta P = P\Delta^\top$ , and  $\Delta^\top \neq \Delta$ .  $\square$

This is the structural reason the obstruction of Proposition 5.8 was not an accident of its proof. The classical covering-graph factorisation — of Bilu and Linial [12], in the form used for signed graphs — decomposes the Laplacian of a double cover into an unsigned and a signed part on the  $\pm 1$  eigenspaces of the deck transformation. That decomposition needs the deck transformation to *commute* with the Laplacian. Here it conjugates the Laplacian to its transpose instead, and the two eigenlattices are exchanged rather than preserved.

The numbers are unambiguous. For  $\phi X174$  at  $k = 11$ ,

$$|\mathcal{K}(D_{11})| = 25\,318\,912, \quad |\mathcal{K}(|D_{11}|)| = 19\,738\,255\,595\,056,$$

and the second is not a multiple of the first; at  $k = 9$  the two orders are 124- and 205-digit integers, again with no divisibility. The directed group is not an extension of anything by  $\mathcal{K}(\Sigma_k)$  and is not a quotient of anything by it. Check (N).

*Remark 6.10* (What  $\rho$  gives instead: a symmetric linking form). The honest formulation is that  $\rho$  makes  $D_k$  into a digraph with a *duality*, not a digraph with a symmetry. Bowen–Franks duality (the proof of Proposition 5.8) says  $\text{coker } M$  and  $\text{coker } M^\top$  are canonically dual for a square integer matrix  $M$  with finite cokernel, and Theorem 6.9 identifies  $\Delta^\top$  with  $P\Delta P$ . Composing gives a perfect pairing on the sandpile group itself:

$$\langle u, v \rangle = (Pu)^\top \Delta^{-\top} v \bmod \mathbb{Z}, \quad \mathcal{K}(D_k) \times \mathcal{K}(D_k) \longrightarrow \mathbb{Q}/\mathbb{Z}.$$

It is well defined — if  $u = \Delta^\top z$  then  $Pu = P\Delta^\top P(Pz) = \Delta(Pz)$ , so  $\langle u, v \rangle = (Pz)^\top v \in \mathbb{Z}$  — and it is *symmetric*, not alternating: inverting  $P\Delta P = \Delta^\top$  gives  $P\Delta^{-\top} = \Delta^{-1}P$ , whence  $\langle u, v \rangle = u^\top P\Delta^{-\top}v = u^\top \Delta^{-1}Pv = \langle v, u \rangle$ . Verified exactly for  $\phi X174$  at  $k = 11$  and  $k = 12$ ; check (D). This answers the first question left open in Remark 5.9.

This matters for how one should look for a double-stranded count. A symmetric linking form has no Lagrangians in the symplectic sense; the relevant notion is a *metabolizer*, a subgroup  $L$  with  $L = L^\perp$ , and a metabolizer forces  $|\mathcal{K}(D_k)| = |L|^2$ . For  $\phi X174$  that already fails:  $|\mathcal{K}(D_{12})| = 33$  and  $|\mathcal{K}(D_{11})| = 25\,318\,912$ , neither a square. So whatever counts double-stranded reconstructions, it is not the order of a self-orthogonal subgroup of  $\mathcal{K}(D_k)$ , and the symmetry of the form — not merely its existence — is the first thing a candidate theory has to respect.

One further negative is worth recording, because it removes the most natural-looking formulation. The sandpile group is a torsor on the arborescences *converging* on a fixed root, and those are what the BEST theorem counts with. One would like to say that a double-stranded count is a sum over the  $\rho$ -orbits of arborescences. There are none:  $\rho$  reverses arcs, so it carries an arborescence converging on  $r$  to one *diverging* from  $\rho(r)$ , and the two families are disjoint. What  $\rho$  gives at tree level is therefore a *bijection*

$$\rho: \{\text{arborescences converging on } r\} \xrightarrow{\sim} \{\text{arborescences diverging from } \rho(r)\},$$

both families having  $|\mathcal{K}(D_k)|$  elements — the tree-level shadow of the duality, not an action. Verified for  $\phi X174$  at  $k = 12$  at three roots, where each family has 33 members and the image lands in the diverging family every time and in the converging one never; check (D). A double-stranded theory must be built from that bijection.

## 6.6 Signed bundle chains do not contract

The single-strand computation for a bacterial genome is only possible because of the splitting theorem (Theorem 3.6): a *chain*  $C = (c_1, \dots, c_s)$  of vertices each of which sends all  $r$  of its out-arcs to the next splits off a factor  $(\mathbb{Z}/r)^{s-1}$  and can be contracted, reducing *E. coli* at  $k = 75$  from 28 579 vertices to a skeleton of 261 (Table 15.1). The question is whether this survives in  $\Sigma_k$ . It does not, and the reason is elementary enough to state exactly.

**Theorem 6.11.** *Let  $\Sigma$  be a signed graph containing a chain  $C = (c_1, \dots, c_s)$ : vertices of degree exactly  $2r$ , joined by  $r$ -bundles to  $c_{t-1}$  and  $c_{t+1}$ , where  $c_0 = a$  and  $c_{s+1} = b$  lie outside  $C$ . Switch so that every chain bundle is positive, and write  $d_t = x_{c_t} - x_{c_{t+1}}$  for  $t = 0, \dots, s$  in  $\text{coker } \Delta_\Sigma$ , where  $x_v$  is the class of the basis vector at  $v$ . Then*

- (a) *the presentation columns at  $c_1, \dots, c_s$  say exactly  $r(d_{t-1} - d_t) = 0$  for  $t = 1, \dots, s$ ; in particular the  $s$  second differences  $\delta_t = x_{c_{t-1}} - 2x_{c_t} + x_{c_{t+1}}$  are killed by  $r$ , and  $rd_0 = rd_1 = \dots = rd_s$ ;*
- (b) *the column at  $a$  in  $\Sigma$  and the column at  $a$  in the contraction  $\Sigma/C$  — which replaces  $C$  by a single  $r$ -bundle  $a-b$  of the product sign — differ by exactly  $sr d_0$ .*

Consequently  $\mathcal{K}(\Sigma) \not\cong (\mathbb{Z}/r)^s \oplus \mathcal{K}(\Sigma/C)$  in general.

*Proof.* (a)  $\Delta_\Sigma$  is symmetric and its column at  $c_t$  has entry  $2r$  at  $c_t$  and  $-r$  at each of  $c_{t-1}, c_{t+1}$ , all other entries zero. The relation it imposes is  $2r x_{c_t} - r x_{c_{t-1}} - r x_{c_{t+1}} = 0$ , which is  $-r(d_{t-1} - d_t) = 0$ . Chaining the  $s$  relations gives  $rd_0 = \dots = rd_s$ .

(b) The column at  $a$  in  $\Sigma$  contains  $-r x_{c_1}$  where the column at  $a$  in  $\Sigma/C$  contains  $-r x_b$ ; all other entries agree, since  $\deg a$  is  $r$  in both. The difference is  $r(x_b - x_{c_1}) = -r \sum_{t=1}^s d_t$ , and by (a) each  $rd_t$  equals  $rd_0$ , so this is  $-sr d_0$ .  $\square$

| $r$ | $s$ | $\mathcal{K}(\Sigma)$                                           | $\mathcal{K}(\Sigma/C)$                                | ratio  | $r^s$ |
|-----|-----|-----------------------------------------------------------------|--------------------------------------------------------|--------|-------|
| 3   | 1   | $(\mathbb{Z}/3)^2 \oplus \mathbb{Z}/8$                          | $\mathbb{Z}/2 \oplus \mathbb{Z}/9$                     | 4.00   | 3     |
| 3   | 2   | $\mathbb{Z}/2 \oplus (\mathbb{Z}/3)^3 \oplus \mathbb{Z}/8$      | $(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/9$                 | 12.00  | 9     |
| 3   | 3   | $(\mathbb{Z}/3)^3 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/9$      | $\mathbb{Z}/2 \oplus \mathbb{Z}/9$                     | 54.00  | 27    |
| 5   | 2   | $(\mathbb{Z}/2)^2 \oplus (\mathbb{Z}/5)^2 \oplus \mathbb{Z}/17$ | $\mathbb{Z}/2 \oplus \mathbb{Z}/4 \oplus \mathbb{Z}/7$ | 30.36  | 25    |
| 5   | 3   | $\mathbb{Z}/4 \oplus (\mathbb{Z}/5)^3 \oplus \mathbb{Z}/11$     | $\mathbb{Z}/2 \oplus \mathbb{Z}/13$                    | 211.54 | 125   |
| 7   | 2   | $\mathbb{Z}/2 \oplus (\mathbb{Z}/7)^2 \oplus \mathbb{Z}/23$     | $\mathbb{Z}/2 \oplus \mathbb{Z}/17$                    | 66.29  | 49    |
| 7   | 3   | $\mathbb{Z}/4 \oplus (\mathbb{Z}/7)^3 \oplus \mathbb{Z}/13$     | $\mathbb{Z}/2 \oplus \mathbb{Z}/17$                    | 524.59 | 343   |
| 11  | 2   | $\mathbb{Z}/2 \oplus (\mathbb{Z}/11)^2 \oplus \mathbb{Z}/64$    | $(\mathbb{Z}/2)^2 \oplus \mathbb{Z}/29$                | 133.52 | 121   |

Table 6.2: An  $r$ -bundle chain of  $s$  internal vertices grafted into a fixed unbalanced base. The  $r$ -rank rises by exactly  $s$  in all eight rows — the  $(\mathbb{Z}/r)^s$  of Theorem 6.11(a) is visible, and Theorem 6.13 says it must be exactly  $s$  — but the order ratio never equals  $r^s$ , and is not even an integer in five rows. Check (C); check (Q) extends the  $p$ -rank column to composite  $r = 4, 6, 9, 10, 12, 15, 21, 25$  and to  $p = 2$ , and to  $p$ -divisible trees, cycles and multi-attachment subgraphs.

Contrast the directed case. There the column at  $c_t$  is  $r(x_{c_t} - x_{c_{t+1}}) = rd_t$ , which kills each *first* difference outright; the chain contributes  $(\mathbb{Z}/r)^{s-1}$  as a direct summand and contracts cleanly. In the signed case only second differences die. The residual first difference  $d_0$  is not torsion; it is coupled to the rest of the graph, and the coupling is the series law of electrical networks —  $s+1$  bundles of conductance  $r$  in series behave like one bundle of conductance  $r/(s+1)$ , which is not an integer multiplicity, so there is no contracted signed graph to compare to.

Table 6.2 separates the two halves of the theorem. What does not survive contraction is the order and the isomorphism type. What survives is the  $r$ -rank, and the reason is a one-line observation that has nothing to do with chains.

**Lemma 6.12** ( $p$ -divisible subgraphs are invisible mod  $p$ ). *Let  $\Sigma$  be a signed multigraph,  $p$  a prime, and  $H$  a set of bundles of  $\Sigma$  every one of which has multiplicity divisible by  $p$ . Then*

$$\text{rank}_{\mathbb{F}_p} \Delta_\Sigma = \text{rank}_{\mathbb{F}_p} \Delta_{\Sigma \setminus H},$$

where  $\Sigma \setminus H$  is  $\Sigma$  with the bundles of  $H$  deleted and the vertex set kept.

*Proof.* Call a vertex *interior* to  $H$  if every bundle at it lies in  $H$ . Its degree is a sum of multiplicities divisible by  $p$ , hence  $\equiv 0$ , and each of its adjacency entries is  $\pm$  a multiplicity in  $H$ , hence  $\equiv 0$ : its whole row and column of  $\Delta_\Sigma$  vanish mod  $p$ , and they vanish in  $\Delta_{\Sigma \setminus H}$  too, that vertex now being isolated. A vertex not interior to  $H$  loses, on passing to  $\Sigma \setminus H$ , exactly the  $H$ -part of its degree and its  $H$ -adjacencies; both are  $\equiv 0$ . So the two matrices are congruent mod  $p$  entry by entry.  $\square$

**Theorem 6.13** (the  $r$ -rank gain, exactly). *Let  $\Sigma$  be connected and unbalanced with an  $r$ -bundle chain  $C$  of  $s$  internal vertices, and let  $\Sigma/C$  be connected and unbalanced as well. Then for every prime  $p$  dividing  $r$ ,*

$$p\text{-rank}(\text{coker } \Delta_\Sigma) - p\text{-rank}(\text{coker } \Delta_{\Sigma/C}) = s,$$

and for every odd prime  $p \mid r$  the same holds with  $\text{coker } \Delta$  replaced by the Reiner–Tseng group  $\mathcal{K}$ . More generally, if  $H$  is any  $p$ -divisible subgraph with  $w$  interior vertices then the  $p$ -rank of  $\text{coker } \Delta$  exceeds that of  $\text{coker } \Delta_{\Sigma \setminus H}$  by exactly  $w$ .

| $k$ | $ V(D_k) $ | $ V(\Sigma_k) $ | $\rho$ -fixed | 7-bundles | longest 7-chain | span (bp)  | $\dim_{\mathbb{F}_7} \ker \Delta_{\Sigma_k}$ |
|-----|------------|-----------------|---------------|-----------|-----------------|------------|----------------------------------------------|
| 31  | 60 548     | 30 274          | 0             | 4919      | 698             | <b>728</b> | 7441                                         |
| 51  | 53 380     | 26 690          | 0             | 4352      | 678             | <b>728</b> | 6581                                         |
| 75  | 48 498     | 24 249          | 0             | 3814      | 654             | <b>728</b> | 5864                                         |

Table 6.3: *E. coli* K-12 MG1655, NC\_000913.3, 4 641 652 bp, at the three odd word lengths of the single-strand study. The longest 7-bundle chain spans the same 728 base pairs at all three  $k$  —  $698 + 31 - 1 = 678 + 51 - 1 = 654 + 75 - 1$  — which is the seven-copy block Chapter 5 matched one-to-one onto the seven ribosomal operons (Corollary 5.6). Check (E).

*Proof.* Both the chain’s bundles in  $\Sigma$  and the single  $a$ – $b$  bundle of  $\Sigma/C$  have multiplicity  $r$ , so Lemma 6.12 applied to each deletes them and leaves the same matrix  $\Delta_{\Sigma'}$ ,  $\Sigma'$  being  $\Sigma$  with the chain and both end bundles removed. Hence  $\text{rank}_{\mathbb{F}_p} \Delta_{\Sigma} = \text{rank}_{\mathbb{F}_p} \Delta_{\Sigma/C}$ , while  $\Delta_{\Sigma}$  is larger by the  $s$  interior vertices, so its nullity mod  $p$  exceeds that of  $\Delta_{\Sigma/C}$  by  $s$ . For a connected unbalanced signed graph  $\Delta$  is positive definite, hence nonsingular, so  $\text{coker } \Delta$  is finite and its  $p$ -rank — the number of invariant factors divisible by  $p$  — equals  $\dim_{\mathbb{F}_p} \text{coker } \Delta \otimes \mathbb{F}_p = \text{nullity}_{\mathbb{F}_p} \Delta$ . For odd  $p$ ,  $\mathcal{K}(\Sigma)$  has index two in  $\text{coker } \Delta_{\Sigma}$  and therefore the same  $p$ -primary part. The general statement is the same argument with  $s$  replaced by the number of interior vertices of  $H$ .  $\square$

*Remark 6.14* (what happens at  $p = 2$ ). For  $p = 2$  the index-two subgroup does matter. Tensoring  $0 \rightarrow \mathcal{K}(\Sigma) \rightarrow \text{coker } \Delta_{\Sigma} \rightarrow \mathbb{Z}/2 \rightarrow 0$  with  $\mathbb{F}_2$  gives  $2\text{-rank } \mathcal{K} = 2\text{-rank coker } \Delta - 1 + \varepsilon$  with  $\varepsilon \in \{0, 1\}$  recording whether the extension splits at 2, so the gain for  $\mathcal{K}$  is  $s + \varepsilon_{\Sigma} - \varepsilon_{\Sigma/C}$ . In the basis  $2e_0, e_0 + e_1, \dots, e_0 + e_{n-1}$  of  $\text{im } \partial$  the chain’s column reduces mod 2 to a single possibly nonzero entry in the row of  $2e_0$ , which bounds the deviation by one. In 5828 connected unbalanced instances with even  $r$  the two  $\varepsilon$  agree and the gain is  $s$ ; that they always agree is *not* proved.

Theorem 6.13 is a partial and negative data point for Question 5.5 of [19], which asks how  $\mathcal{K}(\Sigma_{\varphi})$  depends on  $\mathcal{K}(|\Sigma|)$  as the signature varies and observes only that the number of invariant factors seems to be preserved: here the count of factors divisible by  $p$  is pinned exactly, while the order and the isomorphism type are not.

**Corollary 6.15.** *The reduction that makes  $\mathcal{K}(G_k)$  computable for a bacterial genome — contract every maximal chain, compute on the skeleton, multiply back — has no analogue for  $\mathcal{K}(\Sigma_k)$ .*

## 6.7 *E. coli* K-12: what is left of the seven operons

The construction does what was asked of it structurally.  $\rho$  is free at all three odd word lengths,  $\Sigma_k$  is a signed graph, and the seven operons appear as genuine 7-bundles: 4919 of them at  $k = 31$ , organised into chains of which the longest has 698 vertices. That chain spans  $698 + 31 - 1 = 728$  base pairs, and the corresponding numbers at  $k = 51$  and  $k = 75$  give 728 as well. The agreement with Chapter 5 is literal and not merely numerical: working on  $D_{51}$  rather than on  $\Sigma_{51}$ , and identifying the chain’s seven occurrences against the RefSeq annotation, it reports a longest multiplicity-seven chain of  $s = 678$  spanning 728 bp whose occurrences hit the seven **rrn** operons one each. The quotient neither shortens that chain nor merges it with its own reverse complement, so 678 appears again here. On the structural question — do the five plus-strand and two minus-strand copies become one seven-copy object in the quotient — the answer is yes.

On the quantitative question the answer is no, for the two reasons already proved. Corollary 6.15 says  $\mathcal{K}(\Sigma_k)$  cannot be computed by contraction at this size, and Theorem 6.9 says that even if

it could, it would not count reconstructions. What is computable, and what is the right residue of the operons, is the  $\mathbb{F}_7$ -nullity of the signed Laplacian. The reason it is the right residue is Lemma 6.12 at  $p = 7$ : the union of all 7-bundles is a 7-divisible subgraph, hence invisible over  $\mathbb{F}_7$ , and every vertex interior to it — every vertex whose bundles are all 7-bundles — contributes a zero row and a unit of nullity. So the  $\mathbb{F}_7$ -nullity and the exactness of the chain's 7-rank gain are two readings of one lemma, and the numbers below are an instance of Theorem 6.13 at genome scale. The  $\mathbb{F}_7$ -nullity therefore measures the total length of the seven-copy repeat structure, and decays from 7441 to 5864 as  $k$  rises from 31 to 75 and short seven-copy blocks fall below the word length. The corresponding single-strand figures are 3496, 3201 and 2996: roughly half, as one expects from a construction that cannot see five copies and two copies as seven.

## 6.8 Scope, provenance, and what is open

**Scope.** Everything proved here concerns the *quotient*  $\Sigma_k$  and its undirected shadow. Proposition 6.2, Lemma 6.5, Lemma 6.6, Theorem 6.9 and Theorem 6.11 are proved in full generality. Theorem 6.7 is a combination of Reiner–Tseng Prop. 11.4–11.5 with the connectivity criterion of Proposition 5.4. Theorem 6.8 is *not* ours: it is Reiner and Tseng's Theorem 1.2, and what is new here is only that it is exhibited on a reverse-complement de Bruijn graph, including a non-split 2-primary extension arising from a real genome. Lemma 6.12 and Theorem 6.13 are proved in full; an earlier version of this chapter had the exactness of the  $r$ -rank gain as a computational observation only. Lemma 6.12 is elementary enough that we would not be surprised to find it in the literature on  $p$ -ranks of critical groups, and we claim novelty only for its two consequences here. What remains unproved is the case  $p = 2$  for the Reiner–Tseng group: by Remark 6.14 the gain is  $s + \varepsilon_\Sigma - \varepsilon_{\Sigma/C}$ , and that the two  $\varepsilon$  always agree is verified on 5828 instances and not proved.

**Provenance.** The signed-graph formalism, the signed incidence and Laplacian matrices, balance and switching are Zaslavsky's [88]; the dictionary between bidirected graphs and oriented signed graphs is Edmonds', as recorded in Zaslavsky's glossary [89]. The critical group of a signed graph and the covering sequence used in Section 6.4 are Reiner and Tseng's [75]; the competing rooted convention and the chip-firing model for signed graphs are due to Cho, Dochtermann, Inagaki, Oh, Snustad and Zacovic [19], building on the invertible-matrix chip-firing of Guzmán and Klivans [33]. The spectral form of the double-cover decomposition, which Theorem 6.9 shows is unavailable here, is Bilu and Linial's [12]. The bidirected-graph model of double-stranded assembly is Medvedev, Georgiou, Myers and Brudno's [60]. Sandpile groups of ordinary de Bruijn digraphs are treated by Chan, Hollmann and Pasechnik [24]. We are not aware of prior work computing critical groups of reverse-complement de Bruijn graphs, signed or otherwise; a targeted search found none.

**Open.** Three things. *First*, the residue of Theorem 6.13 at the prime two: show that the index-two extension of Remark 6.14 splits, or fails to split, for  $\Sigma$  and  $\Sigma/C$  together, so that the gain for the Reiner–Tseng group is  $s$  there as well — or exhibit the pair where it is  $s \pm 1$ . *Second*, the series law suggests working with weighted signed graphs over  $\mathbb{Z}[1/(s+1)]$ , where the contraction does exist; whether the resulting group has any combinatorial meaning — whether it counts anything — is unknown to us. *Third*, and most important, Theorem 6.9 closes the quotient route to a double-stranded reconstruction count but does not close the problem. What survives is the symmetric linking form of Remark 6.10, and that is the structure a correct double-stranded theory should use. Two things are already known about it and both are constraints rather than encouragements: the form is symmetric, so there are no Lagrangians in the symplectic sense; and

for  $\phi X174$  the group order is not a square, so there is no metabolizer either. So a count modulo strand exchange cannot be the order of a single distinguished subgroup, and the naive guess — that the answer is  $\sqrt{|\mathcal{K}(D_k)|}$  — is already excluded. Chapter 7 answers the question differently: the count is a lattice sum in which the form does not enter at all (Remark 7.8).

## Chapter 7

# Double-stranded genome assembly as a lattice sum, and the inversion-flip group

**Summary.** Chapters 5 and 6 ended with a question: the sandpile group  $\mathcal{K}(D_k)$  of the reverse-complement double cover counts Eulerian circuits through both strands at once, its deck involution is an anti-automorphism carrying a symmetric linking form and no metabolizer, and what a double-stranded reconstruction count is built from was left open. This chapter answers it. A circular sequence  $T$  is a double-stranded reconstruction of  $S$  at word length  $k$  when  $\text{spec}(T) + \text{spec}(\bar{T}) = \text{spec}(S) + \text{spec}(\bar{S})$ ; writing  $m$  for the arc multiplicities of  $D_k$  and  $c = \text{spec}(T)$ , this says  $c + \rho c = m$ , and with  $f = 2c - m$  it says that  $f$  lies in the  $(-1)$ -eigenlattice  $\Lambda^-$  of  $\rho$  on the circulation lattice of  $D_k$ , in the box  $|f| \leq m$ , in the parity class of  $m$ . The count is therefore a *lattice sum*,  $\#\text{DS}(S, k) = \sum_f N_{\text{seq}}((m + f)/2)$ , each term the BEST count of a multigraph divided by  $\prod_a c(a)!$  — and that division, invisible on one strand where it is a constant, is mandatory here because it varies from point to point. The rank of  $\Lambda^-$  is  $(|A| - |V| + 2 - \#\text{fix}_A - \#\text{fix}_V)/2$  by the Hopf trace formula applied to an arc-reversing involution; it is *not* the number of inverted repeats, and the guess that it is — made in the planning of this chapter — is refuted at  $k = 12$ , where one inverted-repeat pair and one palindrome meet a lattice of rank four. For  $\phi\text{X174}$  the sum gives 10 reconstructions at  $k = 12$  and 864 at  $k = 11$ , against  $|\mathcal{K}(D_k)| = 33$  and 25 318 912, and an independent enumeration of closed walks with the strand-pairing constraint returns the same numbers point by point. The second half concerns moves. A block  $w X \bar{w}$  of  $T$  may be replaced by  $w \bar{X} w$ ; every such flip is a flip between a visit of a vertex  $v$  of  $D_k$  and a later visit of  $\rho(v)$ , and it changes the spectrum by  $-c_B + \rho c_B$ . For  $\phi\text{X174}$  the 864 reconstructions at  $k = 11$  form a *single* component under flips alone — no flip fixes a spectrum, so the same-spectrum repeat exchanges of the single-strand theory are compositions of flips through other classes, of length four, six or eight — and the five inverted-repeat pairs of  $S$  do not suffice: six of the 218 spectrum classes need a flip at a pair  $\{a, \bar{a}\}$  with  $a$  a single-strand repeat, available only after an earlier flip has imported  $\bar{a}$ . Connectivity is a property of the genome and not a theorem: a sequence with repeats and no inverted repeat has a discrete flip graph.

### 7.1 What was asked, and what is true

Part I reads the reconstruction count of a circular sequence  $S$  at word length  $k$  off the critical group of its de Bruijn multidigraph  $G_k(S)$  through the BEST theorem [10, 78]. Chapter 5 passed

to the reverse-complement double cover  $D_k(S)$  of Definition 5.1 and found that everything single-stranded applies to it verbatim but that its involution  $\rho(v) = \bar{v}$  reverses arcs. Chapter 6 proved (Theorem 6.9) that this is fatal for every covering-theoretic route:  $P\Delta P = \Delta^\top$ , the eigenlattices of  $\rho$  are not  $\Delta$ -stable, the sandpile group  $\mathcal{K}(D_k)$  sits in no covering sequence, and what  $\rho$  provides instead is a symmetric linking form on  $\mathcal{K}(D_k)$  which for  $\phi$ X174 has no metabolizer because  $|\mathcal{K}(D_k)|$  is not a square. That chapter closed with the statement that whatever counts double-stranded reconstructions is not the order of a distinguished subgroup of  $\mathcal{K}(D_k)$ , and left open what it is.

The answer is that it is not a group order at all.  $\mathcal{K}(D_k)$  counts Eulerian circuits of  $D_k$ , which are single closed walks through all  $2N$  arcs, that is, through both strands. A double-stranded reconstruction is a circular sequence  $T$  of length  $N$  whose  $(k+1)$ -mer spectrum, together with that of  $\bar{T}$ , is the spectrum of  $S$  together with that of  $\bar{S}$ ; it is a closed walk through *half* the arcs, whose  $\rho$ -image walks the other half. Writing  $c$  for the spectrum of  $T$  and  $m$  for the multiplicities of  $D_k$ , the condition is  $c + \rho c = m$ . That is a linear condition, and with  $f = 2c - m$  it becomes the condition that  $f$  be a  $\rho$ -anti-invariant circulation in the box  $|f| \leq m$  with the parity of  $m$ . The reconstruction count is therefore a sum over the lattice points of a polytope of the sequence count of the multigraph at each point (Theorem 7.6). The rank of the lattice has a closed form (Theorem 7.9), and the sum, evaluated for  $\phi$ X174, gives 10 and 864 where  $|\mathcal{K}(D_k)|$  gives 33 and 25 318 912 (Table 7.2). An enumeration that never mentions the lattice — closed walks of  $D_k$  with the strand-pairing constraint, bucketed by spectrum — returns exactly the connected lattice points, with exactly the predicted count at each (check (E)).

The second half of the chapter is about moves between reconstructions. If  $T$  contains a  $k$ -mer  $w$  and, later, its reverse complement  $\bar{w}$ , the block between and including them may be reverse-complemented in place; this is the inversion of classical genome rearrangement, and it is the only move that changes which strand a stretch of  $T$  is read from. For  $\phi$ X174 the flip graph on all 864 reconstructions at  $k = 11$  is connected (Theorem 7.14), and the way it is connected is informative: no flip ever preserves the spectrum, so the same-spectrum moves of the single-strand theory — exchanging the two copies of a repeat — arise here only as compositions of flips passing through other spectra, always of even length at least four. This is the de Bruijn-graph shadow of the fact, familiar from sorting by reversals [41, 15], that a transposition is a product of reversals; here each reversal must be flanked by an inverted-repeat pair, and the price is four rather than three.

Three corrections to the specification of this chapter are recorded in Remark 7.16: the rank of  $\Lambda^-$  is not the number of inverted repeats; the per-point count must be divided by  $\prod c(a)!$ ; and flip-connectivity is a property of the genome, not a theorem.

Throughout,  $S$  is a circular sequence of length  $N$  over  $\{A, C, G, T\}$ ,  $\bar{S}$  its reverse complement,  $\bar{w}$  that of a word  $w$ , and a *circular sequence* is a word up to rotation. All computations are exact; the scripts `ds_lattice.py` and `ds_flips.py` reproduce every number below and print pass/fail batteries of nine and four checks.

## 7.2 The double cover, its contents, and the spectrum of a reconstruction

**Definition 7.1.**  $D_k = D_k(S)$  is the double cover of Definition 5.1: one vertex per  $k$ -mer occurring at least twice in  $S$  and  $\bar{S}$  together, one arc per occurrence of a  $(k+1)$ -mer in  $S$  or in  $\bar{S}$ , unitig-contracted (Corollary 5.3) so that an arc is a maximal segment between two consecutive repeated  $k$ -mers on one strand. An arc is thus a *labelled* object (a strand and a position); its *content* is

the segment string from the start of its tail  $k$ -mer to the end of its head  $k$ -mer; two arcs with the same content are parallel copies. Let  $V$  be the vertex set,  $A$  the set of contents,  $m: A \rightarrow \mathbb{Z}_{\geq 1}$  the number of arcs of each content,  $\ell(a)$  the number of  $(k+1)$ -mers of the content  $a$ , and  $\partial: \mathbb{Z}^A \rightarrow \mathbb{Z}^V$ ,  $\partial a = \text{head}(a) - \text{tail}(a)$ . The involution  $\rho$  sends a vertex  $w$  to  $\bar{w}$ , a content  $a$  to  $\bar{a}$ , and the arc at position  $p$  of one strand to the arc at the mirrored position of the other.

- Lemma 7.2.** (a)  $\rho$  is a fixed-point-free involution on arcs, an involution on contents with  $m(\bar{a}) = m(a)$  and  $\ell(\bar{a}) = \ell(a)$ , and reverses direction:  $\partial(\bar{a}) = -\rho \partial a$ .
- (b)  $\rho$  is free on vertices if and only if no repeated  $k$ -mer is its own reverse complement, which is automatic for odd  $k$ ; it is free on contents if and only if no content is its own reverse complement, and a content with  $m(a) \geq 2$  is a single  $(k+1)$ -mer, so this is automatic for even  $k$ . On a fixed content  $m(a)$  is even.
- (c)  $D_k$  has one component if  $S$  carries an inverted repeat of length at least  $k$  and two otherwise, the two being the  $S$ -half and the  $\bar{S}$ -half, exchanged by  $\rho$ .

*Proof.* (a) An arc is an occurrence on a strand and  $\rho$  exchanges the strands, so it is free on arcs; its image on contents is reverse complementation, which preserves length and, being a bijection on occurrences, preserves multiplicity. The arc  $u \rightarrow v$  of content  $a$  maps to the arc  $\bar{v} \rightarrow \bar{u}$  of content  $\bar{a}$ . (b) The vertex statement is Proposition 6.2. A content with two parallel copies is a repeated segment, so its interior  $k$ -mers are repeated; but the interior  $k$ -mers of a unitig segment are by construction not repeated, so the segment has no interior  $k$ -mer and is a single  $(k+1)$ -mer, of odd length when  $k$  is even, and no word of odd length equals its reverse complement. If  $a = \bar{a}$  then the occurrences of  $a$  are paired by  $\rho$  with occurrences of  $a$ , in pairs, so  $m(a)$  is even. (c) is Proposition 5.4: all  $k$ -mers of  $S$  are joined by the closed walk  $S$ , all those of  $\bar{S}$  by  $\bar{S}$ , and the two walks meet exactly at  $k$ -mers common to  $S$  and  $\bar{S}$ .  $\square$

Part (b) has a consequence worth stating:  $\rho$  is never free on vertices and contents at once, since palindromic  $k$ -mers need even  $k$  and palindromic  $(k+1)$ -mers need odd  $k$ . For  $\phi X174$ ,  $k = 12$  has one palindromic repeated 12-mer (TGGAGGCCTCCA at 4483) and no fixed content;  $k = 11$  has no fixed vertex and one fixed content, the 12-mer just named, now read as an arc between its two halves. The same word moves from one column of Table 7.1 to the other as  $k$  changes parity.

**Definition 7.3.** A *double-stranded reconstruction* of  $S$  at level  $k$  is a circular sequence  $T$  with

$$\text{spec}(T) + \text{spec}(\bar{T}) = \text{spec}(S) + \text{spec}(\bar{S})$$

as multisets of  $(k+1)$ -mers.  $\#DS(S, k)$  is their number.

**Lemma 7.4.** Let  $T$  be a double-stranded reconstruction. Every  $(k+1)$ -mer of  $T$  is an arc of  $D_k$ ,  $T$  traverses each unitig segment of multiplicity one either once or not at all, and the spectrum of  $T$  is therefore a well-defined vector  $c = \text{spec}(T) \in \mathbb{Z}_{\geq 0}^A$  with  $c + \rho c = m$  and  $\partial c = 0$ . Moreover  $|T| = N$ .

*Proof.* The spectrum identity puts every  $(k+1)$ -mer of  $T$  in the spectrum of  $S$  or  $\bar{S}$ . A non-repeated  $k$ -mer  $w$  of  $D_k$  has one arc in and one arc out; it occurs once in  $S\bar{S}$ , hence once in  $T\bar{T}$ , hence at most once in  $T$ , and if it occurs in  $T$  then the  $(k+1)$ -mers of  $T$  through it are its unique in- and out-arcs, so  $T$  continues along the unitig in both directions. This makes  $c$  constant on unitigs.  $c + \rho c = m$  is the spectrum identity read on contents, since  $\text{spec}(\bar{T}) = \rho \text{spec}(T)$ ;  $\partial c = 0$  because  $T$  is a closed walk. Finally  $\sum_a c(a)\ell(a) + \sum_a c(\bar{a})\ell(\bar{a}) = \sum_a m(a)\ell(a) = 2N$  and the two sums on the left are equal.  $\square$

### 7.3 The lattice sum

**Definition 7.5.**  $Z_1 = \ker \partial \subset \mathbb{Z}^A$  is the circulation lattice of  $D_k$ ,  $\rho$  acts on it by  $(\rho f)(a) = f(\bar{a})$ , and  $\Lambda^+, \Lambda^-$  are its  $(\pm 1)$ -eigenlattices. For a circulation  $c$  with  $0 \leq c \leq m$  let  $G_c$  be the multigraph with  $c(a)$  labelled parallel arcs of content  $a$ ,  $t(c)$  its number of spanning arborescences converging on a root (independent of the root,  $G_c$  being balanced),  $\text{Loc}(c) = \prod_v (d_c^+(v) - 1)!$  over the vertices of positive  $c$ -degree, and

$$N_{\text{seq}}(c) = \frac{t(c) \text{Loc}(c)}{\prod_a c(a)!}.$$

For a circular sequence  $T$  let  $d(T)$  be the number of rotations fixing  $T$ .

**Theorem 7.6** (The lattice sum). *The map  $T \mapsto \text{spec}(T)$  is a surjection from the double-stranded reconstructions of  $S$  at level  $k$  onto*

$$P(S, k) = \{c \in \mathbb{Z}^A : 0 \leq c \leq m, c + \rho c = m, \partial c = 0, \text{supp } c \text{ connected}\},$$

and with  $f = 2c - m$  the set  $P(S, k)$  is the set of  $f \in \Lambda^-$  with  $|f| \leq m$ ,  $f \equiv m \pmod{2}$  and connected support of  $(m + f)/2$ . The fibre over  $c$  has weighted cardinality  $\sum_{\text{spec } T=c} 1/d(T) = N_{\text{seq}}(c)$ . Hence

$$\sum_T \frac{1}{d(T)} = \sum_{f \in \Lambda^- \cap \{|f| \leq m, f \equiv m \pmod{2}\}} N_{\text{seq}}\left(\frac{m + f}{2}\right),$$

where the terms with disconnected support vanish, and when every reconstruction is aperiodic the left side is  $\#DS(S, k)$ .

*Proof.* Lemma 7.4 puts  $\text{spec}(T)$  in  $P(S, k)$ ; the support is connected because  $T$  is one closed walk. Conversely a point  $c \in P(S, k)$  defines a balanced connected multigraph  $G_c$ , which has an Eulerian circuit, and reading the circuit gives a circular sequence  $T$  with  $\text{spec}(T) = c$  and, by  $c + \rho c = m$ , with the spectrum identity. The substitution  $f = 2c - m$  is a bijection between  $\{0 \leq c \leq m\}$  and  $\{|f| \leq m, f \equiv m \pmod{2}\}$ ;  $\partial c = 0$  is  $\partial f = 0$  since  $\partial m = 0$ ; and  $c + \rho c = m$  is  $f + \rho f = 0$ .

For the fibre, the BEST theorem [10, 78] counts the Eulerian circuits of  $G_c$ , as cyclic sequences of its labelled arcs, by  $t(c) \text{Loc}(c)$ . Forgetting the labels sends a circuit to a circular sequence with spectrum  $c$ . Over a fixed  $T$ , a circuit is an assignment of the  $c(a)$  labelled copies of each content to the  $c(a)$  occurrences of that content around  $T$ ,  $\prod_a c(a)!$  assignments in all, and two assignments give the same cyclic labelled sequence exactly when they differ by a rotation of  $T$  onto itself. The rotation group of  $T$ , of order  $d(T)$ , acts freely on assignments, so the fibre has  $\prod_a c(a)!/d(T)$  elements, and summing over  $T$  gives  $t(c) \text{Loc}(c) = \prod_a c(a)! \sum_T 1/d(T)$ . If the support of  $c$  is disconnected then  $G_c$  has no spanning arborescence and  $t(c) = 0$ .  $\square$

*Remark 7.7* (The normalisation is not optional). On one strand there is a single lattice point,  $c = m$ , and  $\prod_a m(a)!$  is a constant; the single-strand notes count in the *arc-rooted* convention  $t(m) \text{Loc}(m)$  and never had to divide. Across the lattice the divisor  $\prod_a c(a)!$  changes from point to point, and the un-normalised sum  $\sum_f t(c_f) \text{Loc}(c_f)$  is  $\sum_T \prod_a c_T(a)!/d(T)$ , which counts each reconstruction with a weight of its own. The two agree whenever  $m \leq 1$ , which is the case at  $k = 12$  for  $\phi X174$ , and disagree as soon as a content has two copies: at  $k = 11$  the un-normalised sum is 2756 and the count is 864 (check (N)). The single-strand number quoted for  $\phi X174$  at  $k = 11$  in Table 1.1,  $|\mathcal{K}(G_{11})| \text{Loc} = 132$ , is the arc-rooted count; the number of circular sequences with the spectrum of  $S$  is  $132/4 = 33$ , and 33 is what appears at the point  $c = \text{spec}(S)$  of Table 7.2.

*Remark 7.8* (Where the linking form went). The symmetric linking form of Remark 6.10 does not enter Theorem 7.6, and the guess that a double-stranded count is a Gauss sum or an isotropic count on  $\mathcal{K}(D_k)$  can be discarded: on  $\mathcal{K}(D_{12}) = \mathbb{Z}/3 \oplus \mathbb{Z}/11$  any non-degenerate quadratic form has a single isotropic vector and a Gauss sum of modulus  $\sqrt{33}$ , while  $\#DS(S, 12) = 10$ . What survives of the form is only the anti-invariance  $f + \rho f = 0$ , which is the linear shadow of  $\rho$  being a duality rather than a symmetry.

**Theorem 7.9** (Rank of the anti-invariant lattice). *Let  $\#\text{fix}_V$  be the number of  $\rho$ -fixed vertices and  $\#\text{fix}_A$  the number of  $\rho$ -fixed contents of  $D_k(S)$ . Then*

$$\text{rank } \Lambda^- = \frac{|A| - |V| + 2 - \#\text{fix}_A - \#\text{fix}_V}{2}, \quad \text{rank } \Lambda^+ = \frac{|A| - |V| - 2 + \#\text{fix}_A + \#\text{fix}_V}{2} + b_0,$$

where  $b_0 \in \{1, 2\}$  is the number of components. The expression does not depend on the compaction:  $|A| - |V|$  and  $\#\text{fix}_A - \#\text{fix}_V$  are the same for the unitig-contracted graph and for the full de Bruijn graph of  $S\bar{S}$ .

*Proof.* Let  $C_1 = \mathbb{Z}^A$ ,  $C_0 = \mathbb{Z}^V$  with the boundary  $\partial$ , and let  $P_A, P_V$  be the permutation matrices of  $\rho$  on contents and vertices. By Lemma 7.2(a),  $\partial P_A = -P_V \partial$ , so  $\varphi = (-P_A, P_V)$  is a chain endomorphism of the cellular chain complex of  $D_k$ , and the Hopf trace formula gives

$$\text{tr}(\varphi_*|H_1) - \text{tr}(\varphi_*|H_0) = \text{tr}(-P_A|C_1) - \text{tr}(P_V|C_0) = -\#\text{fix}_A - \#\text{fix}_V.$$

Here  $H_1 = Z_1$  with  $\varphi_* = -\rho$ , and  $H_0 = \mathbb{Z}^{b_0}$  with  $\varphi_*$  the permutation of components induced by  $\rho$ , whose trace is the number of  $\rho$ -fixed components: 1 when  $b_0 = 1$ , and 0 when  $b_0 = 2$  since then  $\rho$  exchanges the halves (Lemma 7.2(c)). Hence  $\text{tr}(\rho|Z_1) = \#\text{fix}_A + \#\text{fix}_V - \text{tr}(\varphi_*|H_0)$ . Over  $\mathbb{Q}$  the involution  $\rho$  is diagonalisable with eigenvalues  $\pm 1$ , so  $\text{rank } \Lambda^+ - \text{rank } \Lambda^- = \text{tr}(\rho|Z_1)$  while  $\text{rank } \Lambda^+ + \text{rank } \Lambda^- = \text{rank } Z_1 = |A| - |V| + b_0$ . Solving,

$$\text{rank } \Lambda^- = \frac{1}{2}(|A| - |V| + b_0 + \text{tr}(\varphi_*|H_0) - \#\text{fix}_A - \#\text{fix}_V),$$

and  $b_0 + \text{tr}(\varphi_*|H_0) = 2$  in both cases. For the last sentence: unitig contraction is a  $\rho$ -equivariant homotopy equivalence, so  $\text{rank } Z_1$  and the Lefschetz number  $\text{tr}(\varphi_*|H_1) - \text{tr}(\varphi_*|H_0)$  are unchanged, and the latter is  $-\#\text{fix}_A - \#\text{fix}_V$  in either cellular structure.  $\square$

**Corollary 7.10.** *For even  $k$ ,  $\text{rank } \Lambda^- = (|A| - |V| + 2 - \#\{\text{palindromic repeated } k\text{-mers}\})/2$ ; for odd  $k$ ,  $\text{rank } \Lambda^- = (|A| - |V| + 2 - \#\{\text{palindromic } (k+1)\text{-mers of } S\bar{S}\})/2$ . If  $S$  has no inverted repeat of length  $\geq k$  then  $\Lambda^- \cong Z_1(G_k(S))$ .*

*Proof.* The first two statements are Theorem 7.9 with Lemma 7.2(b). If  $D_k$  is disconnected then  $Z_1(D_k) = Z_1(G_k(S)) \oplus Z_1(G_k(\bar{S}))$  with  $\rho$  exchanging the summands, and  $f \mapsto (f, -\rho f)$  identifies  $Z_1(G_k(S))$  with  $\Lambda^-$ .  $\square$

*Remark 7.11* (Not the number of inverted repeats). It is tempting to read  $\text{rank } \Lambda^-$  as the number of independent inversions available to the genome, and both the specification of this chapter and its planning did. Table 7.1 refutes it. At  $k = 12$ ,  $\phi\text{X174}$  carries exactly one inverted-repeat pair of length  $\geq 12$  (ATTAAGCTCATT at 441 against its reverse complement at 1448) and one palindromic repeated 12-mer, and  $\text{rank } \Lambda^- = 4$ . At the other extreme, Corollary 7.10 says that a genome with repeats but no inverted repeat has  $\Lambda^-$  as large as the whole circulation lattice of  $G_k(S)$ , while only two of its box points —  $\text{spec}(S)$  and  $\text{spec}(\bar{S})$  — have connected support. The number that does count inversion classes is the number of connected box points, 8 and 218 in Table 7.2, and it has no closed form we know of.

| $k$ | $ V $ | $\#\text{fix}_V$ | $ A $ | $\#\text{fix}_A$ | $\sum m$ | $b_0$ | $\text{rank } Z_1$ | $\text{rank } \Lambda^-$ |
|-----|-------|------------------|-------|------------------|----------|-------|--------------------|--------------------------|
| 13  | 0     | 0                | 0     | 0                | 0        | —     | —                  | 0                        |
| 12  | 7     | 1                | 14    | 0                | 14       | 1     | 8                  | 4                        |
| 11  | 30    | 0                | 53    | 1                | 60       | 1     | 24                 | 12                       |
| 10  | 125   | 3                | 222   | 0                | 252      | 1     | 98                 | 48                       |
| 9   | 396   | 0                | 685   | 3                | 812      | 1     | 290                | 144                      |

Table 7.1:  $\phi\text{X174}$  (NC\_001422.1, 5386 bp).  $\text{rank } \Lambda^-$  is computed as the corank of the balance matrix on orbit representatives, modulo two large primes, and agrees with Theorem 7.9 in every row. At  $k = 13$  no  $k$ -mer is repeated on either strand and  $D_k$  is the pair of disjoint cycles  $S, \bar{S}$ . Check (L).

| $k$ | box                            | conn. | lattice | brute | un-norm. | one strand    | $ \mathcal{K}(D_k) $ |
|-----|--------------------------------|-------|---------|-------|----------|---------------|----------------------|
| 12  | 10                             | 8     | 10      | 10    | 10       | 4             | 33                   |
| 11  | 528                            | 218   | 864     | 864   | 2756     | 66            | 25 318 912           |
| 10  | rank 48: beyond the enumerator |       |         |       |          | 8 610 708 632 | $3.42 \cdot 10^{35}$ |

Table 7.2: The lattice sum for  $\phi\text{X174}$ . “Box” is  $|\Lambda^- \cap \{f \mid |f| \leq m, f \equiv m\}|$ ; “conn.” the number of points with  $N_{\text{seq}} > 0$ ; “lattice” is  $\#\text{DS}$  from Theorem 7.6; “brute” the number of closed walks of  $D_k$  with usage  $+$   $\rho$  usage  $= m$ , enumerated by backtracking without reference to the lattice; “un-norm.” is  $\sum_f t(c_f) \text{Loc}(c_f)$ ; “one strand” is  $N_{\text{seq}}(S) + N_{\text{seq}}(\bar{S})$ .  $\text{Loc}(D_k) = 1$  at  $k = 12, 11$ , so  $|\mathcal{K}(D_k)|$  is also the Eulerian circuit count of the double cover. Checks (B), (S), (E), (N).

**What the lattice points are.** At  $k = 12$  the eight connected points are  $\text{spec}(S)$  and  $\text{spec}(\bar{S})$ , each with  $N_{\text{seq}} = 2$  (the interleaved two-copy repeat of the single-strand theory), and six spectra with  $N_{\text{seq}} = 1$ . One of the six is the inversion of  $S[441..1459]$  between the inverted-repeat pair. The other five mix that inversion with the repeat exchange and with hairpin turns at the palindrome: for instance the point that drops the two segments  $S[264..452]$  and  $S[1448..2771]$  of  $S$  and takes their reverse complements is the sequence

$$\begin{array}{c} S[441..457] \ S[3205..4483] \ S[4483..264+N] \ S[2760..3205] \ S[457..1448] \\ \overline{S[264..441]} \ \overline{S[1448..2760]}, \end{array}$$

a rearrangement of the whole genome in which two blocks are read from the other strand. At  $k = 11$  the 218 connected points carry  $N_{\text{seq}}$  values 33 (twice), 18, 16 (twice each), 15 (four times), 11 (six), 10 (two), 9 (six), 6 (22), 5 (20), 4 (six), 3 (36), 2 (56) and 1 (54), summing to 864. Every one of the 874 reconstructions at the two levels is aperiodic, so  $d(T) = 1$  throughout and the weighted count is the count (check (A)).

**The brute force.** Check (E) enumerates, by depth-first search on the content multigraph, every closed walk whose usage  $u$  satisfies  $u(a) + u(\bar{a}) \leq m(a)$  along the way and  $= m(a)$  at the end — the walk has exactly  $\sum m/2$  arcs — rooted at its smallest content  $a_0$  and weighted  $1/u(a_0)$ ; the search never sees  $\Lambda^-$ . It returns 90 nodes and 8 spectra at  $k = 12$ , 45 061 nodes and 218 spectra at  $k = 11$ ; the spectra found are exactly the connected box points, and the count at each spectrum is exactly  $N_{\text{seq}}$ .

## 7.4 The inversion flip

**Definition 7.12.** Let  $T$  be a double-stranded reconstruction, read as the closed walk  $a_0 a_1 \cdots a_{L-1}$  of arcs of  $D_k$  with vertex visits  $v_i = \text{tail}(a_i)$ . A *flip* at a pair of visits  $i \neq j$  with  $v_j = \rho(v_i)$  replaces the block  $a_i \cdots a_{j-1}$  (indices cyclic) by  $\bar{a}_{j-1} \cdots \bar{a}_i$ :

$$T' = a_0 \cdots a_{i-1} \bar{a}_{j-1} \cdots \bar{a}_i a_j \cdots a_{L-1}.$$

The *flip graph* has the reconstructions as vertices and the flips as edges.

**Proposition 7.13.** (a)  $T'$  is a double-stranded reconstruction, and every replacement of a block  $w X \bar{w}$  of  $T$  by  $w \bar{X} \bar{w}$ , for a  $k$ -mer  $w$  and a later occurrence of  $\bar{w}$ , is a flip in the sense of Definition 7.12.

- (b)  $\text{spec}(T') = \text{spec}(T) - c_B + \rho c_B$ , where  $c_B$  is the usage of the block. A flip preserves the spectrum if and only if  $c_B = \rho c_B$ .
- (c) Flipping the complementary block gives  $\bar{T}'$ ; flipping the whole cycle at a palindromic visit gives  $\bar{T}$ .
- (d) If  $S$  carries no inverted repeat of length  $\geq k$ , no reconstruction admits a flip: the flip graph is discrete, with  $2N_{\text{seq}}(S)$  vertices.

*Proof.* (a) The block  $a_i \cdots a_{j-1}$  runs from  $v_i$  to  $v_j = \rho(v_i)$ ; its reverse-complement image  $\bar{a}_{j-1} \cdots \bar{a}_i$  runs from  $\rho(v_j) = v_i$  to  $\rho(v_i) = v_j$ , so  $T'$  is a closed walk, and its spectrum is  $\text{spec}(T) - c_B + \rho c_B$ , which still satisfies  $c + \rho c = m$ . For the second statement: the block  $w X \bar{w}$  has reverse complement  $\bar{w} \bar{X} \bar{w} = w \bar{X} \bar{w}$ , so the endpoints are unchanged and the interior  $(k+1)$ -mers are replaced by their reverse complements, which lie in  $\text{spec}(\bar{T})$  and hence in the double-stranded spectrum. It remains to see that  $w$  is a vertex of  $D_k$ :  $w$  occurs in  $T$  and  $\bar{w}$  occurs in  $T$ , so  $w$  occurs in  $T$  and in  $\bar{T}$ , so  $w$  occurs at least twice in  $T\bar{T}$ , hence at least twice in  $S\bar{S}$  by the spectrum identity, hence is a repeated  $k$ -mer, i.e. a vertex. (b) is the computation in (a). (c) The complementary block runs from  $v_j$  to  $v_i = \rho(v_j)$ , and flipping it yields the walk  $\bar{a}_{i-1} \cdots \bar{a}_0 \bar{a}_{L-1} \cdots \bar{a}_j a_j \cdots$ , which as a circular sequence is the reverse complement of  $T'$ ; taking  $i = j$  at a palindromic vertex flips everything. (d) By Lemma 7.2(c)  $D_k$  is then disconnected,  $T$  lies in one half, and  $\rho(v_i)$  lies in the other for every visit, so no pair of visits qualifies; the reconstructions are those of  $S$  and of  $\bar{S}$ .  $\square$

Part (d) says that connectivity of the flip graph is not a theorem: a genome with a multi-copy repeat and no inverted repeat has  $2N_{\text{seq}}(S) > 2$  reconstructions and no move between any of them. What can be asked is whether a *given* genome is flip-connected, and the answer for  $\phi\text{X174}$  is yes, in a strong form.

**Theorem 7.14** ( $\phi\text{X174}$ , computed). *At  $k = 12$  and  $k = 11$ :*

- (a) the flip graph on all reconstructions (10, resp. 864 vertices; 16, resp. 5140 edges) is connected, without the move  $T \mapsto \bar{T}$ ;
- (b) no flip preserves the spectrum:  $c_B \neq \rho c_B$  for every block of every reconstruction; consequently the quotient graph on spectrum classes (8, resp. 218 vertices) is connected, and every class is at flip distance at most 3, resp. 8, from  $\text{spec}(S)$  (histogram 1, 2, 3, 2, resp. 1, 7, 16, 38, 56, 52, 34, 12, 2);

- (c) *two reconstructions with the same spectrum are joined by flip paths of even length at least four: at  $k = 11$  the 6776 ordered same-spectrum pairs have flip distances 4 (3932 pairs), 6 (2556) and 8 (288); at  $k = 12$  all four have distance 4;*
- (d) *the vertex pairs  $\{v, \rho v\}$  realising flips number 3, resp. 14, of which 1, resp. 5, are the inverted-repeat pairs and palindromes of  $S$  itself; restricted to those, flips reach 6 of 8, resp. 212 of 218, classes. The others are pairs  $\{a, \bar{a}\}$  with  $a$  a repeated  $k$ -mer of one strand — **AAACGGCAGAA** against **TTCTGCCGTTT**, for instance — which become flip sites only after an earlier flip has brought  $\bar{a}$  into the reconstruction.*

The theorem is the transcript of `ds_flips.py`: all reconstructions are taken from the brute force of check (E), all flips are generated from Definition 7.12, and each flip is verified to land in the reconstruction set (13 928 instances at  $k = 11$ ).

*Remark 7.15* (Transpositions as products of flips). Part (c) is the de Bruijn-graph form of a classical fact. The two reconstructions of  $\text{spec}(S)$  at  $k = 12$  differ by exchanging the two copies of a repeat, which as a rearrangement of  $S$  is a transposition of the blocks between the copies. In sorting by reversals [15, 41] a transposition is a product of three reversals. Here a reversal is available only when it is flanked by a vertex and its reverse complement, and the shortest product that realises the exchange has four factors; the distances in (c) are all even, which we have not explained. Part (d) shows the mechanism: the exchange needs a flip at  $\{a, \bar{a}\}$  with  $a$  a single-strand repeat, and the only way to bring  $\bar{a}$  into  $T$  is to flip first at an inverted-repeat pair of  $S$ , then at  $\{a, \bar{a}\}$ , and then to undo the first flip.

*Remark 7.16* (Corrections to the specification). Three statements in the specification of this chapter, one of them originating in its planning, are not correct as stated. *First*,  $\text{rank } \Lambda^-$  is not the number of independent inverted repeats of length  $\geq k$  (Remark 7.11); it is given by Theorem 7.9. *Second*, the per-point term is not  $t(c_f) \prod (d^+ - 1)!$  but that quantity divided by  $\prod_a c_f(a)!$  (Remark 7.7); the un-normalised sum is 2756 where the count is 864. *Third*, “the flip graph is connected” and “a same-spectrum transposition is a composition of inversions” are properties of  $\phi X174$  at  $k = 12, 11$ , established by computation, and not theorems: Proposition 7.13(d) gives the general obstruction, and the composition, where it exists, costs four flips rather than the three reversals of the unconstrained model.

## 7.5 Scope, provenance, and what is open

**Scope.** Lemma 7.2, Lemma 7.4, Theorem 7.6, Theorem 7.9, Corollary 7.10 and Proposition 7.13 are proved in full generality for a single circular sequence. Theorem 7.14 is a computation for one genome at two word lengths. The lattice sum is evaluated at  $k = 12$  and  $k = 11$ ; at  $k = 10$  the lattice has rank 48 and the capacity-pruned backtracking enumerator of `ds_lattice.py` produces no point in ten minutes, so only the rank row of Table 7.1 is reported there; Chapters 8 and 9 fill the rest. The count is a count of circular sequences, each weighted by  $1/d(T)$ ; in the data  $d(T) = 1$  throughout.

**Provenance.** The BEST theorem is [10, 78]; its use to count genome reconstructions is [51]; the bidirected-graph model of double-stranded assembly, in which a double-stranded reconstruction is a walk in a bidirected de Bruijn graph, is [60] after [46], and those papers ask for existence and for one solution, not for the count. Sorting by reversals and the transposition-as-three-reversals identity are [15, 41]. The double cover  $D_k$ , its balance and its connectivity criterion are Chapter 5’s;

the anti-automorphism theorem and the symmetric linking form are Theorem 6.9 and Remark 6.10; the single-strand counting convention is Remark 1.1. The Hopf trace formula is standard. We are not aware of prior work expressing a double-stranded reconstruction count as a sum over an eigenlattice of the strand involution, or of prior work on the flip graph of a de Bruijn graph; a targeted search found none.

**Open.** *First*, the number of connected box points — the number of spectrum classes, or equivalently the size of the orbit of  $\text{spec}(S)$  under the flip pseudo-group — has no formula; 8 and 218 are known here, Chapter 8 adds 909 103 210 at  $k = 10$ , and Chapter 10 shows that counting the box points is  $\#P$ -hard in general. *Second*, the genome-scale question: Theorem 7.9 evaluates the rank of  $\Lambda^-$  for *E. coli* in one line from the double cover of Chapter 5, but the sum is not enumerable there, and what is wanted is a transfer-matrix or generating-function evaluation of  $\sum_f N_{\text{seq}}(c_f)$  along the genome, or an Ehrhart-type asymptotic in  $k$ . *Third*, the parity of same-spectrum flip distances in Theorem 7.14(c). *Fourth*, the same lattice with the involution removed and  $m$  split into  $p$  parts is the phasing problem for a polyploid, where the symmetric group on the parts acts on the decompositions rather than on the graph; that is the subject of Chapter 13.



## Chapter 8

# The double-stranded lattice sum by frontier elimination

**Summary.** Chapter 7 wrote the number of double-stranded reconstructions of a circular sequence as a sum over the lattice points of a box,  $\#DS(S, k) = \sum_f N_{\text{seq}}((m + f)/2)$  with  $f$  running over  $\Lambda^- \cap \{|f| \leq m, f \equiv m \pmod{2}\}$ , and enumerated that box by backtracking. The enumeration works at rank  $\Lambda^- = 4$  and 12 and dies at 48:  $\phi X174$  at  $k = 10$  is the blank row of Table 7.2. The obstruction is not the rank. The circulation condition is one linear equation per vertex, and each equation involves only the four endpoints of one  $\rho$ -orbit, so the orbits can be processed in an order while carrying, as state, the partial imbalance at the vertices still open — a product of sparse transfer operators whose cost is governed by the width of the *frontier*, not by the dimension of the lattice. At  $k = 10$ , where the box lies in a 48-dimensional lattice and has more than two billion points, the elimination needs 50 758 states and 8.4 seconds:  $\phi X174$  has exactly 2 481 687 900 box points at  $k = 10$ . Carrying in addition the partition of the frontier into the classes already joined counts the points with connected support, of which there are 909 103 210; both columns Chapter 7 left empty are now filled, and only  $\#DS$  itself is not (Chapter 9 fills it). Two further things follow. Connectivity of  $\text{supp } c$ , carried in Chapter 7 as a separate filter, is not a side condition at all: a balanced digraph has strongly connected weak components, so its out-Laplacian has nullity equal to the number of them, and  $t(G_c)$  — hence  $N_{\text{seq}}(c)$  — vanishes on a disconnected support of its own accord. And the order that works is not the genome’s own: along the reference coordinate the frontier at  $k = 10$  is 62 wide and the elimination does not finish, while a greedy order is 26 wide, which is the interleaving of repeats of Theorem 1.6 seen as a width.

### 8.1 What is being counted

We keep the notation of Chapter 7 exactly:  $D_k = D_k(S)$  is the labelled double cover of Definition 7.1,  $A$  its set of contents,  $m(a)$  the number of labelled arcs of content  $a$ ,  $\rho$  the reverse-complement involution, and  $\Lambda^-$  the  $(-1)$ -eigenlattice of  $\rho$  on the circulations of  $D_k$ . A double-stranded reconstruction is as in Definition 7.3, and by Theorem 7.6

$$\#DS(S, k) = \sum_f N_{\text{seq}}\left(\frac{m + f}{2}\right), \quad N_{\text{seq}}(c) = \frac{t(G_c) \text{Loc}(G_c)}{\prod_a c(a)!}, \quad (8.1)$$

the sum over  $f \in \Lambda^-$  in the box  $|f| \leq m, f \equiv m \pmod{2}$ , with  $G_c, t(G_c)$  and  $\text{Loc}(G_c)$  as there; the division by  $\prod_a c(a)!$  passes from labelled arcs to sequences (Remark 7.7).

Chapter 7 enumerated the box by backtracking over a basis of  $\Lambda^-$  with capacity pruning, and recorded the cost: 10 points at  $k = 12$  where  $\text{rank } \Lambda^- = 4$ , 528 at  $k = 11$  where  $\text{rank } \Lambda^- = 12$ , and nothing at  $k = 10$ , where  $\text{rank } \Lambda^- = 48$  and the search does not return.

## 8.2 The constraint system and its frontier

The point of this chapter is that (8.1) is indexed by the solutions of a very sparse linear system, and that sparsity, not the rank, is what governs the count.

**Definition 8.1** (the orbit variables).  $\rho$  acts on  $A$  without fixed points except for the self-reverse-complementary contents, which carry  $f(a) = 0$  because  $\rho f = -f$ . Choose one representative  $a$  from each free orbit  $\{a, \rho a\}$  and let  $g_j = f(a_j)$ , so  $f(\rho a_j) = -g_j$ . The box condition is  $g_j \in \{-m_j, -m_j + 2, \dots, m_j\}$  with  $m_j = m(a_j)$ , and  $f$  is a circulation exactly when, for every vertex  $v$ ,

$$\sum_j B(v, j) g_j = 0, \quad B(v, j) = [v = t(a_j)] - [v = h(a_j)] - [v = t(\rho a_j)] + [v = h(\rho a_j)]. \quad (8.2)$$

The essential feature is that column  $j$  of  $B$  is supported on at most four vertices — the two endpoints of  $a_j$  and the two of  $\rho a_j$  — whatever the rank of  $B$  may be.

**Definition 8.2** (frontier). Fix an order  $\pi = (j_1, \dots, j_R)$  of the orbits. A vertex  $v$  is *open after step  $i$*  if some  $j_\ell$  with  $\ell \leq i$  and some  $j_\ell$  with  $\ell > i$  both have  $B(v, j_\ell) \neq 0$ . Write  $\mathcal{F}_i(\pi)$  for the set of open vertices and  $w(\pi) = \max_i |\mathcal{F}_i(\pi)|$  for the *width* of  $\pi$ . The minimum of  $w(\pi)$  over all orders is the pathwidth of the hypergraph whose vertices are the vertices of  $D_k$  and whose hyperedges are the supports of the columns of  $B$ .

**Theorem 8.3** (frontier elimination). Let  $\kappa_i(v) = \sum_{\ell > i} |B(v, j_\ell)| m_{j_\ell}$  be the capacity still available at  $v$  after step  $i$ , and let  $X_i = \prod_{v \in \mathcal{F}_i(\pi)} \{-\kappa_i(v), \dots, \kappa_i(v)\}$ . Define  $T_i: \mathbb{Z}^{X_{i-1}} \rightarrow \mathbb{Z}^{X_i}$  by

$$(T_i \phi)(y) = \sum_{\substack{g \in \{-m_{j_i}, \dots, m_{j_i}\} \\ g \equiv m_{j_i} (2)}} \sum_{\substack{x \in X_{i-1} \\ x + gB(\cdot, j_i) \text{ restricts to } y \\ \text{and vanishes at every vertex closing at } i}} \phi(x).$$

Then

$$|\Lambda^- \cap \{|f| \leq m, f \equiv m (2)\}| = (T_R \cdots T_1 \delta_0)(\cdot),$$

and the whole product costs  $O(\sum_i |X_{i-1}| (m_{j_i} + 1) |\mathcal{F}_i|)$  integer operations, with

$$|X_i| \leq \prod_{v \in \mathcal{F}_i} (2\kappa_i(v) + 1) \leq (2 \max_v \kappa_i(v) + 1)^{w(\pi)}.$$

*Proof.* A point of the box is an assignment  $g \in \prod_j \{-m_j, \dots, m_j\}$  of the right parity satisfying (8.2). Process the orbits in the order  $\pi$  and record, after step  $i$ , the partial sums  $x(v) = \sum_{\ell \leq i} B(v, j_\ell) g_{j_\ell}$  at the open vertices. A vertex  $v$  that is not open after step  $i$  receives no further contribution, so (8.2) at  $v$  is decided then and there: it is the condition, imposed in  $T_i$ , that  $x(v) = 0$  at every closing vertex. A vertex not yet touched has  $x(v) = 0$  and need not be recorded. Two assignments of  $g_{j_1}, \dots, g_{j_i}$  with the same restriction to  $\mathcal{F}_i$  therefore admit exactly the same completions, which is the Markov property that makes the transfer operators well defined and their product the count. The capacity bound holds because  $|x(v)| = |\sum_{\ell > i} B(v, j_\ell) g_{j_\ell}| \leq \kappa_i(v)$  for any state that has a completion, so states outside  $X_i$  may be dropped.  $\square$

*Remark 8.4* (what the cost does and does not depend on). Theorem 8.3 contains no reference to  $\text{rank } \Lambda^-$ . The backtracking of Chapter 7 has cost proportional to the number of *nodes* of a search tree of depth  $R$ , which for a lattice of rank  $r$  is at least the number of points and so at least exponential in  $r$ ; the elimination has cost proportional to the number of *states*, and a state is a vector of imbalances on the frontier. At  $k = 10$  these are 2 481 687 900 and 50 758.

**Proposition 8.5** (product-form weights). *Let  $u_j$  be functions on the values of  $g_j$  and  $h_v$  functions on  $\mathbb{Z}_{\geq 0}$ , and*

$$w(c) = \prod_j u_j(g_j) \prod_v h_v(d_c^+(v)).$$

*Then  $\sum_f w(c_f)$  over the box is computed by Theorem 8.3 after enlarging the state at each open vertex by its accumulated out-degree. In particular  $\text{Loc}(G_c)$  and  $\prod_a c(a)!$  are of this form.*

*Proof.*  $d_c^+(v) = \sum_{a: t(a)=v} c(a)$  accumulates one orbit at a time exactly as  $x(v)$  does, and is final when  $v$  closes, where  $h_v$  may be applied;  $u_j$  multiplies into the transfer operator at step  $j$ .  $\text{Loc}$  takes  $h_v(d) = (d-1)!$  and  $u_j = 1$ , while  $\prod_a c(a)!$  takes  $h_v = 1$  and  $u_j(g) = (\frac{m_j+g}{2})!(\frac{m_j-g}{2})!$ .  $\square$

### 8.3 Connectivity is the matrix-tree determinant

Chapter 7 carried, beside the box conditions, a predicate: the support of  $c$  must be connected, or the point contributes nothing. The predicate is real but it is not an extra condition.

**Lemma 8.6** (connectivity is not a side condition). *Let  $c \in \mathbb{Z}_{\geq 0}^A$  be balanced,  $V_c = \text{supp } c$  the set of vertices it meets,  $L_c$  the out-degree Laplacian of  $G_c$  on  $V_c$  and  $\tilde{L}_c$  the matrix  $L_c$  with one row and the corresponding column deleted. Then  $\det \tilde{L}_c = t(G_c)$ , and*

$$\det \tilde{L}_c = 0 \iff G_c \text{ is disconnected.}$$

*Consequently  $N_{\text{seq}}(c) = 0$  whenever  $\text{supp } c$  is disconnected, and the connectivity predicate may be deleted from (8.1) without changing the sum.*

*Proof.*  $\det \tilde{L}_c = t(G_c)$  is the matrix-tree theorem for digraphs [80]. A balanced digraph has  $d^+ = d^-$  at every vertex, so each of its weakly connected components is Eulerian and hence strongly connected; therefore the kernel of  $L_c$  is spanned by the indicators of the components and  $\dim \ker L_c = r$ , the number of them. Deleting one row and one column drops the nullity by at most one, so  $r \geq 2$  forces  $\det \tilde{L}_c = 0$ , while  $r = 1$  gives  $t(G_c) \geq 1$ .  $\square$

*Remark 8.7.* The gain is conceptual rather than computational: as a filter, connectivity is a union-find pass and the determinant is a cubic elimination, so one still tests connectivity first. What Lemma 8.6 removes is the impression that (8.1) is a lattice sum *subject to a combinatorial constraint*. It is a lattice sum, full stop; the constraint is already inside  $N_{\text{seq}}$ . This is also the answer, in the only form it has, to the natural question of an algebraic criterion of connected support avoiding graph traversal: the criterion is the vanishing of a determinant whose entries are linear in  $c$ .

Connectivity can also be carried *inside* the elimination, and then it counts.

**Theorem 8.8** (connected points). *Enlarge the state of Theorem 8.3 by (i) the partition of  $\mathcal{F}_i$  into the classes connected by arcs already chosen, (ii) for each class, whether it meets a chosen arc, and (iii) the number of classes that have left the frontier while carrying a chosen arc. Prune*

| $k$ | $ V $ | contents | orbits | rank $\Lambda^-$ | coordinate width | greedy width |
|-----|-------|----------|--------|------------------|------------------|--------------|
| 12  | 7     | 14       | 7      | 4                | 6                | 6            |
| 11  | 30    | 53       | 26     | 12               | 18               | 10           |
| 10  | 125   | 222      | 111    | 48               | 62               | 26           |

Table 8.1: The width of the two orders. Along the genome the frontier at  $k = 10$  is 62 vertices wide and the elimination does not finish — it passes 1.9 million states and is abandoned; the greedy order is 26 wide and finishes in 8.4 seconds with 50 758 states. Check (O).

whenever (iii) exceeds one, and accept only states with (iii) equal to one and empty frontier. The resulting count is the number of box points with connected support. The state count is at most  $|X_i| \cdot B(|\mathcal{F}_i|) \cdot 2^{|\mathcal{F}_i|} \cdot 2$  with  $B$  the Bell numbers.

*Proof.* A class leaves the frontier only when no vertex of it is open, and after that no arc can join it to anything; so a class that leaves while meeting a chosen arc is a finished connected component of  $\text{supp } c$ , and conversely every component is finished exactly once, at the step where its last vertex closes. A support is connected iff it has exactly one component, which is condition (iii); classes meeting no chosen arc are not vertices of  $\text{supp } c$  and must be ignored, which is what (ii) records. Since the class of a vertex and its flag are determined by the chosen arcs among the processed orbits, the state is again Markov.  $\square$

*Remark 8.9* (the connectivity state is cheaper than it looks). The Bell factor in Theorem 8.8 is a gross overestimate, because the rule “at most one finished component” forbids most partitions from ever arising: at  $k = 12$  and  $k = 11$  the connected elimination peaks at 8 and 136 states against 4 and 20 for the plain one. It is nevertheless the factor that decides the cost at  $k = 10$ , where a frontier of 26 vertices is small for imbalances and large for partitions: 9 749 212 states against 50 758, and seventy-eight minutes against eight seconds.  $B(26)$  is  $5 \times 10^{19}$ , so what is actually reached is a vanishing fraction of the bound.

## 8.4 The genome’s own coordinate is not the good order

Theorem 8.3 turns the problem into the choice of  $\pi$ . There is an obvious candidate, and it is the one the biology suggests: take the orbits in the order of their position along the reference genome, so that the elimination sweeps the chromosome once. It is bounded, and it is not good.

**Proposition 8.10** (the coordinate order). *Order the orbits by the position of their first occurrence on  $S$ . Then  $\mathcal{F}_i$  is contained in the set of repeated  $k$ -mers having occurrences in orbits on both sides of step  $i$ ; in particular, if no more than  $\omega$  repeat classes straddle any coordinate, the elimination runs in  $O(R(2 \max \kappa + 1)^\omega)$  and  $\#DS$ ’s box is counted in time polynomial in  $N$  for fixed  $\omega$ .*

*Proof.* Immediate from Definition 8.2: the vertices met by orbit  $j$  are endpoints of  $a_j$  and  $\rho a_j$ , and a vertex is open only between the first and last orbit meeting it.  $\square$

The hypothesis is the point. Table 8.1 gives the widths.

*Remark 8.11* (why the coordinate order fails). A vertex stays open from its first to its last incident orbit, so the coordinate width at a position  $x$  counts the repeat classes straddling  $x$ . Two things inflate it. First,  $\rho$  pairs an arc of  $S$  with an arc of  $\bar{S}$ , and the two lie at reflected coordinates, so an orbit is not local on the reference at all: every orbit near the origin of  $S$  is also an orbit near the

| $k$ | rank $\Lambda^-$ | box                  |        | connected          |           | % conn. | #DS |
|-----|------------------|----------------------|--------|--------------------|-----------|---------|-----|
|     |                  | points               | states | points             | states    |         |     |
| 13  | 0                | 1                    | —      | 1                  | —         | 100     | 1   |
| 12  | 4                | 10                   | 4      | 8                  | 8         | 80.0    | 10  |
| 11  | 12               | 528                  | 20     | 218                | 136       | 41.3    | 864 |
| 10  | 48               | <b>2 481 687 900</b> | 50 758 | <b>909 103 210</b> | 9 749 212 | 36.6    | —   |

Table 8.2: The blank row of Table 7.2, filled in both of its counting columns. The rows for  $k = 12$  and  $k = 11$  reproduce that table exactly, by the elimination and independently by the backtracking enumerator; the box count at  $k = 10$  is confirmed by pinning one coordinate and summing the elimination over its values. “states” is the peak number of frontier states: 8.4 seconds for the box, 78 minutes and 10 GB for the connected count. Checks (T), (K), (W), (M).

end of  $\bar{S}$ . Second, and this is the substance, inverted repeats of a real sequence *interleave* — that is the content of the interleaving theorem (Theorem 1.6), where the crossing pattern of repeats is exactly what the sandpile group of the de Bruijn graph records. Interleaved repeats are nested intervals crossing each other, and a family of pairwise crossing intervals forces every one of them to be open at the crossing point. The coordinate width is therefore a measure of interleaving, and a genome with interleaved inverted repeats has no narrow sweep. The greedy order abandons the chromosome and follows the constraint hypergraph instead.

## 8.5 $\phi X174$ at $k = 10$

$\phi X174$  at  $k = 10$  has 125 vertices, 222 contents, 111 free  $\rho$ -orbits and rank  $\Lambda^- = 48$ ; the box  $\Lambda^- \cap \{|f| \leq m, f \equiv m \pmod{2}\}$  contains exactly

$$2\,481\,687\,900$$

points. Three remarks on the number. It is exact and integral throughout — the elimination adds integers and never divides. It is confirmed independently: pinning one orbit variable to each of its admissible values in turn and summing the resulting eliminations, for two different orbits, returns the same total, and the two orbits have disjoint schedules. And it is not reachable by enumeration:  $2.48 \times 10^9$  points at the rate at which Chapter 7 enumerates 528 of them would be several months, and the relevant comparison is that the whole count took 8.4 seconds because it never listed a point.

*Remark 8.12* (the connected fraction). The proportion of box points with connected support falls with  $k$ : 100%, 80%, 41.3%, 36.6% at  $k = 13, 12, 11, 10$ . This is what one expects and it is worth saying why. A point chooses one arc from each free  $\rho$ -orbit, so the support of  $c$  has close to  $|A|/2$  arcs on  $|V|$  vertices whatever the point; at  $k = 10$  that is about 111 arcs on 125 vertices, which is sparse enough that a random choice usually falls apart. What the elimination shows is that 36.6% of them do not, and that the surviving set is still of size  $9 \times 10^8$  — so the failure to reach #DS at  $k = 10$  is not for want of a small enough index set.

*Remark 8.13* (what is still missing, exactly). Of the three counting columns of Table 8.2 only #DS is now empty at  $k = 10$ , and it is empty for a reason of a different kind from the other two. Those were questions of state size: Theorem 8.3 settles the first in seconds and Theorem 8.8 the second in an hour and a quarter.

| key | claim                                                                                                                                                                                |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (D) | the double cover rebuilt from the definition reproduces $ V $ , $\#\text{fix}_V$ , $ A $ , $\#\text{fix}_A$ , $\sum m$ and $\text{rank } \Lambda^-$ of Table 7.1 at $k = 10, 11, 12$ |
| (T) | the frontier elimination and the backtracking enumerator return the same box count, 10 and 528                                                                                       |
| (C) | on all 538 box points at $k = 12, 11$ : $\text{supp } c$ connected $\iff \det \tilde{L}_c \neq 0$ — Lemma 8.6                                                                        |
| (W) | summing $N_{\text{seq}}$ over the box returns 8 and 218 connected points and $\#\text{DS} = 10$ and 864                                                                              |
| (O) | coordinate width 6, 18, 62 against greedy width 6, 10, 26; at $k = 10$ the coordinate order does not finish                                                                          |
| (K) | $\phi X174$ at $k = 10$ : 2 481 687 900 box points, and the same total after pinning each of two orbits                                                                              |
| (N) | the connectivity-class elimination returns Chapter 7's 8 and 218                                                                                                                     |
| (M) | $\phi X174$ at $k = 10$ : 909 103 210 box points with connected support                                                                                                              |

Table 8.3: The eight checks. All pass. Check (M) takes about eighty minutes and 10 GB; the flag `--quick` skips it and the rest of the battery runs in six minutes.

The  $\#\text{DS}$  column is a question of principle. By Proposition 8.5 the elimination sums any weight that factors into per-orbit and per-vertex terms, and two of the three factors of  $N_{\text{seq}}$  do:  $\text{Loc}(G_c)$  is a product over vertices of a function of the out-degree, and  $\prod_a c(a)!$  a product over orbits. The third,  $t(G_c)$ , does not factor over any bounded frontier — it is a determinant of order up to  $|V_c|$ , a global function of the whole support — and it is the one that carries the count. Evaluating it point by point is cheap per point and hopeless in aggregate:  $9.09 \times 10^8$  determinants of order up to 125. So  $\#\text{DS}(\phi X174, 10)$  remains out of reach of the elimination (Chapter 9 obtains it point by point), and it is now clear that what stands in the way is the arborescence factor and not the lattice — the index set is known exactly, and it is the summand that cannot be carried. Table 7.2 gives the two ends of the interval it lies in: the single-strand count  $N_{\text{seq}}(S) + N_{\text{seq}}(\bar{S}) = 8\,610\,708\,632$  is a lower bound, since  $\text{spec}(S)$  and  $\text{spec}(\bar{S})$  are two of the connected points, and  $\text{ec}(D_{10}) = 1.368 \times 10^{36}$  is an upper one.

## 8.6 Verification

`ds_transfer_matrix.py` is standalone — Python and the standard library, no `numpy`, no `sympy` and no other script of the book — and rebuilds  $D_k$  from the definition rather than importing it, so that check (D) is a genuine cross-check of Table 7.1 and not a tautology. Exact integer arithmetic throughout; the determinants of Lemma 8.6 are computed over  $\mathbb{Q}$  with `fractions`.

## 8.7 Scope, provenance, and what is open

**Scope.** Theorem 8.3, Proposition 8.5, Lemma 8.6 and Theorem 8.8 hold for any circular  $S$  and any  $k$ . Proposition 8.10 is a bound, and Table 8.1 shows it is not a good one on real sequence. Section 8.5 is a computation on one named genome and claims nothing beyond it. Nothing here improves (8.1) itself, which is Chapter 7's.

**Provenance.** The lattice sum,  $\Lambda^-$ , its rank by the Hopf trace formula, and the  $k = 12$  and  $k = 11$  values are Chapter 7; the double cover and its balance are Chapter 5, its deck involution as an anti-automorphism Theorem 6.9, the interleaving of repeats Theorem 1.6. The BEST theorem is [10, 78] and the matrix-tree theorem for digraphs is [80]. Frontier elimination is not new: it is the

standard dynamic-programming-over-a-path-decomposition, and in the form used in Theorem 8.8 — carrying a partition of the frontier so as to count connected subgraphs — it is the frontier-based search of [44], itself descended from Knuth’s SIMPATH [49]; the width parameter is pathwidth [14]. What is specific here is the observation that the double-stranded reconstruction count is indexed by a system whose columns have support four, so that the frontier is narrow for reasons having nothing to do with rank  $\Lambda^-$ , and the identification of the coordinate order as the wrong one. For counting lattice points in a polytope of *fixed* dimension there is Barvinok’s algorithm [6]; it does not apply here, the dimension being the thing that grows.

**Open.** *First*,  $\#DS(\phi X174, 10)$ , for which something must be done about  $t(G_c)$  (Chapter 9 computes it by a different route); the natural attempt is to carry a partial Schur complement of  $L_c$  on the frontier, which is a matrix state and not a scalar one, and whether that can be made to fit is the real question this chapter leaves. An order chosen to keep partitions few rather than imbalances small would help both it and Theorem 8.8, whose cost at  $k = 10$  is entirely the partition factor. *Second*, the width of a bacterial genome. The constraint hypergraph of *E. coli* at  $k = 31$  falls apart into many small blocks, one per repeat cluster, and the elimination factors over them; whether the largest block stays narrow at the word lengths where the double cover is interesting is not known to us. *Third*, whether the coordinate width of Proposition 8.10 is, for a random sequence with a prescribed inverted-repeat structure, a computable function of that structure — it would make “how hard is double-stranded assembly here” a statistic of the genome.



## Chapter 9

# The double-stranded lattice sum as a fermionic partition function

**Summary.** Chapter 8 counted the lattice points of the double-stranded box of  $\phi X174$  at  $k = 10$ , and those with connected support, but not the number  $\#DS$  of double-stranded reconstructions. The obstacle is the arborescence factor  $t(G_c)$ , a determinant of the whole support. We show that it is a fermionic partition function with one *even* factor per arc, so that it can be accumulated arc by arc in any order. A reduced determinant needs a root, and the support moves with the point. Both difficulties are removed together by the reverse-complement symmetry  $c \mapsto m - c$ , which fixes the summand and halves the box along any orbit of odd multiplicity, where one fixed vertex is always in the support. The weight is made integral by trading  $\prod_a c(a)!^{-1}$  for binomials and one constant. Everything is checked exactly against Chapter 7 at  $k = 11, 12$  and on random covers. Frontier elimination of the fermionic weight is exact but does not reach  $k = 10$ , because its state is linear and a single history already fills up to  $3^w$  of its coordinates. Nor does the positive (arborescence) form of the same weight: it has one state per history, but merges too few of them. The number is obtained instead by visiting all 1 240 843 950 points of the half-box in compiled code, contracting forced arcs so that only a small determinant is left at each point, and working modulo five primes. The run reproduces both of Chapter 8's counts, halved, and agrees with the exact arborescence elimination on  $k = 10$  sub-boxes. The result is

$$\#DS(\phi X174, 10) = 31\,925\,753\,246\,212,$$

the last blank entry of Table 7.2.

### 9.1 What is left to count

We keep the notation of Chapters 7 and 8:  $D_k$  is the labelled double cover of Definition 7.1,  $A$  its set of arc contents,  $m(a)$  the number of labelled arcs of content  $a$ ,  $\rho$  the reverse-complement involution, and  $\Lambda^-$  its  $(-1)$ -eigenlattice on the circulations. By Theorem 7.6

$$\#DS(S, k) = \sum_f N_{\text{seq}}\left(\frac{m+f}{2}\right), \quad N_{\text{seq}}(c) = \frac{t(G_c) \text{Loc}(G_c)}{\prod_a c(a)!}, \quad (9.1)$$

the sum over  $f \in \Lambda^-$  in the box  $|f| \leq m$ ,  $f \equiv m \pmod{2}$ , with  $G_c$ ,  $t(G_c)$  and  $\text{Loc}(G_c)$  as there.

Chapter 8 observed that the constraint defining the sum is one linear equation per vertex, each involving only the four endpoints of one  $\rho$ -orbit, and replaced Chapter 7's backtracking by a

*frontier elimination*: process the orbits in a good order carrying, as state, the partial imbalance at the vertices still open. At  $k = 10$ , where  $\text{rank } \Lambda^- = 48$ , that counts the 2 481 687 900 points of the box in seconds with a frontier 26 wide and 50 758 states, and, carrying in addition the partition of the frontier into the classes already joined, the 909 103 210 points whose support is connected.

It did not reach #DS. The reason is entirely in  $t(G_c)$ : the number of spanning arborescences is the determinant of a reduced Laplacian of the whole support — of order 125 at  $k = 10$  — and a determinant is not a product of local scalars. A frontier state that is a number, or a partition, cannot carry it.

The point of this chapter is that it *is* a product of local *fermions*.

## 9.2 The determinant as a product over arcs

Let  $V$  be the vertex set,  $r \in V$  a distinguished *root*, and let  $\Lambda = \Lambda(\bar{\psi}_v, \psi_v : v \in V \setminus \{r\})$  be the Grassmann algebra on two odd generators per non-root vertex; we put  $\bar{\psi}_r = \bar{\psi}_r = 0$ . For  $c \in \mathbb{Z}_{\geq 0}^A$  let  $L_c$  be the out-Laplacian of  $G_c$ ,  $L_c(v, v) = d_c^+(v) - (\text{loops at } v)$  and  $L_c(v, u) = -\sum_{a: t(a)=v, h(a)=u} c(a)$  for  $u \neq v$ , and let

$$A_c = L_c + \text{diag}(\mathbf{1}_{v \notin \text{supp } c}), \quad \text{supp } c = \{v : d_c^+(v) > 0\}, \quad (9.2)$$

so that  $A_c$  agrees with  $L_c$  on  $\text{supp } c$  and is the identity off it. Write  $\tilde{A}_c^r$  for  $A_c$  with the row and the column of  $r$  deleted.

**Proposition 9.1** (arc factorisation). *In  $\Lambda$ ,*

$$P_c = \prod_{a \in A} \left(1 - c(a) \bar{\psi}_{t(a)}(\psi_{t(a)} - \psi_{h(a)})\right) \cdot \prod_{v \notin \text{supp } c} (1 - \bar{\psi}_v \psi_v) = (-1)^{|V|-1} \det \tilde{A}_c^r \cdot \Psi, \quad (9.3)$$

where  $\Psi = \bar{\psi}_{v_1} \psi_{v_1} \bar{\psi}_{v_2} \psi_{v_2} \cdots$  runs over  $V \setminus \{r\}$  in any order. Every factor in (9.3) has even degree, so the product may be formed in any order whatsoever, one arc at a time.

*Proof.* Each factor is  $1 + x$  with  $x$  a linear combination of  $\bar{\psi}_v \psi_u$ , so  $x^2 = 0$  and  $1 + x = e^x$ ; and  $\bar{\psi}_v^2 = 0$  makes the product of the factors belonging to one tail  $v$  equal to  $1 - \bar{\psi}_v \sum_u A_c(v, u) \psi_u$ , the loops cancelling between the two terms of  $\psi_{t(a)} - \psi_{h(a)}$ . Hence  $P_c = \prod_{v \neq r} (1 - \bar{\psi}_v (A_c \psi)_v) = \exp(-\sum_{v,u} \bar{\psi}_v A_c(v, u) \psi_u)$ , the Berezin integrand of  $\det \tilde{A}_c^r$ . Expanding,  $P_c = \sum_{\sigma} \prod_v (-A_c(v, \sigma v)) \bar{\psi}_{v_1} \psi_{\sigma v_1} \bar{\psi}_{v_2} \psi_{\sigma v_2} \cdots$ , the sum over the bijections  $\sigma$  of  $V \setminus \{r\}$ . Each block  $\bar{\psi}_v \psi_{\sigma v}$  is even, so the blocks commute and the vertex order chosen for  $\Psi$  is immaterial; and interchanging two of the  $\psi$ 's moves an odd generator across a block of odd length, which costs  $-1$ , so the monomial equals  $\text{sgn}(\sigma) \Psi$ . The  $(-1)^{|V|-1}$  is the one minus sign per row.  $\square$

*Remark 9.2.* The constant  $(-1)^{|V|-1}$  in (9.3) is attached to the *interleaved* order  $\bar{\psi}_{v_1} \psi_{v_1} \bar{\psi}_{v_2} \psi_{v_2} \cdots$ . Two interleaved orders differ by a permutation of the even blocks  $\bar{\psi}_v \psi_v$  and so give the same constant, but listing all the  $\bar{\psi}$  first and all the  $\psi$  afterwards does not: that costs a shuffle sign which, when the two lists are ordered differently — as they are in an implementation where  $\bar{\psi}_v$  and  $\psi_v$  have different lifetimes — varies from instance to instance. This is not a scruple; it is the one place where an implementation of (9.3) silently returns  $-\text{\#DS}$ .

The point of Proposition 9.1 for the frontier is this. The factor of the arc  $a$  involves only  $\bar{\psi}_{t(a)}$ ,  $\psi_{t(a)}$  and  $\psi_{h(a)}$ , and it is even. So the whole determinant can be accumulated in Chapter 8's orbit order, one  $\rho$ -orbit at a time, and what has to be carried across the frontier is: for each open vertex

$v$ , whether  $\bar{\psi}_v$  has already been spent and whether  $\psi_v$  has already been taken. Two bits per vertex, not a set partition. Section 9.6 explains why this turns out to be the more expensive of the two alphabets.

### 9.3 The root, and why the box is halved

A reduced determinant needs a root, and  $\text{supp } c$  moves with  $c$ : no vertex of  $D_k$  lies in every support. The diagonal completion (9.2) takes care of the vertices that are missing, and the following lemma says what  $\tilde{A}_c^r$  then computes.

**Lemma 9.3.** *Let  $c$  be a circulation with  $c \neq 0$ . If  $r \in \text{supp } c$  then  $\det \tilde{A}_c^r = t(G_c)$ , independently of which  $r \in \text{supp } c$  is taken; and it vanishes exactly when  $\text{supp } c$  is disconnected. If  $r \notin \text{supp } c$  then  $\det \tilde{A}_c^r = 0$ .*

*Proof.*  $c$  is a circulation, so  $d_c^- = d_c^+$  and no arc of  $G_c$  leaves  $\text{supp } c$  or enters it:  $A_c$  is block diagonal,  $L_c$  on  $\text{supp } c$  and the identity off it. Deleting  $r \in \text{supp } c$  leaves  $\det$  equal to the corresponding reduced Laplacian of  $G_c$ , which is  $t(G_c)$  by the matrix-tree theorem;  $G_c$  is balanced, so its weak components are strongly connected and the arborescence count does not depend on the root, and the nullity of  $L_c$  on  $\text{supp } c$  is the number of those components (Lemma 8.6). Deleting  $r \notin \text{supp } c$  leaves the whole singular block  $L_c$  on  $\text{supp } c$  intact, so the determinant is 0.  $\square$

So a fixed root is legitimate on any part of the box on which it lies in the support. Such a part is produced, for free, by the symmetry of the problem.

**Lemma 9.4** ( $\rho$ -symmetry).  *$c \mapsto m - c$  is an involution of the set of circulations with  $c + \rho c = m$ , and  $N_{\text{seq}}(m - c) = N_{\text{seq}}(c)$ .*

*Proof.*  $(m - c)(a) = c(\rho a)$ , so  $a \mapsto \rho a$ ,  $v \mapsto \bar{v}$  carries  $G_{m-c}$  isomorphically to  $G_c$  with all arcs reversed, because  $t(\rho a) = \bar{h(a)}$  and  $h(\rho a) = \bar{t(a)}$ . Hence  $d_{m-c}^+(v) = d_c^-(\bar{v}) = d_c^+(\bar{v})$ , so  $\text{Loc}$  is unchanged;  $\prod_a c(a)!$  is unchanged because the multiset of values is permuted; and  $t(G_{m-c})$  is the number of spanning *in*-trees of  $G_c$ , which for a balanced digraph equals the number of out-trees.  $\square$

**Corollary 9.5.** *Let  $j_0$  be a  $\rho$ -orbit with  $m_{j_0}$  odd. Then  $f_{j_0} \neq 0$  on the box, the involution of Lemma 9.4 exchanges  $\{f_{j_0} > 0\}$  with  $\{f_{j_0} < 0\}$ , and*

$$\#DS(S, k) = 2 \sum_{f: f_{j_0} > 0} N_{\text{seq}}\left(\frac{m+f}{2}\right).$$

*On that half-box  $c(a_{j_0}) \geq 1$ , so  $t(a_{j_0}) \in \text{supp } c$  for every point of it and the single vertex  $r = t(a_{j_0})$  serves as root throughout.*

At  $k = 10$ , 96 of the 111 orbits of  $\phi X174$  have odd multiplicity, so the corollary applies; it halves the work and removes the root from the list of things the sweep has to decide.

### 9.4 What the sweep has to carry besides the fermions

Two more quantities in (9.1) are not arc-local.

**Loc.**  $(d_c^+(v) - 1)!$  needs the whole out-degree of  $v$  at once. The partial out-degree is therefore carried as well — but only on the sub-frontier of the vertices whose out-arcs are not all decided, which at  $k = 10$  is 18 wide against the 26 of the frontier itself, and is narrower still under an order chosen with it in mind. It is spent, and dropped, at the step at which the last out-arc of  $v$  is decided; if that out-degree is 0 the vertex is outside  $\text{supp } c$  and the factor  $(1 - \bar{\psi}_v \psi_v)$  of (9.3) is applied instead.

**Integrality.**  $\prod_a c(a)!$  has no reason to divide anything along the way. It is removed from the sweep altogether: on a free orbit  $c(a_j) + c(\rho a_j) = m_j$ , so

$$\prod_a \frac{1}{c(a)!} = \frac{\prod_j \binom{m_j}{c(a_j)}}{\prod_j m_j! \cdot \prod_{\rho a=a} (m_a/2)!}, \quad (9.4)$$

whose numerator is a product of integer weights, one per orbit, and whose denominator is a constant of  $D_k$ . The sweep therefore runs over  $\mathbb{Z}$  and divides once, at the end.

## 9.5 The sweep

The order of the orbits is Chapter 8's greedy order, re-tuned: the elements the frontier now has to carry are one  $\psi$  per vertex over its whole window, weighted for the imbalance it travels with, and one  $\bar{\psi}$  per vertex over its out-arc window, weighted for the partial out-degree. A short local search on top of a randomised greedy brings  $\phi X174$  at  $k = 10$  from 42 elements to 35.

The state of the sweep is

$$(\text{imbalance } b_v)_{v \text{ open}} \times (\text{partial out-degree})_{v \text{ out-open}} \times (\text{which } \bar{\psi}_v, \psi_v \text{ are spent}),$$

and its value is an integer. Each vertex is given a *slot* for the length of its window, by colouring the windows as intervals; slots are reused, so the layout is fixed once and for all and nothing has to be relabelled from step to step. The imbalances live in fixed-width fields of one machine word, biased so that an empty field reads zero, and the capacity bound of Theorem 8.3 —  $|b_v|$  at most the capacity still to come — is tested on all fields at once by two additions and two masks, which also forces a vertex leaving the frontier to balance. The out-arc window of  $v$  sits inside its full window, so  $\bar{\psi}_v$  and  $\psi_v$  can share  $v$ 's slot; giving them the two bits  $2s$  and  $2s + 1$  of that slot makes the generator order interleaved, which by Remark 9.2 is what the constant of Proposition 9.1 requires. The signs are then local: multiplying on the right by a generator costs  $(-1)$  to the number of spent generators above it, and integrating one out when its window closes costs  $(-1)$  to the number of spent generators below it, both read off the bit mask; the permutation between the order in which the generators leave and the canonical one is a single sign computed once.

Finally, the half-box may be cut into disjoint sub-boxes by pinning the values of a few early orbits, and the sums added. This changes nothing but the peak memory.

The sweep reproduces  $\#DS = 10$  at  $k = 12$  and 864 at  $k = 11$  from the half-box, with one fixed root and one division at the end. It also agrees with the brute-force sum on twelve random covers. At  $k = 11$  it peaks at 560 states.

## 9.6 Why the frontier stops short of $k = 10$

The frontier state of Section 9.5 is *linear*. At a fixed imbalance it is a vector in the exterior algebra of the open generators, and summing over histories adds vectors. The alphabet is small, but the

vectors are not sparse. A boundary vertex  $v$  whose out-window is still open, and which already carries an arc to an open vertex  $u$ , contributes the factor  $1 - \bar{\psi}_v \psi_v + \bar{\psi}_v \psi_u$ . A single history with  $s$  such vertices therefore already has up to  $3^s$  non-zero coordinates, before any two histories are added. At  $k = 10$  the tuned order still has 35 open generators at its widest, and the weighted width estimate of the order search is  $2^{86.8}$ . The cancellations of the Berezin integral occur only at the end; along the way, the linear state pays for all the signed terms that will cancel.

The obvious alternative is the positive expansion of the same determinant,  $t(G_c) = \sum_T \prod_{a \in T} c(a)$  over in-arborescences  $T$  of  $\text{supp } c$  rooted at  $r$ . It needs no signs, and it puts *one* state on each history.

**Proposition 9.6** (arborescence elimination). *Process the orbits in order. Call a vertex pending until its last out-arc has been decided, and let every pending vertex be the root of its own class. Carry, for each open vertex, its imbalance. For a pending vertex, also carry its candidate arcs so far as a multiset over the classes their heads belong to; its partial out-degree is their total. For a vertex that is no longer pending, carry the class it belongs to. When the last out-arc of a pending vertex  $v \neq r$  has been decided, do one of two things. If its out-degree is 0, declare it outside  $\text{supp } c$ . Otherwise let it choose a candidate class  $K$  other than its own, with weight the multiplicity towards  $K$  times  $(d_c^+(v) - 1)!$ , and merge its class into  $K$ . The root never chooses; it contributes  $(d_c^+(r) - 1)!$ . Then the sum of the weights over all runs equals  $\sum_c \prod_j \binom{m_j}{c(a_j)} t(G_c) \text{Loc}(G_c)$  over the half-box.*

*Proof.* A choice is an arc from  $v$  into a class other than  $v$ 's own. Following the choices from any vertex eventually reaches a pending vertex or  $r$ , so a cycle in the chosen arcs would have a last vertex to choose; that vertex would have found its head already in its own class. Conversely, a choice into a different class never closes a cycle. So the runs that survive correspond exactly to the in-arborescences of  $\text{supp } c$  to  $r$ , each counted with  $\prod_{a \in T} c(a)$ . A component of the support that does not contain  $r$  has a last pending vertex whose candidates all lie in its own class, and so it contributes nothing.  $\square$

It is exact. It reproduces 10 and 864, the random covers, and itself as a sum over sub-boxes. At  $k = 10$  it agrees with the point-by-point sum of  $N_{\text{seq}}$  on five random sub-boxes of the half-box. But it merges too few histories. Those five sub-boxes hold between 12 and 166 points, with 30 to 36 of the 111 orbits pinned, and they already peak at between 55 890 and 623 000 states. The candidate multisets and class pointers are strictly finer than the connectivity partition with which Chapter 8 counted the connected points, and that elimination already needed 9.7 million states. Neither frontier reaches the whole half-box on a 16 GB machine.

## 9.7 Point by point

What does reach it is the least clever method, made cheap at each point. The half-box has 1 240 843 950 points: half of Chapter 8's count, by Corollary 9.5. The support of a point has about 126 arcs on about 110 vertices, so most vertices have out-degree 1, and the determinant is almost entirely forced.

**Lemma 9.7** (forced arcs). *Let  $G$  be a balanced multidigraph,  $r$  a vertex, and  $v \neq r$  a vertex all of whose out-arcs (there are  $w \geq 1$  of them) end at one vertex  $u \neq v$ . Then  $t_r(G) = w t_r(G/vu)$ , where  $G/vu$  identifies  $v$  with  $u$ , deletes the  $w$  arcs  $v \rightarrow u$  and redirects the arcs into  $v$  to  $u$ .*

*Proof.* Every in-arborescence to  $r$  contains exactly one out-arc of  $v$ , hence one of the  $w$  arcs to  $u$ . Contracting it is a bijection onto the in-arborescences of  $G/vu$  to  $r$ ; the other copies of  $v \rightarrow u$ , and the arcs from  $u$  to  $v$ , become loops, which no arborescence uses.  $\square$

Applying the lemma until no vertex qualifies leaves a reduced Laplacian of small order. At each point, then:

1. the support is tested for connectivity by union-find, and discarded if it is not connected (Lemma 9.3);
2.  $t(G_c)$  is computed by Lemma 9.7 followed by Gaussian elimination on what is left;
3.  $\text{Loc}(G_c)/\prod_a c(a)!$  is read from tables.

All of this is done modulo the five largest primes below  $2^{31}$ , so every product fits in a machine word. The points are visited by Chapter 7's imbalance-bounded backtracking in Chapter 8's greedy order, compiled with `numba`. The first 35 levels are expanded into 28 512 prefixes, which are shuffled and processed in parallel. The half-sum is recovered by the Chinese remainder theorem. This is exact: the half-sum is at most half of the upper bound  $1.368 \cdot 10^{36}$  of Remark 8.13, far below the product of the five primes (about  $2^{155}$ ).

The run is checked in three ways, all exactly:

1. At  $k = 12$  and  $k = 11$  it returns 5 and 264 half-box points, 4 and 109 of them connected, and  $\#DS = 10$  and 864; on twelve random covers it agrees with the brute force.
2. On  $k = 10$  sub-boxes it agrees with the arborescence elimination of Proposition 9.6. That is a second, unrelated algorithm, and the sub-boxes are built from the same order, windows and root as a full run would use.
3. At  $k = 10$  it counts exactly half of Chapter 8's box points and exactly half of its connected points (Corollary 9.5; the involution preserves connectivity and has no fixed point on the box).

## 9.8 $\phi\mathbf{X174}$ at $k = 10$

The root is vertex 20, the tail of orbit 33, which has multiplicity 1. The enumeration visits

$$\begin{aligned} 1\,240\,843\,950 &= \tfrac{1}{2} \cdot 2\,481\,687\,900 \text{ points, of which} \\ 454\,551\,605 &= \tfrac{1}{2} \cdot 909\,103\,210 \text{ have connected support,} \end{aligned}$$

with no overflow of the work arrays. It takes about thirty-five minutes on sixteen threads. The sum of  $N_{\text{seq}}$  over the half-box, doubled, is

$\#DS(\phi\mathbf{X174}, 10) = 31\,925\,753\,246\,212 \approx 3.19 \cdot 10^{13}.$

This lies inside the bounds  $8\,610\,708\,632 \leq \#DS \leq 1.368 \cdot 10^{36}$  of Remark 8.13: about 3 700 times the lower bound and twenty-three orders of magnitude below the upper one. Averaged over the 909 103 210 points with connected support, a point carries about 35 118 circular sequences. At  $k = 11$  the average is  $864/218 \approx 4$ .

So at word length 10,  $\phi\mathbf{X174}$  is one of about  $3.2 \cdot 10^{13}$  circular sequences with the same double-stranded 10-mer spectrum. The lattice has rank 48, but the count is not astronomical: the box has  $2.5 \cdot 10^9$  points, and each carries a modest number of Eulerian circuits.

| $k$ | rank $\Lambda^-$ | box points    | connected   | #DS                |
|-----|------------------|---------------|-------------|--------------------|
| 12  | 4                | 10            | 8           | 10                 |
| 11  | 12               | 528           | 218         | 864                |
| 10  | 48               | 2 481 687 900 | 909 103 210 | 31 925 753 246 212 |

Table 9.1: Table 7.2, completed. The two counting columns before the last at  $k = 10$  are Chapter 8's; the last is this chapter's.

## Ancillary files

`ds_fermion.py` is standalone: Python 3.10+ and the standard library. It rebuilds the double cover from its definition. It checks Proposition 9.1 against the reduced determinant point by point on  $\phi X174$  at  $k = 12$  and on twelve small random covers, and it checks Lemma 9.3, Lemma 9.4 and Corollary 9.5 on the enumerable boxes. It reproduces  $\#DS = 10$  and 864 with the fermionic sweep and with the arborescence elimination, and it checks the latter, and Lemma 9.7, on  $k = 10$  sub-boxes: seven checks in about ten minutes.

`ds_count.py` needs `numpy` and `numba`, and imports `ds_fermion.py`. It is the enumeration of Section 9.7 with its three checks, and it computes  $\#DS(\phi X174, 10)$ . The run takes about thirty-five minutes on sixteen threads and resumes from a checkpoint.



## Chapter 10

# Why there is no closed formula for $\#DS$ : double-stranded reconstruction is $\#P$ -hard

**Summary.** Single-stranded reconstruction counts are given by one determinant, the BEST theorem. The double-stranded count  $\#DS(S, k)$  of Chapter 7 is a sum of such determinants over the points of a lattice in a box. Chapters 8 and 9 evaluated it for  $\phi X174$  by frontier elimination and by visiting  $1.24 \cdot 10^9$  points. It is natural to ask for a formula that avoids the sum: a Gauss sum over a discriminant form, a Pfaffian, anything computable in polynomial time. We show that no such formula exists unless  $FP = \#P$ . From every connected loopless graph  $G$  of maximum degree 4 with even degrees, we build in polynomial time a circular DNA sequence  $S$  of length linear in  $|E(G)|$  and an even  $k$  such that

$$\#DS(S, k) = 2 \varepsilon(G),$$

where  $\varepsilon(G)$  is the number of Eulerian circuits of  $G$ . The number of Eulerian circuits is  $\#P$ -complete for 4-regular graphs (Ge and Štefankovič, after Brightwell and Winkler). The mechanism is simple: palindromic  $k$ -mers make the reverse-complement involution act trivially on vertices, so that choosing a strand for each  $\rho$ -orbit is the same as orienting an edge. Two corollaries follow. The number of box points, Chapter 8's first column, is the number of Eulerian orientations of  $G$ , which is  $\#P$ -complete as well. And on this class the generation conjecture of Chapter 11 (that transpositions at repeated  $k$ -mers, inversions at inverted repeats and global reverse complementation connect all double-stranded reconstructions) holds, by Kotzig's theorem. All identities are verified exactly on 23 graphs by `bio19_verify.py`. The lattice sum, and the frontier and enumeration methods used to evaluate it, are therefore not an artefact of the approach; they are what the problem requires.

### 10.1 The question

For one strand the count of sequences with a given  $(k + 1)$ -mer spectrum is a determinant times factorials [10]. For two strands, Theorem 7.6 gives

$$\#DS(S, k) = \sum_{f \in \Lambda^-, |f| \leq m, f \equiv m \pmod{2}} N_{\text{seq}}\left(\frac{m+f}{2}\right)$$

Chapter 8 counted the 2 481 687 900 points of the box for  $\phi X174$  at  $k = 10$ , and Chapter 9 evaluated the sum, 31 925 753 246 212. Each step needed an exponential-size object: a frontier of width 26,

or a visit to every point. A proposal for this chapter asked for a “meshless” formula instead. It would be a Gauss sum built on the symmetric linking form of the double cover’s sandpile group (Remark 6.10), or a Pfaffian in a Clifford algebra, evaluated in time polynomial in the number of vertices. We prove that such a formula cannot exist unless every problem in #P is solvable in polynomial time.

## 10.2 Eulerian circuits and their orientations

Let  $G$  be a connected loopless multigraph whose degrees are all even, with at least three edges. An *Eulerian circuit* is a cyclic sequence of its edges, traversing each once, considered up to rotation and reversal. Let  $\varepsilon(G)$  be their number. An *Eulerian orientation* directs each edge so that every vertex has equal in- and out-degree. For a balanced digraph  $O$  with labelled arcs,  $\text{ec}(O) = \tau(O) \prod_v (d_v^+ - 1)!$  counts its Eulerian circuits up to rotation [10].

**Lemma 10.1.** 
$$\sum_{O \text{ Eulerian orientation of } G} \text{ec}(O) = 2 \varepsilon(G).$$

*Proof.* An Eulerian circuit of  $G$ , traversed in one of its two directions, orients every edge. It is balanced at every vertex, since each visit enters and leaves once, and it gives a directed Eulerian circuit of that orientation. Conversely, a directed Eulerian circuit of an orientation, forgetting the arrows, is a traversed Eulerian circuit of  $G$ . So the left side counts pairs (circuit, direction). The two directions of a circuit are different directed circuits, because they traverse the first edge oppositely.  $\square$

**Theorem 10.2** ([17, 34]). *Computing  $\varepsilon(G)$  is #P-complete, even for 4-regular graphs.*

Counting the Eulerian orientations of a graph is also #P-complete [62].

## 10.3 Genomes whose double cover is a given graph

Let  $G$  have vertex set  $\{1, \dots, n\}$ ,  $m \geq 3$  edges, and all degrees in  $\{2, 4\}$ . Fix an even word length  $k = 2h$  and a spacer length  $\ell$ . The construction has three steps.

1. To each vertex  $x$  assign a palindrome  $P_x = u_x \text{rc}(u_x)$  with  $|u_x| = h$ .
2. Compute an Eulerian circuit  $x_0 \xrightarrow{e_1} x_1 \xrightarrow{e_2} \dots \xrightarrow{e_m} x_m = x_0$  of  $G$ . Spell

$$S = P_{x_0} \Sigma_1 P_{x_1} \Sigma_2 \dots P_{x_{m-1}} \Sigma_m \quad (\text{circular}),$$

with one spacer  $\Sigma_i$  of length  $\ell$  per traversal.

3. Assign the boundary letters of the spacers.  $P_x$  occurs  $\deg(x)$  times in  $S$  and  $\text{rc } S$  together. The letters that follow it are the first letters of the outgoing spacers and the complements of the last letters of the incoming ones. At the  $j$ -th visit to  $x$  ( $j = 0, 1$ ), let the outgoing spacer begin with  $\text{AC}[j]$  and the incoming one end with  $\text{CA}[j]$ . The letters following the occurrences of  $P_x$  are then A, G for a vertex of degree 2 and A, C, G, T for a vertex of degree 4, pairwise different. The interior letters of the spacers are free.

**Lemma 10.3** (realization). *Suppose that in  $S \cup \text{rc } S$  the only  $k$ -mers occurring more than once are the  $P_x$ , the  $P_x$  are distinct, and no  $P_x \Sigma P_y$  is its own reverse complement. Then the double cover  $D_k(S)$  has the following properties:*

1. its vertices are the  $n$  palindromes  $P_x$ , so  $\bar{v} = v$  for every vertex;
2. its contents are the words  $P_{x_{i-1}}\Sigma_i P_{x_i}$  and their reverse complements, one  $\rho$ -orbit of multiplicity  $m = 1$  per edge of  $G$ , with no self-complementary content;
3. the quotient of  $D_k(S)$  by  $\rho$  is  $G$ .

*Proof.* The vertices of  $D_k$  are the repeated  $k$ -mers (Definition 7.1);  $P_x$  occurs  $\deg(x) \geq 2$  times. The contents are the stretches between consecutive vertex occurrences, i.e. the  $P\Sigma P$  words of  $S$  and their reverse complements in  $\text{rc } S$ . Each occurs once, so it has multiplicity 1. The orbit of the  $i$ -th one joins  $P_{x_{i-1}}$  and  $P_{x_i}$ , which are fixed by  $\rho$ .  $\square$

**Lemma 10.4** (the hypotheses can be met in polynomial time). *There are  $h, \ell = O(\log m)$  and a deterministic algorithm, polynomial in  $m$ , that chooses the words  $u_x$  and the spacer interiors so that the hypotheses of Lemma 10.3 hold.*

*Proof sketch.* Choose the free letters uniformly at random, with  $\ell \geq k$ . A *bad event* is one of three things: two occurrences in  $S \cup \text{rc } S$  of equal  $k$ -mers that are not both occurrences of one  $P_x$ ;  $P_x = P_y$  for some  $x \neq y$ ; or a self-complementary  $P\Sigma P$ . Since  $\ell \geq k$ , a window of length  $k$  is an occurrence of some  $P_x$ , or it lies inside one spacer, or it straddles one junction between a palindrome and a spacer. In the last two cases it contains at least  $h - 2$  free letters from a single source: half of one palindrome, or the interior of one spacer.

Two windows that differ only in fixed letters cannot be equal, by the boundary assignment. This is the reason for it: without it the  $(k + 1)$ -mers  $P_x c$  would repeat with probability  $1 - 4!/4^4 \approx 0.91$  at a vertex of degree 4. Otherwise, equality of two windows is a system of equations  $a_i = b_i$  or  $a_i = \bar{b}_i$  between letters. The complement case arises when one window lies in  $\text{rc } S$ . An equation  $a = \bar{a}$  has no solution over  $\{A, C, G, T\}$ , and the remaining equations link the free letters in paths and cycles. So the system fixes at least half of them, and its probability is at most  $4^{-(h-2)/2}$ . There are at most  $(2m(k + \ell))^2$  pairs of windows. With  $h \geq 2 \log_4(8m^2(k + \ell)^2) + 2$  the expected number of bad events is below  $1/2$ .

The conditional probability of each bad event, given some letters, is computed exactly from the same system of equations. So the method of conditional expectations fixes the letters one at a time, keeping the expected number of bad events below 1. This is deterministic and polynomial.  $\square$

The verification script uses random interiors and checks the hypotheses, retrying on failure. With  $k = 12$  and  $\ell = 18$  it never needed a retry on the graphs below.

## 10.4 The count

**Theorem 10.5.** *For  $S$  and  $k$  as in Section 10.3,*

$$\#\text{DS}(S, k) = 2\varepsilon(G),$$

*and the box points of Theorem 7.6 are in bijection with the Eulerian orientations of  $G$ .*

*Proof.* Every orbit has  $m = 1$ . So a point of the box chooses, for each orbit  $\{a, \rho a\}$ , exactly one of  $a$  and  $\rho a$ . Since all vertices are palindromic,  $a: P_x \rightarrow P_y$  and  $\rho a: P_y \rightarrow P_x$  are the two orientations of the edge  $\{x, y\}$  of  $G$ . A point is a circulation iff the chosen orientation is balanced, i.e. Eulerian. At it,  $c(a) \in \{0, 1\}$ , so the factorials of  $N_{\text{seq}} = \tau \prod (d^+ - 1)! / \prod c(a)!$  disappear and  $N_{\text{seq}} = \text{ec}(O)$  (Theorem 7.6). Summing gives Lemma 10.1.  $\square$

| graph             | $n$ | $ E $ | $ S $ | $\varepsilon(G)$ | Eulerian orientations | $\#DS(S, 12)$ |
|-------------------|-----|-------|-------|------------------|-----------------------|---------------|
| 4-regular         | 3   | 6     | 180   | 16               | 10                    | 32            |
| 4-regular         | 4   | 8     | 240   | 44               | 16                    | 88            |
| 4-regular         | 5   | 10    | 300   | 112              | 28                    | 224           |
| 4-regular         | 6   | 12    | 360   | 328              | 42                    | 656           |
| 4-regular, simple | 5   | 10    | 300   | 132              | 24                    | 264           |
| 4-regular, simple | 6   | 12    | 360   | 372              | 38                    | 744           |
| 4-regular, simple | 7   | 14    | 420   | 1052             | 60                    | 2104          |
| degrees 2, 4      | 6   | 10    | 300   | 36               | 18                    | 72            |
| degrees 2, 4      | 4   | 6     | 180   | 6                | 6                     | 12            |

Table 10.1: A selection of the 23 graphs of checks (E), (R), (D), (K).  $\varepsilon(G)$  is found by exhaustive enumeration of trails.  $\#DS$  comes from Chapter 7’s lattice sum on the double cover built from the definition, and it equals  $2\varepsilon(G)$  in every case. So does the closure of  $S$  under the moves of Conjecture 11.2. The box points equal the Eulerian orientations in every case.

**Corollary 10.6.** *Given a circular DNA sequence  $S$  and an even  $k$ , computing  $\#DS(S, k)$  is #P-hard, and so is computing the number of points of Chapter 7’s box. In particular, no algorithm, closed formula or “meshless” expression evaluates  $\#DS$  in time polynomial in  $|S|$  unless  $FP = \#P$ .*

*Proof.* The map  $G \mapsto (S, k)$  is polynomial and  $|S| = m(k + \ell) = O(m \log m)$ . Divide by 2 and apply Theorem 10.2. For the box, use [62].  $\square$

Table 10.1 shows a selection of the verified cases. In every case, the double cover of the constructed genome, built from the definition, has exactly  $n$  palindromic vertices,  $|E|$  orbits of multiplicity 1 and no self-complementary content. In check (S), 4-regular graphs with  $n = 4, \dots, 8$  give genomes of 240–480 bp, linear in  $|E|$ . On them  $\#DS$  grows from 80 to 6 008.

## 10.5 Two by-products

**Chapter 11’s conjecture on this class.** Conjecture 11.2 of Chapter 11 states that transpositions, inversions at inverted repeats, and reverse complementation connect  $DS(S, k)$ . Here every vertex is its own complement. So an inversion between two visits of  $P_x$  (a flip in the sense of Definition 7.12) reverses the closed stretch of the circuit between them, which is a  $\kappa$ -transformation. Kotzig proved that the  $\kappa$ -transformations connect all Eulerian circuits of a connected graph [50, 30]. Reversal of the whole circuit supplies the direction.

**Corollary 10.7.** *Conjecture 11.2 holds for every genome of Section 10.3.*

Check (K) confirms it: the closure of  $S$  under inversions and reverse complementation alone is all of  $DS(S, k)$  in all 23 cases.

**Both counting columns are hard.** Chapter 8 counted the points of the box (2 481 687 900 for  $\phi X174$  at  $k = 10$ ) and Chapter 9 the reconstructions. By Theorem 10.5 the first count is the number of Eulerian orientations of a graph, and the second twice its number of Eulerian circuits. Both are #P-complete in general.

## 10.6 What this does and does not exclude

- *Excluded:* any method polynomial in  $|S|$  that is exact on all inputs. That includes a Pfaffian, a determinant, or a Gauss sum with polynomially many polynomially computable terms. Unless  $\text{FP} = \#\text{P}$ , none of them exists. The proposed Gauss-sum expansion  $\#\text{DS} = |K|^{-1} \sum_{\gamma} \mathcal{G}(\gamma) \Phi(\gamma)$  fails for a second, independent reason. The summation domain is the box, not a group, so Poisson summation over the finite group  $K(D_k)$  does not collapse it. Written as a Fourier expansion on  $K$ , the  $\Phi(\gamma)$  carry the entire difficulty.
- *Not excluded:* algorithms exponential in a structural parameter. Chapter 8's frontier elimination is exponential in the frontier width and linear otherwise. Genomes of bounded width are tractable.  $\phi\text{X174}$  at  $k = 10$  has width 26, which is why it was feasible.
- *Not excluded:* efficient approximation. Whether  $\#\text{DS}$  admits a fully polynomial randomized approximation scheme is open; so is the corresponding question for undirected Eulerian circuits.
- *Odd  $k$ .* The construction needs palindromic  $k$ -mers, which exist only for even  $k$ . For odd  $k$  no vertex is its own complement, and the quotient of  $D_k$  is the signed graph of Chapter 6. The strand choices are then orientations of a signed graph, and the count is one of Eulerian circuits of a bidirected graph. We expect hardness there too, but do not prove it.

### Ancillary file

`bio19_verify.py` (standard library; imports `ds_fermion.py` of Chapter 9 and `bio17_verify.py` of Chapter 11) runs checks (E), (R), (D), (K) and (S) in two seconds. It writes its transcript to `bio19_output.txt`.



## Part III

# Spectral fibres and rearrangements



## Chapter 11

# Inversions at inverted repeats and the double-stranded fibre

**Summary.** Two circular sequences with the same  $k$ -mer spectrum differ by transpositions of segments bounded by repeated words: this is the theorem of Ukkonen and Pevzner. Double-stranded data do not see the spectrum of one strand, only the sum of the spectra of both strands. The corresponding set of sequences is the double-stranded fibre  $\text{DS}(S, k)$  counted in Chapters 7, 8 and 9. Chapter 7 identified the move that the second strand adds, the *inversion flip*. Write  $T = P w Z' \text{rc}(w) R$  with  $w$  a  $k$ -mer. Replacing the segment  $w Z' \text{rc}(w)$  by its reverse complement preserves the double-stranded spectrum. For  $\phi\text{X174}$  at  $k = 11, 12$  the flips alone connect the fibre. But flips alone cannot connect it when the genome has no inverted repeat. We add the lattice picture: a flip moves the point of Chapter 7 by  $-2(1 - \rho)z$ , where  $z$  is a path from a vertex to its complement. Three results follow. (i) Transpositions and flips, together with global reverse complementation, generate the whole fibre in every case we can close. The cases are twenty-four random sequences, checked against exhaustive enumeration and against Chapter 7's lattice sum, and  $\phi\text{X174}$  at  $k = 12$  and  $k = 11$ , where the closure of the genome is exactly  $\#\text{DS} = 10$  and 864. Transpositions alone reach 2 and 33 sequences there, and adding reverse complementation reaches 4 and 66. Adding transpositions removes Chapter 7's obstruction, and we state the general case as a conjecture. (ii) The *inversion lattice* is always all of  $\Lambda^-$ . The cycle part  $(1 - \rho)Z_1$  has index

$$[\Lambda^- : (1 - \rho)Z_1] = \begin{cases} 2 & \text{if } D_k \text{ has no palindromic vertex and no self-complementary arc,} \\ 1 & \text{otherwise,} \end{cases}$$

by a Tate-cohomology computation, confirmed on 600 random covers. When the index is 2, the missing class is exactly an inversion vector. (iii)  $\phi\text{X174}$  at  $k = 10$  carries 31 inversions flanked by inverted repeats of length 10. Random walks of 20 000 moves stay inside the fibre of 31 925 753 246 212 reconstructions and visit about 5 000 of its lattice points. The inversion-group programme proposed for this chapter (Artin groups, Garside normal forms) is set aside, and we say why.

### 11.1 The question

Let  $S$  be a circular DNA sequence and  $k$  a word length. In the single-stranded world, the sequences with the same  $(k + 1)$ -mer spectrum as  $S$  are the Eulerian circuits of one graph. Any two of them are related by *transpositions*: exchanges of segments that start and end at repeated  $k$ -mers [81, 68].

A sequencer that cannot tell the strands apart sees only

$$\text{spec}(T) + \text{spec}(\text{rc } T) = \text{spec}(S) + \text{spec}(\text{rc } S),$$

and the set  $\text{DS}(S, k)$  of circular  $T$  satisfying this is larger. Chapter 7 wrote its size as a sum over the points of a lattice  $\Lambda^-$  in a box, Chapter 8 counted the box, and Chapter 9 computed  $\#\text{DS}(\phi\text{X174}, 10) = 31\,925\,753\,246\,212$ . A transposition never leaves the lattice point of  $T$ . So the question of this chapter is *which moves carry one lattice point to another*, and whether together with transpositions they connect the fibre.

The move itself is the *inversion flip* of Definition 7.12: an inversion between two copies of an inverted repeat. It is an operation every geneticist knows. In the human genome it is the classical mechanism of recurrent inversions, via non-allelic homologous recombination between inverted low-copy repeats [77], and here it is the combinatorial content of the second strand. Chapter 7 proved two things about it. The flips preserve the double-stranded spectrum, and a genome without inverted repeats admits none, so its flip graph is discrete. It also computed that for  $\phi\text{X174}$  at  $k = 12$  and  $k = 11$  the flips alone connect all 10 and 864 reconstructions. This chapter adds three things. It combines flips with transpositions, which removes that obstruction and gives a general conjecture. It computes the lattice spanned by flip vectors (the Tate dichotomy). And it walks the fibre at  $k = 10$ .

## 11.2 Setting

We keep the notation of Chapter 7.  $D_k$  is the labelled double cover of Definition 7.1, whose arcs are the *contents*;  $\rho$  is reverse complementation on vertices ( $v \mapsto \bar{v}$ ) and on contents, a content  $a: t \rightarrow h$  going to  $\rho a: \bar{h} \rightarrow \bar{t}$ . A reconstruction  $T \in \text{DS}(S, k)$  is an Eulerian circuit of the multigraph with  $c(a)$  copies of each content, where  $c + \rho c = m$ , and  $\#\text{DS}(S, k)$  is the lattice sum of Theorem 7.6 over the box points  $f = 2c - m$  of  $\Lambda^-$ . Here  $(\rho z)(a) = z(\rho a)$  on 1-chains. A vertex is *palindromic* if  $\bar{v} = v$ , and a content is *self-complementary* if  $\rho a = a$ .

## 11.3 Inversions at inverted repeats

**Theorem 11.1** (flips preserve the double-stranded spectrum Proposition 7.13(a)). *Let  $T$  be circular and  $w$  a  $k$ -mer with an occurrence of  $w$  and a later occurrence of  $\text{rc}(w)$ , so that, read from the first,  $T = ZR$  with  $Z = wZ'\text{rc}(w)$ . Let  $T' = \text{rc}(Z)R$ . Then  $\text{spec}_{k+1}(T') + \text{spec}_{k+1}(\text{rc } T') = \text{spec}_{k+1}(T) + \text{spec}_{k+1}(\text{rc } T)$ .*

*Proof.*  $\text{rc}(Z) = w\text{rc}(Z')\text{rc}(w)$  begins with  $w$  and ends with  $\text{rc}(w)$ , as  $Z$  does. A  $(k+1)$ -mer of  $T$  that straddles a junction between  $Z$  and  $R$  meets  $Z$  in at most  $k$  letters, and those letters lie inside the flanking copy of  $w$  or  $\text{rc}(w)$ . So it is also a  $(k+1)$ -mer of  $T'$ . The  $(k+1)$ -mers inside  $Z$  are replaced by their reverse complements, so  $\text{spec}(T') = \text{spec}(T) - \text{spec}(Z) + \text{rc spec}(Z)$ . Applying  $\text{rc}$  and adding gives the claim.  $\square$

This is Proposition 7.13(a), with the same proof; we repeat it only to fix notation. The move of this section is a *flip* in the sense of Definition 7.12.

On the double cover, the two copies  $w, \text{rc}(w)$  are vertices  $v, \bar{v}$ . The segment  $Z$  is a path  $z$  of contents from  $v$  to  $\bar{v}$ , and  $\rho z$  is again a path from  $v$  to  $\bar{v}$ , because  $\rho$  reverses paths and exchanges their ends. The inversion replaces  $z$  by  $\rho z$ :

$$c' = c - z + \rho z, \quad f' = f - 2(1 - \rho)z. \quad (11.1)$$

|                                                 | T  | T + R | T + I | T + I + R |
|-------------------------------------------------|----|-------|-------|-----------|
| random cases whose closure is all of DS (of 24) | 0  | 2     | 23    | 24        |
| $\phi$ X174, $k = 12$ (#DS = 10)                | 2  | 4     | 10    | 10        |
| $\phi$ X174, $k = 11$ (#DS = 864)               | 33 | 66    | 864   | 864       |

Table 11.1: Closures of  $S$  under the move families: T transpositions, R reverse complement, I inversions at inverted repeats. For  $\phi$ X174 the entries are the sizes of the closures. At  $k = 12$  and  $k = 11$  the full closure visits 8 and 218 lattice points, exactly the box points with connected support found by Chapter 7 and Chapter 8.

The vector  $(1 - \rho)z$  is a cycle, since  $z$  and  $\rho z$  have the same ends, and it is  $\rho$ -antisymmetric. So it lies in  $\Lambda^-$ . An inversion changes the strand of every content of  $Z$ . A transposition changes no multiplicity, so  $f' = f$ .

Check (I) of `bio17_verify.py` tests Theorem 11.1 directly on strings. All 156 inversions at inverted repeats of the random test sequences preserve the double-stranded spectrum. So do all 2, 9 and 31 such inversions of  $\phi$ X174 itself at  $k = 12$ , 11 and 10. As a control, only 64 of 960 inversions of random, unflanked segments preserve it, and those are chance coincidences at small  $k$ .

## 11.4 Which moves generate the fibre

We represent a reconstruction as a cyclic sequence of contents and apply three families of moves:

- **T** (transpositions): at a vertex visited three times, exchange two of the returning loops; or, at two vertices  $u \neq w$  visited alternately  $u w u w$ , exchange the two  $u \rightarrow w$  segments;
- **I**: the inversions of Theorem 11.1, at every pair of visits to  $v$  and  $\bar{v}$ ;
- **R**: global reverse complementation.

The closure of  $S$  under a family is computed by breadth-first search, and every state is checked to be a valid reconstruction.

**Three counts agree.** On twenty-four random sequences of length 16–33 with  $k = 3$ –5, chosen so that at least two lattice points carry reconstructions, three quantities coincide in every case. The first is the set of circular strings with the double-stranded spectrum of  $S$ , found by exhaustive search. The second is Chapter 7’s lattice sum. The third is the closure of  $S$  under T, I and R. The fibres range from 4 to 1872 sequences (check (C)).

For  $\phi$ X174 these closures refine Theorem 7.14, which showed that at  $k = 12$  and  $k = 11$  the flips *alone*, without transpositions and without reverse complementation, already connect the fibre. What transpositions add is the general case: Proposition 7.13(d) shows that a genome without inverted repeats admits no flip at all. Table 11.1 separates the contributions. Transpositions alone give the single-stranded fibre of  $S$ :  $N_{\text{seq}}$  at its own lattice point, 2 and 33 for  $\phi$ X174. Reverse complementation doubles this. The inversions are what carry the closure across the lattice. At  $k = 11$  they alone, with transpositions, reach all 864 reconstructions and all 218 lattice points with connected support. In one random case of twenty-four, T and I reach only part of the fibre and R is needed as well. By Proposition 7.13(c), reverse complementation is itself a flip only at a palindromic visit. We have not checked whether the exceptional case lacks one.

**Conjecture 11.2** (double-stranded Ukkonen–Pevzner). *For every circular  $S$  and every  $k$ ,  $\text{DS}(S, k)$  is connected under transpositions at repeated  $k$ -mers, inversions at inverted repeats of length  $k$ , and global reverse complementation.*

Within one lattice point the conjecture is the classical theorem, and on the palindromic genomes of Section 10.3 it holds by Kotzig’s theorem (Section 10.5). The open part is the passage between lattice points: whether any two box points with connected support are joined by a chain of inversions that are realizable, i.e. whose flanking copies  $v, \bar{v}$  are actually visited by the current circuit. Section 11.5 shows that nothing is lost at the level of the lattice.

## 11.5 The inversion lattice

By (11.1) the inversion vectors  $(1 - \rho)z$ , for all paths  $z$  from a vertex to its complement, generate a sublattice  $M \subseteq \Lambda^-$ . The differences of two such paths with the same ends are cycles, so  $M$  contains  $(1 - \rho)Z_1$ . The question is whether  $M$ , or already  $(1 - \rho)Z_1$ , is all of  $\Lambda^-$ . The group  $\langle \rho \rangle \cong \mathbb{Z}/2$  acts, and the answer is a Tate-cohomology computation. Recall that for  $\mathbb{Z}/2$  acting on a module  $A$  one has  $\hat{H}^0(A) = \ker(1 - \rho)/\text{im}(1 + \rho)$  and  $\hat{H}^1(A) = \ker(1 + \rho)/\text{im}(1 - \rho)$  [16]. In particular  $\Lambda^-/(1 - \rho)Z_1 = \hat{H}^1(Z_1)$ .

**Theorem 11.3** (the Tate dichotomy). *Let  $D_k$  be connected. Then*

$$\Lambda^-/(1 - \rho)Z_1 \cong \begin{cases} \mathbb{Z}/2 & \text{if } D_k \text{ has no palindromic vertex and no self-complementary content,} \\ 0 & \text{otherwise.} \end{cases}$$

*In every case  $M = \Lambda^-$ . In the first case the non-trivial class is represented by  $(1 - \rho)p$  for any path  $p$  from a vertex  $v$  to  $\bar{v}$ .*

*Proof.* Let  $C_1$  be the free abelian group on the contents and  $C_0$  that on the vertices, with  $\partial a = h(a) - t(a)$ . Because  $\rho a$  runs from  $\bar{h}$  to  $\bar{t}$ , one has  $\partial \rho = -\rho \partial$ . So  $\partial$  is equivariant when  $C_0$  carries the twisted action  $\sigma = -\rho$ . Let  $B_0 = \partial C_1$ . Since  $D_k$  is connected,  $B_0$  is the group of 0-chains of total degree 0. The sequence  $0 \rightarrow Z_1 \rightarrow C_1 \rightarrow B_0 \rightarrow 0$  is exact and equivariant, and its long exact sequence in Tate cohomology reads

$$\hat{H}^0(C_1) \longrightarrow \hat{H}^0(B_0) \xrightarrow{\delta} \hat{H}^1(Z_1) \longrightarrow \hat{H}^1(C_1).$$

$C_1$ . The contents fall into free orbits  $\{a, \rho a\}$ , each a copy of  $\mathbb{Z}[\mathbb{Z}/2]$  with vanishing Tate cohomology, and self-complementary contents, each a copy of the trivial module  $\mathbb{Z}$ , with  $\hat{H}^0 = \mathbb{Z}/2$  and  $\hat{H}^1 = 0$ . So  $\hat{H}^1(C_1) = 0$ , and  $\hat{H}^0(C_1) = (\mathbb{Z}/2)^{\#\text{self-compl.}}$ , generated by the classes of the self-complementary contents.

$B_0$ .  $\ker(1 - \sigma)$  consists of the  $x$  with  $x(\bar{v}) = -x(v)$ . Such an  $x$  vanishes at palindromic vertices, and is spanned by the  $e_v - e_{\bar{v}}$ . Now  $\text{im}(1 + \sigma)$  consists of the  $y - \rho y$  with  $y \in B_0$ . If  $p$  is a palindromic vertex, then  $e_v - e_{\bar{v}} = (1 - \rho)(e_v - e_p)$  with  $e_v - e_p \in B_0$ , so  $\hat{H}^0(B_0) = 0$ . If there is none, any  $y$  with  $(1 - \rho)y = e_v - e_{\bar{v}}$  has the form  $e_v + s$  with  $\rho s = s$ , and  $s$  has even total degree, because its values pair up over orbits of size two. So  $y \notin B_0$ . Hence  $\hat{H}^0(B_0) = \mathbb{Z}/2$ , generated by the common class of all  $e_v - e_{\bar{v}}$ ; they differ by  $(1 - \rho)(e_v - e_u)$  with  $e_v - e_u \in B_0$ .

*The map.* A self-complementary content  $a: t \rightarrow h$  has  $h = \bar{t}$ , so  $\partial a = e_{\bar{t}} - e_t$ , the generator of  $\hat{H}^0(B_0)$ . So if a self-complementary content exists,  $\hat{H}^0(C_1) \rightarrow \hat{H}^0(B_0)$  is onto and  $\hat{H}^1(Z_1) = 0$ . If none exists and there is no palindromic vertex,  $\hat{H}^1(Z_1) \cong \hat{H}^0(B_0) = \mathbb{Z}/2$ . If there is a palindromic vertex,  $\hat{H}^0(B_0) = 0$  and again  $\hat{H}^1(Z_1) = 0$ .

| double cover                                          | covers | $[\Lambda^- : (1 - \rho)Z_1]$ | $[\Lambda^- : M]$ |
|-------------------------------------------------------|--------|-------------------------------|-------------------|
| no palindromic vertex, no self-complementary content  | 60     | 2 in all 60                   | 1                 |
| self-complementary content, no palindromic vertex     | 241    | 1                             | 1                 |
| palindromic vertex, no self-complementary content     | 299    | 1                             | 1                 |
| $\phi X174$ , $k = 10$ (3 palindromic vertices)       | 1      | 1                             | 1                 |
| $\phi X174$ , $k = 11$ (1 self-complementary content) | 1      | 1                             | 1                 |
| $\phi X174$ , $k = 12$ (1 palindromic vertex)         | 1      | 1                             | 1                 |

Table 11.2: Indices computed by Smith normal form on 600 random connected double covers (lengths 14–59,  $k = 2$ –7; 36 disconnected covers skipped) and on  $\phi X174$ . The ranks of both lattices always equal rank  $\Lambda^-$ : 48, 12 and 4 for  $\phi X174$ .

*The representative.* The connecting map sends the class of  $x = e_{\bar{v}} - e_v$  to the class of  $z - \rho z$ , for any  $z \in C_1$  with  $\partial z = x$ : then  $\partial(\rho z) = -\rho x = x$ , so  $z - \rho z \in Z_1$ . A path  $p$  from  $v$  to  $\bar{v}$  is such a  $z$ . So  $(1 - \rho)p$  represents the non-trivial class, and  $M = \Lambda^-$ .  $\square$

Table 11.2 confirms the theorem exactly on every cover tested. The dichotomy has a transparent meaning. A palindromic vertex, or a self-complementary content, is a place where the two strands touch. Without one, flipping closed cycles,  $(1 - \rho)Z_1$ , can only reach a sublattice of index 2, and the missing half needs an open path from a word to its complement: a genuine inversion.

## 11.6 $\phi X174$ at $k = 10$

At  $k = 10$  the fibre has 31 925 753 246 212 elements on  $1\,240\,843\,950 \times 2$  box points (Chapter 9). It cannot be closed, but it can be walked. The genome carries 31 inversions at inverted repeats of length 10. The count on the cover is 32. The extra one comes from a copy of  $\text{rc}(w)$  at position 4482 and a copy of  $w$  at 4484. Their two flanks overlap, and the string-level count excludes that case. One inversion from  $\phi X174$  reaches 23 other lattice points. Two random walks of 20 000 moves each, one without and one with global reverse complementation, stay inside the fibre at every step: each state is checked to be a valid reconstruction, and the final one also at the level of strings. They visit 4 907 and 5 049 distinct lattice points, and none of the sampled sequences repeats. That is consistent with Conjecture 11.2 but does not test it: at this scale the question is not whether the walk can move, but whether it can reach every one of the  $454\,551\,605 \times 2$  points with connected support.

## 11.7 Biological reading

Theorem 11.1 says exactly which rearrangements double-stranded  $k$ -mer data cannot see: transpositions between direct repeats, and inversions between inverted repeats, each at least  $k$  long. These are the two classical products of non-allelic homologous recombination: deletions, duplications and translocations between direct repeats, and inversions between inverted ones [77]. The first class preserves the content of one strand and the second only the content of both. Any strand-agnostic encoder that depends on  $k$ -mer content alone is blind to both. That includes a reverse-complement-invariant local DNA encoder of window  $k + 1$ , in the sense of Section 12.7. A short-read assembler faces the same limit. The inversion lattice being all of  $\Lambda^-$  means that, algebraically, these two mechanisms already account for every direction in which the double-stranded fibre extends.

## 11.8 On the proposed inversion-group programme

This chapter was specified with a different programme: a group  $G_{\text{inv}}$  generated by adjacent exchanges and segment inversions, embedded in an Artin–Tits group, shown to be Garside, with the minimum inversion distance read off a Garside normal form. We set it aside for three reasons.

- The group generated by signed permutations of  $n$  oriented segments is the finite hyperoctahedral group  $B_n$ , of order  $2^n n!$ . It is a Coxeter group, not an Artin group, and it does not embed in one as the specification intended. Its word problem is trivial.
- The minimum number of inversions between two signed orders, the reversal distance, is computed in polynomial time by Hannenhalli and Pevzner [41]. Word length in  $B_n$  with respect to its Coxeter generators is a different quantity, and it does not model inversions of arbitrary segments.
- The object that is new, and that belongs to algebraic genomics, is not the group of all inversions but the inversions that preserve the double-stranded spectrum. Those are the ones flanked by inverted repeats, and they are what this chapter studies.

### Ancillary file

`bio17_verify.py` (standard library; imports `ds_fermion.py` of Chapter 9) runs checks (I), (C), (G), (L), (H) and (W). These are the inversion theorem on strings; the three counts; the closures; the inversion lattice on  $\phi\text{X174}$  and random covers; the Tate dichotomy on 600 random covers; and the walks at  $k = 10$ . The whole battery takes under two minutes.

## Chapter 12

# What a local encoder cannot see: spectral fibres of the human proteome and the representation of perturbagens

*Joint work with Zhengyi Chen, published separately; included with the co-author's agreement.*

**Summary.** UniPert-G2CP (Li et al., *Cell* 189, 2026) represents genetic perturbagens by their protein sequences and chemical perturbagens by ECFP4 fingerprints, and shows that its protein encoder outperforms composition-type encoders such as pseudo amino acid composition (PseAAC). We look at this comparison from the side of algebraic genomics, i.e. the combinatorics of sequences with prescribed local content. An encoder is *local of order  $k$*  if it depends on a sequence only through its multiset of  $k$ -mers. Composition,  $k$ -mer composition, PseAAC with lag  $\lambda \leq k - 1$ , and every mean-pooled convolutional encoder of window  $k$  are local. So is ECFP, in the molecular-graph sense. A local encoder is constant on the *spectral fibre* of a sequence, the set of sequences with the same  $k$ -mer multiset. The size of the fibre is an Eulerian-trail count, which we compute exactly by the BEST theorem after contracting forced arcs, for all 20 431 reviewed human proteins. Three findings follow. (i) *Separation is not resolution.* No two distinct human proteins share a  $k$ -mer spectrum for any  $k \geq 2$ , so every local encoder of order  $\geq 2$  can separate the reference proteome. Yet the median protein shares its 3-mer spectrum with  $2^{239}$  other sequences and its 4-mer spectrum with  $2^{19}$ . The spectral resolution  $\kappa$ , the least  $k$  beyond which the spectrum determines the sequence, has median 5 and exceeds 10 for 854 proteins. (ii) *The unresolved order lives in duplication-driven families.* The largest  $\kappa$  occur in the hominoid-specific segmental-duplication families: NBPF, with median  $\kappa = 46$  and up to 951, then GOLGA6, NPIP and POTE. They also occur in keratin-associated proteins, collagens and mucins. KRAB zinc-finger proteins are 3% of the proteome but 41% of the proteins with  $\kappa > 10$ . (iii) *Explicit collisions.* For 12 human proteins we construct a different sequence, altered at up to 15% of its positions, with the same 25-spectrum. It receives the identical 68-dimensional PseAAC vector (with  $\lambda = 24$ , the setting of the UniPert benchmark) and the identical output from a window-25 convolutional encoder. On the DNA side, which UniPert names as its limitation, a strand-agnostic encoder is constant on the double-stranded reconstruction set. For bacteriophage  $\phi$ X174 at  $k = 10$  that set has 31 925 753 246 212 elements.

## 12.1 Introduction

UniPert-G2CP [82] is a two-stage framework for perturbation modelling. UniPert places genetic and chemical perturbagens in one embedding space. A compound enters as an ECFP4 fingerprint [74]. A gene enters as its protein: mean-pooled ESM-2 embeddings [29] are refined by a graph network over a Smith–Waterman similarity graph of 19 187 human proteins. G2CP then pretrains a perturbation-effect predictor on CRISPR screens and fine-tunes it on compound screens. In the genetic benchmarks (their Fig. 3B, C), UniPert is compared with ESM, OntoProtein and PseAAC. PseAAC was computed with `propy`, 68-dimensional, with lag  $\lambda = 24$ ; this is Chou’s amphiphilic form [22, 23].

The question we ask is structural: *what can an encoder of a given form not distinguish?* For the family of encoders built from local statistics the answer is exact, and it is a question of algebraic genomics. Chapters 7–9 count the circular sequences that share a double-stranded  $k$ -mer spectrum with a genome, by a lattice sum over the circulations of a de Bruijn double cover. For single-stranded linear sequences the same count is a single BEST determinant [10]. It can be computed for every protein of the human proteome, and it measures precisely what a local encoder discards.

Section 12.2 sets up local encoders and their fibres, and proves the counting formula. Section 12.3 computes fibres and spectral resolution for the proteome. Section 12.4 locates the unresolved order in protein families. Section 12.5 constructs explicit collisions for PseAAC and for convolutional encoders. Section 12.6 treats ECFP, and Section 12.7 DNA perturbagens. Section 12.8 states what the results do and do not imply for UniPert-G2CP.

## 12.2 Local encoders and spectral fibres

Let  $\Sigma$  be an alphabet,  $S \in \Sigma^L$  and  $2 \leq k \leq L$ . The  $k$ -spectrum  $\text{spec}_k(S)$  is the multiset of the  $m = L - k + 1$  substrings of length  $k$  of  $S$ . The *fibre* of  $S$  is

$$F_k(S) = \{T \in \Sigma^L : \text{spec}_k(T) = \text{spec}_k(S)\}, \quad N_k(S) = |F_k(S)|.$$

We let  $F_1(S)$  be the set of rearrangements of  $S$ , so that  $N_1(S) = L! / \prod_a n_a!$ .

**Definition 12.1.** An encoder  $\phi: \Sigma^* \rightarrow \mathbb{R}^d$  is *local of order  $k$*  if  $\phi(T) = \phi(S)$  whenever  $\text{spec}_k(T) = \text{spec}_k(S)$  and  $T, S$  have the same first  $(k - 1)$ -mer.

The first  $(k - 1)$ -mer is part of the definition only for bookkeeping. If the first and last  $(k - 1)$ -mers of  $S$  differ, then every  $T \in F_k(S)$  has the same first and last  $(k - 1)$ -mers as  $S$ , because they are the unique vertices of unequal in- and out-degree below.

**Proposition 12.2** (local encoders). *The following are local of order  $k$ :*

1. *composition, and  $j$ -mer composition for  $j \leq k$ ;*
2. *Chou’s pseudo amino acid composition, in the original and in the amphiphilic form, with any property scales, any weight and any lag  $\lambda \leq k - 1$ ;*
3. *every encoder of the form  $\phi(S) = \frac{1}{n} \sum_i f(W_i)$ , where  $W_1, \dots, W_n$  are the windows of length  $k$  of  $S$  padded with  $k - 1$  sentinel symbols at each end and  $f$  is arbitrary. This includes every convolutional network of receptive field  $k$  followed by mean or sum pooling.*

*Proof.* All three are functions of multisets of short substrings. For (1), the  $j$ -mer multiset is the multiset of  $j$ -prefixes of the  $k$ -mers, together with the  $j$ -mers inside the last  $(k-1)$ -mer. For (2), PseAAC is a function of the composition and of the correlation factors  $\tau_j = \frac{1}{L-j} \sum_i \Theta(S_i, S_{i+j})$ ,  $j \leq \lambda$ . Here  $\tau_j$  depends only on the multiset of pairs  $(S_i, S_{i+j})$ , i.e. on the first and last letters of the  $(j+1)$ -mers. For (3), the padded windows are the  $k$ -mers of  $S$  together with windows that meet the sentinels, and those contain only the first or the last  $k-1$  letters.  $\square$

Being local is a property of the *form* of an encoder, not of its training. A mean-pooled transformer such as ESM-2 is not local, because attention couples all positions. An alignment score against a reference set is not local either.

**Theorem 12.3** (fibre size). *Let  $G_k(S)$  be the multigraph on the  $(k-1)$ -mers of  $S$  with one arc  $a[1..k-1] \rightarrow a[2..k]$  for each occurrence of each  $k$ -mer  $a$ . Let  $\mu(a)$  be its multiplicity,  $s$  and  $t$  the first and last  $(k-1)$ -mers of  $S$ , and  $d^+$  the out-degree. If  $s \neq t$ , add one arc  $t \rightarrow s$ . Write  $\tau_s$  for the number of spanning in-arborescences rooted at  $s$ , with parallel arcs counted separately. Then*

$$N_k(S) = \frac{m^{[s=t]} \cdot \tau_s \cdot \prod_v (d^+(v) - 1)!}{\prod_a \mu(a)!}, \quad (12.1)$$

where  $[s=t]$  is 1 if  $s=t$  and 0 otherwise. The whole numerator is formed before the division.

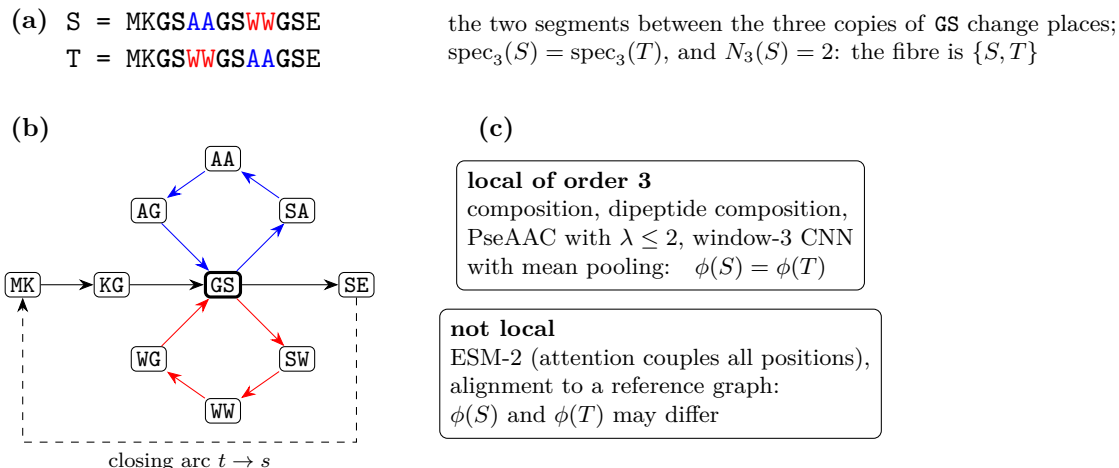
*Proof.* Label the arcs. The strings with  $k$ -spectrum  $\text{spec}_k(S)$  correspond to the Eulerian trails of  $G_k(S)$ , each string to  $\prod_a \mu(a)!$  labelled trails. If  $s \neq t$ , the trails run from  $s$  to  $t$  and correspond to the Eulerian circuits of the augmented graph that begin with the added arc. By the BEST theorem [10] there are  $\tau_s \prod_v (d^+(v) - 1)!$  of them. If  $s = t$ , the trails are closed. Every labelled circuit gives  $m$  labelled closed trails, one for each starting arc, so the count is  $m \tau_s \prod_v (d^+(v) - 1)!$ . The factor  $m$  must be applied before dividing by  $\prod \mu(a)!$ . Dividing first does not give an integer, e.g. for  $S = \text{AAA}$ ,  $k = 2$ .  $\square$

*Remark 12.4* (relation to the BEST theorem). For a connected balanced multigraph  $G$  with labelled arcs, the BEST theorem counts Eulerian circuits up to rotation:  $\text{ec}(G) = \tau_r(G) \prod_v (d^+(v) - 1)!$ , the same for every root  $r$  [10]; this is (1.1). Formula (12.1) is ec of the closed-up graph with two corrections. The factor  $m^{[s=t]}$  restores the starting point that “up to rotation” forgets. The division by  $\prod_a \mu(a)!$  removes the labels, because a  $k$ -mer occurring  $\mu$  times gives  $\mu!$  labelled copies of the same string. The same constant appears in the count  $N_{\text{seq}}(c) = t(G_c) \text{Loc}(G_c) / \prod_a c(a)!$  of Theorem 7.6 for circular sequences.

When  $s = t$ , the fibre also contains the rotations that start elsewhere. The encoders of Proposition 12.2 are constant on the subset that starts at  $s$ . In the human proteome,  $s = t$  occurs for a handful of proteins and only at  $k \leq 3$ .

**Lemma 12.5** (forced arcs, Lemma 9.7). *If a vertex  $v \neq r$  has all its out-arcs ( $w \geq 1$  of them) ending at one vertex  $u \neq v$ , then  $\tau_r(G) = w \tau_r(G/vu)$ . Here  $G/vu$  merges  $v$  into  $u$ , deletes the arcs  $v \rightarrow u$  and redirects the arcs into  $v$ .*

For  $k \geq 4$  almost every  $(k-1)$ -mer of a protein occurs once, and the lemma reduces  $\tau$  to the determinant of a small reduced Laplacian. The fibre moves are also classical. Two strings with the same first  $(k-1)$ -mer have the same  $k$ -spectrum iff one is obtained from the other by a sequence of transpositions of segments bounded by repeated  $(k-1)$ -mers [81, 68]. A local encoder of order  $k$  is therefore blind to exactly those rearrangements: swaps of segments that begin and end in repeated  $(k-1)$ -mers. Figure 12.1 shows the smallest interesting case.



every vertex but **GS** has one out-arc and is contracted;  
 $d^+(\text{GS}) = 3$ ,  $\tau = 1$ , so  $N_3 = \tau(3 - 1)! = 2$

Figure 12.1: A spectral fibre in miniature. (a) Two sequences that differ by a transposition of the segments between three copies of the repeated 2-mer **GS**. (b) Their common de Bruijn graph for  $k = 3$ . The fibre size follows from Theorem 12.3 after contracting forced arcs (Lemma 12.5). (c) Every local encoder of order 3 assigns them the same vector. In the human proteome the same mechanism acts on repeat copies 25 residues long and longer (Table 12.4).

**Definition 12.6.** The *spectral resolution* of  $S$  is  $\kappa(S) = \min\{k : N_{k'}(S) = 1 \text{ for all } k' \geq k\}$ . It is at most two more than the length of the longest repeated substring of  $S$ .

## 12.3 The human proteome

We take the 20 431 reviewed human entries of UniProt [83], with 11 418 237 residues, median length 415 and maximum 34 350 (titin). For every protein and every  $k$  from 1 up to the point where no  $(k - 1)$ -mer repeats, we compute  $N_k$  by Theorem 12.3 and Lemma 12.5. The computation is exact whenever the contracted Laplacian has order at most 150. Larger orders occur only at small  $k$  in long proteins, where  $\log_2 N_k$  is in the thousands. There the determinant is taken in floating point, and the logarithms of the factorials come from  $\log \Gamma$ . Against exact values on 200 random sequences, the worst relative error of that path is  $6.5 \cdot 10^{-16}$ . The formula itself is checked against exhaustive enumeration in 21 187 cases (alphabets of size 2–4, lengths up to 12). The contraction is checked against the full determinant on 400 random sequences. The whole run takes 18 minutes on 16 processes.

Table 12.1 summarizes the fibres. From  $k = 5$  on, the median of  $\log_2 N_k$  is 0, i.e.  $N_k = 1$ : more than half of all human proteins (53.5%) are already determined by their 5-mer spectrum, and the median describes the resolved majority. The tail is what matters, and it is in the second and last columns (Figure 12.2a). The spectral resolution has median  $\kappa = 5$  (Figure 12.2b). Its 90% and 99% quantiles are 7 and 20, and its maximum is 951 (NBPF20). The cumulative shares are:

$$\begin{array}{lll} \kappa \leq 4 : 10.0\%, & \kappa \leq 5 : 53.5\%, & \kappa \leq 6 : 83.1\%, \\ \kappa \leq 8 : 93.4\%, & \kappa \leq 10 : 95.8\%, & \kappa \leq 25 : 99.4\%. \end{array}$$

| $k$ | proteins with $N_k > 1$ | share   | median $\log_2 N_k$ | largest $\log_2 N_k$ |
|-----|-------------------------|---------|---------------------|----------------------|
| 1   | 20 431                  | 100.00% | 1628.4              | 140 656.7 (TTN)      |
| 2   | 20 422                  | 99.96%  | 1080.5              | 136 673.4 (TTN)      |
| 3   | 20 357                  | 99.64%  | 239.4               | 114 168.6 (TTN)      |
| 4   | 18 391                  | 90.02%  | 18.9                | 55 107.2 (TTN)       |
| 5   | 9 504                   | 46.52%  | 0                   | 13 489.5 (TTN)       |
| 6   | 3 445                   | 16.86%  | 0                   | 6 247.9 (MUC3B)      |
| 8   | 1 346                   | 6.59%   | 0                   | 3 715.5 (MUC3B)      |
| 10  | 854                     | 4.18%   | 0                   | 2 252.5 (MUC3B)      |
| 12  | 614                     | 3.01%   | 0                   | 1 269.2 (MUC3B)      |
| 16  | 327                     | 1.60%   | 0                   | 404.2 (MUC4)         |
| 20  | 194                     | 0.95%   | 0                   | 255.4 (MUC2)         |
| 25  | 114                     | 0.56%   | 0                   | 164.2 (MUC2)         |
| 30  | 81                      | 0.40%   | 0                   | 108.8 (MUC2)         |
| 40  | 50                      | 0.25%   | 0                   | 63.1 (MUC2)          |
| 50  | 37                      | 0.18%   | 0                   | 37.2 (NBPF20)        |

Table 12.1: Spectral fibres of the 20 431 reviewed human proteins.  $N_k$  is the number of sequences with the same  $k$ -mer multiset.

**Separation is not resolution.** Among the 20 353 distinct sequences, exactly one pair shares its amino acid composition. No two share their  $k$ -spectrum for  $k = 2, 3, 4$ . So even dipeptide composition separates every protein of the reference proteome from every other, and nothing in the *training set* of a protein encoder forces it to look further. What the reference set does not contain is the fibre: at  $k = 3$  the median protein has  $2^{239}$  spectral twins, and at  $k = 4$  still  $2^{19}$ . Those are exactly the inputs on which out-of-distribution generalization is tested: rearranged, recombined or unseen sequences. On them a local encoder of order 3 or 4 is constant by construction, whatever it learned. Separating a finite reference set and resolving sequence space are different properties, and benchmarks on the reference set test only the first.

**Proteins named in UniPert-G2CP.** Table 12.2 lists proteins that appear in the UniPert-G2CP analyses: the unseen single-gene targets CEP55, AURKA and AURKB, the combination CBL+CNN1, the kinesin/FOXN1 example, PD-1/PD-L1, the GABA<sub>A</sub> receptor, and ESR1. All have  $\kappa$  between 5 and 11. The kinesin CENPE is the least resolved, with 226 bits of unresolved order at  $k = 5$ .

## 12.4 Where the unresolved order lives

Fibres grow with length at small  $k$ : a long protein repeats many short words by chance, so  $\log_2 N_3/L$  is largest for titin (3.32 bits per residue), the mucins, nesprins and plakins. The spectral resolution does not grow with length in this way. It is large only if the protein contains long repeats that interleave or recur, so it is the family-level statistic. Figure 12.3b shows this directly. Proteins of every length reach  $\kappa \leq 10$ , while the large values belong to particular families at moderate lengths.

Table 12.3 ranks families by median  $\kappa$ . The top of the list is not a random sample of repetitive proteins. The first three families and POTE arose by segmental duplication in the hominoid lineage. The NBPF proteins carry the Olduvai (DUF1220) domain, whose copy number expanded dramatically in humans [64], and NBPF20 has  $\kappa = 951$ . The next group consists of tandem-repeat

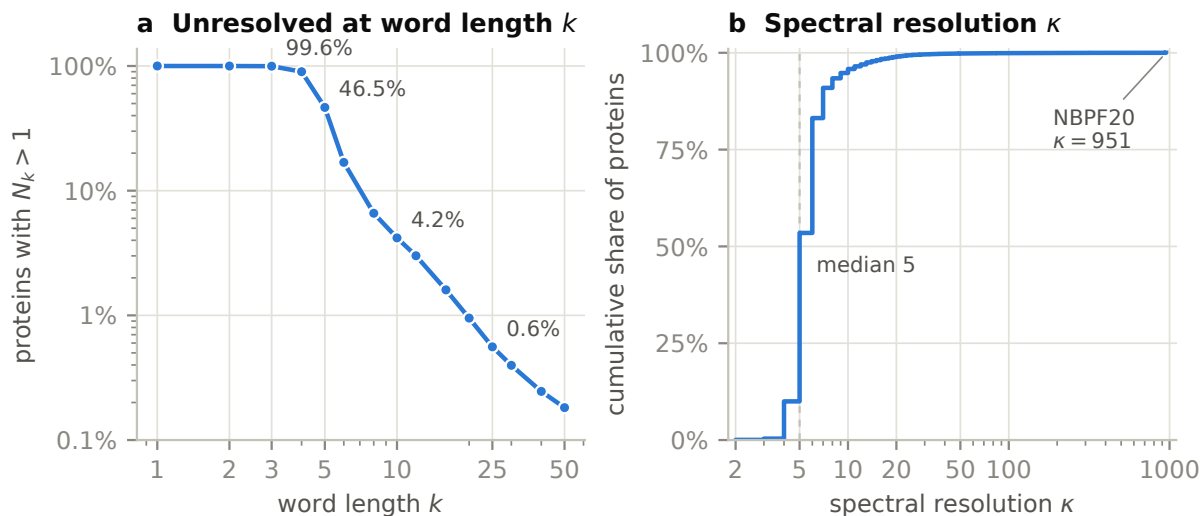


Figure 12.2: Spectral resolution of the 20 431 reviewed human proteins. (a) Share of proteins whose  $k$ -spectrum does not determine them ( $N_k > 1$ ), on logarithmic axes. (b) Cumulative distribution of the spectral resolution  $\kappa$ . The dashed line marks the median, 5.

families whose repeat units are identical or nearly identical: keratin-associated proteins, collagens, and the Krueppel C2H2 zinc-finger proteins. The latter are dominated by the KRAB zinc-finger proteins, which expanded with the transposable elements they control [42].

The zinc fingers are the clearest case. The 604 proteins whose family annotation contains “C2H2-type zinc-finger” are 3.0% of the proteome. Yet 348 of them have  $N_{10} > 1$ , i.e. 41% of the 854 proteins that remain unresolved at  $k = 10$ . ZNF91 and ZNF208, which have  $\kappa = 51$  and 95, are typical. These are also the proteins whose *specificity* lies in which finger sits where: the DNA motif a KZFP recognizes is read off the order of its fingers. UniPert’s Fig. 2G singles out the zinc-finger family as one its embedding clusters well. That is consistent with our finding: a composition-type encoder sees these proteins as nearly interchangeable, while an alignment graph and a contextual language model do not.

## 12.5 Explicit collisions

For 114 human proteins  $N_{25} > 1$ , so PseAAC with  $\lambda = 24$ , the setting of the UniPert benchmark, is not injective on their fibres. To make this concrete, we take 12 of them spread over lengths 187–2078. For each, we draw a random Eulerian trail with the same start by Hierholzer’s algorithm, and keep it if it spells a different sequence. For every pair we then check exactly (in rational arithmetic) three things: the 25-spectra coincide; the 68-dimensional amphiphilic PseAAC vectors coincide; and the outputs of a random 8-channel integer convolutional network with window 25 and mean pooling coincide. The PseAAC vectors were computed with the Kyte–Doolittle and Hopp–Woods scales, but by Proposition 12.2 the scales are irrelevant. The pairs are far from identical (Table 12.4). Figure 12.4 shows where they differ. The changes come in contiguous blocks, the transposed segments, and in the zinc-finger proteins these blocks span several fingers.

The pairs changed at two positions deserve a remark. There, two copies of a repeat unit differ at one residue each, and the fibre-mate swaps the two variants between the copies. A local encoder

| protein       | $L$  | $\kappa$ | $\log_2 N_3$ | $\log_2 N_4$ | $\log_2 N_5$ | $\log_2 N_6$ |
|---------------|------|----------|--------------|--------------|--------------|--------------|
| ESR1          | 595  | 5        | 408.2        | 33.7         | 0            | 0            |
| CEP55         | 464  | 6        | 361.9        | 43.5         | 1.6          | 0            |
| AURKA         | 403  | 5        | 234.1        | 18.9         | 0            | 0            |
| AURKB         | 344  | 5        | 152.2        | 6.6          | 0            | 0            |
| CBL           | 906  | 7        | 878.8        | 90.8         | 15.2         | 1.6          |
| CNN1          | 297  | 7        | 109.1        | 13.1         | 5.0          | 1.6          |
| CENPE         | 2701 | 11       | 5298.0       | 1352.8       | 226.3        | 48.3         |
| FOXM1         | 763  | 6        | 773.9        | 78.3         | 3.0          | 0            |
| PDCD1 (PD-1)  | 288  | 5        | 128.3        | 9.2          | 0            | 0            |
| CD274 (PD-L1) | 290  | 6        | 92.0         | 6.2          | 1.0          | 0            |
| GABRA1        | 456  | 5        | 265.1        | 19.4         | 0            | 0            |

Table 12.2: Proteins discussed in UniPert-G2CP [82].  $\log_2 N_k$  is the number of bits of order that a local encoder of order  $k$  cannot see.

| family (UniProt)             | members | median $\kappa$ | origin of the repeats                    |
|------------------------------|---------|-----------------|------------------------------------------|
| NBPF                         | 16      | 46              | Olduvai (DUF1220) domain                 |
| GOLGA6                       | 17      | 19              | amplification, hominoid-specific         |
| NPIP                         | 20      | 16              | segmental duplication, hominoid-specific |
| KRTAP type 4                 | 12      | 14              | segmental duplication, hominoid-specific |
| Krueppel C2H2 zinc finger    | 536     | 13              | cysteine-rich pentapeptide repeats       |
| KRTAP types 5, 10            | 11, 13  | 13              | tandem C2H2 fingers                      |
| POTE                         | 13      | 12              | cysteine-rich repeats                    |
| KRTAP types 28, 9            | 10, 8   | 11, 10          | segmental duplication, primate-specific  |
| collagens (fibrillar, FACIT) | 11, 9   | 10              | cysteine-rich repeats                    |
|                              |         |                 | Gly-X-Y triplets                         |
| all proteins                 | 20 431  | 5               |                                          |

Table 12.3: Families with at least eight members, ranked by median spectral resolution.

of order 25 cannot tell *which copy carries which variant*. In the zinc-finger pairs, whole fingers change places (10–15% of positions). The alignment score drops to 0.77–0.85 of the self-score, so the two sequences are far apart for an alignment-based encoder, and yet they are indistinguishable for PseAAC.

## 12.6 The chemical side: ECFP

ECFP is the molecular analogue of a  $k$ -mer spectrum [74]. After  $r$  rounds, each atom carries an identifier of its radius- $r$  environment. The fingerprint is the *set* of identifiers of all atoms, hashed into 2048 bits, with multiplicities discarded in the binary form that UniPert uses ( $r = 2$ ). It is therefore local in the graph sense. Molecules with the same radius-2 environment set receive the same fingerprint. The refinement is the Weisfeiler–Lehman colour refinement, whose blind spots (regular and symmetric graphs, rings of equal local structure) are well documented [87]. Binarization makes the fibre larger than its sequence counterpart, since it forgets how many atoms share an environment. The radius also fixes a horizon. A radius-2 environment reaches two bonds

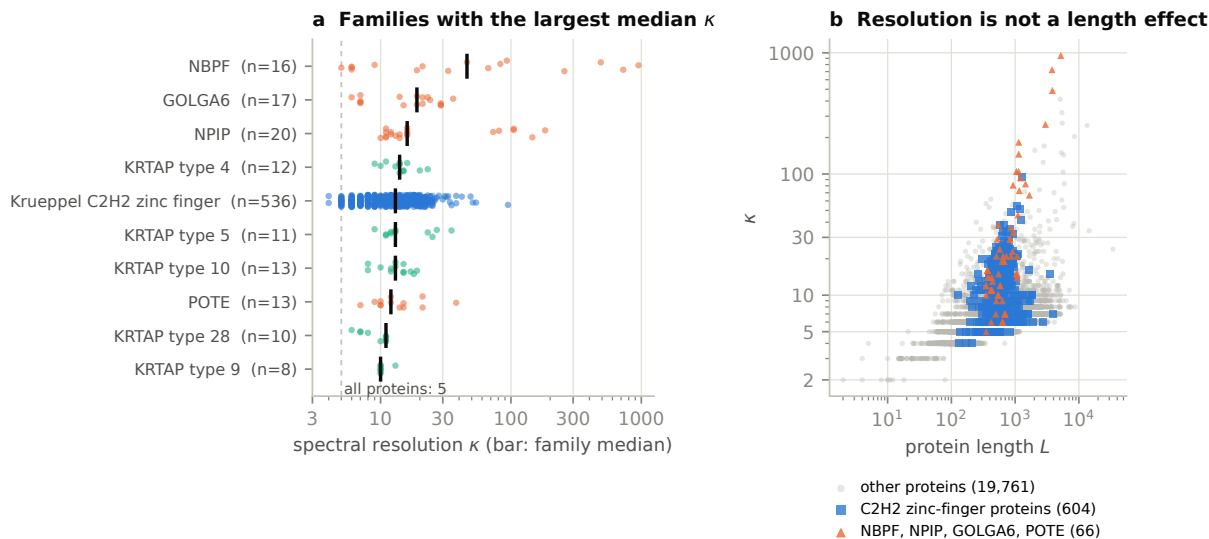


Figure 12.3: Where the unresolved order lives. (a) The ten families (at least eight members) with the largest median  $\kappa$ . Each dot is a protein, and the bar is the family median. Orange: hominoid/primate-specific segmental-duplication families; blue: Krueppel C2H2 zinc-finger proteins; green: keratin-associated proteins. The dashed line is the proteome median. (b)  $\kappa$  against protein length for all proteins. The two highlighted groups are shown in their own colours and marker shapes.

from its centre, so no single identifier spans more than four bonds. Two molecules that assemble the same environments into different long-range arrangements (the order of substituents along a chain, the placement of equal units around a macrocycle) therefore lie in the same fibre. This is the exact analogue of a transposition between repeated  $(k - 1)$ -mers. The default ECFP also carries no stereochemistry, so stereoisomers (enantiomers, *cis/trans* isomers) always share a fingerprint unless chirality flags are switched on. Both are *molecular fibres* in the sense of this chapter: sets of distinct perturbagens that a local encoder cannot tell apart, whatever it was trained on. We do not compute molecular fibres here. We note only that in UniPert-G2CP both perturbagen types would have been local, had the protein encoder been PseAAC: the gain on the genetic side comes precisely from giving up locality.

## 12.7 DNA perturbagens and the double cover

UniPert-G2CP names as its limitation that genetic perturbations outside protein-coding genes cannot be represented, and points to DNA language models. DNA brings one more symmetry. A regulatory element, or a CRISPR target, has no preferred strand, so a DNA encoder that is to be strand-agnostic must satisfy  $\phi(S) = \phi(\bar{S})$  for the reverse complement  $\bar{S}$ . A local encoder with this property depends only on the *double-stranded* spectrum  $\text{spec}_k(S) + \text{spec}_k(\bar{S})$ . Its fibre is the set  $\text{DS}(S, k)$  of double-stranded reconstructions.

**Proposition 12.7.** *A strand-agnostic local encoder of order  $k$  is constant on  $\text{DS}(S, k)$ . For circular  $S$ ,*

$$|\text{DS}(S, k)| = \sum_f N_{\text{seq}}\left(\frac{m+f}{2}\right),$$

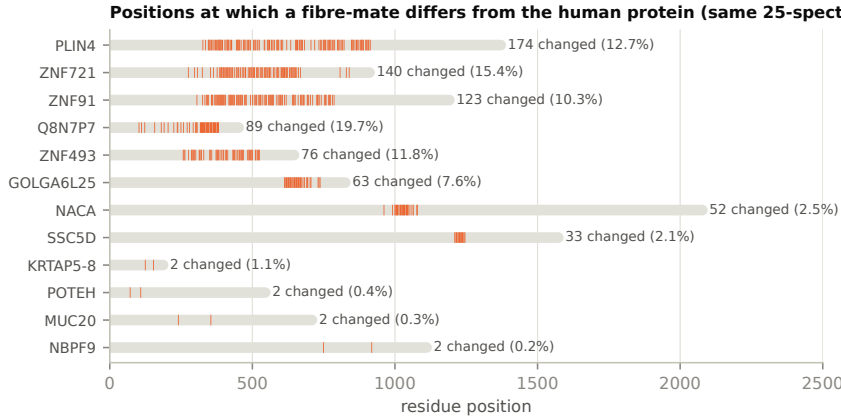


Figure 12.4: The twelve fibre-mates of Table 12.4. Each grey bar is a human protein, and orange marks the positions at which its fibre-mate differs. Each pair has the same 25-spectrum, the identical 68-dimensional PseAAC vector and the identical output from a window-25 convolutional encoder.

*the sum running over the points  $f$  of the  $(-1)$ -eigenlattice of the reverse-complement involution on the circulations of the double cover  $D_k$ , inside the box  $|f| \leq m$  (Theorem 7.6).*

This fibre is not given by a single determinant. It is a lattice sum of determinants, and it is much harder to compute. For the 5386-bp genome of  $\phi$ X174 at  $k = 10$ , the lattice has rank 48 and the box has 2 481 687 900 points, of which 909 103 210 have connected support (Chapter 8). The number of double-stranded reconstructions is

$$|\text{DS}(\phi\text{X174}, 10)| = 31\,925\,753\,246\,212$$

(Chapter 9). A strand-agnostic local DNA encoder of window 10 assigns all of them the same vector. The design choice “make the encoder reverse-complement invariant” is sound, but it has a price. For local encoders that price is exactly the passage from a sequence fibre to a double-stranded fibre.

## 12.8 What this does and does not imply

Table 12.5 sets our results against the statements of UniPert-G2CP. Three limitations should be stated plainly.

First, a fibre is a set of *sequences*, not of proteins. Most fibre members are not biologically plausible, and a model trained on real proteins may never meet them. The fibres that matter are those reached by real processes, and here one must distinguish. Exon skipping, exon duplication and domain deletion change the  $k$ -mer content: they add or remove  $k$ -mers. So they move a sequence *out of* its fibre, and a local encoder does register them. What a local encoder cannot register is a change of order at constant content. Such changes include: gene conversion or unequal crossing-over that swaps variants between repeat copies; a variant assigned to the wrong copy of a repeat; and a reordering of identical or near-identical units, such as fingers in a KZFP, Olduvai domains in an NBPF protein, or cysteine-rich repeats in a KRTAP. These are the transpositions of Section 12.2, and they are concentrated in the families of Table 12.3. Whether such events have

| protein   | $L$  | positions changed | $SW(S, T)/SW(S, S)$ | PseAAC, CNN |
|-----------|------|-------------------|---------------------|-------------|
| KRTAP5-8  | 187  | 2 (1.1%)          | 0.984               | identical   |
| Q8N7P7    | 452  | 89 (19.7%)        | 0.814               | identical   |
| POTEH     | 545  | 2 (0.4%)          | 0.994               | identical   |
| ZNF493    | 646  | 76 (11.8%)        | 0.824               | identical   |
| MUC20     | 709  | 2 (0.3%)          | 0.996               | identical   |
| GOLGA6L25 | 826  | 63 (7.6%)         | 0.944               | identical   |
| ZNF721    | 911  | 140 (15.4%)       | 0.769               | identical   |
| NBPF9     | 1111 | 2 (0.2%)          | 0.997               | identical   |
| ZNF91     | 1191 | 123 (10.3%)       | 0.846               | identical   |
| PLIN4     | 1371 | 174 (12.7%)       | 0.815               | identical   |
| SSC5D     | 1573 | 33 (2.1%)         | 0.981               | identical   |
| NACA      | 2078 | 52 (2.5%)         | 0.986               | identical   |

Table 12.4: A human protein  $S$  and a different sequence  $T$  with the same 25-spectrum and the same start. SW is a Smith–Waterman local alignment score (match 2, mismatch  $-1$ , gap  $-2$ ). PseAAC is the 68-dimensional amphiphilic form with  $\lambda = 24$ ; CNN is an 8-channel window-25 integer network with mean pooling.

phenotypic consequences that a perturbation model should predict is open. When they do, a local encoder cannot predict them.

Second, we have not evaluated UniPert, ESM or any trained model. Our statements about PseAAC and convolutional encoders are theorems about their form. The comparison with UniPert’s benchmark offers a mechanism consistent with their ranking, not a test of it. Such a test is easy to set up: a *fibre-mate benchmark*. It would pair each protein with a sequence from its  $k$ -fibre (for example, a repeat-copy swap), and ask whether an encoder separates the pair and whether a predictor assigns them different effects.

Third, the spectral resolution measures what is *invisible* to local statistics, not what is *relevant* to a phenotype. Whether finger order in a KZFP changes a perturbation phenotype is a biological question. Our analysis says only that a local encoder cannot represent the answer.

## Data and code availability

`spectral_fibres.py` (Python 3.10+, standard library; `numpy` only for the floating-point determinants) downloads the reviewed human proteome from UniProt. It then runs the five checks (formula, contraction, proteome, collisions, fibre-mates) and writes the per-protein results (`fibre_results.json`). The transcript is `spectral_fibres_output.txt`. The double-stranded count of Section 12.7 is reproduced by the ancillary files of Chapter 9.

| UniPert-G2CP                                                      | what the fibre analysis adds                                                                                                                                                                    |
|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| UniPert outperforms PseAAC, ESM, OntoProtein in genetic OOD tasks | PseAAC ( $\lambda = 24$ ) is constant on 25-fibres: provably identical vectors for 12 explicit human protein pairs differing at up to 15% of positions; 114 proteins have non-trivial 25-fibres |
| sequence-based encoders “risk losing local information”           | composition-type encoders lose <i>global order</i> , not local content: median $2^{239}$ twins at $k = 3$ , $2^{19}$ at $k = 4$ ; median resolution $\kappa = 5$                                |
| reference set of 19 187 proteins                                  | every local encoder of order $\geq 2$ already separates the reference proteome; separation of the reference set is not evidence of resolution                                                   |
| zinc-finger family clustered well (Fig. 2G)                       | KZFPs are 3% of proteins but 41% of those with $\kappa > 10$ ; their function is in finger order, which local encoders cannot see                                                               |
| ESR1 domains and mutants (Fig. 6)                                 | $\kappa(\text{ESR1}) = 5$ ; a point substitution changes at most $k$ of the $k$ -mers, and it is visible to any local encoder of order $\leq \kappa$                                            |
| ECFP4 for compounds                                               | ECFP is a local (radius-2, binarized) encoder; its blind spots are those of Weisfeiler–Lehman refinement                                                                                        |
| limitation: non-coding perturbations                              | a strand-agnostic local DNA encoder is constant on the double-stranded reconstruction set, $3.19 \cdot 10^{13}$ sequences for $\phi\text{X174}$ at $k = 10$                                     |

Table 12.5: The results of this chapter set against the corresponding statements of UniPert-G2CP.



## Chapter 13

# Polyploid phasing as a $p$ -fold decomposition of a balanced flow, and the homogeneous space of alleles

**Summary.** The  $k$ -mer spectrum of  $p$  homologous chromosomes sequenced together is the arc multiplicity vector  $m = \sum_i \text{spec}(H_i)$  of one pooled de Bruijn multigraph, and phasing is the recovery of the summands. We show that the labelled phasings — ordered  $p$ -tuples of circular sequences whose spectra add to  $m$  — are counted by a sum over the ordered decompositions  $m = c_1 + \cdots + c_p$  into connected balanced points of the box  $[0, m]$  of the circulation lattice, each term the product of the per-sequence BEST counts  $N_{\text{seq}}(c_i) = t(c_i) \prod (d^+ - 1)! / \prod c_i(a)!$ . This is the lattice sum of Chapter 7 with the strand involution removed and one constraint replaced by a  $p$ -fold partition. The symmetric group  $\mathfrak{S}_p$  permutes the parts of a decomposition; it does *not* act on the pooled graph, and the natural map from the disjoint union of the haplotype graphs to the pooled graph is a covering only when the haplotypes share no  $k$ -mer, so there is no monodromy in the sense of covering spaces. What replaces it is a homogeneous space: at a biallelic site carried by  $a$  of the  $p$  haplotypes the local choice is an  $a$ -subset of the labels,  $\mathfrak{S}_p / (\mathfrak{S}_a \times \mathfrak{S}_{p-a})$ , and for  $b$  sites in distinct  $(k+1)$ -windows on a repeat-free reference the labelled count is  $\prod_j \binom{p}{a_j}$  exactly, which  $|\text{Hom}(\pi_1, \mathfrak{S}_p)| = (p!)^b$  does not reproduce. Sites within one  $(k+1)$ -window merge into a single multi-allelic site whose alleles are the distinct local haplotypes. Unlabelled phasings are multisets, counted by  $\prod_c \binom{N_{\text{seq}}(c) + r_c - 1}{r_c}$  over the multiset of spectra, equal to the labelled count divided by  $p!$  exactly when the parts are pairwise distinct. Dropping the requirement that all haplotypes have the reference length exposes the copy-number ambiguity through repeats: the pooled spectrum of two copies of  $\phi$ X174 is also the pooled spectrum of two circles of 2496 and 8276 base pairs, and of two tandem-dimer circles, whose weight in the sum is  $1/2$ . Every count is verified on  $\phi$ X174 with synthetic substitutions at  $p = 2$  and  $p = 3$  by an enumeration of walks that does not use the lattice.

### 13.1 What was asked, and what is true

Chapter 7 showed that a double-stranded reconstruction of a circular sequence  $S$  is a closed walk through half the arcs of the reverse-complement double cover whose image under the strand involution  $\rho$  walks the other half, and that the count is a sum over the box points of the  $(-1)$ -eigenlattice of  $\rho$  on circulations. The involution was the obstacle: it reverses arcs, it is a duality rather than a symmetry of the digraph, and all the machinery of covering graphs is unavailable.

Polyploid phasing looks at first like the case where that obstacle is absent. The pooled  $k$ -mer spectrum of  $p$  homologous circular chromosomes  $H_1, \dots, H_p$  is the arc multiplicity vector  $m = \sum_i \text{spec}(H_i)$  of the pooled de Bruijn multigraph, and a phasing is a way of writing  $m$  as a sum of  $p$  spectra of circular sequences. Exchanging two haplotypes is a permutation of the summands; it does not reverse anything. One is tempted to say that the symmetric group  $\mathfrak{S}_p$  acts on the pooled graph as a genuine automorphism group, that the haplotypes are the sheets of a  $p$ -fold cover, and that the phasing space is the space of flat  $\mathfrak{S}_p$ -connections  $H^1(G_k, \mathfrak{S}_p) = \text{Hom}(\pi_1(G_k), \mathfrak{S}_p)$  on the pooled graph, with a transposition at each heterozygous site. The specification of this chapter said so, and so did its planning.

None of that is so, and the correct statements are simpler.  $\mathfrak{S}_p$  acts on the set of *decompositions* of  $m$ , not on the graph: the pooled graph has no distinguished symmetry exchanging haplotypes, because the haplotypes are different words and the graph does not remember which arc came from which. The natural map from the disjoint union of the haplotype graphs to the pooled graph is a covering map only when no  $k$ -mer is shared, in which case the pool is the disjoint union and there is nothing to phase (Proposition 13.7). And the local structure at a biallelic site is not a group but a homogeneous space: if  $a$  of the  $p$  haplotypes carry one allele and  $p - a$  the other, the choice at the site is which  $a$  labels carry it, a point of  $\mathfrak{S}_p / (\mathfrak{S}_a \times \mathfrak{S}_{p-a})$  of cardinality  $\binom{p}{a}$ ; only for  $p = 2$ , or at a site where all  $p$  alleles differ, is this the group. For  $b$  sites in distinct  $(k+1)$ -windows on a repeat-free reference the labelled count is exactly  $\prod_j \binom{p}{a_j}$  (Theorem 13.8), and  $(p!)^b$  is wrong already at  $p = 3, b = 1, a = 1$ : the answer is 3, not 6.

What survives, and is the subject of Section 13.2, is the lattice sum. The labelled phasings are counted exactly by

$$\# \text{Lab}(H_1 + \dots + H_p, k) = \sum_{\substack{c_1 + \dots + c_p = m \\ c_i \in Z_1 \cap [0, m], \text{ supp } c_i \text{ connected}}} \prod_{i=1}^p N_{\text{seq}}(c_i),$$

with  $N_{\text{seq}}$  the per-sequence BEST count of Theorem 7.6; the unlabelled phasings by a multiset formula over the unordered decompositions (Section 13.4); and when the parts are not required to have the reference length the sum exposes the copy-number ambiguity through repeats, including tandem-dimer circles whose weight is  $1/2$  (Section 13.5). Section 13.6 records the verification on  $\phi\text{X174}$ .

Throughout, sequences are circular over  $\{A, C, G, T\}$ , a *circular sequence* is a word up to rotation, and  $d(T)$  is the number of rotations fixing  $T$ . All computations are exact integer or rational arithmetic; the script `polyploid_phasing.py` reproduces every number below and prints a six-line pass/fail battery.

## 13.2 The pooled multigraph and the phasing lattice

**Definition 13.1.** Let  $H_1, \dots, H_p$  be circular sequences of lengths  $N_1, \dots, N_p$ . The *pooled de Bruijn multigraph*  $G_k = G_k(H_1 + \dots + H_p)$  has one vertex per  $k$ -mer occurring in some  $H_i$  and one arc per occurrence of a  $(k+1)$ -mer in some  $H_i$ . A vertex is a *branch vertex* if it has at least two distinct successors or at least two distinct predecessors in the pool; the *compacted* pooled graph has the branch vertices as vertices and the maximal unitigs between them as arcs. An arc is labelled by its haplotype and position; its *content* is the unitig string,  $A$  is the set of contents,  $m(a)$  the number of arcs of content  $a$ ,  $\ell(a)$  its number of  $(k+1)$ -mers, and  $\partial: \mathbb{Z}^A \rightarrow \mathbb{Z}^V$  the boundary.  $Z_1 = \ker \partial$  is the circulation lattice.

The compaction criterion differs from the one used for a single genome in Lemma 15.1, where a  $k$ -mer is a branch vertex if and only if it is repeated. In a pool every shared  $k$ -mer is repeated, so that criterion would contract nothing; the criterion of Definition 13.1 contracts exactly the  $k$ -mers through which every haplotype must pass in the same way. The per-sequence count below is intrinsic and does not depend on which compaction is used.

**Definition 13.2.** As in Section 7.3, for  $c \in \mathbb{Z}_{\geq 0}^A$  balanced with connected support let  $G_c$  be the multigraph with  $c(a)$  labelled parallel arcs of content  $a$ ,  $t(c)$  its number of spanning arborescences converging on a root,  $\text{Loc}(c) = \prod_v (d_c^+(v) - 1)!$  over the vertices of positive  $c$ -degree, and

$$N_{\text{seq}}(c) = \frac{t(c) \text{Loc}(c)}{\prod_a c(a)!} = \sum_{\text{spec } T=c} \frac{1}{d(T)} \in \mathbb{Q}_{>0},$$

the second equality being Theorem 7.6.  $N_{\text{seq}}(c) = 0$  if the support is disconnected. The *length* of  $c$  is  $|c| = \sum_a c(a)\ell(a)$ .

**Lemma 13.3.** *Let  $T$  be a circular sequence all of whose  $(k+1)$ -mers occur in the pool. Then  $T$  traverses each unitig of the compacted pooled graph a well-defined number of times, so  $\text{spec}(T)$  is a vector  $c \in \mathbb{Z}_{\geq 0}^A$  with  $\partial c = 0$  and connected support, and  $|c|$  is the length of  $T$ .*

*Proof.* An interior  $k$ -mer  $w$  of a unitig has exactly one successor and one predecessor in the pool. Every  $(k+1)$ -mer of  $T$  through  $w$  is a  $(k+1)$ -mer of the pool, hence is the unique in- or out-arc of  $w$ ; so each passage of  $T$  through  $w$  continues along the unitig in both directions, and the number of passages is the same at every  $k$ -mer of the unitig.  $\square$

**Definition 13.4.** A *labelled phasing* of the pool at level  $k$  is an ordered  $p$ -tuple  $(T_1, \dots, T_p)$  of circular sequences with  $\sum_i \text{spec}(T_i) = m$  as multisets of  $(k+1)$ -mers; its *weight* is  $\prod_i 1/d(T_i)$ . An *unlabelled phasing* is the multiset  $\{T_1, \dots, T_p\}$ . An *ordered decomposition* of  $m$  is a  $p$ -tuple  $(c_1, \dots, c_p)$  of balanced vectors with connected support,  $0 \leq c_i \leq m$  and  $\sum c_i = m$ ;  $\mathcal{D}_p(m)$  is their set. The *equal-length*phasings and decompositions are those with  $|T_i| = N_1$ , resp.  $|c_i| = N_1$ , for all  $i$ .

**Theorem 13.5** (The phasing lattice). *The map  $(T_1, \dots, T_p) \mapsto (\text{spec } T_1, \dots, \text{spec } T_p)$  is a surjection from the labelled phasings onto  $\mathcal{D}_p(m)$ , and the weighted number of labelled phasings is*

$$\#\text{Lab}(H_1 + \dots + H_p, k) = \sum_{(c_1, \dots, c_p) \in \mathcal{D}_p(m)} \prod_{i=1}^p N_{\text{seq}}(c_i).$$

*The same holds for equal-length phasings with the sum restricted to equal-length decompositions. In particular the count is an integer whenever every part of every decomposition is the spectrum of aperiodic sequences only.*

*Proof.* Lemma 13.3 sends each  $T_i$  to a balanced connected  $c_i \geq 0$ , and  $\sum \text{spec } T_i = m$  is  $\sum c_i = m$ , which also gives  $c_i \leq m$ . Conversely each part of a decomposition is balanced and connected, so  $G_{c_i}$  has an Eulerian circuit, which read as a sequence has spectrum  $c_i$ ; the  $p$  sequences together have spectrum  $m$ . The fibre over  $(c_1, \dots, c_p)$  is the product of the fibres over the  $c_i$ , each of weighted cardinality  $N_{\text{seq}}(c_i)$  by Definition 13.2. Lengths are preserved by Lemma 13.3.  $\square$

**Remark 13.6.** The lattice is the same object as in Chapter 7: there  $p = 2$ ,  $c_2 = \rho c_1$  is forced, and the second part is the strand image of the first, so  $\mathcal{D}_2(m)$  collapses to the box points of the anti-invariant lattice. Here nothing ties the parts together except  $\sum c_i = m$ , and  $\mathfrak{S}_p$  acts on  $\mathcal{D}_p(m)$  by permuting them. The single-strand theory is  $p = 1$ , where  $\mathcal{D}_1(m) = \{m\}$  and the sum is one term.

**Proposition 13.7** (No covering). *Let  $\pi: \sqcup_i G_k(H_i) \rightarrow G_k(H_1 + \dots + H_p)$  send each vertex and arc of a haplotype graph to the same  $k$ -mer or occurrence in the pool. Then  $\pi$  is a surjective homomorphism of multidigraphs, and it is a covering map if and only if no  $k$ -mer occurs in two different haplotypes, in which case  $\pi$  is an isomorphism. In particular, when  $p \geq 2$  and the haplotypes share a  $k$ -mer,  $\mathfrak{S}_p$  acts on  $\mathcal{D}_p(m)$  but there is no  $p$ -sheeted cover of the pooled graph with deck group  $\mathfrak{S}_p$ , and no action of  $\mathfrak{S}_p$  on the pooled graph is induced by relabelling haplotypes.*

*Proof.* A covering map is a local isomorphism: the out-arcs of a vertex  $\tilde{w}$  upstairs map bijectively onto the out-arcs of  $\pi(\tilde{w})$ . If a  $k$ -mer  $w$  occurs in  $H_i$  and in  $H_j$ ,  $i \neq j$ , then  $w$  has two preimages  $\tilde{w}_i, \tilde{w}_j$ , the out-degree of  $\tilde{w}_i$  is the multiplicity of  $w$  in  $H_i$ , and the out-degree of  $w$  is the sum of the multiplicities of  $w$  in all haplotypes, which strictly exceeds it. If no  $k$ -mer is shared,  $\pi$  is a bijection on vertices and on arcs. For the last sentence: an action of  $\mathfrak{S}_p$  on the pooled graph induced by relabelling would have to fix every vertex and every arc, since the pool does not record labels, and the only such action is trivial.  $\square$

The honest picture is therefore this. The pooled graph carries the flow  $m$ ; a phasing is a decomposition of the flow into  $p$  closed walks; the labels are a decoration of the decomposition and  $\mathfrak{S}_p$  permutes the decoration. The question “how many phasings” is a question about the fibres of  $\mathcal{D}_p(m) \rightarrow \mathcal{D}_p(m)/\mathfrak{S}_p$ , and it is answered locally by homogeneous spaces (Section 13.3) and globally by a multiset count (Section 13.4).

### 13.3 The homogeneous fibre, and linkage

Fix a circular *reference*  $R$  of length  $N$  and let each  $H_i$  be obtained from  $R$  by substitutions at a set of positions, the *sites*; at a site  $s$  let the alleles be the letters occurring there across the  $H_i$  and  $a_s(\alpha)$  the number of haplotypes carrying allele  $\alpha$ . A site is *biallelic* if two letters occur, and we then write  $a_s$  for the count of the minor allele. Two sites are in the same  $(k+1)$ -*window* if some  $(k+1)$ -mer position of  $R$  contains both, i.e. their circular distance is at most  $k$ .

**Theorem 13.8** (Independent sites on a repeat-free reference). *Suppose  $R$  has no repeated  $k$ -mer, the sites lie in pairwise distinct  $(k+1)$ -windows, and no  $k$ -mer of any  $H_i$  overlapping a site coincides with a  $k$ -mer of  $R$  or with one overlapping another site. Then the equal-length ordered decompositions of  $m$  are in bijection with  $\prod_s \mathfrak{S}_p / (\prod_\alpha \mathfrak{S}_{a_s(\alpha)})$ , every part has  $N_{\text{seq}} = 1$ , and*

$$\#\text{Lab} = \prod_s \binom{p}{a_s(\alpha_1), a_s(\alpha_2), \dots} = \prod_{s \text{ biallelic}} \binom{p}{a_s} \quad \text{when all sites are biallelic.}$$

*Proof.* Under the hypotheses the compacted pooled graph is a cycle of  $N$  base pairs in which each site  $s$  is replaced by a *bubble*: the  $k$ -mer of  $R$  ending just before  $s$  is a branch vertex  $u_s$ , the  $k$ -mer starting just after  $s$  is a branch vertex  $v_s$ , and between them run parallel unitigs, one per allele  $\alpha$  at  $s$ , each of length  $k+1$  in  $(k+1)$ -mers and of multiplicity  $a_s(\alpha)$ ; the  $k$ -mers strictly between sites are shared by all haplotypes and contract into unitigs of multiplicity  $p$ . There are no other branch vertices because  $R$  has no repeats and the allele  $k$ -mers are new. A closed walk in this graph has a winding number  $w \geq 1$  around the cycle and length exactly  $wN$ , since all branches of a bubble have the same length. A part of an equal-length decomposition has length  $N$ , hence  $w = 1$ , hence passes each bubble exactly once through one branch; its spectrum is the 0/1 vector selecting one allele per site,  $G_c$  is a simple cycle, and  $N_{\text{seq}}(c) = 1$ . An ordered decomposition is thus a  $p$ -tuple of allele selections, one per label, with  $a_s(\alpha)$  labels selecting  $\alpha$  at each site: exactly the product

over sites of the sets of ordered partitions of the labels into classes of sizes  $a_s(\alpha)$ , which are the homogeneous spaces stated, of cardinalities the multinomial coefficients. Theorem 13.5 gives the count.  $\square$

**Corollary 13.9.** *Under the hypotheses of Theorem 13.8 the pooled graph has  $\pi_1$  free of rank  $\sum_s (n_s - 1)$ , where  $n_s$  is the number of alleles at  $s$ , and  $|\text{Hom}(\pi_1, \mathfrak{S}_p)| = (p!)^{\text{rank } \pi_1}$ , which equals  $\#\text{Lab}$  only when  $p \leq 2$  or every site has  $p$  distinct alleles.*

*Proof.* The cycle rank of a cycle with bubbles is the number of extra branches. For  $p = 2$  and biallelic sites both counts are  $2^b$ ; for  $p \geq 3$  and a biallelic site with  $a_s = 1$  the site contributes  $p$  to  $\#\text{Lab}$  and  $p!$  to the other.  $\square$

**Theorem 13.10** (Linkage). *Let a set  $B$  of sites lie in one  $(k+1)$ -window, with the hypotheses of Theorem 13.8 otherwise in force for  $B$  as a whole (no  $k$ -mer overlapping  $B$  is a  $k$ -mer of  $R$  or overlaps another site). Then in the compacted pooled graph  $B$  is a single bubble whose branches are the distinct local haplotypes — the distinct restrictions of  $H_1, \dots, H_p$  to  $B$  — with multiplicity the number of haplotypes carrying each, and in Theorem 13.8 the block  $B$  counts as one site whose alleles are its local haplotypes. In particular two biallelic sites within distance  $k$  carried in only two combinations contribute  $\binom{p}{a}$  and not  $\binom{p}{a'}\binom{p}{a''}$ .*

*Proof.* Some  $(k+1)$ -mer position of  $R$  contains all of  $B$ , so every haplotype has a  $(k+1)$ -mer through the whole block, and two haplotypes with the same local restriction have the same  $k$ -mers throughout the window while two with different restrictions differ in that  $(k+1)$ -mer. The  $k$ -mers overlapping  $B$  but not containing all of it may be shared between haplotypes that agree on the sites they contain; this merges vertices inside the bubble but creates no new closed walk of length  $N$ , because a walk of length  $N$  has winding number one and its passage through the bubble is determined by the  $(k+1)$ -mer it uses at the position containing all of  $B$ . So the equal-length parts through the block are the local haplotypes, and the count is the multinomial in their multiplicities.  $\square$

For sites that are pairwise within distance  $k$  but not all inside one  $(k+1)$ -window, or a chain of sites each within  $k$  of the next, the bubble structure is a chain of partially merged bubbles and we do not have a closed form; see the open questions.

*Remark 13.11* (With repeats). When  $R$  has repeats the argument of Theorem 13.8 gives a lower bound. Each allele selection  $c$  —  $\text{spec}(R)$  with one allele chosen at each site — is still a part of length  $N$ , and  $G_c$  is isomorphic to  $G_{\text{spec } R}$  with the bubbles replaced by single unitigs of the same length, so  $N_{\text{seq}}(c) = N_{\text{seq}}(R)$ ; the ordered allele selections therefore contribute  $\prod_s \binom{p}{a_s} N_{\text{seq}}(R)^p$  to  $\#\text{Lab}$ . Other equal-length decompositions can exist only if  $p \text{ spec}(R)$  admits a balanced connected sub-vector of length  $N$  that is not an allele selection, which requires a coincidence among the lengths of the sub-circles of  $R$  through its repeats. For  $\phi\text{X174}$  at  $k = 12$  no such coincidence occurs and the bound is the count at every tested configuration (Table 13.1).

## 13.4 The unlabelled multiset formula and the $\mathfrak{S}_p$ -action

**Proposition 13.12.**  $\mathfrak{S}_p$  acts on  $\mathcal{D}_p(m)$  by permuting the parts; the stabiliser of  $(c_1, \dots, c_p)$  is  $\prod_c \mathfrak{S}_{r_c}$ , where  $r_c$  is the number of parts equal to  $c$ . If every sequence with spectrum a part of the

decomposition is aperiodic, the number of unlabelled phasings whose spectra form the multiset  $\{c^{r_c}\}$  is

$$\prod_c \binom{N_{\text{seq}}(c) + r_c - 1}{r_c},$$

and the total number of unlabelled phasings is the sum of this over the orbits  $\mathcal{D}_p(m)/\mathfrak{S}_p$ . It equals  $\#\text{Lab}/p!$  exactly when every decomposition has pairwise distinct parts.

*Proof.* An unlabelled phasing with spectrum multiset  $\{c^{r_c}\}$  is, for each  $c$ , a multiset of  $r_c$  sequences drawn from the  $N_{\text{seq}}(c)$  sequences with spectrum  $c$  (integers, by aperiodicity); multisets of size  $r$  from a set of size  $n$  number  $\binom{n+r-1}{r}$ . The labelled phasings over an orbit number  $|\text{orbit}| \cdot \prod_c N_{\text{seq}}(c)^{r_c} = (p!/\prod_c r_c!) \prod_c N_{\text{seq}}(c)^{r_c}$ , and  $\binom{n+r-1}{r} = n^r/r!$  only for  $r \leq 1$ .  $\square$

The distinction is not academic. For three haplotypes  $S, S', S$  differing at one site, the three ordered decompositions all have the same multiset of spectra, and the unlabelled count is  $N_{\text{seq}}(S) \binom{N_{\text{seq}}(S)+1}{2}$ , which is 6 for  $\phi\text{X174}$  at  $k = 12$ , against a labelled count of 24;  $24/6 = 4 \neq 6$ .

*Remark 13.13* (Periodic parts). When a part  $c$  is  $2c'$  for a circulation  $c'$  and  $G_{c'}$  is Eulerian, the sequences with spectrum  $c$  include the tandem dimers  $UU$  of the sequences  $U$  with spectrum  $c'$ , which have  $d(UU) = 2$ , and more generally every sequence with spectrum  $c$  may be periodic, in which case  $N_{\text{seq}}(c)$  is a half-integer or worse. The weighted count of Theorem 13.5 remains correct, the multiset formula does not apply to such parts, and the script reports them separately. In the data they arise only when the equal-length constraint is dropped, and they are the dimer circles of Section 13.5.

## 13.5 Copy-number variation across repeats

The equal-length constraint encodes the substitution-only model. Without it, the sum of Theorem 13.5 ranges over all decompositions of the pooled flow into closed walks, and for a reference with repeats it contains decompositions in which a block bounded by two copies of a repeat is carried once by one haplotype and three times by the other, or in which the two haplotypes are two circles of different sizes. These are the copy-number and excision–integration ambiguities of the pooled spectrum, and the lattice sum enumerates them.

For  $\phi\text{X174}$  at  $k = 12$  the single-strand graph is the interleaved two-copy repeat of Proposition 2.5,  $a X_1 b Y_1 a X_2 b Y_2$ : two branch 12-mers  $a, b$  and four unitigs, with  $N_{\text{seq}}(S) = 2$ . The sub-circles  $aX_i bY_j$  have lengths 2496, 2890, 2638 and 2748, in two complementary pairs summing to  $N = 5386$ . The pool  $S + S$  has  $m = 2$  on each unitig, and its ordered decompositions with connected parts number 17: one with two parts of length  $N$ , and sixteen of the shapes

$$(2496, 8276), (2890, 7882), (2638, 8134), (2748, 8024), (4992, 5780), (5276, 5496)$$

and their transposes. The first four are a sub-circle  $aX_i bY_j$  of length  $\ell$  and the complementary circle of length  $2N - \ell$ , which carries the two unitigs of the sub-circle once and the other two twice: a sub-circle excised from one copy of the genome and its content integrated into the other. The last two have *both* parts periodic:  $4992 = 2 \cdot 2496$  and  $5780 = 2 \cdot 2890$ , the tandem dimers of a complementary pair of sub-circles, each with  $d(T) = 2$  and weight  $1/2$ . With one substitution added the decompositions number 24, of which 22 have unequal lengths (Table 13.3). A pooled spectrum thus does not distinguish two identical circular chromosomes from two dimer circles of complementary halves, and the count records the fact with the weight it deserves.

| configuration               | $ V $ | $ A $ | rank $Z_1$ | box | conn. | decomp. | #Lab | #Unl |
|-----------------------------|-------|-------|------------|-----|-------|---------|------|------|
| 1 site                      | 4     | 7     | 4          | 26  | 25    | 2       | 8    | 4    |
| 2 sites                     | 6     | 10    | 5          | 40  | 39    | 4       | 16   | 8    |
| 3 sites                     | 8     | 13    | 6          | 56  | 55    | 8       | 32   | 16   |
| 5 sites                     | 12    | 19    | 8          | 122 | 121   | 32      | 128  | 64   |
| 600, 606 linked + 2 sites   | 8     | 13    | 6          | 56  | 55    | 8       | 32   | 16   |
| $\{600, 606\}$ vs $\{600\}$ | 4     | 7     | 4          | 26  | 25    | 2       | 8    | 4    |

Table 13.1: Diploid pools, equal length. “decomp.” is the number of equal-length ordered decompositions; the prediction is  $2^b$  for  $b$  linkage blocks,  $\#Lab = 2^b N_{\text{seq}}(S)^2$  and  $\#Unl = 2^{b-1} N_{\text{seq}}(S)^2$ , met in every row. The walk enumeration reproduces  $N_{\text{seq}}$  on every spectrum it finds and the same  $\#Lab$ ; the explicit unordered pairs number 4, 8, 16, 64, 16, 4. Checks (P), (L), (W), (U).

| configuration                            | $ V $ | $ A $ | rank $Z_1$ | box | conn. | decomp. | #Lab | #Unl |
|------------------------------------------|-------|-------|------------|-----|-------|---------|------|------|
| three sites: $a = 1, a = 2$ , triallelic | 8     | 14    | 7          | 248 | 247   | 54      | 432  | 72   |
| one site, haplotypes $S, S', S$          | 4     | 7     | 4          | 68  | 67    | 3       | 24   | 6    |

Table 13.2: Triploid pools, equal length. First row: sites 600 (minor allele in one haplotype), 1700 (in two) and 2900 (three alleles);  $\binom{3}{1}\binom{3}{2}3! = 54$  decompositions, all with pairwise distinct parts,  $\#Lab = 54 \cdot 2^3$  and  $\#Unl = \#Lab/6$ ;  $|\text{Hom}(\pi_1, \mathfrak{S}_3)|$  would be  $6^3 = 216$ . Second row: three decompositions with the same multiset of spectra,  $\#Unl = N_{\text{seq}}(S) \binom{N_{\text{seq}}(S)+1}{2} = 6 \neq 24/6$ . Check (T).

## 13.6 Verification and data

All configurations use  $\phi X174$  (NC\_001422.1, 5386 bp) as the reference at  $k = 12$ , where  $N_{\text{seq}}(S) = 2$ , with substitutions at positions 600, 1700, 2900, 3900, 4900 (pairwise distance  $> k$ , none within  $k$  of a repeat  $k$ -mer) and, for linkage, at 606. Each row of Tables 13.1–13.3 is computed twice: from the lattice, by enumerating the box  $Z_1 \cap [0, m]$  with the balance constraint and evaluating  $N_{\text{seq}}$  at every point; and by an enumeration of closed walks  $T_1$  in the content multigraph (rooted at the smallest content  $a_0$ , weighted  $1/u(a_0)$ , usage bounded by  $m$ ) that never sees the lattice, followed for  $p = 2$  by an explicit enumeration of the partner walks  $T_2$  and a count of the unordered pairs  $\{T_1, T_2\}$  as sets.

**The six checks.** (P) the pooled compacted graph is balanced and gains exactly two branch vertices per site away from repeats; (L)  $b$  unlinked biallelic sites give  $2^b$  equal-length decompositions and  $2^b N_{\text{seq}}(S)^2$  labelled phasings, and two sites within  $k$  are one site; (W) for  $p = 2$  the walk enumeration reproduces  $N_{\text{seq}}$  on every spectrum and  $\sum_{T_1} N_{\text{seq}}(m - \text{spec } T_1)$  equals the lattice sum; (U) the unordered phasings enumerated as sets of explicit walk pairs equal the multiset formula; (T) for  $p = 3$  the decompositions number  $\prod_s \binom{3}{a_s}$ ,  $3!$  at the triallelic site, and the unlabelled count is the multiset count; (C) free lengths admit copy-number decompositions through the repeat, and the equal-length unlabelled count of  $S + S$  is  $\binom{N_{\text{seq}}(S)+1}{2} = 3$ . All six pass, in under a second.

| configuration                    | box | conn. | decomp. | unequal | periodic | #Lab |
|----------------------------------|-----|-------|---------|---------|----------|------|
| diploid, 1 site, all lengths     | 26  | 25    | 24      | 22      | 4        | 66   |
| homozygous $S + S$ , all lengths | 19  | 18    | 17      | 16      | 4        | 33   |

Table 13.3: Without the equal-length constraint. “unequal” is the number of ordered decompositions whose parts have different lengths; “periodic” the number with a periodic part, each counted with weight  $1/2$  per such part. Length profiles for  $S + S$ : one  $(N, N)$ , and  $(2496, 8276)$ ,  $(2890, 7882)$ ,  $(2638, 8134)$ ,  $(2748, 8024)$ ,  $(4992, 5780)$ ,  $(5276, 5496)$  twice each. The explicit pair enumeration for the diploid row finds 34 unordered pairs: 32 on the ten decompositions without a periodic part, matching the multiset formula, and two pairs of dimer circles. Check (C).

## 13.7 Scope, provenance, and what is open

**Scope.** Theorem 13.5, Proposition 13.7 and Proposition 13.12 hold for arbitrary circular haplotypes. Theorem 13.8, Corollary 13.9 and Theorem 13.10 assume a repeat-free reference and isolated sites; Remark 13.11 is a lower bound for references with repeats, met with equality in all data. The model is the pooled  $k$ -mer spectrum alone: no reads, no read pairs, no coverage, no errors. Reads that span sites constrain the decompositions further and are the whole content of the haplotype assembly problem [54, 7, 4]; what is counted here is the ambiguity that remains before any read is used, which is the analogue for phasing of the reconstruction count of [51] for assembly.

**Provenance.** The BEST theorem is [10, 78], its per-sequence normalisation and the notation  $N_{\text{seq}}$  are Theorem 7.6, and the single-genome decomposition of  $\phi\text{X174}$  at  $k = 12$  is Proposition 2.5. Haplotype assembly as a combinatorial problem on sites is [54]; the cycle-basis formulation of [4] is the closest in spirit to the circulation lattice here, though it lives on a graph of sites and reads rather than on the de Bruijn graph. Polyploid phasing methods are surveyed in [32]. We are not aware of prior work counting phasings through the de Bruijn graph, or of the homogeneous-space description of the local fibre; a targeted search found none.

**Open.** *First*, the count for a chain of sites each within  $k$  of the next but not all in one window (partially merged bubbles). *Second*, the equality in Remark 13.11: characterise the references for which  $p \text{ spec}(R)$  has no balanced connected sub-vector of length  $N$  beyond the allele selections. *Third*, the copy-number decompositions at genome scale, where the lattice has the rank of  $Z_1$  of the pooled skeleton and the sum is not enumerable; a transfer-matrix evaluation along the genome is the natural tool, as Chapter 8 shows for the double-stranded sum. *Fourth*, the effect of reads: a read spanning sites  $s, s'$  is a linear constraint on the decomposition, and the phasings compatible with a read set are a sub-lattice section; whether the standard phasing objectives (minimum error correction, maximum fragments cut) have an expression in these terms is unexplored.

## Chapter 14

# The pan-genome sheaf sees nothing: the recombination group of a pan-genome graph, and its torsion

**Summary.** A pan-genome graph is the pooled de Bruijn multigraph  $G$  of strains  $H_1, \dots, H_p$  together with the strain walks, so the decomposition of the pooled flow into strains, which Chapter 13 had to count, is here given. The natural cellular sheaf on  $G$  — stalk the free abelian group on the strains through a vertex, restriction maps the projections — is the direct sum of the constant sheaves on the strain subgraphs  $G_i$ : its cohomology is  $\mathbb{Z}^p$  and  $\bigoplus_i H^1(G_i)$ , and the integer cokernel of its sheaf Laplacian is  $\bigoplus_i (\mathbb{Z} \oplus \mathcal{K}(|G_i|))$ . It contains no pan-genomic information. What does is the *recombination group*  $\mathcal{R} = Z_1(G) / \sum_i Z_1(G_i)$ , the circulations of the pooled graph modulo those that live within single strains: the cycles that exist only by switching strain. The Mayer–Vietoris spectral sequence of the cover  $\{G_i\}$  identifies  $\mathcal{R}$  with the first homology of the nerve of intersection components,  $C_q = \bigoplus_{|\sigma|=q+1} H_0(G_\sigma)$ . For two strains  $\mathcal{R}$  is free of rank  $c(G_1 \cap G_2) - 1$  — the number of shared blocks minus one, which on a repeat-free reference with  $b$  bubbles is  $b - 1$  and on  $\phi$ X174, whose two-copy repeat glues shared blocks together, is smaller. For three or more strains  $\mathcal{R}$  can have torsion: six synthetic strains whose intersection pattern is the six-vertex triangulation of the projective plane have  $\mathcal{R} = \mathbb{Z}/2$ , while the octahedron gives 0 and the seven-vertex torus  $\mathbb{Z}^2$ . A torsion class is a mosaic cycle twice of which, but not itself, is a sum of within-strain cycles; no single strain and no pair can see it. Six  $\phi$ X174 isolates from GenBank, strand- and origin-normalised, give  $\mathcal{R} = \mathbb{Z}^{38}$  with all 63 intersections non-empty and no torsion. Every group is computed twice, by a Smith form of the inclusion of the strain cycle lattices and from the nerve, and the two agree in all 28 pools. On complete *E. coli* chromosomes — where the cycle lattice is far out of reach but the nerve is not —  $\mathcal{R}$  is free, and freely so at every prime at once: the Čech boundary  $d_2$  reduces to the identity under  $\pm 1$  pivots, which certifies that all its elementary divisors are 1. Torsion needs a tight triangulated surface, and a real pan-genome nerve, whose strains share their core in tens of thousands of separate blocks, is the opposite shape. The certificate holds on every pool from two to eight chromosomes, up to  $\mathcal{R} = \mathbb{Z}^{117\,848}$  on a nerve with 1.7 million two-cells, and does so even where  $d_2$  has a large kernel; the rank of  $\mathcal{R}$  saturates as strains are added, measuring the diversity of the set rather than its size.

### 14.1 What was asked, and what is true

The plan for this book listed a chapter on “cellular sheaves on pan-genome graphs” and ranked it last, with the warning that the sheaf might turn out to be a relabelling of what Chapter 13

already had. This chapter carries out the construction and reports that the warning was right in one direction and wrong in another. The cellular sheaf that a pan-genome graph carries naturally is trivial in a precise sense (Proposition 14.2); but the question the sheaf was meant to ask — what does the pooled graph know that no single strain knows — has a clean answer that is not a relabelling of anything, and it comes with an integer invariant that can be torsion.

A pan-genome graph is a sequence graph together with the walks of the individual genomes through it [35, 28]. In the de Bruijn setting of this book, that is the pooled multigraph  $G = G_k(H_1 + \dots + H_p)$  of Chapter 13 with the decomposition  $m = \sum_i \text{spec}(H_i)$  known. Chapter 13 counted the decompositions (Theorem 13.5); here there is one, and the object of interest is the relation between the strain subgraphs  $G_i$  and their union. The cellular sheaf  $\mathcal{F}$  that records, at each vertex and arc, which strains pass through it is the sheaf-theoretic encoding of that relation, and by the spectral theory of cellular sheaves [37] one would compute its cohomology and its sheaf Laplacian. Proposition 14.2 says both decompose strain by strain:  $\mathcal{F}$  is the direct sum of the constant sheaves on the  $G_i$ , and everything one computes from it is a list of per-strain quantities. The sheaf sees no interaction because the projections that define it never mix strains.

The interaction lives one step further along. The circulation lattice  $Z_1(G)$  of the pooled graph contains the circulation lattices  $Z_1(G_i)$  of the strains, and the quotient

$$\mathcal{R}(H_1, \dots, H_p; k) = Z_1(G) / \sum_{i=1}^p Z_1(G_i)$$

is the group of cycles of the pan-genome that cannot be assembled from cycles of individual genomes — the cycles a walk can only close by switching from one strain to another at a shared  $k$ -mer, which is to say the recombination cycles. Section 14.3 computes it. The Mayer–Vietoris spectral sequence of the cover of  $G$  by the  $G_i$  identifies  $\mathcal{R}$  with the first homology of a small chain complex built from the connected components of the intersections  $G_{i_0} \cap \dots \cap G_{i_q}$  (Theorem 14.4). Two consequences follow. For two strains,  $\mathcal{R}$  is free of rank one less than the number of shared blocks (Corollary 14.5). For three or more,  $\mathcal{R}$  can have torsion, and Proposition 14.6 exhibits six strains with  $\mathcal{R} = \mathbb{Z}/2$ . Section 14.4 verifies all of it, and computes  $\mathcal{R}$  for six deposited  $\phi$ X174 genomes.

Throughout, strains are circular sequences over  $\{A, C, G, T\}$ ,  $k$  is fixed, and the pooled graph is unitig-compacted with the branch criterion of Definition 13.1 (a  $k$ -mer is a vertex if it has two distinct successors or two distinct predecessors in the pool); a *content* is a unitig string,  $A$  the set of contents, and  $Z_1(G) \subset \mathbb{Z}^A$  the circulation lattice. All computations are exact; the script `pangenome_sheaf.py` reproduces every number below and prints a five-line pass/fail battery.

## 14.2 The pan-genome sheaf

**Definition 14.1.** Let  $G_i \subseteq G$  be the subgraph of contents and vertices used by strain  $i$ , and for a vertex or content  $x$  let  $S(x) = \{i : x \in G_i\}$ . The *pan-genome sheaf*  $\mathcal{F}$  has stalks  $\mathcal{F}(x) = \mathbb{Z}^{S(x)}$  and, for a content  $a$  with endpoint  $v$ , the restriction map  $\mathcal{F}(v) \rightarrow \mathcal{F}(a)$  the coordinate projection  $\mathbb{Z}^{S(v)} \rightarrow \mathbb{Z}^{S(a)}$  (note  $S(a) \subseteq S(v)$ ). Its coboundary  $\delta: \bigoplus_v \mathcal{F}(v) \rightarrow \bigoplus_a \mathcal{F}(a)$  is  $(\delta x)_a = x_{\text{head}(a)}|_{S(a)} - x_{\text{tail}(a)}|_{S(a)}$ , and its sheaf Laplacian is  $L_{\mathcal{F}} = \delta^T \delta$ .

**Proposition 14.2.**  $\mathcal{F} \cong \bigoplus_{i=1}^p \mathbb{Z}_{G_i}$ , the direct sum of the constant sheaves  $\mathbb{Z}$  supported on the strain subgraphs. Consequently  $H^0(G; \mathcal{F}) \cong \mathbb{Z}^p$ ,  $H^1(G; \mathcal{F}) \cong \bigoplus_i H^1(G_i; \mathbb{Z})$ ,  $L_{\mathcal{F}}$  is block diagonal

with blocks the graph Laplacians of the undirected shadows  $|G_i|$ , and

$$\text{coker } L_{\mathcal{F}} \cong \bigoplus_{i=1}^p (\mathbb{Z} \oplus \mathcal{K}(|G_i|)),$$

$\mathcal{K}(|G_i|)$  the critical group of the undirected multigraph  $|G_i|$ .

*Proof.* The coordinate  $i$  of every stalk is present exactly on the cells of  $G_i$ , and the restriction maps are the identity on that coordinate wherever both cells carry it and zero otherwise; that is the constant sheaf  $\mathbb{Z}$  on the subcomplex  $G_i$ , extended by zero. Direct sums of sheaves have direct sums of cochain complexes, so cohomology and Laplacian decompose; the cochain complex of  $\mathbb{Z}_{G_i}$  is the cellular cochain complex of the graph  $G_i$ , whose  $H^0$  is  $\mathbb{Z}$  for connected  $G_i$  (a strain walk is connected) and whose Laplacian is the graph Laplacian of  $|G_i|$ , with cokernel  $\mathbb{Z} \oplus \mathcal{K}(|G_i|)$ .  $\square$

Check (F) computes  $\text{coker } L_{\mathcal{F}}$  for a diploid with three substitutions and for three strains, obtaining  $\mathbb{Z}/38 \oplus \mathbb{Z}/38 \oplus \mathbb{Z}^2$  and  $(\mathbb{Z}/84)^3 \oplus \mathbb{Z}^3$  respectively, and confirms the block decomposition by a Smith form of the assembled Laplacian. The numbers 38 and 84 are the critical groups of the strains' own shadows; they would be the same if the strains were sequenced separately. There is nothing pan-genomic in the sheaf, and no choice of stalk dimensions rescues this as long as the restriction maps are projections: a sheaf whose restriction maps never mix coordinates is a direct sum of coordinate sheaves.

### 14.3 The recombination group

**Definition 14.3.**  $\mathcal{R} = Z_1(G) / \sum_i Z_1(G_i)$ , where  $Z_1(G_i) = Z_1(G) \cap \mathbb{Z}^{A_i}$  is the circulation lattice of the subgraph  $G_i$ . For a subset  $\sigma \subseteq \{1, \dots, p\}$  write  $G_\sigma = \bigcap_{i \in \sigma} G_i$  and let  $C_q = \bigoplus_{|\sigma|=q+1} H_0(G_\sigma)$ , the free abelian group on the connected components of the  $(q+1)$ -fold intersections, with the Čech differential  $\partial(\sigma, c) = \sum_j (-1)^j (\sigma \setminus \sigma_j, c')$ ,  $c'$  the component of  $G_{\sigma \setminus \sigma_j}$  containing  $c$ . We call  $(C_\bullet, \partial)$  the *nerve of components*  $\mathcal{N}$ .

**Theorem 14.4.**  $\mathcal{R} \cong H_1(\mathcal{N})$ .

*Proof.* The subgraphs  $G_i$  are subcomplexes covering the one-dimensional CW complex  $G$ , so the Mayer–Vietoris spectral sequence of the cover [16, VII.4] applies:  $E_{q,r}^1 = \bigoplus_{|\sigma|=q+1} H_r(G_\sigma) \Rightarrow H_{q+r}(G)$ , with  $d^1$  the Čech differential. Since intersections of graphs are graphs,  $H_r(G_\sigma) = 0$  for  $r \geq 2$ , and the  $E^2$  page has two rows,  $E_{q,0}^2 = H_q(\mathcal{N})$  and  $E_{q,1}^2 = H_q(\bigoplus_\sigma H_1(G_\sigma))$ . The only differentials that can touch total degree one are  $d^2: E_{2,0}^2 \rightarrow E_{0,1}^2$  and the vanishing  $d^2: E_{1,0}^2 \rightarrow E_{-1,1}^2$ , so  $E_{1,0}^\infty = E_{1,0}^2 = H_1(\mathcal{N})$  and  $E_{0,1}^\infty = E_{0,1}^2 / d^2(E_{2,0}^2)$ . The filtration of  $H_1(G)$  has  $F_0 = E_{0,1}^\infty$  equal to the image of  $\bigoplus_i H_1(G_i) \rightarrow H_1(G)$  (the edge homomorphism) and  $H_1(G)/F_0 \cong E_{1,0}^\infty$ . For a graph  $H_1 = Z_1$ , and the image of  $\bigoplus_i Z_1(G_i)$  in  $Z_1(G)$  is  $\sum_i Z_1(G_i)$ .  $\square$

**Corollary 14.5** (Two strains). *For  $p = 2$  with  $G_1, G_2$  connected and  $G_1 \cap G_2 \neq \emptyset$ ,  $\mathcal{R}$  is free of rank  $c(G_1 \cap G_2) - 1$ , the number of connected components of the shared subgraph minus one.*

*Proof.*  $\mathcal{N}$  has  $C_0 = \mathbb{Z}^2$ ,  $C_1 = \mathbb{Z}^{c(G_{12})}$  and  $C_2 = 0$ ;  $\partial$  sends every component to  $e_2 - e_1$ , so  $H_1(\mathcal{N}) = \ker \partial$  is the sublattice of coefficient sum zero.  $\square$

On a repeat-free reference with  $b$  substitution sites in distinct  $(k+1)$ -windows, the shared subgraph is the  $b$  stretches between the bubbles and  $\mathcal{R} \cong \mathbb{Z}^{b-1}$ . When the reference has a repeat

| pool ( $k = 12$ )                            | $ V $ | $ A $ | $\text{rank } Z_1(G)$ | $\text{rank } \sum Z_1(G_i)$ | $c(G_1 \cap G_2)$ | $\mathcal{R}$  |
|----------------------------------------------|-------|-------|-----------------------|------------------------------|-------------------|----------------|
| $\phi\text{X174}$ , 1 substitution           | 4     | 7     | 4                     | 4                            | 1                 | 0              |
| $\phi\text{X174}$ , 2 substitutions          | 6     | 10    | 5                     | 4                            | 2                 | $\mathbb{Z}$   |
| $\phi\text{X174}$ , 3 substitutions          | 8     | 13    | 6                     | 5                            | 2                 | $\mathbb{Z}$   |
| $\phi\text{X174}$ , 5 substitutions          | 12    | 19    | 8                     | 6                            | 3                 | $\mathbb{Z}^2$ |
| $\phi\text{X174}$ , sites 600, 606 linked +1 | 6     | 10    | 5                     | 4                            | 2                 | $\mathbb{Z}$   |
| repeat-free reference, 2 sites               | 4     | 6     | 3                     | 2                            | 2                 | $\mathbb{Z}$   |
| repeat-free reference, 3 sites               | 6     | 9     | 4                     | 2                            | 3                 | $\mathbb{Z}^2$ |
| repeat-free reference, 4 sites               | 8     | 12    | 5                     | 2                            | 4                 | $\mathbb{Z}^3$ |

Table 14.1: Two strains.  $\mathcal{R}$  is free of rank  $c(G_1 \cap G_2) - 1$  in every row; on the repeat-free reference (a random 3000-mer) that is the number of bubbles minus one, on  $\phi\text{X174}$  the repeat glues shared stretches and lowers it. Check (B).

the shared stretches that contain copies of the same repeated  $k$ -mer are joined through it and  $c(G_1 \cap G_2)$  drops: for  $\phi\text{X174}$  at  $k = 12$ , whose two repeated 12-mers  $a, b$  each occur twice, five substitutions leave three shared components and  $\mathcal{R} = \mathbb{Z}^2$ , three substitutions leave two and  $\mathcal{R} = \mathbb{Z}$  (Table 14.1). A repeat inside a strain thus reduces the number of independent recombination cycles between strains: the cycle through the repeat is already a within-strain cycle.

**Proposition 14.6** (Torsion). *Let  $T$  be a triangulated closed surface with vertex set  $\{1, \dots, p\}$ , and construct strains as follows: a random segment  $F_\tau$  for each face  $\tau$  and  $E_\varepsilon$  for each edge  $\varepsilon$ ; strain  $i$  reads, for each edge  $\varepsilon = ij$  at  $i$  with faces  $\tau, \tau'$ , the word  $F_\tau E_\varepsilon F_{\tau'} S F_{\tau'} E_\varepsilon F_\tau S'$  with unique spacers  $S, S'$ . Then for  $k$  larger than any accidental shared  $k$ -mer the nerve of components is the simplicial chain complex of  $T$ , and  $\mathcal{R} \cong H_1(T; \mathbb{Z})$ . In particular the six-vertex triangulation of the projective plane gives  $\mathcal{R} = \mathbb{Z}/2$ , the octahedron gives 0, and the seven-vertex torus gives  $\mathbb{Z}^2$ .*

*Proof.*  $G_i \cap G_j$  for an edge  $ij$  consists of  $E_{ij}$ , the two face segments at  $ij$ , and the four junction arcs between them, which both strains contain because each reads the two junction directions; it is connected. For a face  $ijk$ ,  $G_i \cap G_j \cap G_k = F_{ijk}$ , a segment; for a non-face triple and for all larger subsets the intersection is empty, since every segment is shared by the vertices of exactly one simplex and every junction arc by exactly two strains. So  $C_\bullet$  is the chain complex of  $T$  and Theorem 14.4 applies.  $H_1(\mathbb{RP}^2; \mathbb{Z}) = \mathbb{Z}/2$ ,  $H_1(S^2) = 0$ ,  $H_1(T^2) = \mathbb{Z}^2$ .  $\square$

The torsion class is concrete. In the projective-plane pool there is a circulation  $z$  of the pooled graph such that  $2z$  is a sum of cycles each lying within a single strain but  $z$  itself is not;  $z$  is a mosaic that closes up only by using every strain, and its double decomposes. No pair of strains detects it, because for pairs  $\mathcal{R}$  is free (Corollary 14.5); no strain detects it, because within a strain there is nothing to detect. It is an invariant of the pan-genome as a whole.

## 14.4 Verification and data

Every group is computed twice: as the cokernel, via Smith normal form, of the matrix whose columns are fundamental cycles of the subgraphs  $G_i$  expressed in  $\mathbb{Z}^A$  (torsion of  $\mathbb{Z}^A / \sum \mathbb{Z} \cdot Z_1(G_i)$  equals that of  $\mathcal{R}$  because  $Z_1(G)$  is saturated in  $\mathbb{Z}^A$ ), and as  $H_1(\mathcal{N})$  from the component structure of the intersections. The two agree in every one of the 28 pools, which is check (T) and part of the others.

| surface pool ( $k = 16$ )            | strains | $ V $ | $ A $ | $\text{rank } Z_1(G)$ | $\text{rank } \sum Z_1(G_i)$ | $\mathcal{R} = H_1(\mathcal{N})$ |
|--------------------------------------|---------|-------|-------|-----------------------|------------------------------|----------------------------------|
| octahedron ( $S^2$ )                 | 6       | 114   | 190   | 77                    | 77                           | 0                                |
| seven-vertex torus ( $T^2$ )         | 7       | 186   | 319   | 134                   | 132                          | $\mathbb{Z}^2$                   |
| hemi-icosahedron ( $\mathbb{RP}^2$ ) | 6       | 138   | 233   | 96                    | 96                           | $\mathbb{Z}/2$                   |

Table 14.2: Proposition 14.6. In each pool the nerve of components is verified to be exactly the designed triangulation (one component per simplex, none elsewhere). At  $k = 12$  the  $\approx 10^4$  random 12-mers of a pool produce accidental shared  $k$ -mers and spurious components;  $k = 16$  removes them. Check (S).

|   | accession   | substitutions after normalisation |    |    |    |    |    |  |
|---|-------------|-----------------------------------|----|----|----|----|----|--|
| 1 | LC786485.1  | 0                                 | 5  | 27 | 26 | 24 | 33 |  |
| 2 | NC_001422.1 | 5                                 | 0  | 26 | 25 | 23 | 32 |  |
| 3 | MH378443.1  | 27                                | 26 | 0  | 19 | 24 | 48 |  |
| 4 | MH378442.1  | 26                                | 25 | 19 | 0  | 22 | 47 |  |
| 5 | MH378441.1  | 24                                | 23 | 24 | 22 | 0  | 45 |  |
| 6 | JQ764988.2  | 33                                | 32 | 48 | 47 | 45 | 0  |  |

Table 14.3: Six  $\phi$ X174 genomes of length 5386 from GenBank, brought to the strand and origin of NC\_001422.1 (one was deposited rotated). Pooled at  $k = 12$ :  $|V| = 104$ ,  $|A| = 158$ ,  $\text{rank } Z_1(G) = 55$ ,  $\text{rank } \sum Z_1(G_i) = 17$ ,  $\mathcal{R} = \mathbb{Z}^{38}$ , no torsion; all 63 intersections are non-empty, with 4 to 27 components for pairs and 41 for the full intersection. Check (D).

**The checks.** In `pangenome_sheaf.py`: (F) the sheaf Laplacian’s cokernel is the direct sum of the strains’  $\mathbb{Z} \oplus \mathcal{K}(|G_i|)$ ; (B) two strains, eight pools,  $\mathcal{R}$  free of rank  $c(G_1 \cap G_2) - 1$ ; (S) the three surface pools, with the designed nerve; (T) twelve random three-strain pools from a shared segment library, Smith form and nerve agreeing, no torsion; (D) the six isolates. All pass in seven seconds. In `real_pangenome.py`, for Section 14.5: (A) the hashed branch criterion against brute force on random 20 kbp fragment pools; (V) the  $\mathbb{F}_p$  detector against the three surface pools, where it returns the  $\mathbb{Z}/2$  of the projective plane; (E) the complete chromosomes of Table 14.4.

## 14.5 A real pan-genome

Theorem 14.4 replaces a lattice quotient by the homology of the nerve, and the nerve needs only the *connected components* of the shared subgraphs. Components are cheap: the recombination group is therefore computable on complete bacterial chromosomes, where  $\mathcal{R}$  itself, as a quotient of a cycle lattice of rank  $10^4$  or more, is not. This section carries that out.

**The pool.** Complete *Escherichia coli* chromosomes were taken from GenBank and brought to the strand of K-12 MG1655 by  $k$ -mer overlap; a circular genome’s  $k$ -mer set is rotation invariant, so only the strand has to be fixed. The pooled graph is compacted by the criterion of Definition 13.1: a  $k$ -mer is a vertex when it has two distinct successors or two distinct predecessors *in the pool*. Branch vertices are found by hashing on the pairs ( $k$ -mer, next base) and ( $k$ -mer, previous base) and then re-tested on the exact strings, so a collision can only propose a candidate that the string pass discards; check (A) audits the whole criterion against brute force on random 20 kbp fragment pools.

| strains ( $k = 31$ ) | branch vtx | contents | $C_1$   | $C_2$     | rank $d_2$ | $\mathcal{R}$           | certified |
|----------------------|------------|----------|---------|-----------|------------|-------------------------|-----------|
| MG1655, DH10B        | 1 509      | 2 343    | 86      | 0         | 0          | $\mathbb{Z}^{85}$       | yes       |
| + O157:H7 Sakai      | 63 245     | 95 168   | 58 454  | 28 820    | 28 820     | $\mathbb{Z}^{29\,632}$  | yes       |
| + CFT073             | 137 375    | 207 028  | 166 011 | 134 401   | 99 561     | $\mathbb{Z}^{66\,447}$  | yes       |
| + IAI39              | 180 256    | 271 937  | 260 369 | 277 947   | 172 507    | $\mathbb{Z}^{87\,858}$  | yes       |
| + S88                | 214 033    | 323 073  | 409 007 | 588 230   | 304 507    | $\mathbb{Z}^{104\,495}$ | yes       |
| + 536                | 237 525    | 358 660  | 575 831 | 1 053 725 | 459 769    | $\mathbb{Z}^{116\,056}$ | yes       |
| + UTI89              | 241 592    | 364 819  | 744 598 | 1 683 865 | 626 743    | $\mathbb{Z}^{117\,848}$ | yes       |

Table 14.4: The recombination group of eight complete *E. coli* chromosomes, about 40 Mbp, each row adding one strain to the one above. In every row the unit-pivot reduction of  $d_2$  completes, so by Proposition 14.7  $\mathcal{R}$  is free at every prime, and the rank, being  $\dim C_1 - \text{rank } d_1 - \text{rank } d_2$  with both ranks computed by unimodular integer elimination, is exact. The two K-12 strains share 86 blocks and give  $\text{rank } \mathcal{R} = 86 - 1$ , which is Corollary 14.5 on real data. From the fourth strain on  $d_2$  is far from injective — at eight strains 626 743 of 1 683 865 — which costs the certificate nothing: a kernel makes columns vanish during the reduction, it does not stop the surviving pivots being units. Check (E).

**A certificate, not a sample.** Torsion is a statement about every prime, and one cannot test them all. Two devices avoid having to.

**Proposition 14.7** (freeness certificate). *Suppose the Čech boundary  $d_2: C_2 \rightarrow C_1$  can be reduced by integer elimination in which every pivot is  $\pm 1$ . Then all its elementary divisors are 1,  $\text{im } d_2$  is a direct summand of  $C_1$ , and  $\mathcal{R} = \ker d_1 / \text{im } d_2$  is free.*

*Proof.* A  $\pm 1$  pivot is a unimodular row and column operation, so a complete reduction exhibits  $d_2$  as  $\begin{pmatrix} I \\ 0 \end{pmatrix}$  up to unimodular change of basis; its elementary divisors are all 1 and  $C_1 / \text{im } d_2$  is free. A finitely generated submodule of a finitely generated free  $\mathbb{Z}$ -module is free, and  $\mathcal{R} = \ker d_1 / \text{im } d_2 \subseteq C_1 / \text{im } d_2$ .  $\square$

*Remark 14.8* (a torsion detector without a Smith form). When the reduction does not complete, the prime-by-prime test is still cheap.  $H_0(\mathcal{N})$  is free, so the universal coefficient theorem gives  $\dim_{\mathbb{F}_p} H_1(\mathcal{N}; \mathbb{F}_p) = \text{rank } \mathcal{R} + t_p(\mathcal{R})$  with  $t_p$  the  $p$ -torsion rank: any excess of the  $\mathbb{F}_p$  dimension over the rational rank is  $p$ -torsion, and both are ranks of sparse matrices. Check (V) runs this detector on the three surface pools of Table 14.2, where the answer is known exactly, and it does return the  $\mathbb{Z}/2$  of the projective plane — so a null result on real data is a result and not a blind spot.

**The answer.**  $\mathcal{R}$  is free, with a certificate, on every pool from two to eight complete chromosomes (Table 14.4). This settles the first open question of an earlier version of this chapter for those strains, and the reason is visible in the same table. A torsion class needs a tight two-dimensional cycle: the projective-plane pool of Proposition 14.6 has exactly one component per simplex, so its nerve is a triangulated surface. A real pan-genome nerve is the opposite shape. Each pair of divergent strains shares its core genome in tens of thousands of blocks — 86 for two K-12 strains, 28 817 and 29 551 for K-12 against O157:H7 — and each triple intersection is more fragmented still, so the nerve is wide and sparse, every elementary divisor of  $d_2$  is 1, and  $H_1$  is a large free group. The synthetic construction of Section 14.3 was necessary precisely because real data does not produce that shape, and the honest reading of Proposition 14.6 is that torsion is realisable in principle rather than observed in *E. coli*.

**The rank saturates.** One thing in Table 14.4 is not about torsion. The successive increments of rank  $\mathcal{R}$  are 29 547, 36 815, 21 411, 16 637, 11 561, 1 792: after the third strain each new chromosome contributes less than the one before, and the eighth contributes an order of magnitude less than the fourth, on a nerve whose two-cells meanwhile grew to 1.7 million. The pooled graph agrees — the last strain adds 4 067 branch vertices where the fifth added 42 881. The ordering is not neutral: the last four additions (CFT073, S88, 536, UTI89, with IAI39 between them) are extraintestinal pathogenic strains, so each arrives with most of its mosaic structure already represented, and the two K-12 strains at the top are a pair, not a sample. Read as an observation on these eight genomes rather than a law, it says that rank  $\mathcal{R}$  is a measure of the *diversity* of a strain set and not of its size, which is what a count of independent strain-switching cycles ought to be.

*Remark 14.9* (a failed certificate is a statement about the elimination). Proposition 14.7 is one-sided: it certifies freeness when the reduction completes and says nothing when it does not, because  $d_2$  may have all its elementary divisors equal to one with no  $\pm 1$  pivot surviving to the end of one particular elimination. That is not a theoretical caveat. An earlier version of this computation chose each pivot by scanning the 64 shortest live columns, and stalled at four strains with 70 374 columns left and at five with 190 814, from which we could conclude only — via Remark 14.8 — that there is no 2- and no 3-torsion. Taking instead the globally shortest column, and within it the row meeting fewest columns, produces less fill-in, the entries never leave  $\{0, \pm 1\}$ , and every column is eliminated: those two rows of Table 14.4 became certificates, and the three below them became feasible at all. The ranks 99 561 and 172 507 are the same under both orderings, which is what says the two runs compute the same map. Two consequences for the arithmetic. Because the unit-pivot phase is a unimodular column reduction over  $\mathbb{Z}$  it is valid modulo every prime with its pivots still units, so when it *does* stall,  $\text{rank}_{\mathbb{F}_p} d_2$  is the number of pivots plus the rank of the residual block alone; and the ordering is worth more than the arithmetic, the five-strain reduction taking 4 seconds where the scan took 14 080.

**What the single-strand convention costs here.** Of the eight strains, IAI39 matched the reference on 2331 of its sampled 31-mers forward against 1234 reverse, where every other strain ran between 6835 against 234 and 2970 against 109. That is a large inversion, and to the single-strand pooled graph an inverted segment is strain-specific sequence: real homology is lost and the nerve is cut into more components than it should be. The rank of  $\mathcal{R}$  reported here is therefore an overcount for divergent strains, in a way a bidirected pooled graph would repair. The freeness is not affected, being a statement about the shape of the nerve.

## 14.6 Scope, provenance, and what is open

**Scope.** Proposition 14.2, Theorem 14.4, Corollary 14.5 and Proposition 14.7 hold for arbitrary strains; Proposition 14.6 is a construction; Section 14.5 is a computation on named genomes and claims nothing beyond them. The pooled graph is single-stranded, as in Chapter 13; the reverse-complement version would pool double covers (Chapter 5) and is not attempted. “Recombination” is used in the graph sense — a cycle that switches strain — and no claim is made about the biological mechanism that produced any particular mosaic.

**Provenance.** Cellular sheaves on graphs and their Laplacians are [37]; pan-genome and variation graphs are [35, 28], and the bubble/snarl decomposition of such graphs, which is the combinatorial shadow of the components of pairwise intersections, is [66]. The Mayer–Vietoris spectral sequence

for a cover by subcomplexes is standard, e.g. [16, VII.4]; Theorem 14.4 is its two-row case read off for graphs. We are not aware of prior work defining the quotient  $Z_1(G)/\sum Z_1(G_i)$  for a pan-genome graph or observing that it can have torsion; a targeted search found none.

**Open.** *First*, torsion in a real pan-genome. Section 14.5 answers this for the *E. coli* chromosomes pooled there —  $\mathcal{R}$  is free, with a certificate — and explains why: the nerve of a real pan-genome is wide and sparse, nothing like the triangulated surface a torsion class needs. What is open is whether any real pan-genome has one. A plausible place to look is a set of strains whose shared blocks are *few and large*, so that the nerve is small enough to close up: closely related strains with a handful of recombination breakpoints, rather than the divergent ones used here. *Second*, the relation of  $H_1(\mathcal{N})$  to the snarl tree of [66] (used for elimination in Chapter 16), which organises the pairwise components hierarchically. *Third*, a bidirected pooled graph, so that an inversion is homology rather than strain-specific sequence; by the last paragraph of Section 14.5 this changes the rank of  $\mathcal{R}$  for divergent strains, and the anti-automorphism theorem (Theorem 6.9) says the repair cannot be a quotient by the strand involution. *Fourth*, a sheaf that does see interaction: the restriction maps would have to mix strains, and the natural candidate is the sheaf of local decompositions of the pooled flow, whose stalk at a bubble is the homogeneous space of Theorem 13.8; it is not a sheaf of abelian groups.

## Part IV

# Algorithmic implementations on genomes



## Chapter 15

# The bundle-chain decomposition of a bacterial genome: *Escherichia coli* K-12 at five $k$ -mer lengths

**Summary.** Part I computed the sandpile group of a de Bruijn graph on a 5386 bp phage. This chapter computes it on a 4641652 bp bacterium, *Escherichia coli* str. K-12 substr. MG1655 (NC\_000913.3), at  $k = 31, 51, 75, 100, 150$ . The computation is possible because of one observation that costs a line: in the convention used throughout this book every vertex of  $G_k$  satisfies  $d^+(v) = d^-(v) = \text{mult}(v)$ , so a vertex survives unitig contraction *if and only if its  $k$ -mer is repeated*. The 4.6-million-vertex graph is therefore never built; the compacted graph is read off the repeated  $k$ -mers directly, and the whole pipeline runs in about 13 seconds. Three findings. *One:* the irreducible part of the ambiguity of a 4.6 Mbp genome is a digraph on 115 vertices at  $k = 150$  — the chain  $4641652 \rightarrow 23933 \rightarrow 115$  — and its group is computed exactly. *Two:* the splitting is extremely lopsided; at  $k = 150$  the length-driven summand has order  $10^{10439}$  and the arrangement summand  $10^{49.7}$ , so 99.5% of the exponent is removable by longer  $k$ -mers and 0.5% is removable by no  $k$  at all, while the irreducible part still grows by  $10^{344}$  as  $k$  falls to 31. *Three:* the seven ribosomal operons do *not* form a multiplicity-seven repeat family, at any  $k$ . Five lie on one strand and two on the other, and the single-strand theory of Part I sees a repeat only between co-oriented copies; the two minus-strand operons give the longest clean spine in the genome (4030 bp at  $k = 150$ ) and the five plus-strand ones a separate, fragmented family. This is a quantitative measure of what a strand-aware theory has to supply, and the double cover of Chapter 5 supplies it. Six checks, all exact, including a rerun of the whole fast pipeline against the phage decomposition of Chapter 3.

### 15.1 Introduction

For a circular sequence  $S$ ,  $G_k(S)$  is the de Bruijn multigraph of Section 1.2; by the BEST theorem the number of reconstructions of  $S$  from its  $k$ -mer spectrum is  $|\mathcal{K}(G_k)| \prod_v (d^+(v) - 1)!$  (Proposition 1.3), and the splitting theorem (Theorem 3.6) writes

$$\mathcal{K}(G_k(S)) \cong \bigoplus_{\text{chains } c} (\mathbb{Z}/r_c)^{\ell_c-1} \oplus \mathcal{K}(\text{Sk}),$$

a *length-driven* summand that longer  $k$ -mers remove and an *arrangement* summand, the sandpile group of the repeat-adjacency skeleton, that no  $k$  removes. Part I computed everything on  $\varphi$ X174,

5386 bp. The question this chapter answers is what the decomposition looks like on a genome three orders of magnitude larger.

**What is known.** Assemblers have compacted de Bruijn graphs to unitigs for twenty years [72, 46, 9], and the repeat graph is the central object of the long-read assembler Flye [71, 31]; the repeat content of *E. coli* K-12 — seven *rrn* operons, some fifty IS elements in a dozen families, several hundred REP/BIME palindromic elements — has been catalogued since the genome was published [13]. What is new here is not the repeat catalogue but the algebra attached to it: the exact invariant factors of the arrangement group, and the exact split between the part of the ambiguity that a longer read length removes and the part it cannot.

## 15.2 Why a 4.6 Mbp genome is tractable

**Lemma 15.1** (Branch vertices are the repeated  $k$ -mers). *In  $G_k(S)$ ,  $d^+(v) = d^-(v) = \text{mult}(v)$ , the number of occurrences of the  $k$ -mer  $v$  in the circular sequence. Consequently a vertex has in- and out-degree one — i.e. is smoothed away by the unitig contraction of Lemma 1.4 — if and only if its  $k$ -mer occurs exactly once.*

*Proof.* Each position  $i$  contributes exactly one arc, from  $S[i..i+k)$  to  $S[i+1..i+k+1)$ . The arcs leaving  $v$  are indexed by the positions at which  $v$  occurs, and the arcs entering  $v$  by the positions at which  $v$  occurs shifted by one; both sets have  $\text{mult}(v)$  elements.  $\square$

This is immediate, and it is the whole algorithm. The compacted graph is obtained without constructing  $G_k$ : let  $R = (p_1 < \dots < p_m)$  be the positions at which a repeated  $k$ -mer starts. Then the compacted multidigraph has one vertex per distinct repeated  $k$ -mer and one arc  $S[p_t..p_t+k) \rightarrow S[p_{t+1}..p_{t+1}+k)$  for each  $t$  cyclically, because the walk leaving a branch vertex runs through unique  $k$ -mers until the next repeat position. For *E. coli* this replaces 4 641 652 vertices by between 23 933 and 34 272 (Table 15.1), and the decomposition of Chapter 3 then runs in one pass, since the chain-step relation is a partial injection (Lemma 3.5).

Finding the repeated  $k$ -mers is the only step that touches the whole genome. We hash each of the  $N$  positions to 64 bits, take multiplicities with a sort, and then — this is what keeps the result exact — recover the actual  $k$ -mer *strings* at the positions whose hash repeats and group by string. A hash collision can therefore only add a candidate that is discarded; it cannot manufacture a repeat. No collision occurred at any  $k$  (check (H)).

## 15.3 Results

*Remark 15.2* (The shape of the answer). Three things are worth reading off Table 15.1.

*The skeleton is small.* At  $k = 150$  the entire arrangement ambiguity of a 4.6 Mbp genome is carried by a digraph on 115 vertices:  $4\,641\,652 \rightarrow 23\,933$  by unitig contraction,  $23\,933 \rightarrow 115$  by bundle-chain contraction. The second step is the one the theory supplies, and it is the larger reduction in proportion.

*The split is lopsided.* At  $k = 150$  the length-driven summand has order  $10^{10439.18}$  and the arrangement summand  $10^{49.71}$ : 99.53% of the exponent of  $|\mathcal{K}(G_{150})|$  is removable by reading longer  $k$ -mers, and 0.47% is not removable by any  $k$ . The percentage is not the interesting part —  $10^{49.7}$  is still an enormous number of globally distinct threadings, and it is the part that only long reads, mate pairs or a reference can resolve.

| $k$ | distinct $k$ -mers | branch | arcs    | chains | skeleton   | $\log_{10}  \mathcal{K}(\text{Sk}) $ | $\log_{10}  \mathcal{K}(G_k) $ |
|-----|--------------------|--------|---------|--------|------------|--------------------------------------|--------------------------------|
| 150 | 4 594 348          | 23 933 | 71 237  | 107    | <b>115</b> | 49.71                                | 10 488.89                      |
| 100 | 4 588 709          | 26 631 | 79 574  | 166    | 175        | 71.11                                | 11 690.57                      |
| 75  | 4 584 636          | 28 579 | 85 595  | 243    | 261        | 99.80                                | 12 563.08                      |
| 51  | 4 579 164          | 31 011 | 93 499  | 342    | 366        | 143.17                               | 13 716.83                      |
| 31  | 4 570 869          | 34 272 | 105 055 | 683    | 870        | 393.75                               | 15 350.54                      |

Table 15.1: *E. coli* K-12 MG1655 under unitig contraction followed by bundle-chain contraction. “branch” is the number of repeated  $k$ -mers (Lemma 15.1) and equals the vertex count of the compacted graph. All structural entries exact;  $\log_{10} |\mathcal{K}(\text{Sk})|$  by float LU, and exact as a group at  $k = 150, 100$  (Table 15.2).

| $k$ | $\mathcal{K}(\text{Sk})$ , the arrangement group                                                                                                                                                                                                                                                                                                                                                |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 150 | $(\mathbb{Z}/2)^{17} \oplus (\mathbb{Z}/3)^9 \oplus (\mathbb{Z}/4)^{12} \oplus (\mathbb{Z}/5)^{10} \oplus (\mathbb{Z}/7)^3 \oplus (\mathbb{Z}/8)^4 \oplus (\mathbb{Z}/9)^3 \oplus (\mathbb{Z}/16)^3$<br>$\oplus \mathbb{Z}/27 \oplus (\mathbb{Z}/32)^2 \oplus \mathbb{Z}/49 \oplus \mathbb{Z}/21308303$                                                                                         |
| 100 | $(\mathbb{Z}/2)^{29} \oplus (\mathbb{Z}/3)^{16} \oplus (\mathbb{Z}/4)^{15} \oplus (\mathbb{Z}/5)^{15} \oplus (\mathbb{Z}/7)^2 \oplus (\mathbb{Z}/8)^4 \oplus (\mathbb{Z}/9)^2 \oplus (\mathbb{Z}/16)^6$<br>$\oplus \mathbb{Z}/25 \oplus \mathbb{Z}/27 \oplus (\mathbb{Z}/32)^2 \oplus \mathbb{Z}/49 \oplus \mathbb{Z}/81 \oplus \mathbb{Z}/128 \oplus \mathbb{Z}/512 \oplus \mathbb{Z}/3448649$ |

Table 15.2: The arrangement group, exactly, where the skeleton is small enough for the  $p$ -adic Smith form (check (K)). At  $k = 75, 51, 31$  the skeleton has 261, 366 and 870 vertices and only the order is given; a 100- to 400-digit determinant could not be factored in any case.

*The irreducible part grows fast as  $k$  falls.* From  $k = 150$  to  $k = 31$  the arrangement summand grows from  $10^{49.71}$  to  $10^{393.75}$ , a factor of  $10^{344}$ , while the skeleton grows from 115 to 870 vertices. Shortening the read length does not merely leave more length-driven ambiguity; it creates genuinely new arrangement ambiguity, because repeats shorter than the old  $k$  enter the graph and connect parts of the skeleton that were previously unrelated.

By the BEST theorem the number of circular sequences with exactly the  $k$ -mer spectrum of this genome is  $|\mathcal{K}(G_k)| \prod_v (d^+(v) - 1)!$ , which is  $10^{20735.14}$  at  $k = 150$  and  $10^{31748.76}$  at  $k = 31$ .

## 15.4 The ribosomal operons, and a correction

*E. coli* K-12 has seven ribosomal RNA operons, each about 5 kb, and the natural expectation — the one recorded in the specification of this chapter — is a repeat family of multiplicity seven. *No such family exists, at any  $k$ .*

The reason is structural, not numerical. The NCBI feature table places five operons on the  $+$  strand (at 223 771, 3 941 808, 4 035 531, 4 166 659, 4 208 147) and two on the  $-$  strand (at 2 726 069 and 3 423 423). The theory of Part I is single-strand: two copies are a repeat of  $G_k(S)$  only if they are co-oriented *in*  $S$ . The seven operons therefore cannot form one family. They split by strand, and the computation confirms the split exactly (check (R)):

- the **longest bundle chain in the whole genome**, at every  $k$  tested, is the multiplicity-*two* spine of the two minus-strand operons —  $s = 3881$  at  $k = 150$  (span 4030 bp),  $s = 4412$  at  $k = 51$  (span 4462 bp) — annotated 16S, 23S and 5S ribosomal RNA. Two copies annotated on the same (minus) strand are co-oriented with each other, hence a direct repeat in  $S$ ;
- the five plus-strand operons give the multiplicity-*five* family, whose longest chain is the 16S gene (span 775 bp), with copies at 223 997, 3 942 034, 4 035 757, 4 166 885 and 4 208 373 — all

five operons;

- the five are not identical to one another either, so the rRNA region additionally throws off chains of multiplicity three and four, one per subset of copies that agree.

Multiplicity seven does occur in this genome, but it is IS5A, not rRNA (five chains, 88  $k$ -mers at  $k = 31$ ), as are multiplicities eight and nine. The other long chains are all insertion sequences: IS186A ( $\times 3$ , 1345 bp), IS2D ( $\times 3$ , 1331 bp), IS3B ( $\times 2$ , 1260 bp), IS3A ( $\times 3$ , 1255 bp), IS5U ( $\times 3$ , 1202 bp). At  $k = 31$  the highest multiplicities are 22 and 23 with spans of 32 to 35 bp: the REP/BIME extragenic palindromic elements, which are shorter than 51 and so vanish completely from the graph at every larger  $k$  tested.

*Remark 15.3* (What this measures). A strand-aware theory — the open problem of Section 2.8, item (a) — should see all seven operons as one family, and the IS copies that sit in inverted orientation. The double cover of Chapter 5 does so for the operons (Corollary 5.6), and Chapter 6 shows what the bidirected quotient can and cannot add. The  $5 + 2$  split is a quantitative statement of what the single-strand restriction costs on a real genome: not a footnote, but the dominant feature of the largest repeat family present.

## 15.5 Verification

The script `ecoli_sandpile.py` runs the six checks of Table 15.3 in about 13 seconds under half a gigabyte; its output is archived as `ecoli_sandpile_output.txt`. The genome and the feature table are cached beside it and re-fetched from the NCBI E-utilities API if absent.

| key | check                                                                                      | mode          |
|-----|--------------------------------------------------------------------------------------------|---------------|
| (V) | the fast path reproduces the $\varphi$ X174 decomposition of Table 3.1, $k = 9, \dots, 12$ | exact         |
| (H) | hashing finds candidates, strings decide; no collision survives                            | exact         |
| (C) | the compacted graph is balanced with $d^+(v) = \text{mult}(v)$                             | exact         |
| (B) | maximal bundle chains in one pass; Lemma 3.5 injectivity asserted on 1541 chains           | exact         |
| (K) | the arrangement group, exactly for skeletons of $\leq 200$ vertices                        | exact/numeric |
| (R) | no rRNA chain has multiplicity 7; the longest chain is the minus-strand pair               | exact         |

Table 15.3: The verification battery, 6/6.

Check (V) deserves a word. The pipeline used here is not the one used in Chapter 3: it never builds  $G_k$ , it contracts chains in a single pass instead of iteratively, and it takes the Smith form  $p$ -adically instead of by Euclidean elimination. Run on  $\varphi$ X174 it returns, at  $k = 9, 10, 11, 12$ , the same branch counts (2, 10, 47, 146), chain counts (1, 2, 8, 28), skeleton sizes (1, 8, 37, 110), length-driven exponents and the same exact skeleton groups — including  $\mathbb{Z}/4 \oplus \mathbb{Z}/1076338579$  at  $k = 10$  and the seventeen elementary divisors at  $k = 9$ . Three independent implementations of the same theorem agreeing on a genome is the only protection available when the target genome is too large to check by any other route.

## 15.6 Scope

*Exact.* Every structural quantity — repeated  $k$ -mers, branch counts, the multiset of  $(r_c, \ell_c)$ , skeleton sizes, genomic coordinates, the length-driven summand — is exact integer work, as is  $\mathcal{K}(\text{Sk})$  at  $k = 150$  and  $k = 100$  and the location of every repeat family against the feature table.

*Numeric.*  $\log_{10} |\mathcal{K}(\text{Sk})|$  at  $k = 75, 51, 31$  is a float LU determinant (`slogdet`) of a 260- to 869-dimensional matrix; the relative error in the logarithm is at the level of machine precision times the dimension, but it is not an exact integer and is not claimed to be.

*Single-strand.* As in Part I; Chapter 5 lifts the restriction. Section 15.4 is the measurement of what that costs here.

*Open.* (a) The two exact arrangement groups each carry one large prime factor (21 308 303 at  $k = 150$ , 3 448 649 at  $k = 100$ ) that we do not interpret; as in Section 3.7 that is what the sandpile group of a generic digraph looks like, and no topological reading is claimed. (b) The 870-vertex skeleton at  $k = 31$  is out of exact reach; a modular Smith form would settle it, and nothing in principle prevents that. (c) Whether the  $10^{344}$  growth of the arrangement summand between  $k = 150$  and  $k = 31$  has a law in terms of the repeat-length distribution.

## Provenance

Lemma 15.1 is elementary and is stated because it is what makes the computation possible, not because it is deep; everything else in this chapter is computation on public data. The genome is NC\_000913.3 and the annotation is the NCBI feature table for the same accession, both downloaded by the script and cached with it. The expectation of a multiplicity-seven rRNA family, which Section 15.4 corrects, was the premise of the specification of this chapter, and is recorded here because the correction is the useful part.



## Chapter 16

# The BEST determinant along the snarl tree: Schur complements, boundary forests, and linear-time counting

**Summary.** The number of sequences that spell a given single-stranded genome graph is a determinant, by the BEST theorem:  $\text{ec}(G) = t_r(G) \prod_v (d_v^+ - 1)!$ , where  $t_r(G) = \det \tilde{L}$  is a reduced Laplacian. Genome and variation graphs are locally tangled but globally chain-like, organized by their snarl tree. A proposal for this chapter was to count Eulerian circuits by contracting a “snarl tensor network” along that tree. What survives is simpler and exact. Eliminating the interior  $I$  of a snarl is taking a Schur complement:  $\det \tilde{L} = \det \tilde{L}_{II} \cdot \det(\tilde{L}/\tilde{L}_{II})$ . The Schur complement is again a reduced Laplacian, of a weighted digraph on the snarl’s boundary. And  $\det \tilde{L}_{II}$  counts the spanning forests of the interior rooted into the boundary. The “boundary tensor” of a snarl is thus an  $|B| \times |B|$  Laplacian and nothing more, because the determinant is multiplicative. Eliminating snarls in the order of the tree keeps the front bounded and the work linear in the size of the graph. On synthetic genome graphs of up to 39 714 vertices the front stays at 8, and the time per vertex stays at 9–11  $\mu\text{s}$ . The snarl-order determinant agrees with dense Bareiss elimination, with forced-arc contraction, and with minimum-degree elimination, the last modulo two primes. On the de Bruijn graphs of  $\phi\text{X174}$  the front is 196 at  $k = 7$  and 2 at  $k = 12$ . None of this helps the double-stranded count, which is  $\#P$ -hard (Chapter 10). The proposal’s claims about polyploid phasing and specific human loci are not addressed.

### 16.1 The question

A single-stranded genome graph  $G$  is a balanced multidigraph. Its vertices are the  $k$ -mers and its arcs the  $(k + 1)$ -mers, with multiplicity. The circular sequences it spells are its Eulerian circuits, and the BEST theorem counts them [10] (Proposition 1.3):

$$\text{ec}(G) = t_r(G) \prod_v (d_v^+ - 1)!, \quad t_r(G) = \det \tilde{L},$$

where  $\tilde{L}$  is the out-Laplacian with the row and column of a root  $r$  deleted. For one strand, then, counting is already polynomial. The issue is size: a pangenome graph has millions of vertices, and a dense determinant is cubic.

Genome graphs are not dense. Variation graphs decompose into *snarls*, subgraphs separated from the rest by a pair of boundary nodes, nested in a snarl tree [66]. The proposal for this chapter was to turn each snarl into a “local transition tensor” and contract a tree tensor network, with bond dimensions truncated by singular value decomposition. We keep the geometry and replace the machinery. The right object is the Schur complement, it needs no truncation, and it is exact.

## 16.2 Schur complements of Laplacians

Split the non-root vertices into an interior  $I$  and the rest  $B$ , and write  $\tilde{L}$  in blocks.

**Proposition 16.1** (elimination). *If  $\tilde{L}_{II}$  is invertible, then*

$$\det \tilde{L} = \det \tilde{L}_{II} \cdot \det S, \quad S = \tilde{L} / \tilde{L}_{II} = \tilde{L}_{BB} - \tilde{L}_{BI} \tilde{L}_{II}^{-1} \tilde{L}_{IB}.$$

*$S$  is again a reduced Laplacian: its off-diagonal entries are  $\leq 0$  and its row sums are  $\geq 0$ , the deficit being the weight of arcs to the root.*

*Proof.* The determinant formula is the block factorization.  $\tilde{L}$  is a nonsingular M-matrix with non-negative row sums. Both properties pass to Schur complements, which is the Kron reduction of network theory [26]. Concretely, row  $b$  of  $S$  is row  $b$  of  $\tilde{L}$  with each arc into  $I$  replaced by the distribution of its first exit from  $I$ . That distribution is non-negative and has total mass at most 1.  $\square$

**Proposition 16.2** (boundary forests).  *$\det \tilde{L}_{II}$  is the number of spanning forests of  $I$  in which every vertex of  $I$  chooses one out-arc and every tree is rooted outside  $I$ , i.e. in  $B \cup \{r\}$ .*

*Proof.* This is the all-minors matrix-tree theorem for the principal minor on  $I$  [20]. Choosing one out-arc per vertex of  $I$  gives a functional graph, and the choice is counted iff it has no cycle inside  $I$ .  $\square$

**Lemma 16.3** (pivots). *If  $G$  is strongly connected, every principal minor of  $\tilde{L}$  is positive. Hence Gaussian elimination of  $\tilde{L}$  in any order of vertices has positive pivots, and  $\det \tilde{L}$  is their product.*

*Proof.* By Proposition 16.2, the minor on a set  $J$  counts forests of  $J$  rooted outside  $J$ . Strong connectivity gives every vertex of  $J$  a path out of  $J$ , hence at least one such forest.  $\square$

So a snarl with boundary  $B$  contributes a scalar  $\det \tilde{L}_{II}$ , the number of its boundary-rooted forests, and hands its parent an  $|B| \times |B|$  Laplacian. That Laplacian is the snarl’s “boundary tensor”. It is a matrix, not a higher tensor, because the determinant is multiplicative. Its size is the boundary size, and there is nothing to truncate.

## 16.3 Elimination along the snarl tree

Order the vertices so that every snarl’s interior comes before its boundary, children before parents: a post-order of the snarl tree. Eliminating a vertex  $v$  costs one division and  $|\text{row}(v)| \cdot |\text{col}(v)|$  updates, where the row and column are those of the current matrix. The *front* at  $v$  is  $w_v = |\text{row}(v)| + |\text{col}(v)|$ .

**Theorem 16.4.** *Elimination in snarl-tree order computes  $t_r(G)$  exactly with  $O(\sum_v w_v^2)$  field operations. If every snarl has at most  $b$  boundary vertices and at most  $s$  interior vertices, then  $w_v = O(b + s)$ , and the cost is  $O(|V|(b + s)^2)$ , linear in  $|V|$ .*

| snarls | haplotypes | vertices | arcs | max front | $t_r(G)$                              |
|--------|------------|----------|------|-----------|---------------------------------------|
| 10     | 2          | 74       | 126  | 4         | 10 digits; three methods agree        |
| 20     | 3          | 162      | 360  | 6         | 37 digits; three methods agree        |
| 40     | 3          | 311      | 691  | 7         | 79 digits; three methods agree        |
| 60     | 4          | 516      | 1365 | 8         | 159 digits; snarl order = contraction |

Table 16.1: Exact arborescence counts on synthetic genome graphs. The three methods are snarl-order elimination, dense Bareiss (skipped above 320 vertices) and forced-arc contraction.

| snarls | vertices | arcs   | front (snarl) | front (min-deg.) | $\mu s$ / vertex | agree mod $p_1, p_2$ |
|--------|----------|--------|---------------|------------------|------------------|----------------------|
| 500    | 3 981    | 8 638  | 8             | 6                | 9.2              | yes                  |
| 1 500  | 11 949   | 26 197 | 8             | 6                | 9.3              | yes                  |
| 5 000  | 39 714   | 86 967 | 8             | 7                | 10.7             | yes                  |

Table 16.2: Linear time on synthetic genome graphs (3 haplotypes),  $p_1 = 2^{31} - 1$ ,  $p_2 = 2\,147\,483\,629$ .

*Proof.* When an interior vertex is eliminated, its row and column involve only vertices of its own snarl and that snarl’s boundary. Earlier snarls have been reduced to Schur complements on boundaries shared with it. Proposition 16.1 and Lemma 16.3 give correctness.  $\square$

**Arithmetic.** Over  $\mathbb{Q}$  the entries of the Schur complements are rationals whose size grows along the chain. For exact integers on large graphs we eliminate modulo several primes and recombine. A zero pivot modulo a prime has probability about  $|V|/p$ , and it is detected. Two independent orders, snarl order and minimum degree, give two independent checks.

## 16.4 Computations

All numbers are produced by `bio14_verify.py`, which writes `bio14_output.txt`. The battery runs in 44 seconds.

**The identities.** Proposition 16.1 is checked on 60 random Eulerian digraphs with random interiors, including the sign conditions on  $S$ . Proposition 16.2 is checked on 40 random interiors against exhaustive enumeration of out-arc choices. The BEST theorem is checked on 40 random digraphs with up to 12 arcs against exhaustive enumeration of Eulerian circuits.

**Synthetic genome graphs.** A chain of snarls between boundary vertices  $b_i$  is traversed by 2–4 haplotypes, as one closed walk. Each snarl has an interior pool of four vertices and a nested bubble with its own pool. The snarl order eliminates the nested interior, the nested boundary, the snarl interior, and then the entry vertex. Table 16.1 compares it with dense Bareiss elimination and with forced-arc contraction (Lemma 9.7, as implemented for Chapter 12). All three are exact and agree.

**Scaling.** Table 16.2 runs the snarl order and the minimum-degree order modulo two primes on graphs of up to 39 714 vertices. They agree modulo both primes. The front stays at 8 in snarl order and at 6–7 in minimum-degree order, and the time per vertex is essentially constant.

| $k$ | vertices | arcs  | max front | $\log_{10} t_r(G)$ |
|-----|----------|-------|-----------|--------------------|
| 7   | 4 115    | 5 386 | 196       | 336.5              |
| 8   | 4 928    | 5 386 | 80        | 128.2              |
| 10  | 5 339    | 5 386 | 12        | 12.6               |
| 12  | 5 384    | 5 386 | 2         | 0.3                |

Table 16.3: Exact arborescence counts of the single-stranded de Bruijn graphs of  $\phi$ X174, by minimum-degree Schur elimination and by forced-arc contraction.

**$\phi$ X174.** The de Bruijn graph of the circular genome at word length  $k$  has one vertex per distinct  $k$ -mer and 5 386 arcs. It has no snarl tree given in advance, so we use minimum-degree order. Table 16.3 compares it with forced-arc contraction; the two agree exactly. The front measures how tangled the graph is. At  $k = 7$ , where most 7-mers recur by chance, it is 196, and the elimination takes 35 seconds in exact rationals. At  $k = 12$  it is 2, and  $t_r = 2$ .

## 16.5 What this does not do

- *It does not touch the double-stranded count.*  $\#DS$  is a sum of determinants over the points of a lattice in a box, and it is  $\#P$ -hard (Corollary 10.6). A Schur complement is a function of one matrix. It does not commute with a sum over lattice points, which is why the frontier eliminations of Chapters 8 and 9 carry combinatorial or fermionic states instead of boundary matrices.
- *It does not change the complexity class of the single-stranded count*, which BEST already places in polynomial time. It turns a cubic dense computation into a linear sparse one when the snarl width is bounded. The  $\phi$ X174 graphs at small  $k$  show that the width need not be small.
- *No truncation.* Singular-value truncation of bond dimensions, as proposed, would make the count approximate. That defeats the purpose of an exact count, and it is unnecessary: the exact bond dimension is the boundary size.
- *Not addressed:* phasing constraints for diploid and polyploid genomes, and measurements on the human MHC or on centromeric repeats, as proposed. We have no graphs for those loci, and phasing is a different problem from counting.

## Ancillary file

The script `bio14_verify.py` runs checks (M), (F), (B), (E), (P) and (S) and writes `bio14_output.txt`. It needs only the standard library and imports two earlier scripts: `ds_fermion.py` of Chapter 9 and `spectral_fibres.py` of Chapter 12.

## Chapter 17

# The isoform decomposition polytope of a splicing graph, and a BEST identity for its integer points

**Summary.** Junction and exon read counts of a gene are an integer flow  $w$  on its splicing graph, a directed acyclic graph from the transcription start to the polyadenylation site, and the transcript isoforms are  $s$ - $t$  paths. The abundances compatible with  $w$  form the fibre polytope  $\Theta(w) = \{\theta \geq 0 : \sum_j \theta_j \mathbf{1}_{P_j} = w\}$ . We prove that its dimension is the number of  $s$ - $t$  paths in the support of  $w$  minus  $|E_w| - |V_w| + 2$ , because  $w$  lies in the relative interior of the cone spanned by those paths and the path-arc matrix has the rank of the flow space; for alternative events in series with  $k_1, \dots, k_r$  branches this is  $\prod k_i - \sum (k_i - 1) - 1$ , so deconvolution from junction counts is unique for one event and fails already for two cassette exons, and for Dscam ( $12 \times 48 \times 33 \times 2$ ) the 38 016 isoforms sit in a 37 924-dimensional polytope. Every minimum-support decomposition, with real or integer weights, is a vertex of  $\Theta(w)$ ; but the vertices of  $\Theta(w)$  are not integral in general — three parallel-arc pairs in series and  $w \equiv 1$  have a vertex with four weights  $\frac{1}{2}$  beside an integer minimum of two paths — so the extreme rays of the flow cone, the vertices of the fibre, and integrality are three different things. The new algebraic content is an identity. Adding  $F$  return arcs  $t \rightarrow s$  to the multigraph  $G_w$ ,  $F$  the flow value, the per-sequence BEST count of the augmented graph equals  $\sum_{\theta} (F - 1)! / \prod_j \theta_j!$  over the integer points of  $\Theta(w)$ : a determinant computes the multinomial-weighted partition function of the integer decompositions, of which the plain number of integer points has no such expression. The identity and the vertex statements are verified exactly on 238 flows, and no instance with a real minimum below the integer minimum was found among 232 random small flows.

### 17.1 What was asked, and what is true

This book reads assembly ambiguity off the critical group of a de Bruijn graph (Chapter 1), double-stranded ambiguity off the box points of an eigenlattice of the strand involution (Chapter 7), and polyploid phasing off the  $p$ -fold decompositions of a pooled flow (Chapter 13). The common object is the decomposition of a balanced integer flow into closed walks. Transcript quantification is the same object with the circle cut open: a splicing graph is a DAG with a source and a sink [36], read counts are a flow, and the isoforms are  $s$ - $t$  paths; the question “which isoforms, in what abundance” is the question of decomposing the flow into paths, and the ambiguity is the polytope of such decompositions.

The specification of this chapter asked for a proof that “minimum flow decompositions correspond to convex combinations of integer lattice points on the extreme rays of the flow polytope”. That sentence runs three distinct objects together, and it is worth separating them because two of the three are elementary and the third is false as stated. The extreme rays of the cone of  $s$ – $t$  flows of a DAG are the path indicators, and the vertices of the polytope of flows of fixed value are the paths scaled by the value: this is the total unimodularity of the incidence matrix and nothing else. The polytope that matters for a *given*  $w$  is the fibre  $\Theta(w)$  of the path–arc map over  $w$ , whose vertices are the decompositions with linearly independent support; every minimum-support decomposition is one of them (Proposition 17.5). But the path–arc matrix is not totally unimodular, and  $\Theta(w)$  has fractional vertices for integer  $w$  (Proposition 17.6); the integer decompositions are the lattice points of  $\Theta(w)$ , of which the vertices are in general not a subset. Minimum flow decomposition is NP-hard [38], and nothing here changes that.

What is true and, we believe, new is in Sections 17.2 and 17.4. The dimension of  $\Theta(w)$  is exactly the number of paths in the support of  $w$  minus the rank of the flow space of that support (Theorem 17.3); this is the precise form of the non-identifiability of isoform deconvolution from junction counts, a phenomenon known since [57, 39], and it gives the closed form  $\prod k_i - \sum (k_i - 1) - 1$  for events in series and the number 37 924 for Dscam. And the BEST theorem, applied to the flow closed up by  $F$  return arcs, evaluates the multinomial-weighted sum over the integer points of  $\Theta(w)$  as a determinant (Theorem 17.7): the same normalisation by  $\prod c(a)!$  that Theorem 7.6 needed for double strands (Remark 7.7) appears here as the passage from cyclic sequences of isoforms to abundance vectors. The script `isoform_polytope.py` verifies every statement and reproduces every number; it prints a seven-line pass/fail battery.

## 17.2 The splicing graph, its flows, and the fibre polytope

**Definition 17.1.** A *splicing graph* is a directed acyclic multigraph  $G = (V, E)$  with a source  $s$  and a sink  $t$  in which every arc lies on some  $s$ – $t$  path. An *isoform* is an  $s$ – $t$  path;  $\mathcal{P}$  is the set of isoforms and  $A \in \{0, 1\}^{E \times \mathcal{P}}$  the path–arc matrix. A *flow* is  $w \in \mathbb{Z}_{\geq 0}^E$  with  $\partial w = F(\delta_t - \delta_s)$ ,  $F$  its *value*;  $E_w = \text{supp } w$ ,  $V_w$  the vertices touched,  $\mathcal{P}(w)$  the isoforms inside  $E_w$ , and

$$\Theta(w) = \left\{ \theta \in \mathbb{R}_{\geq 0}^{\mathcal{P}(w)} : \sum_{P \in \mathcal{P}(w)} \theta_P \mathbf{1}_P = w \right\}.$$

$\Theta(w)$  is non-empty by flow decomposition (a non-negative flow on a DAG is a non-negative combination of  $s$ – $t$  paths, and an integer flow of integer paths), and bounded since  $\theta_P \leq F$ . It is the polytope of abundance vectors compatible with the counts.

**Lemma 17.2.**  $\text{rank } A = |E| - |V| + 2$ .

*Proof.* The columns of  $A$  lie in  $\mathcal{F} = \{x \in \mathbb{Q}^E : \partial x \in \mathbb{Q}(\delta_t - \delta_s)\}$ , which has dimension  $\dim \ker \partial + 1 = |E| - |V| + 2$  since  $G$  is connected. They span it: let  $x \in \mathcal{F}$  and let  $x_0 = \sum_P \mathbf{1}_P$ , which is positive on every arc because every arc lies on a path; for  $M$  large,  $x + Mx_0$  is a non-negative flow, hence a non-negative combination of paths, so  $x$  is in the span.  $\square$

**Theorem 17.3** (Dimension of the isoform polytope).  $\dim \Theta(w) = |\mathcal{P}(w)| - (|E_w| - |V_w| + 2)$ . In particular the abundances are determined by the counts if and only if the isoforms inside the support of  $w$  are linearly independent as arc vectors, i.e.  $|\mathcal{P}(w)| = |E_w| - |V_w| + 2$ .

*Proof.* The subgraph  $G_w = (V_w, E_w)$  is a splicing graph in its own right (every arc of positive flow lies on a path of positive flow, by flow decomposition), so Lemma 17.2 gives  $\text{rank } A_w = |E_w| - |V_w| + 2$  for its path-arc matrix. The cone  $C_w = \{x \in \mathcal{F}_w : x \geq 0\}$  is the cone generated by  $\mathcal{P}(w)$ , and its relative interior is  $\{x \in \mathcal{F}_w : x_a > 0 \ \forall a \in E_w\}$ , because every coordinate  $x_a$  is a non-trivial linear form on  $\mathcal{F}_w$  (some path uses  $a$ ). Since  $w > 0$  on  $E_w$ ,  $w \in \text{relint } C_w$ , and a point in the relative interior of a finitely generated cone is a strictly positive combination of all the generators. So  $\Theta(w)$  contains a point with full support, its affine hull is the whole affine space  $\{\theta : A_w \theta = w\}$ , and the dimension is  $|\mathcal{P}(w)| - \text{rank } A_w$ .  $\square$

**Corollary 17.4** (Events in series). *If  $G$  is a chain of  $r$  alternative events, the  $i$ -th offering  $k_i$  branches between consecutive junctions, then for a flow of full support*

$$\dim \Theta(w) = \prod_{i=1}^r k_i - \sum_{i=1}^r (k_i - 1) - 1,$$

*which is 0 if and only if at most one  $k_i$  exceeds 1. Two cassette exons in series give  $4 - 2 - 1 = 1$ ; three give 4; a cassette, two pairs of mutually exclusive exons, a second cassette and a four-way mutually exclusive block give  $64 - 8 = 56$ ; the four blocks of Dscam [76], with 12, 48, 33, 2 alternatives, give  $38016 - 92 = 37924$ .*

*Proof.* Paths multiply,  $|\mathcal{P}| = \prod k_i$ ; branches with  $\ell$  arcs have  $\ell - 1$  interior vertices, so  $|E| - |V| + 2 = \sum_i k_i - r - 1 + 2 = \sum_i (k_i - 1) + 1$  independently of the branch lengths. The last claim is Theorem 17.3.  $\square$

The identifiability statement in Theorem 17.3 is the rank condition of [39] specialised to junction-and-exon counts; the dimension formula makes it quantitative, and Corollary 17.4 says that for any gene with two independent alternative events the counts do not determine the abundances, whatever the depth of sequencing. Table 17.1 lists the models and their dimensions.

## 17.3 Vertices, minimum decompositions, and integrality

**Proposition 17.5.** *The vertices of  $\Theta(w)$  are the decompositions whose support is a linearly independent set of path vectors. Every decomposition of minimum support — among real decompositions, or among integer ones — is a vertex.*

*Proof.* The first sentence is the characterisation of basic feasible solutions of  $\{\theta \geq 0 : A\theta = w\}$ . For the second, let  $\theta$  have minimum support and suppose the columns of its support are dependent,  $Av = 0$  with  $v \neq 0$  supported in  $\text{supp } \theta$ . Then  $\theta + \tau v \in \Theta(w)$  for  $\tau$  in a closed interval containing 0 in its interior, and at an endpoint of that interval some coordinate of  $\text{supp } \theta$  vanishes, giving a decomposition of smaller support; if  $\theta$  and  $v$  are integral, the endpoints can be taken at integer  $\tau$  (the first coordinate to vanish does so at  $\tau = -\theta_j/v_j$ , and one may stop at the last integer before it, or note that  $v$  may be scaled to make the endpoint integral without changing the argument). Contradiction.  $\square$

**Proposition 17.6** (A fractional vertex). *Let  $G$  consist of four vertices  $s = v_0, v_1, v_2, v_3 = t$  with two parallel arcs  $a_i, a'_i$  from  $v_{i-1}$  to  $v_i$ , and  $w \equiv 1$  (value 2). Then  $\Theta(w)$  has 8 isoforms, dimension 4, four integer points, and six vertices of which two are fractional:  $\theta = \frac{1}{2}$  on the four paths  $(a_1, a'_2, a_3), (a_1, a_2, a'_3), (a'_1, a_2, a_3), (a'_1, a'_2, a'_3)$  is a vertex, while the minimum decompositions are the two integer points with two paths each.*

*Proof.* The four listed paths cover every arc exactly twice, so  $\theta$  is feasible; their arc vectors are independent, because a relation  $\alpha P_1 + \beta P_2 + \gamma P_3 + \delta P_4 = 0$  read on  $a_1, a_2, a_3$  gives  $\alpha + \beta = \beta + \gamma = \alpha + \gamma = 0$ , hence  $\alpha = \beta = \gamma = 0$  and then  $\delta = 0$ . So  $\theta$  is a vertex by Proposition 17.5, and it is not integral. The  $3 \times 3$  submatrix of the path–arc matrix on those three arcs and three paths has determinant 2, which is where the halves come from. The counts 8, 4, 4, 6, 2 are the transcript of the script.  $\square$

So the lattice points of  $\Theta(w)$  and its vertices are different sets, each containing the minimum decompositions. Whether the integer minimum can exceed the real minimum we do not know: the script compared the two on 232 random flows on small DAGs with 3 to 12 isoforms and found no instance where they differ, and one instance with a fractional vertex. Minimum flow decomposition being NP-hard [38], with the bound  $|E| - |V| + 2$  on the number of paths needed [85] and exact solvers in practice [45, 27], nothing in this section is intended as an algorithm.

## 17.4 The BEST identity

Close the flow up. Let  $\hat{G}_w$  be the multigraph with  $w(a)$  labelled parallel copies of each arc  $a \in E_w$  and  $F$  labelled *return arcs*  $t \rightarrow s$ . It is balanced and connected, so it has Eulerian circuits, and the per-sequence count of Theorem 7.6 applies:

$$N_{\text{seq}}(\hat{G}_w) = \frac{t(\hat{G}_w) \text{Loc}(\hat{G}_w)}{F! \prod_a w(a)!} = \sum_T \frac{1}{d(T)},$$

the sum over the circular sequences  $T$  of  $\hat{G}_w$  — cyclic sequences of arc contents using each content with its multiplicity — weighted by the order  $d(T)$  of the rotation group of  $T$ ; here  $t$  is the number of spanning arborescences converging on a root and  $\text{Loc} = \prod_v (d^+(v) - 1)!$ .

**Theorem 17.7** (BEST identity for isoform decompositions).

$$N_{\text{seq}}(\hat{G}_w) = \sum_{\substack{\theta \in \Theta(w) \\ \theta \in \mathbb{Z}^{\mathcal{P}(w)}}} \frac{(F-1)!}{\prod_P \theta_P!}.$$

*Proof.* A circular sequence of  $\hat{G}_w$  uses the  $F$  return arcs, and between two consecutive returns it walks from  $s$  to  $t$  along arcs of the DAG; a walk in a DAG repeats no vertex, so it is an isoform. Thus  $T$  is a cyclic sequence  $(P_{j_1}, \dots, P_{j_F})$  of isoforms, its arc usage is  $\sum_i \mathbf{1}_{P_{j_i}} = w$ , and the multiplicity vector  $\theta_P = \#\{i : P_{j_i} = P\}$  is an integer point of  $\Theta(w)$  with  $\sum_P \theta_P = F$ . Conversely every integer point  $\theta$  and every cyclic arrangement of the multiset  $\{P^{\theta_P}\}$  gives such a  $T$ . For a fixed  $\theta$  the cyclic arrangements, weighted by  $1/d(T)$ , number  $\frac{1}{F} \cdot \frac{F!}{\prod_P \theta_P!}$ : the cyclic group of order  $F$  acts on the  $F!/\prod \theta_P!$  linear arrangements, and  $\sum_{\text{orbits}} 1/|\text{stab}| = |\text{set}|/|\text{group}|$ .  $\square$

The left side is a determinant times factorials; the right side is a sum over the lattice points of a polytope of the multinomial coefficient of each point, i.e. the number of linear orders of  $F$  transcripts with those abundances. Theorem 17.7 is thus a closed form for the “partition function” of the integer decompositions with the natural combinatorial weight, and it is the isoform analogue of the fact that BEST counts genome reconstructions: there the whole spectrum is one closed walk, here it is a cyclic word in the isoforms. The plain number of integer points  $|\Theta(w) \cap \mathbb{Z}^{\mathcal{P}}|$ , which is what a uniform prior over decompositions would need, is a lattice-point count with no determinantal formula, and Theorem 17.7 says precisely which weighting is determinantal.

| model                                              | $ V $ | $ E $ | isoforms | $ E  -  V  + 2$ | $\dim \Theta$ | unique? |
|----------------------------------------------------|-------|-------|----------|-----------------|---------------|---------|
| cassette exon (skipping)                           | 3     | 3     | 2        | 2               | 0             | yes     |
| mutually exclusive exon pair                       | 4     | 4     | 2        | 2               | 0             | yes     |
| alternative 5' splice site                         | 2     | 2     | 2        | 2               | 0             | yes     |
| two cassettes in series                            | 5     | 6     | 4        | 3               | 1             | no      |
| three cassettes in series                          | 7     | 9     | 8        | 4               | 4             | no      |
| tropomyosin-like (five events)                     | 16    | 22    | 64       | 8               | 56            | no      |
| three parallel-arc pairs                           | 4     | 6     | 8        | 4               | 4             | no      |
| <i>Dscam</i> ( $12 \times 48 \times 33 \times 2$ ) | 100   | 190   | 38 016   | 92              | 37 924        | no      |

Table 17.1: Ranks and dimensions. The tropomyosin-like model is a pair of mutually exclusive exons, a cassette, a second pair, a second cassette and a four-way mutually exclusive block in series. The rank of the path-arc matrix equals  $|E| - |V| + 2$  in every row (over  $\mathbb{Q}$  for the small models, modulo  $2^{31} - 1$  for the  $190 \times 38\,016$  Dscam matrix, 12 s). Checks (R), (D), (U), (S).

| flow                 | $F$ | paths | dim | int. pts | vert. | frac. | MFD | $t$                 | $N_{\text{seq}}$ |
|----------------------|-----|-------|-----|----------|-------|-------|-----|---------------------|------------------|
| two cassettes        | 6   | 4     | 1   | 3        | 2     | 0     | 3   | 216                 | 50               |
| three parallel pairs | 2   | 8     | 4   | 4        | 6     | 2     | 2   | 8                   | 4                |
| tropomyosin-like     | 10  | 24    | 18  | 714      | 358   | 90    | 4   | $3.6288 \cdot 10^9$ | 22 680 000       |
| three cassettes      | 7   | 4     | 1   | 2        | 2     | 0     | 3   | 57 624              | 35               |

Table 17.2: Flows: two cassettes with  $w = (3, 3, 3, 2, 4, 2)$ ; three parallel pairs with  $w \equiv 1$ ; the tropomyosin-like model with a flow generated from five random isoforms of weights 1 to 3; three cassettes with a random flow. “paths” is  $|\mathcal{P}(w)|$ ; “MFD” the minimum support, equal for real and integer weights in every row;  $t$  the arborescence count of  $\hat{G}_w$ ;  $N_{\text{seq}}$  its per-sequence count, which equals  $\sum_{\theta} (F - 1)! / \prod \theta_P!$  over the integer points in every row. In the tropomyosin-like row the flow was generated from five isoforms and the minimum decomposition uses four. Checks (V), (B).

*Remark 17.8.* On one strand,  $\prod w(a)!$  is a constant and Part I counts arc-rooted circuits without it (Remark 1.1). Here, as in the double-stranded and polyploid sums, the divisor varies with the point: for the tropomyosin-like flow of Table 17.2 the arborescence count is  $t = 3\,628\,800\,000$ , and only after multiplying by Loc and dividing by  $F! \prod_a w(a)!$  does one obtain  $N_{\text{seq}} = 22\,680\,000$ , the quantity that matches the lattice sum.

## 17.5 Verification and data

**The seven checks.** (R) rank  $A = |E| - |V| + 2$  on the seven small models; (D) the dimension formula and its series form; (U) uniqueness for one event, dimension one for two cassettes; (S) Dscam; (V) every minimum-support integer decomposition is a vertex, and the fractional vertex of Proposition 17.6 exists; (B) the BEST identity on the four flows of Table 17.2 and on 232 random flows on DAGs with 4 to 6 vertices; (I) real versus integer minimum on those 232 flows — never different — with one fractional vertex among them. All pass; the run takes about three and a half minutes, almost all of it in the random search.

## 17.6 Scope, provenance, and what is open

**Scope.** Lemma 17.2, Theorem 17.3, Corollary 17.4, Proposition 17.5 and Theorem 17.7 are proved for arbitrary splicing graphs; Proposition 17.6 is one graph. The model is the exact flow: no read-length effects, no paired reads, no noise, and no length normalisation of abundances; reads spanning several junctions are additional linear constraints that cut  $\Theta(w)$  by hyperplanes and lower its dimension, which is how phasing information enters [39]. Minimum flow decomposition is treated only to the extent of Section 17.3.

**Provenance.** Splicing graphs are [36]; the non-identifiability of isoform deconvolution is [57, 39]; the  $|E| - |V| + 2$  bound on the number of paths in a decomposition and the greedy algorithms are [85]; NP-hardness of minimum flow decomposition is [38]; practical exact solvers are [45, 27]; the Dscam isoform count is [76]; the BEST theorem is [10, 78] and its per-sequence normalisation is Theorem 7.6. The tropical geometry of statistical models, which the planning of this chapter once proposed to bring in, is [69] and is not used: nothing in the polytope  $\Theta(w)$  is tropical. We are not aware of prior work stating the dimension of the isoform fibre in closed form, or of the BEST identity of Theorem 17.7; a targeted search found neither.

**Open.** *First*, whether the integer minimum can exceed the real minimum for an integer flow on a DAG. *Second*, a formula or complexity classification for the number of integer points of  $\Theta(w)$ , as opposed to the multinomial-weighted sum of Theorem 17.7. *Third*, the effect of multi-junction reads on the dimension: each read spanning a set  $J$  of junctions with allele pattern  $\pi$  adds the constraint that the total abundance of isoforms with pattern  $\pi$  on  $J$  be the observed count, and the dimension after all such constraints is the true residual ambiguity of a sequencing design. *Fourth*, the analogue of Theorem 17.7 with the return arcs weighted, which would give a generating function in  $F$  rather than a single number.

# Epilogue: what is open

Each problem below is left open by a specific chapter, and each is stated in the terms of that chapter.

**1. Generation of the double-stranded fibre.** Are any two double-stranded reconstructions of a circular sequence joined by transpositions at repeated  $k$ -mers, inversions at inverted repeats of length  $k$ , and global reverse complementation? This is true within one lattice point (Ukkonen, Pevzner), true on every case that could be closed, and true on the palindromic class of Chapter 10 by Kotzig's theorem (Chapter 11). The open part is the passage between lattice points: whether any two box points with connected support are joined by *realizable* inversions.

**2. Odd word length.** Chapter 10's reduction needs palindromic  $k$ -mers, which exist only for even  $k$ . For odd  $k$  the quotient of the double cover is a signed graph without fixed vertices (Chapter 6), and a reconstruction is an Eulerian circuit of a bidirected graph. Is the count still #P-hard?

**3. Approximation.** Does the double-stranded count admit a fully polynomial randomized approximation scheme? The same question for Eulerian circuits of undirected graphs is open, and Chapter 10 shows that an answer for genomes would answer it for 4-regular graphs.

**4. Parity of flip distances.** In Chapter 7, reconstructions with the same spectrum are joined by flip paths of even length only, at  $k = 11$  and  $k = 12$  for  $\phi X174$ . We have no explanation.

**5. Beyond two copies.** The sandpile group of a repeat skeleton with two-copy repeats is presented by an interlace matrix (Chapter 3). For three or more copies no pairwise presentation exists (Chapter 4). Is there a presentation by data of higher order, and what is it?

**What is not on this list.** The specifications of several chapters proposed further directions that the chapters themselves do not support: Massey products on graphs, whose second cohomology vanishes; a sheaf formalism for phasing, which leaves its complexity unchanged; and a gauge theory of chromatin, which has no observable that can be derived. Thermodynamic phase transitions and anyonic representations of sequences were also suggested, but no chapter defines the objects they would be about. We list them here only to say that, in the form proposed, they are not open problems of this book. A problem belongs on the list above only once it can be stated in the terms of a chapter.



# Guide to chapters and verification scripts

Every chapter was first written as a paper with its own verification script. The scripts, their reference transcripts and their input data live in the repository <https://github.com/Ruqing1963/algebraic-genomics>, archived under DOI 10.5281/zenodo.22943487, in the folder listed below. In the second column, “Bio  $n$ ” is the number of the paper in its series; the numbers 15, 16 and 18 were withdrawn before being written. `build_all.py` recompiles every paper and reruns every script: `--quick` in about half an hour, `--full` including the long computations. The last column gives the checks of the reference transcript.

| chapter                                       | paper  | folder papers/ and script                                  | checks |
|-----------------------------------------------|--------|------------------------------------------------------------|--------|
| 1 Critical groups of de Bruijn graphs         | Bio 1  | Ch01_Bio_01_Critical_Groups/dbg_sandpile.py                | 30     |
| 2 The sandpile group as a coordinate system   | Bio 2  | Ch02_Bio_02_Rotor_Routing/rotor_assembly.py                | 8      |
| 3 The sandpile group splits along the repeats | Bio 3  | Ch03_Bio_03_Repeat_Splitting/repeat_splitting.py           | 7      |
| 4 Multi-copy repeats                          | Bio 4  | Ch04_Bio_04_Multicopy_Repeats/multicopy.py                 | 4      |
| 5 The reverse-complement double cover         | Bio 6  | Ch05_Bio_06_RC_Double_Cover/rc_double_cover.py             | 6      |
| 6 The quotient is a signed graph              | Bio 7  | Ch06_Bio_07_Signed_Quotient/bidirected_sandpile.py         | 9      |
| 7 Double-stranded assembly as a lattice sum   | Bio 8  | Ch07_Bio_08_DS_Lattice_Sum/ds_lattice.py, ds_flips.py      | 9, 4   |
| 8 Frontier elimination                        | Bio 12 | Ch08_Bio_12_Frontier_Elimination/ds_transfer_matrix.py     | 8      |
| 9 A fermionic partition function              | Bio 13 | Ch09_Bio_13_Fermionic_Partition/ds_fermion.py, ds_count.py | 7, 4   |
| 10 Double-stranded reconstruction is #P-hard  | Bio 19 | Ch10_Bio_19_SharpP_Hardness/bio19_verify.py                | 5      |
| 11 Inversions at inverted repeats             | Bio 17 | Ch11_Bio_17_Inversions_Fibre/bio17_verify.py               | 6      |
| 12 What a local encoder cannot see            | joint  | Ch12_Spectral_Fibres/spectral_fibres.py                    | 5      |
| 13 Polyploid phasing                          | Bio 9  | Ch13_Bio_09_Polyploid_Phasing/polyploid_phasing.py         | 6      |

| chapter                                             | paper  | folder papers/ and script                                                | checks |
|-----------------------------------------------------|--------|--------------------------------------------------------------------------|--------|
| 14 The recombination group of a pan-genome          | Bio 10 | Ch14_Bio_10_Pangenome_Sheaf/<br>pangenome_sheaf.py,<br>real_pangenome.py | 5, 3   |
| 15 The bundle-chain decomposition of <i>E. coli</i> | Bio 5  | Ch15_Bio_05_Ecoli_Decomposition/<br>ecoli_sandpile.py                    | 6      |
| 16 The BEST determinant along the snarl tree        | Bio 14 | Ch16_Bio_14_Snarl_Schur/<br>bio14_verify.py                              | 6      |
| 17 The isoform decomposition polytope               | Bio 11 | Ch17_Bio_11_Isoform_Polytope/<br>isoform_polytope.py                     | 7      |

# Build record

The repository was built with `python build_all.py --quick` on 24 September 2026 (Python 3.12.4, Windows 11), after the chapter folders were renumbered. The build compiles every paper twice with `pdflatex` and reruns its verification scripts; a script passes when it exits normally and its final line reports all of its checks passed.

|                                                    |        |
|----------------------------------------------------|--------|
| items passed                                       | 37     |
| items failed                                       | 0      |
| items skipped (run only with <code>--full</code> ) | 4      |
| wall-clock time                                    | 24 min |

All 17 papers compile with no error and no undefined reference. Every script run in quick mode passes all of its checks. Two of them, Chapter 8 (`ds_transfer_matrix.py`) and Chapter 9 (`ds_count.py`), run a subset of their checks in quick mode (7 of 8, 2 of 4).

**Not rerun in this build.** The four items that run only with `--full` are the signed-quotient battery of Chapter 6, the eight-strain pan-genome of Chapter 14, and the proteome computation and its figures in Chapter 12. The long computations of Chapters 8 and 9 were also run only in their quick form. These are the connected count at  $k = 10$  and the enumeration of the 1 240 843 950 half-box points. For all of them the numbers in the text come from the reference transcripts in the repository, which were produced when the chapters were written. A `--full` build, of about four hours, reruns them.



# Bibliography

- [1] R. Arratia, B. Bollobás, D. Coppersmith and G. B. Sorkin, *Euler circuits and DNA sequencing by hybridization*, Discrete Appl. Math. **104** (2000), 63–96. doi:10.1016/s0166-218x(00)00190-6.
- [2] R. Arratia, B. Bollobás and G. B. Sorkin, *The interlace polynomial of a graph*, J. Combin. Theory Ser. B **92** (2004), 199–233. doi:10.1016/j.jctb.2004.03.003.
- [3] S. F. Altschul and B. W. Erickson, *Significance of nucleotide sequence alignments: a method for random sequence permutation that preserves dinucleotide and codon usage*, Mol. Biol. Evol. **2** (1985), 526–538. doi:10.1093/oxfordjournals.molbev.a040370.
- [4] D. Aguiar and S. Istrail, *HapCompass: a fast cycle basis algorithm for accurate haplotype assembly of sequence data*, J. Comput. Biol. **19** (2012), no. 6, 577–590. doi:10.1089/cmb.2012.0084.
- [5] D. Alar, M. Celaya, L. D. García Puente, S. Henson and A. Wheeler, *The sandpile group of a thick cycle graph*, Electron. J. Graph Theory Appl. **10** (2022), no. 2. doi:10.5614/ejgta.2022.10.2.20.
- [6] A. I. Barvinok, *A polynomial time algorithm for counting integral points in polyhedra when the dimension is fixed*, Math. Oper. Res. **19** (1994), 769–779. doi:10.1287/moor.19.4.769.
- [7] V. Bansal and V. Bafna, *HapCUT: an efficient and accurate algorithm for the haplotype assembly problem*, Bioinformatics **24** (2008), no. 16, i153–i159. doi:10.1093/bioinformatics/btn298.
- [8] G. Bresler, M. Bresler and D. Tse, *Optimal assembly for high throughput shotgun sequencing*, BMC Bioinformatics **14** (Suppl. 5) (2013), S18. doi:10.1186/1471-2105-14-s5-s18.
- [9] R. Chikhi, A. Limasset and P. Medvedev, *Compacting de Bruijn graphs from sequencing data quickly and in low memory*, Bioinformatics **32** (2016), i201–i208. doi:10.1093/bioinformatics/btw279.
- [10] T. van Aardenne-Ehrenfest, N. G. de Bruijn, *Circuits and trees in oriented linear graphs*, Simon Stevin **28** (1951) 203–217.
- [11] H. Bidkhori and S. Kishore, *A bijective proof of a theorem of Knuth*, Combin. Probab. Comput. **20** (2011), 11–25. doi:10.1017/s0963548310000192.
- [12] Y. Bilu and N. Linial, *Lifts, discrepancy and nearly optimal spectral gap*, Combinatorica **26** (2006), no. 5, 495–519. doi:10.1007/s00493-006-0029-7.

- [13] F. R. Blattner *et al.*, *The complete genome sequence of Escherichia coli K-12*, Science **277** (1997), 1453–1462. doi:10.1126/science.277.5331.1453.
- [14] H. L. Bodlaender, *A partial  $k$ -arboretum of graphs with bounded treewidth*, Theoret. Comput. Sci. **209** (1998), 1–45. doi:10.1016/s0304-3975(97)00228-4.
- [15] V. Bafna and P. A. Pevzner, *Genome rearrangements and sorting by reversals*, SIAM J. Comput. **25** (1996), no. 2, 272–289. doi:10.1137/s0097539793250627.
- [16] K. S. Brown, *Cohomology of Groups*, Graduate Texts in Mathematics 87, Springer (1982).
- [17] G. R. Brightwell and P. Winkler, *Counting Eulerian circuits is  $\#P$ -complete*, in: Proc. ALNEX/ANALCO 2005, SIAM, 2005, pp. 259–262.
- [18] P. Creed and M. Cryan, *The number of Euler tours of random directed graphs*, Electron. J. Combin. **20** (2013), P13. doi:10.37236/2377.
- [19] M. Cho, A. Dochtermann, R. Inagaki, S. Oh, D. Snustad and B. Zacovic, *Chip-firing and critical groups of signed graphs*, preprint, arXiv:2306.09315v2 (2024). doi:10.1137/24m1655287.
- [20] S. Chaiken, *A combinatorial proof of the all minors matrix tree theorem*, SIAM J. Algebraic Discrete Methods **3** (1982) 319–329. doi:10.1137/0603033.
- [21] W. Chen and T. Schedler, *Concrete and abstract structure of the sandpile group for thick trees with loops*, preprint, arXiv:math/0701381 (2007).
- [22] K.-C. Chou, *Prediction of protein cellular attributes using pseudo-amino acid composition*, Proteins **43** (2001) 246–255. doi:10.1002/prot.1072.
- [23] K.-C. Chou, *Using amphiphilic pseudo amino acid composition to predict enzyme subfamily classes*, Bioinformatics **21** (2005) 10–19. doi:10.1093/bioinformatics/bth466.
- [24] S. H. Chan, H. D. L. Hollmann and D. V. Pasechnik, *Sandpile groups of generalized de Bruijn and Kautz graphs and circulant matrices over finite fields*, J. Algebra **421** (2015), 268–295. doi:10.1016/j.jalgebra.2014.08.029.
- [25] C. J. Colbourn, W. J. Myrvold and E. Neufeld, *Two algorithms for unranking arborescences*, J. Algorithms **20** (1996), 268–281. doi:10.1006/jagm.1996.0014.
- [26] F. Dörfler, F. Bullo, *Kron reduction of graphs with applications to electrical networks*, IEEE Trans. Circuits Syst. I **60** (2013) 150–163. doi:10.1109/tcsi.2012.2215780.
- [27] F. H. C. Dias, L. Williams, B. Mumey and A. I. Tomescu, *Fast, flexible, and exact minimum flow decompositions via ILP*, in: Research in Computational Molecular Biology (RECOMB 2022), Lecture Notes in Computer Science **13278**, Springer, 2022, pp. 230–245. doi:10.1007/978-3-031-04749-7\_14.
- [28] J. M. Eizenga, A. M. Novak, J. A. Sibbesen, S. Heumos, A. Ghaffaari, G. Hickey, X. Chang, J. D. Seaman, R. Rounthwaite, J. Ebler, M. Rautiainen, S. Garg, B. Paten, T. Marschall, J. Sirén and E. Garrison, *Pangenome graphs*, Annu. Rev. Genomics Hum. Genet. **21** (2020), 139–162. doi:10.1146/annurev-genom-120219-080406.

- [29] Z. Lin et al., *Evolutionary-scale prediction of atomic-level protein structure with a language model*, Science **379** (2023) 1123–1130. doi:10.1126/science.ade2574.
- [30] H. Fleischner, *Eulerian Graphs and Related Topics*, Part 1, Vol. 1, Annals of Discrete Mathematics 45, North-Holland (1990).
- [31] M. Kolmogorov, J. Yuan, Y. Lin and P. A. Pevzner, *Assembly of long, error-prone reads using repeat graphs*, Nat. Biotechnol. **37** (2019), 540–546. doi:10.1038/s41587-019-0072-8.
- [32] S. Garg, *Computational methods for chromosome-scale haplotype reconstruction*, Genome Biology **22** (2021), 101. doi:10.20944/preprints202101.0116.v1.
- [33] J. Guzmán and C. Klivans, *Chip firing on general invertible matrices*, SIAM J. Discrete Math. **30** (2016), no. 2, 1115–1127. doi:10.1137/15m1034404.
- [34] Q. Ge, D. Štefankovič, *The complexity of counting Eulerian tours in 4-regular graphs*, Algorithmica **63** (2012) 588–601. doi:10.1007/s00453-010-9463-4.
- [35] E. Garrison, J. Sirén, A. M. Novak, G. Hickey, J. M. Eizenga, E. T. Dawson, W. Jones, S. Garg, C. Markello, M. F. Lin, B. Paten and R. Durbin, *Variation graph toolkit improves read mapping by representing genetic variation in the reference*, Nature Biotechnology **36** (2018), 875–879. doi:10.1038/nbt.4227.
- [36] S. Heber, M. Alekseyev, S.-H. Sze, H. Tang and P. A. Pevzner, *Splicing graphs and EST assembly problem*, Bioinformatics **18** (2002), suppl. 1, S181–S188. doi:10.1093/bioinformatics/18.suppl\_1.s181.
- [37] J. Hansen and R. Ghrist, *Toward a spectral theory of cellular sheaves*, J. Appl. Comput. Topol. **3** (2019), 315–358. doi:10.1007/s41468-019-00038-7.
- [38] T. Hartman, A. Hassidim, H. Kaplan, D. Raz and M. Segalov, *How to split a flow?*, in: Proc. IEEE INFOCOM 2012, pp. 828–836. doi:10.1109/infcom.2012.6195830.
- [39] D. Hiller, H. Jiang, W. Xu and W. H. Wong, *Identifiability of isoform deconvolution from junction arrays and RNA-Seq*, Bioinformatics **25** (2009), no. 23, 3056–3059. doi:10.1093/bioinformatics/btp544.
- [40] A. E. Holroyd, L. Levine, K. Mészáros, Y. Peres, J. Propp and D. B. Wilson, *Chip-firing and rotor-routing on directed graphs*, in: In and Out of Equilibrium 2, Progress in Probability **60**, Birkhäuser, 2008, pp. 331–364. doi:10.1007/978-3-7643-8786-0\_17.
- [41] S. Hannenhalli and P. A. Pevzner, *Transforming cabbage into turnip: polynomial algorithm for sorting signed permutations by reversals*, J. ACM **46** (1999), no. 1, 1–27. doi:10.1145/300515.300516.
- [42] M. Imbeault, P.-Y. Helleboid, D. Trono, *KRAB zinc-finger proteins contribute to the evolution of gene regulatory networks*, Nature **543** (2017) 550–554. doi:10.1038/nature21683.
- [43] J. Jonsson, *On the number of Euler trails in directed graphs*, Math. Scand. **90** (2002), 191–214. doi:10.7146/math.scand.a-14370.

- [44] J. Kawahara, T. Inoue, H. Iwashita and S. Minato, *Frontier-based search for enumerating all constrained subgraphs with compressed representation*, IEICE Trans. Fundamentals **E100-A** (2017), 1773–1784. doi:10.1587/transfun.e100.a.1773.
- [45] K. Kloster, P. Kuinke, M. P. O’Brien, F. Reidl, F. S. Villaamil, B. D. Sullivan and A. van der Poel, *A practical fpt algorithm for flow decomposition and transcript assembly*, in: Proc. Workshop on Algorithm Engineering and Experiments (ALENEX 2018), SIAM, 2018, pp. 75–86. doi:10.1137/1.9781611975055.7.
- [46] J. D. Kececioğlu and E. W. Myers, *Combinatorial algorithms for DNA sequence assembly*, Algorithmica **13** (1995), 7–51. doi:10.1007/bf01188580.
- [47] D. Kandel, Y. Matias, R. Unger and P. Winkler, *Shuffling biological sequences*, Discrete Appl. Math. **71** (1996), 171–185. doi:10.1016/s0166-218x(97)81456-4.
- [48] D. E. Knuth, *Oriented subtrees of an arc digraph*, J. Combin. Theory **3** (1967), 309–314. doi:10.1016/s0021-9800(67)80101-7.
- [49] D. E. Knuth, *The Art of Computer Programming, Volume 4A: Combinatorial Algorithms, Part 1*, Addison-Wesley, 2011, §7.1.4. doi:10.5860/choice.48-6329.
- [50] A. Kotzig, *Eulerian lines in finite 4-valent graphs and their transformations*, in Theory of Graphs (Proc. Colloq. Tihany, 1966), Academic Press (1968) 219–230.
- [51] C. Kingsford, M. C. Schatz and M. Pop, *Assembly complexity of prokaryotic genomes using short reads*, BMC Bioinformatics **11** (2010), 21. doi:10.1186/1471-2105-11-21.
- [52] M. Las Vergnas, *Le polynôme de Martin d’un graphe eulérien*, Ann. Discrete Math. **17** (1983), 397–411. doi:10.1016/s0304-0208(08)73415-7.
- [53] J. Lauri, *On a determinant formula for enumerating Euler trails in a class of digraphs*, Rend. Sem. Mat. Messina Ser. II **5** (1999), 75–90.
- [54] G. Lancia, V. Bafna, S. Istrail, R. Lippert and R. Schwartz, *SNPs problems, complexity, and algorithms*, in: Algorithms — ESA 2001, Lecture Notes in Computer Science **2161**, Springer, 2001, pp. 182–193. doi:10.1007/3-540-44676-1\_15.
- [55] L. Levine, *Sandpile groups and spanning trees of directed line graphs*, J. Combin. Theory Ser. A **118** (2011), 350–364. doi:10.1016/j.jcta.2010.04.001.
- [56] G. Loukides, S. P. Pissis et al., *Beyond the BEST theorem: fast assessment of Eulerian trails*, in: Fundamentals of Computation Theory (FCT 2021), Lecture Notes in Comput. Sci. **12867**, Springer, 2021, pp. 162–175. doi:10.1007/978-3-030-86593-1\_11.
- [57] V. Lacroix, M. Sammeth, R. Guigó and A. Bergeron, *Exact transcriptome reconstruction from short sequence reads*, in: Algorithms in Bioinformatics (WABI 2008), Lecture Notes in Computer Science **5251**, Springer, 2008, pp. 50–63. doi:10.1007/978-3-540-87361-7\_5.
- [58] N. Macris and J. V. Pulé, *An alternative formula for the number of Euler trails for a class of digraphs*, Discrete Math. **154** (1996), 301–305. doi:10.1016/0012-365x(94)00333-e.
- [59] P. Medvedev and M. Brudno, *Maximum likelihood genome assembly*, J. Comput. Biol. **16** (2009), 1101–1116. doi:10.1089/cmb.2009.0047.

- [60] P. Medvedev, K. Georgiou, G. Myers and M. Brudno, *Computability of models for sequence assembly*, in: Algorithms in Bioinformatics (WABI 2007), Lecture Notes in Computer Science, Springer, 2007, pp. 289–301. doi:10.1007/978-3-540-74126-8\_27.
- [61] C. Merino, I. Moffatt and S. Noble, *The critical group of a combinatorial map*, Combinatorial Theory **5** (2025), no. 3; arXiv:2308.13342. doi:10.5070/c65365550.
- [62] M. Mihail, P. Winkler, *On the number of Eulerian orientations of a graph*, Algorithmica **16** (1996) 402–414. doi:10.1007/s004539900057.
- [63] N. Nagarajan and M. Pop, *Parametric complexity of sequence assembly: theory and applications to next generation sequencing*, J. Comput. Biol. **16** (2009), 897–908. doi:10.1089/cmb.2009.0005.
- [64] M. O’Bleness et al., *Evolutionary history and genome organization of DUF1220 protein domains*, G3 **2** (2012) 977–986. doi:10.1534/g3.112.003061.
- [65] N. Obscura Acosta and A. I. Tomescu, *Simplicity in Eulerian circuits: uniqueness and safety*, Inf. Process. Lett. **183** (2024), 106421. doi:10.1016/j.ipl.2023.106421.
- [66] B. Paten, J. M. Eizenga, Y. M. Rosen, A. M. Novak, E. Garrison, G. Hickey, *Superbubbles, ultrabubbles, and cacti*, J. Comput. Biol. **25** (2018) 649–663. doi:10.1089/cmb.2017.0251.
- [67] V. B. Priezzhev, D. Dhar, A. Dhar and S. Krishnamurthy, *Eulerian walkers as a model of self-organized criticality*, Phys. Rev. Lett. **77** (1996), 5079–5082. doi:10.1103/physrevlett.77.5079.
- [68] P. A. Pevzner, *DNA physical mapping and alternating Eulerian cycles in colored graphs*, Algorithmica **13** (1995) 77–105. doi:10.1007/bf01188582.
- [69] L. Pachter and B. Sturmfels, *Tropical geometry of statistical models*, Proc. Natl. Acad. Sci. USA **101** (2004), no. 46, 16132–16137. doi:10.1073/pnas.0406010101.
- [70] D. Perkinson, N. Salter and T. Xu, *A note on the critical group of a line graph*, Electron. J. Combin. **18** (2011), P124. doi:10.37236/611.
- [71] P. A. Pevzner, H. Tang and G. Tesler, *De novo repeat classification and fragment assembly*, Genome Res. **14** (2004), 1786–1796. doi:10.1101/gr.2395204.
- [72] P. A. Pevzner, H. Tang and M. S. Waterman, *An Eulerian path approach to DNA fragment assembly*, Proc. Natl. Acad. Sci. USA **98** (2001), 9748–9753. doi:10.1073/pnas.171285098.
- [73] J. G. Propp and D. B. Wilson, *How to get a perfectly random sample from a generic Markov chain and generate a random spanning tree of a directed graph*, J. Algorithms **27** (1998), 170–217. doi:10.1006/jagm.1997.0917.
- [74] D. Rogers, M. Hahn, *Extended-connectivity fingerprints*, J. Chem. Inf. Model. **50** (2010) 742–754. doi:10.1021/ci100050t.
- [75] V. Reiner and D. Tseng, *Critical groups of covering, voltage and signed graphs*, Discrete Math. **318** (2014), 10–40; arXiv:1301.2977. doi:10.1016/j.disc.2013.11.008.

- [76] D. Schmucker, J. C. Clemens, H. Shu, C. A. Worby, J. Xiao, M. Muda, J. E. Dixon and S. L. Zipursky, *Drosophila Dscam is an axon guidance receptor exhibiting extraordinary molecular diversity*, *Cell* **101** (2000), no. 6, 671–684. doi:10.1016/s0092-8674(00)80878-8.
- [77] P. Stankiewicz, J. R. Lupski, *Genome architecture, rearrangements and genomic disorders*, *Trends Genet.* **18** (2002) 74–82. doi:10.1016/s0168-9525(02)02592-1.
- [78] C. A. B. Smith and W. T. Tutte, *On unicursal paths in a network of degree 4*, *Amer. Math. Monthly* **48** (1941), 233–237. doi:10.1080/00029890.1941.11991103.
- [79] L. Tóthmész, *On canonical sandpile actions of embedded graphs*, *Advances in Combinatorics* (2026), no. 6; arXiv:2504.05275. doi:10.19086/aic.2026.6.
- [80] W. T. Tutte, *The dissection of equilateral triangles into equilateral triangles*, *Proc. Cambridge Philos. Soc.* **44** (1948), 463–482. doi:10.1017/s030500410002449x.
- [81] E. Ukkonen, *Approximate string-matching with q-grams and maximal matches*, *Theoret. Comput. Sci.* **92** (1992) 191–211. doi:10.1016/0304-3975(92)90143-4.
- [82] Y. Li, M. Zeng, J. Zhu, L. Liu, F. Wang, L. Huang, F. Yang, M. Li, J. Yao, *UniPert-G2CP bridges genetic and chemical screens from molecular representation to phenotype modeling*, *Cell* **189** (2026) 4946–4963. doi:10.1016/j.cell.2026.06.005.
- [83] The UniProt Consortium, *UniProt: the Universal Protein Knowledgebase in 2023*, *Nucleic Acids Res.* **51** (2023) D523–D531. doi:10.1093/nar/gkac1052.
- [84] M. Jiang, J. Anderson, J. Gillespie and M. Mayne, *uShuffle: a useful tool for shuffling biological sequences while preserving the k-let counts*, *BMC Bioinformatics* **9** (2008), 192. doi:10.1186/1471-2105-9-192.
- [85] B. Vatinlen, F. Chauvet, P. Chrétienne and P. Mahey, *Simple bounds and greedy algorithms for decomposing a flow into a minimal set of paths*, *European J. Oper. Res.* **185** (2008), no. 3, 1390–1401. doi:10.1016/j.ejor.2006.05.043.
- [86] D. B. Wilson, *Generating random spanning trees more quickly than the cover time*, in: *Proc. 28th ACM Symp. Theory of Computing* (1996), pp. 296–303. doi:10.1145/237814.237880.
- [87] K. Xu, W. Hu, J. Leskovec, S. Jegelka, *How powerful are graph neural networks?*, *International Conference on Learning Representations* (2019).
- [88] T. Zaslavsky, *Signed graphs*, *Discrete Appl. Math.* **4** (1982), no. 1, 47–74; erratum **5** (1983), 248. doi:10.1016/0166-218x(83)90047-1.
- [89] T. Zaslavsky, *Glossary of signed and gain graphs and allied areas*, *Electron. J. Combin., Dynamic Survey #DS9* (1998).